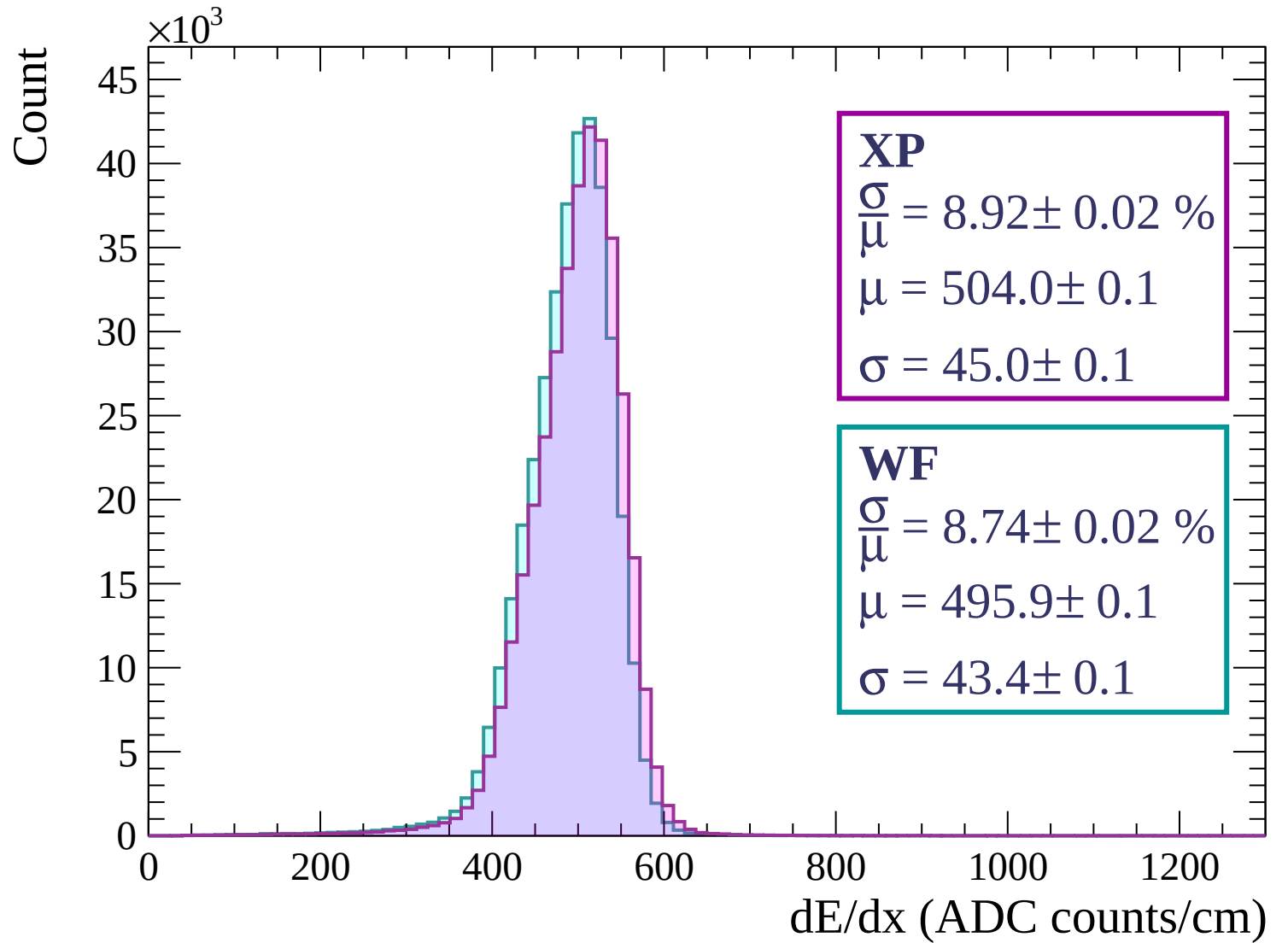
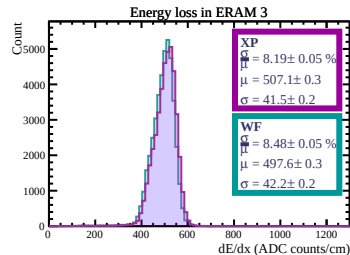
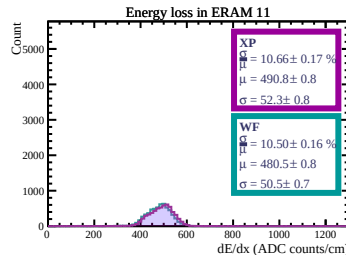
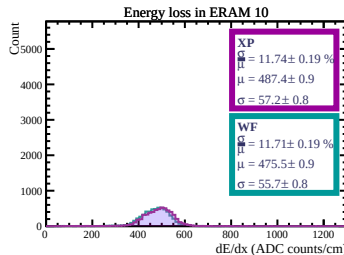
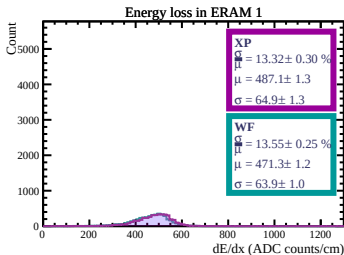
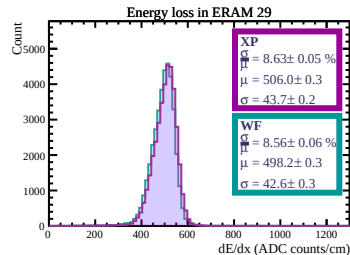
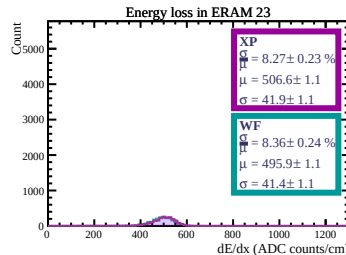
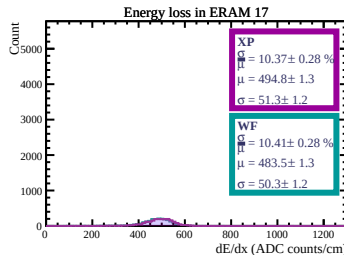
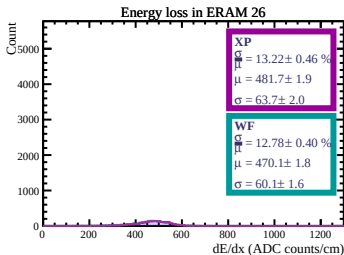
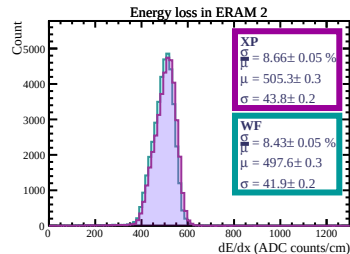
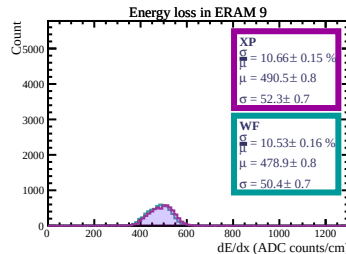
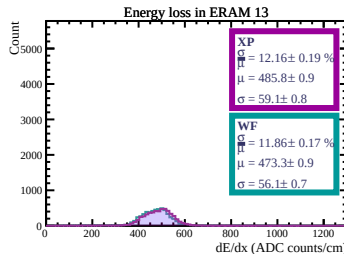
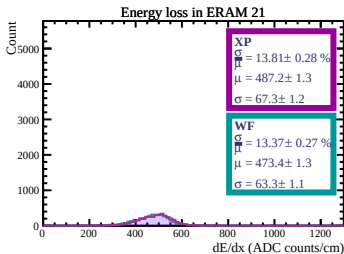
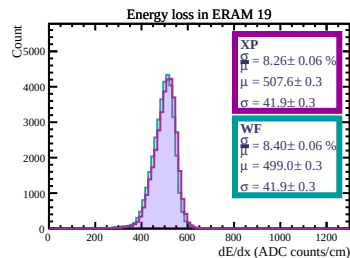
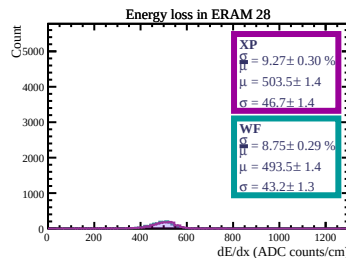
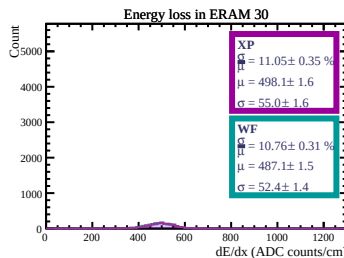
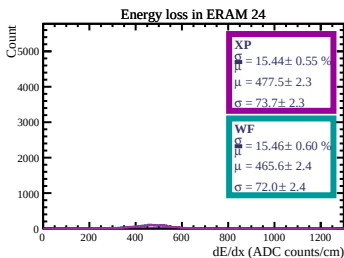
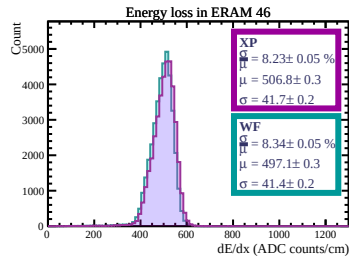
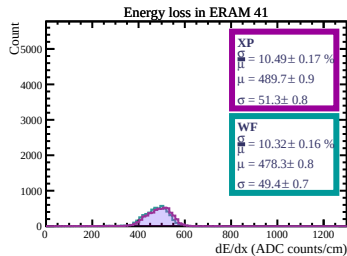
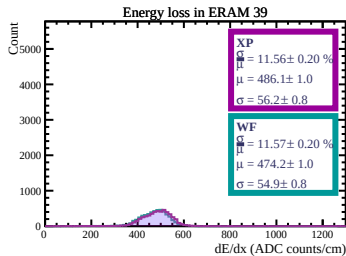
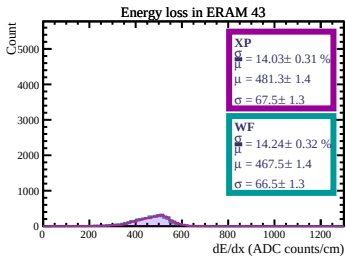
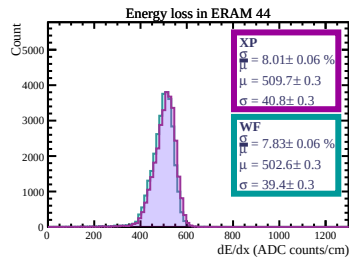
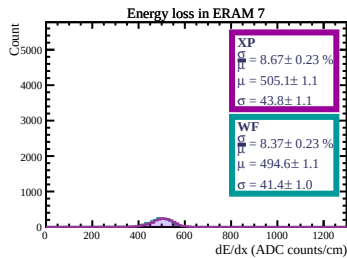
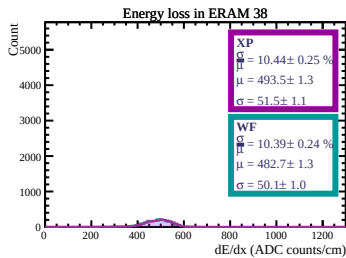
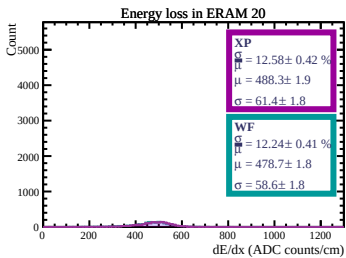
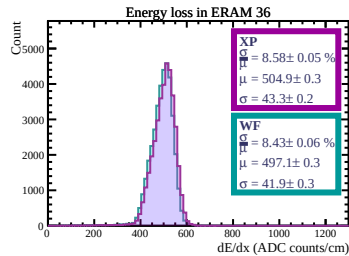
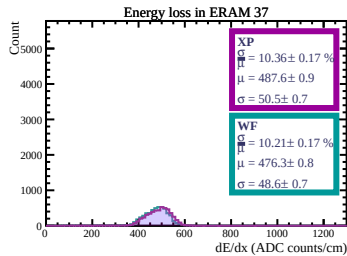
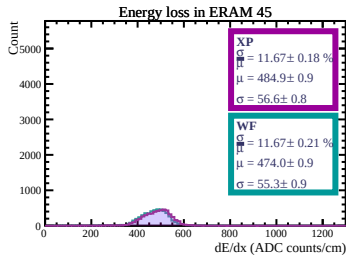
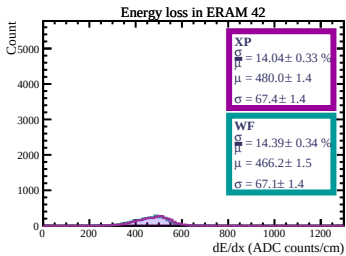
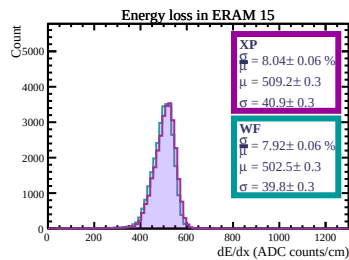
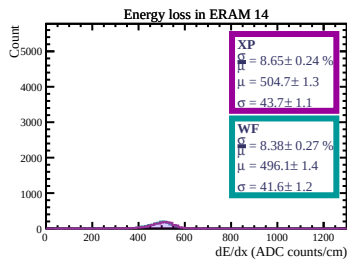
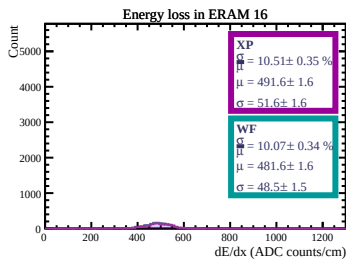
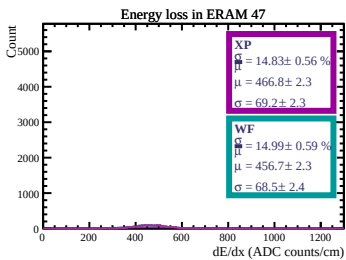
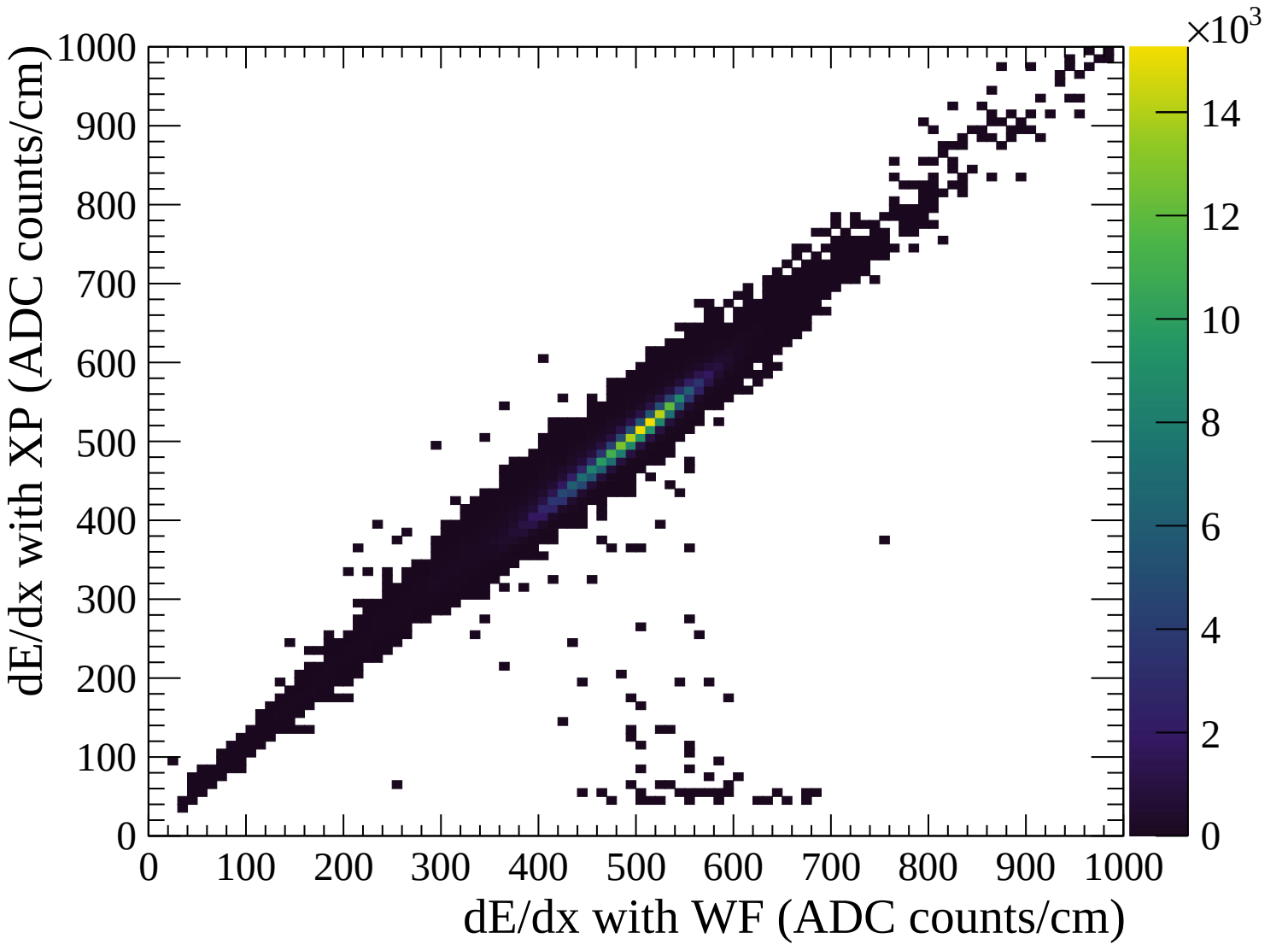
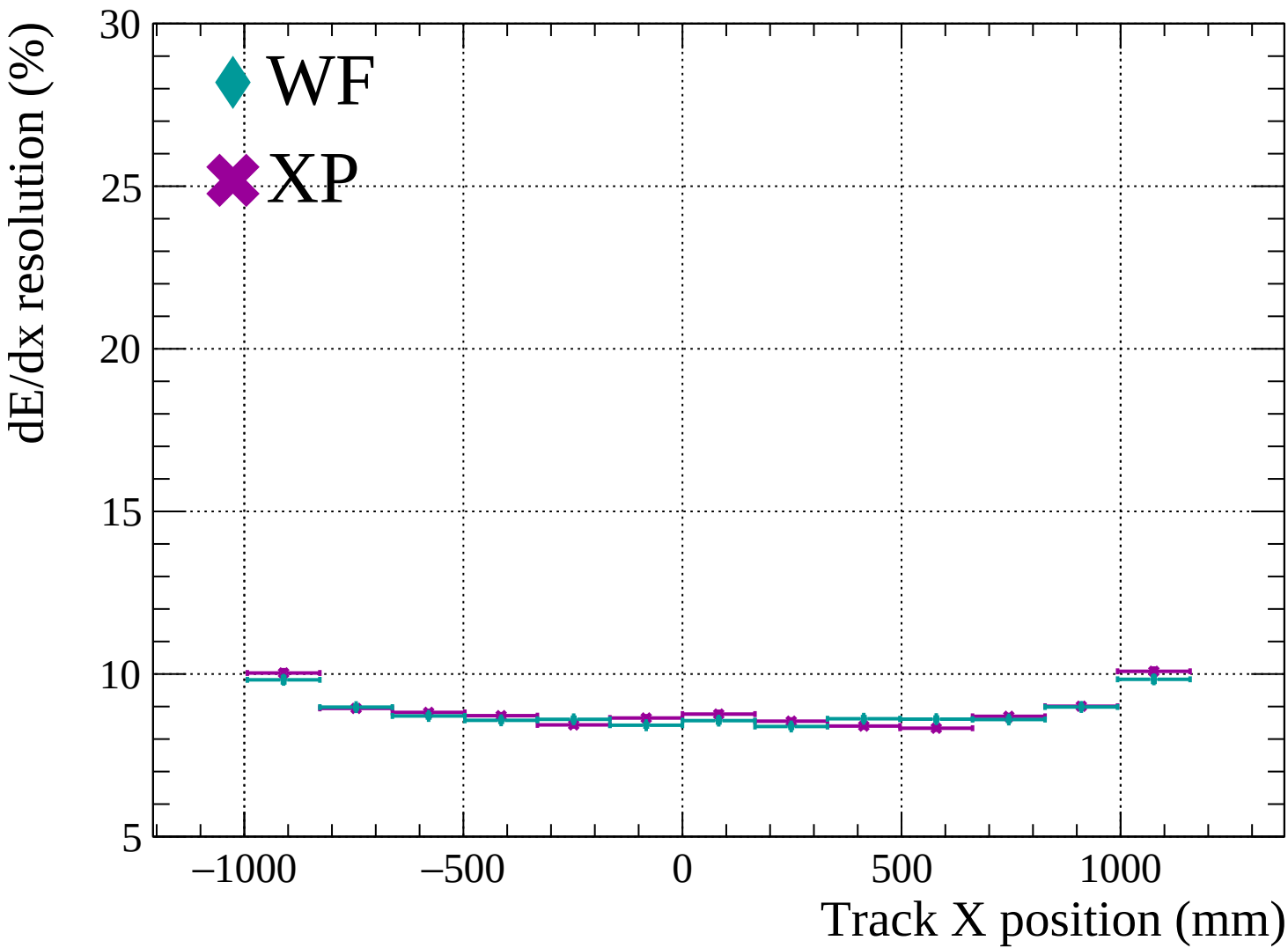


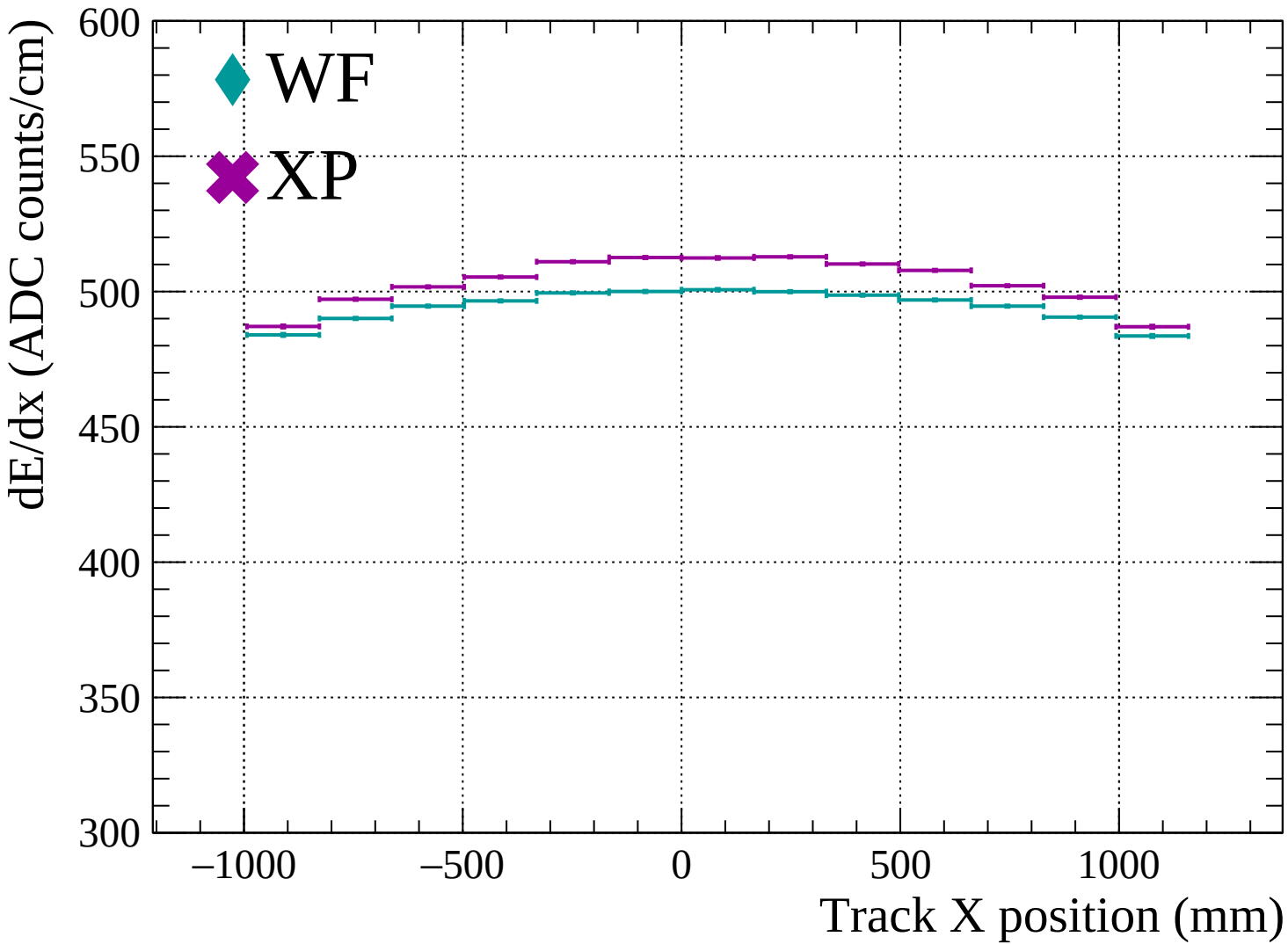


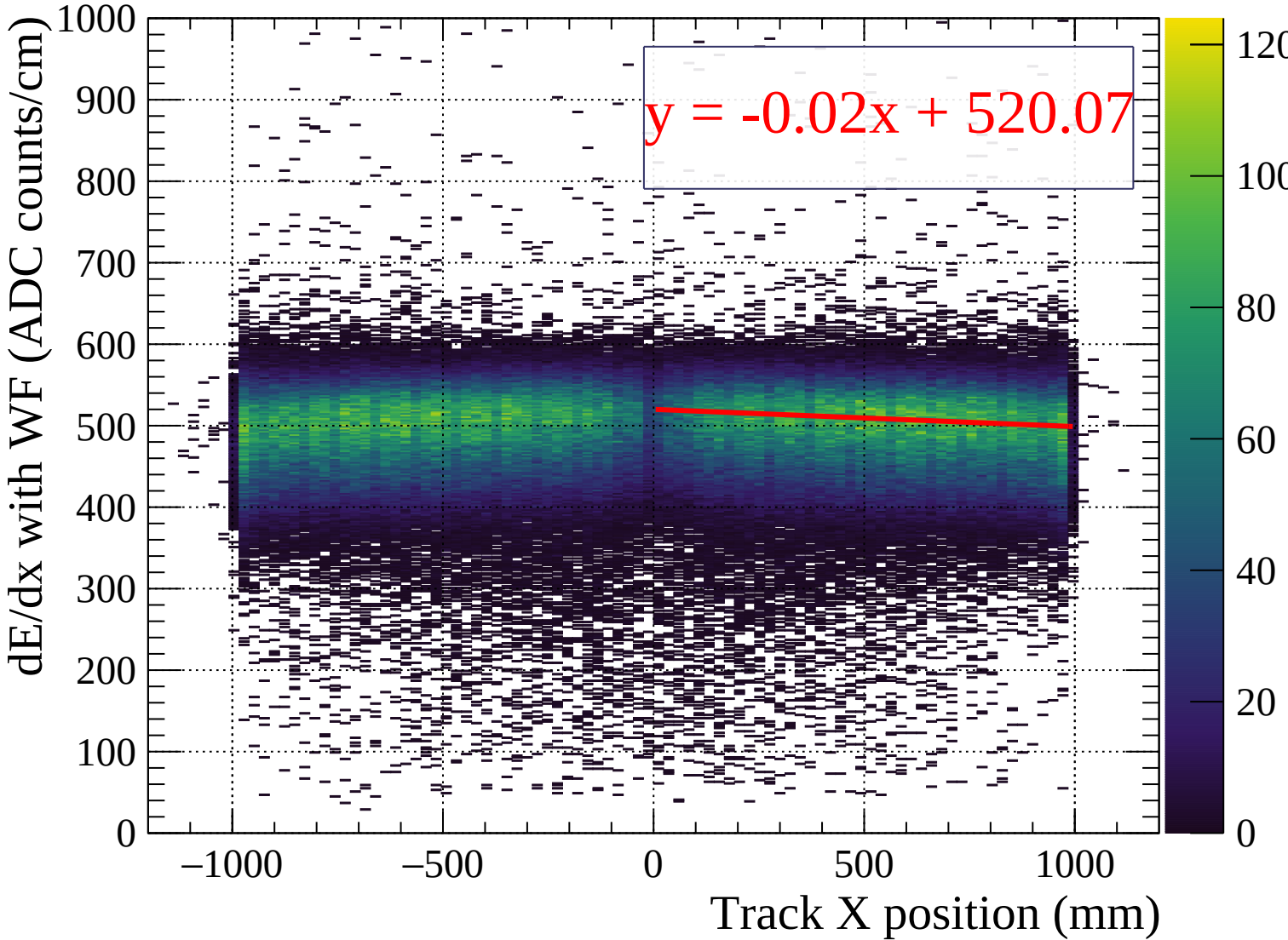


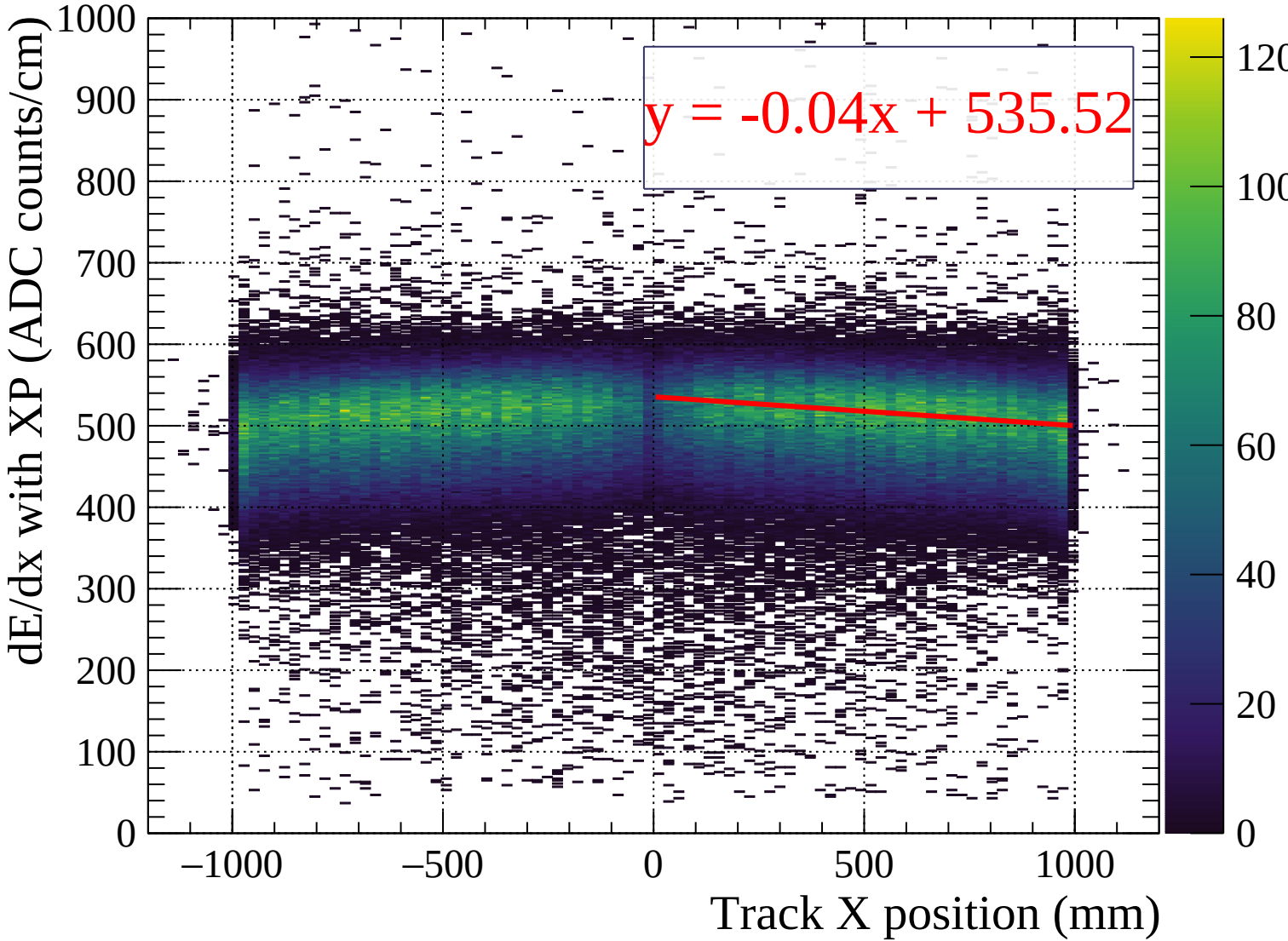


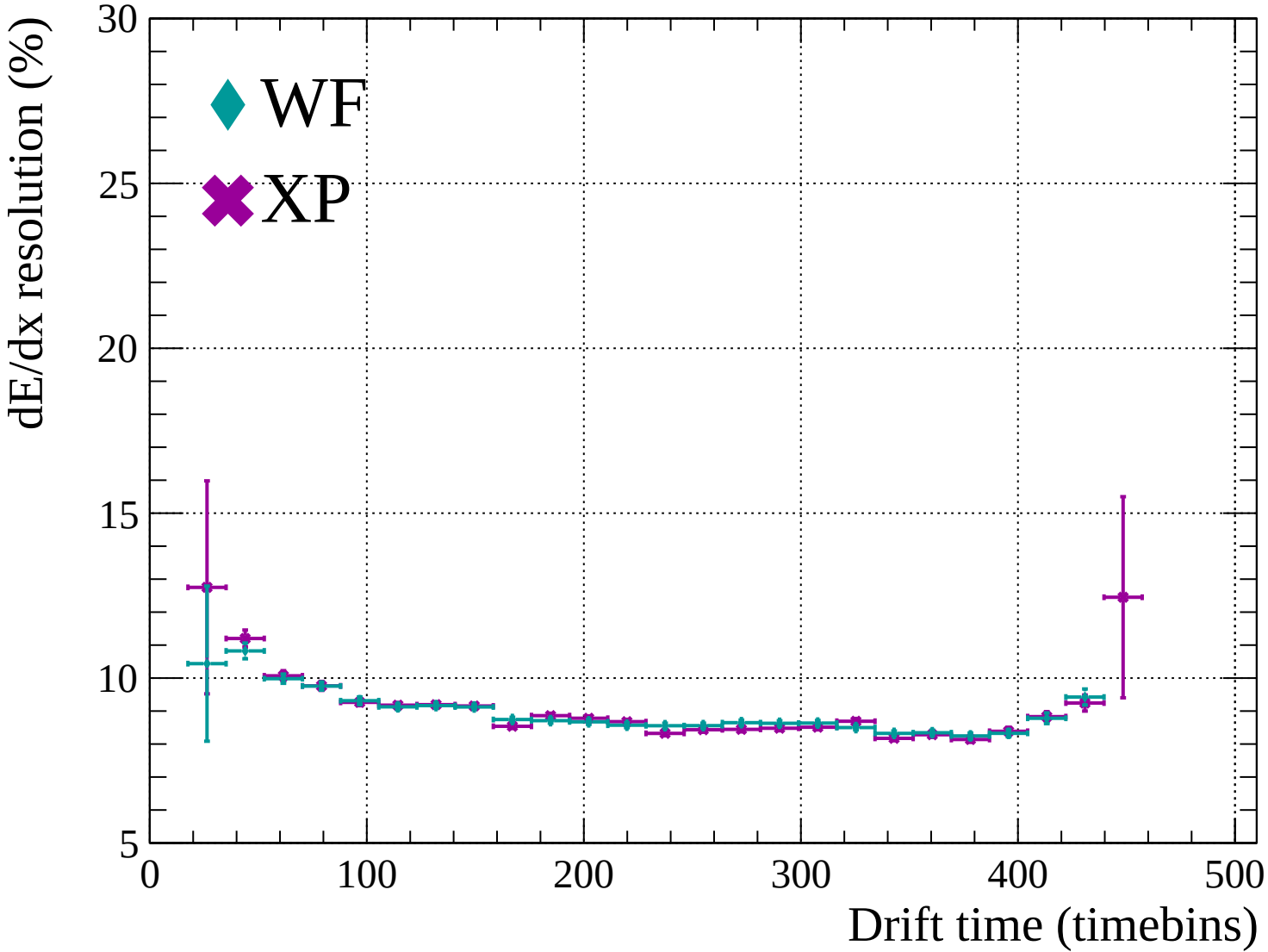


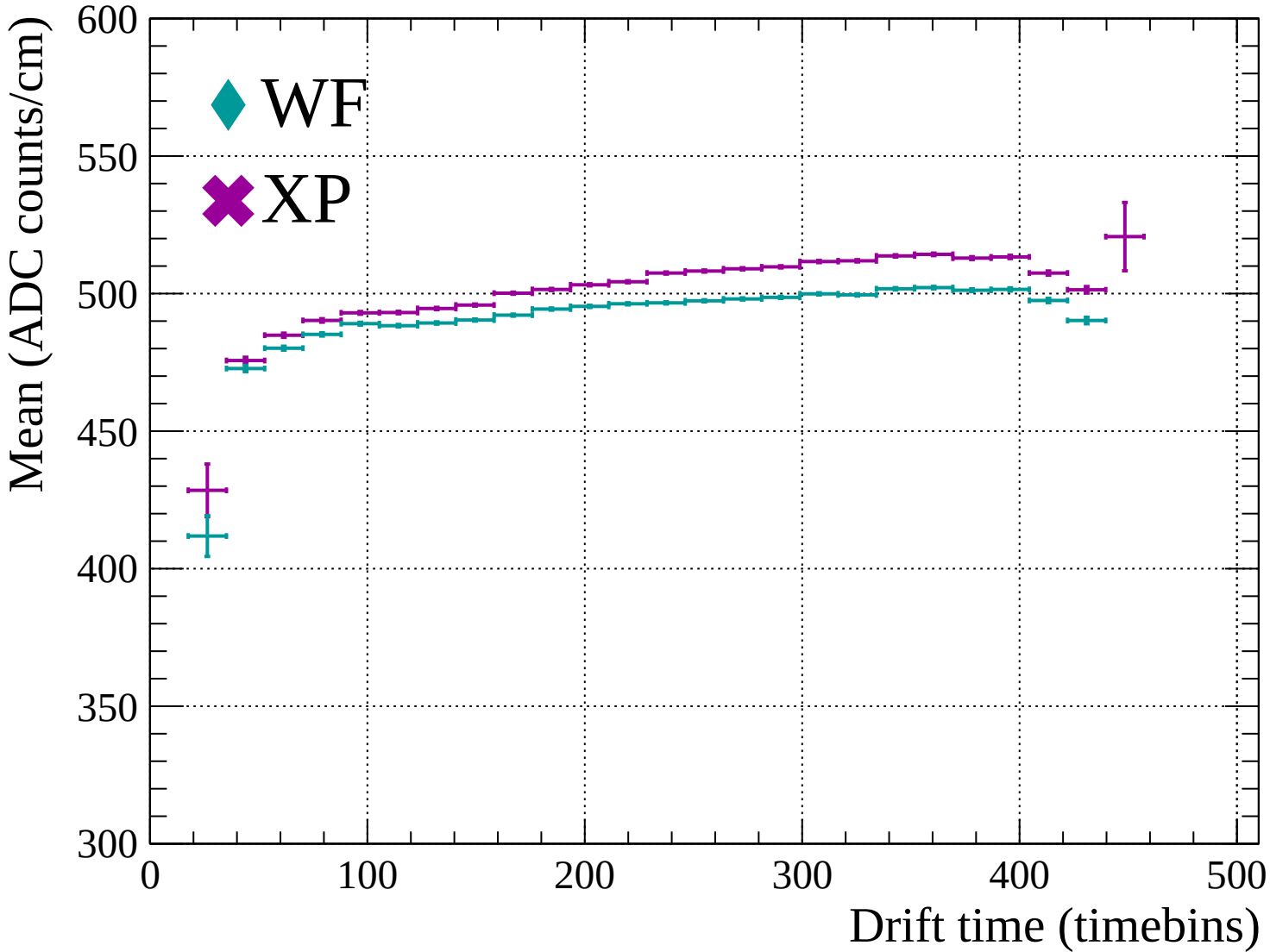


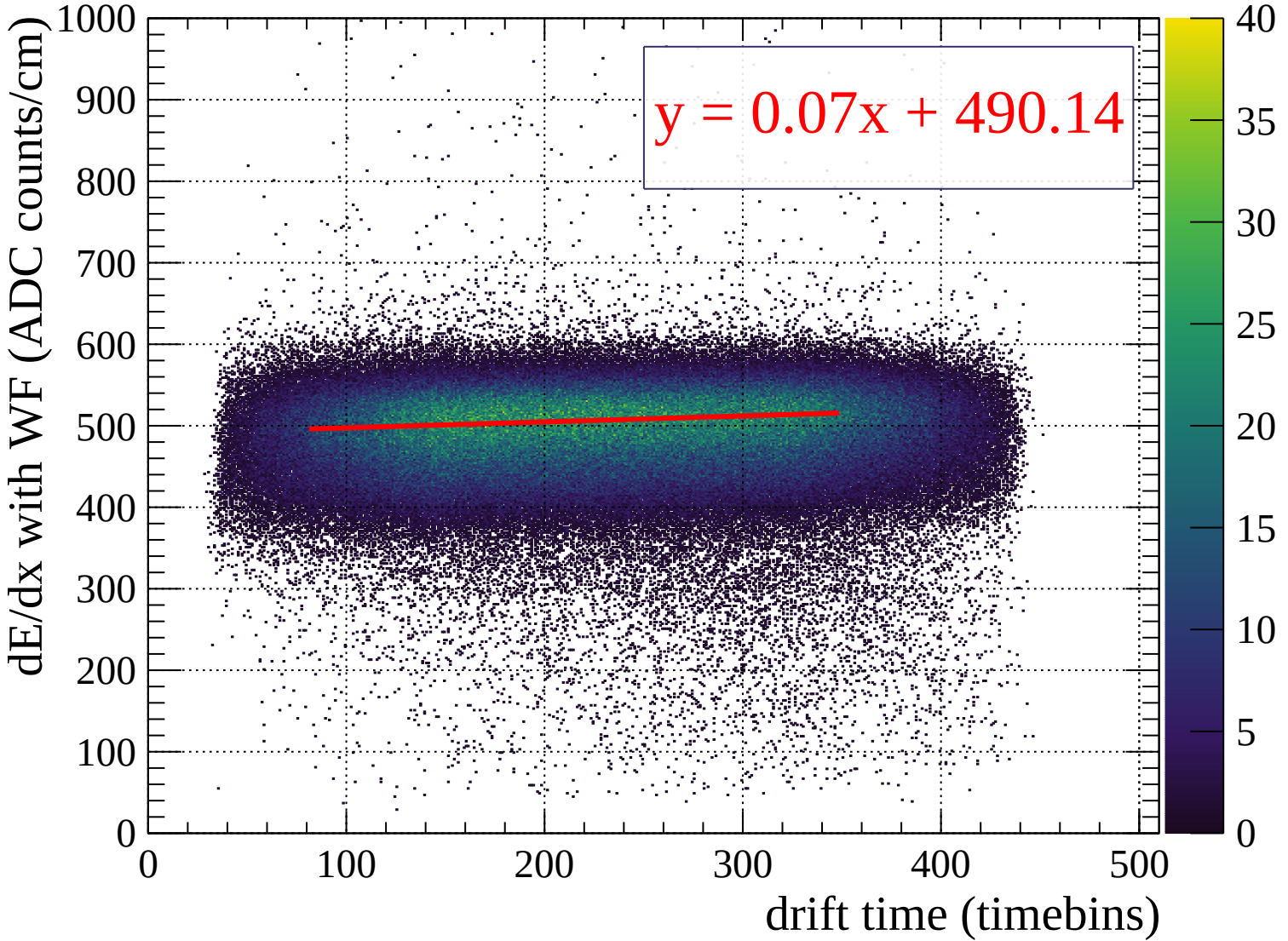


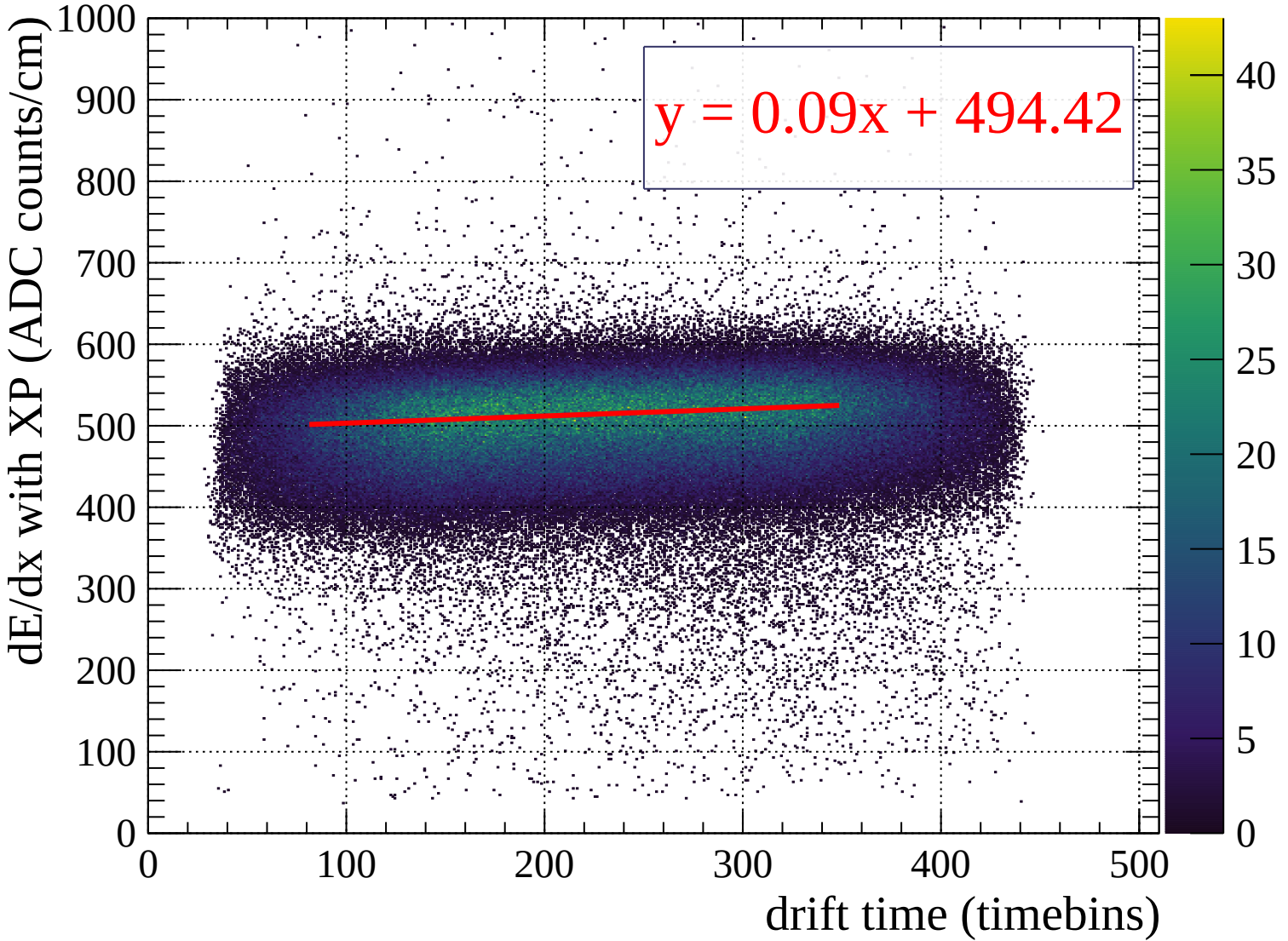


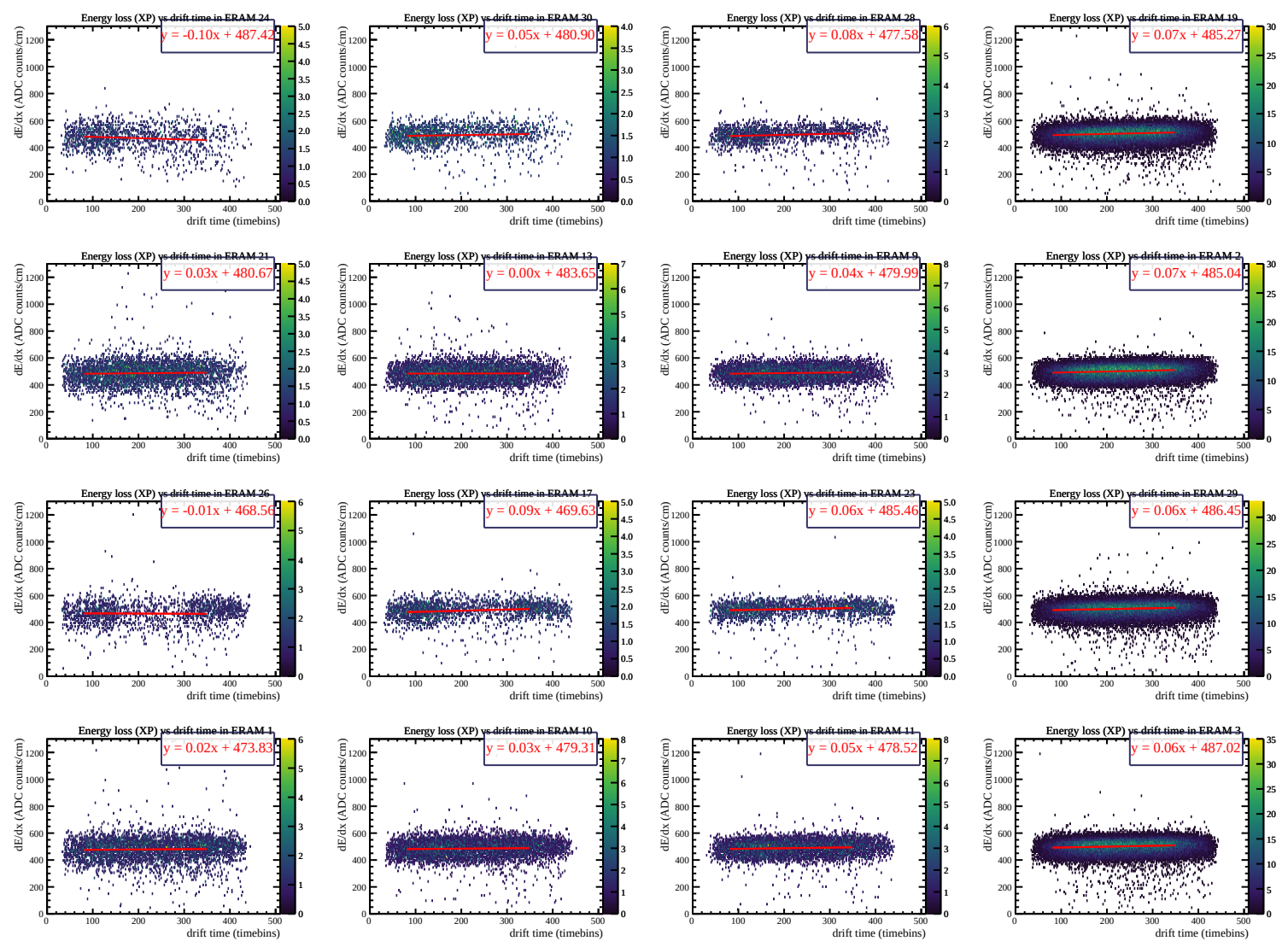


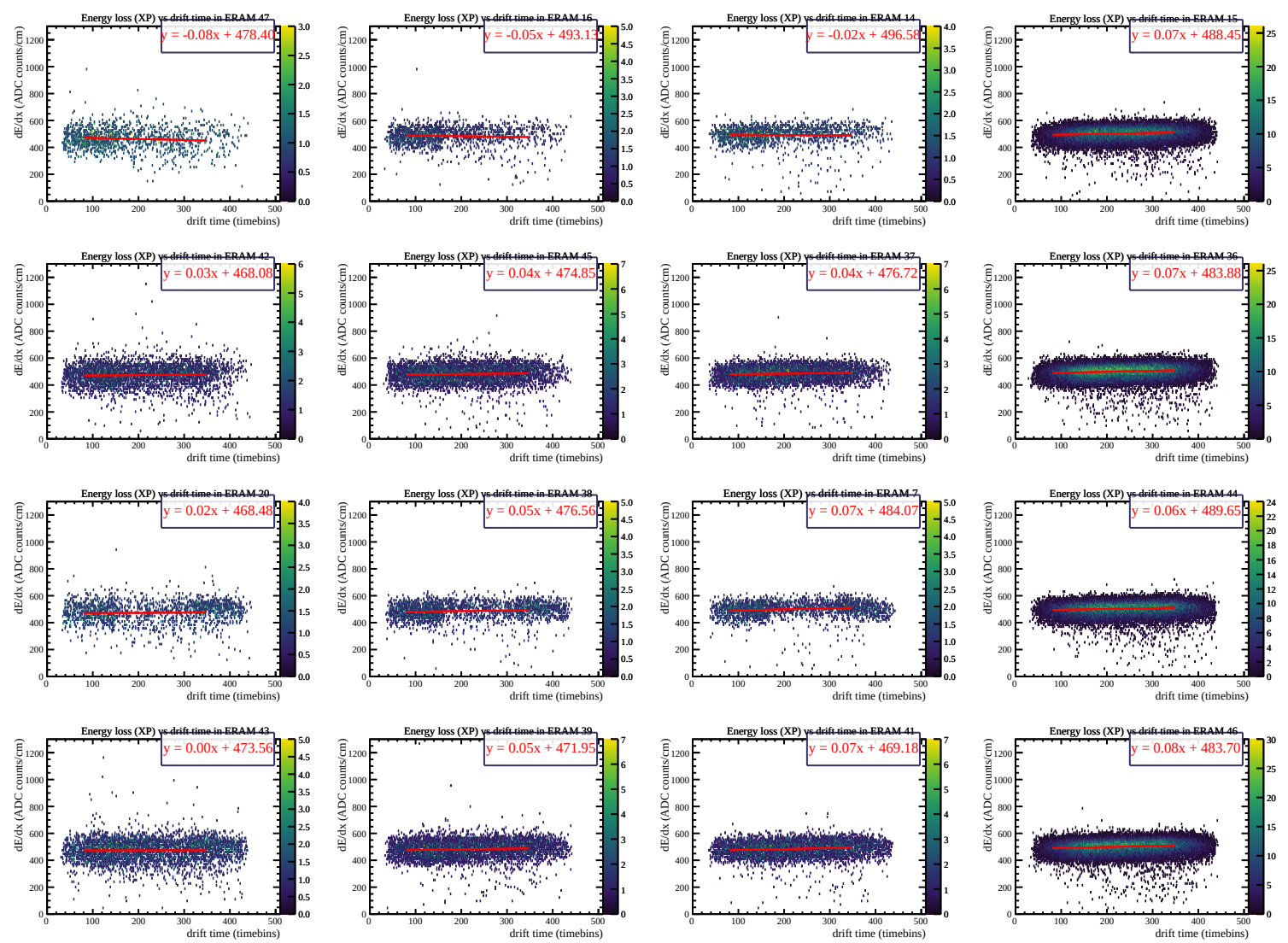


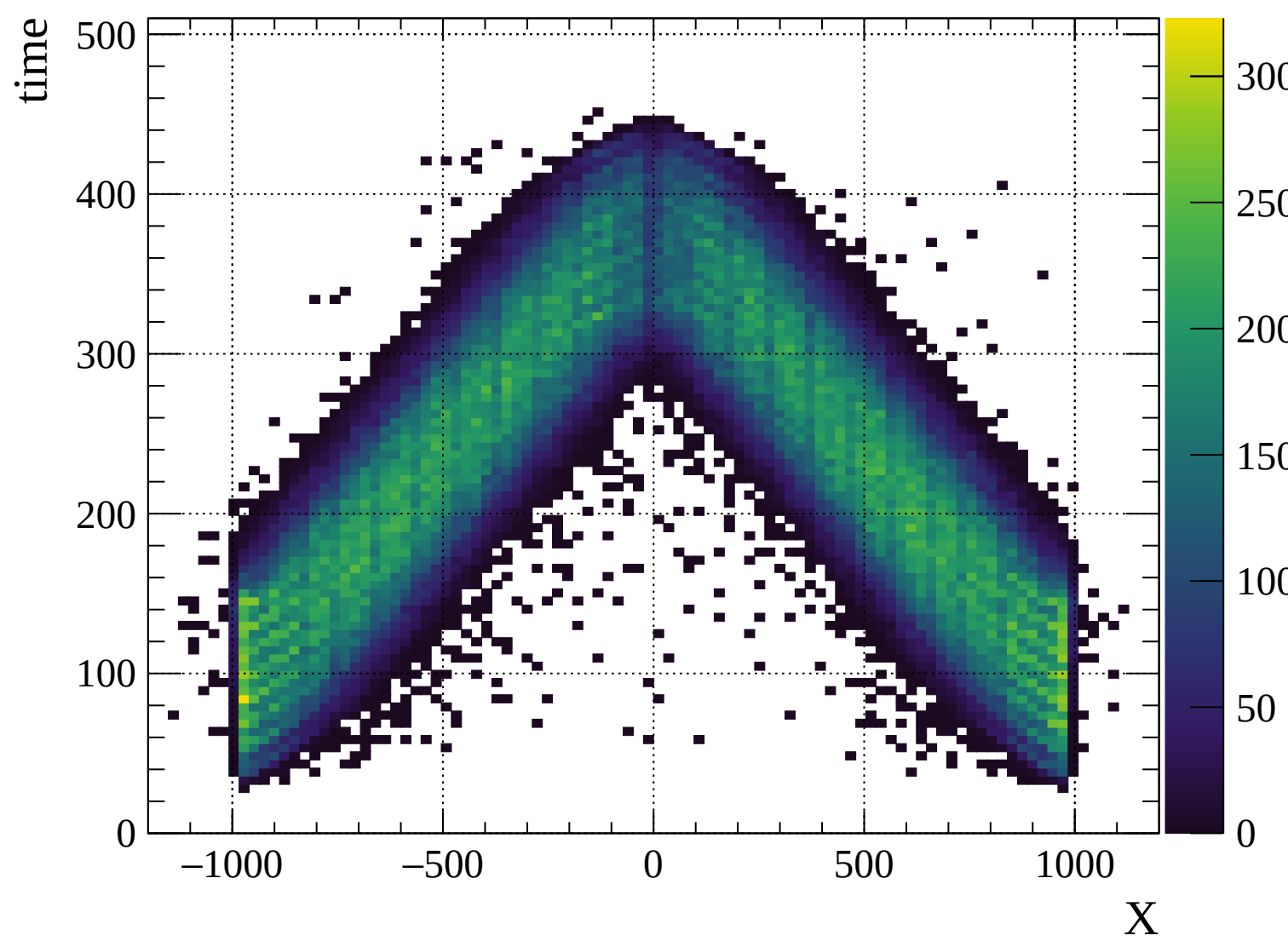


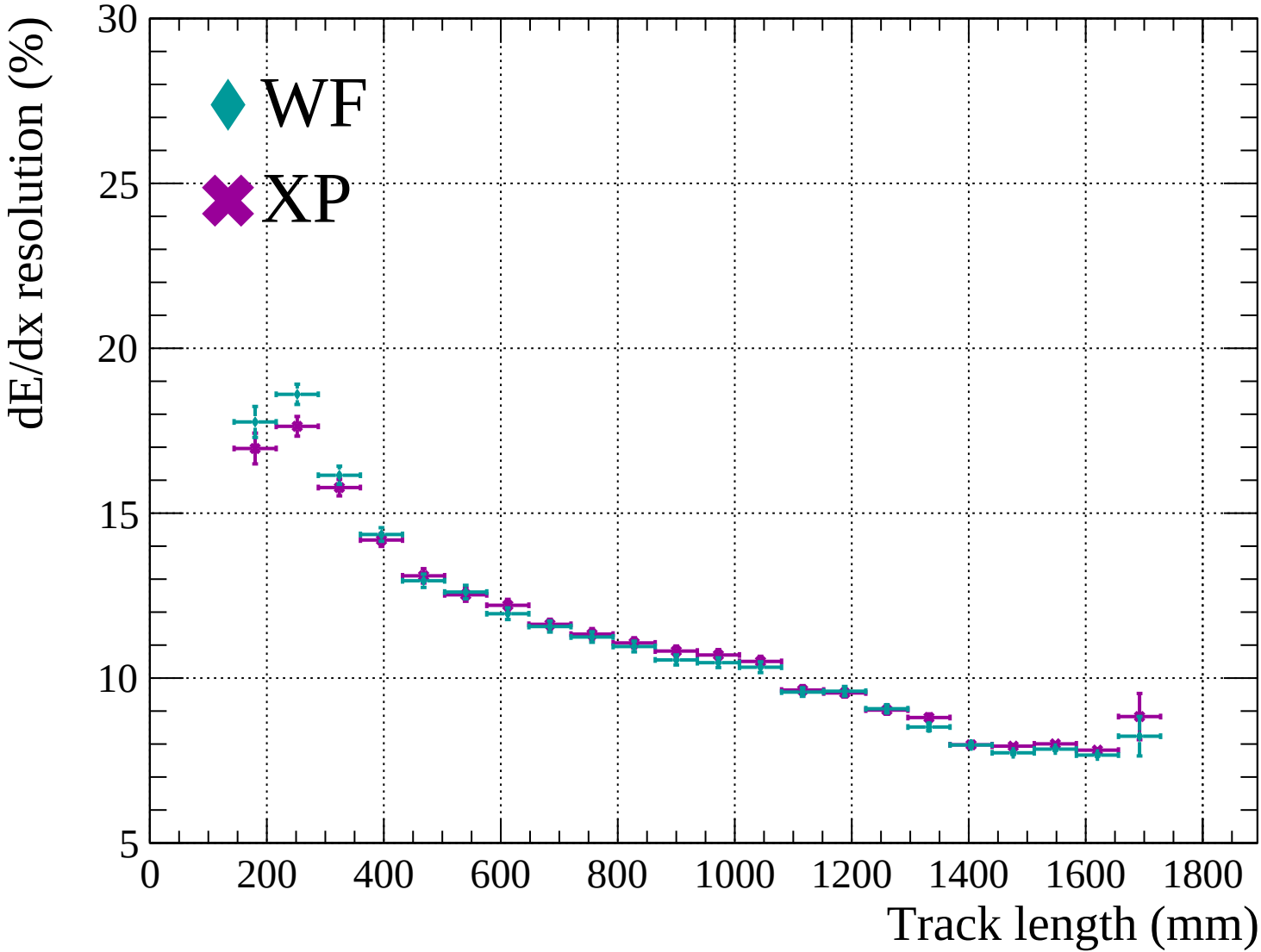


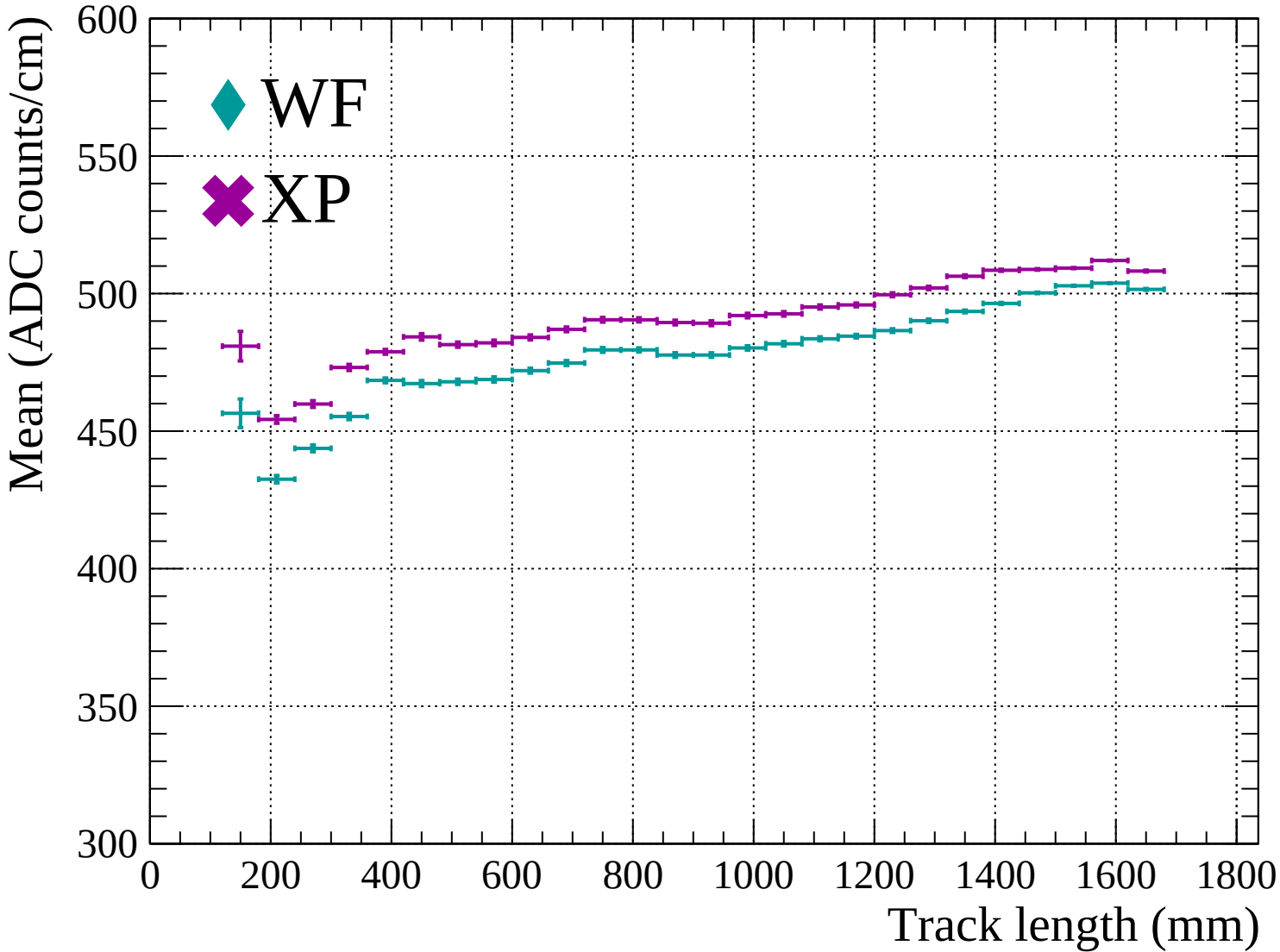


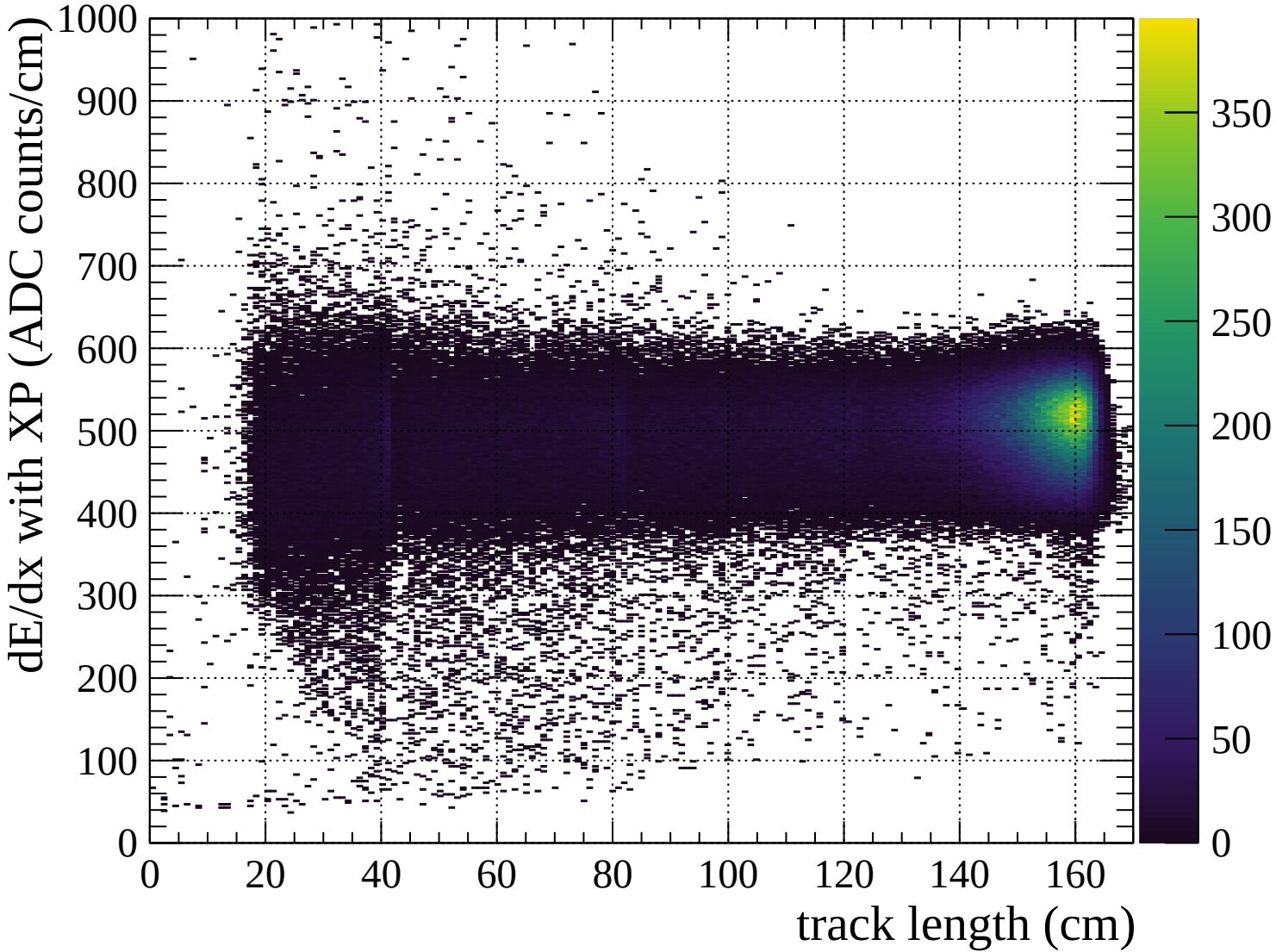


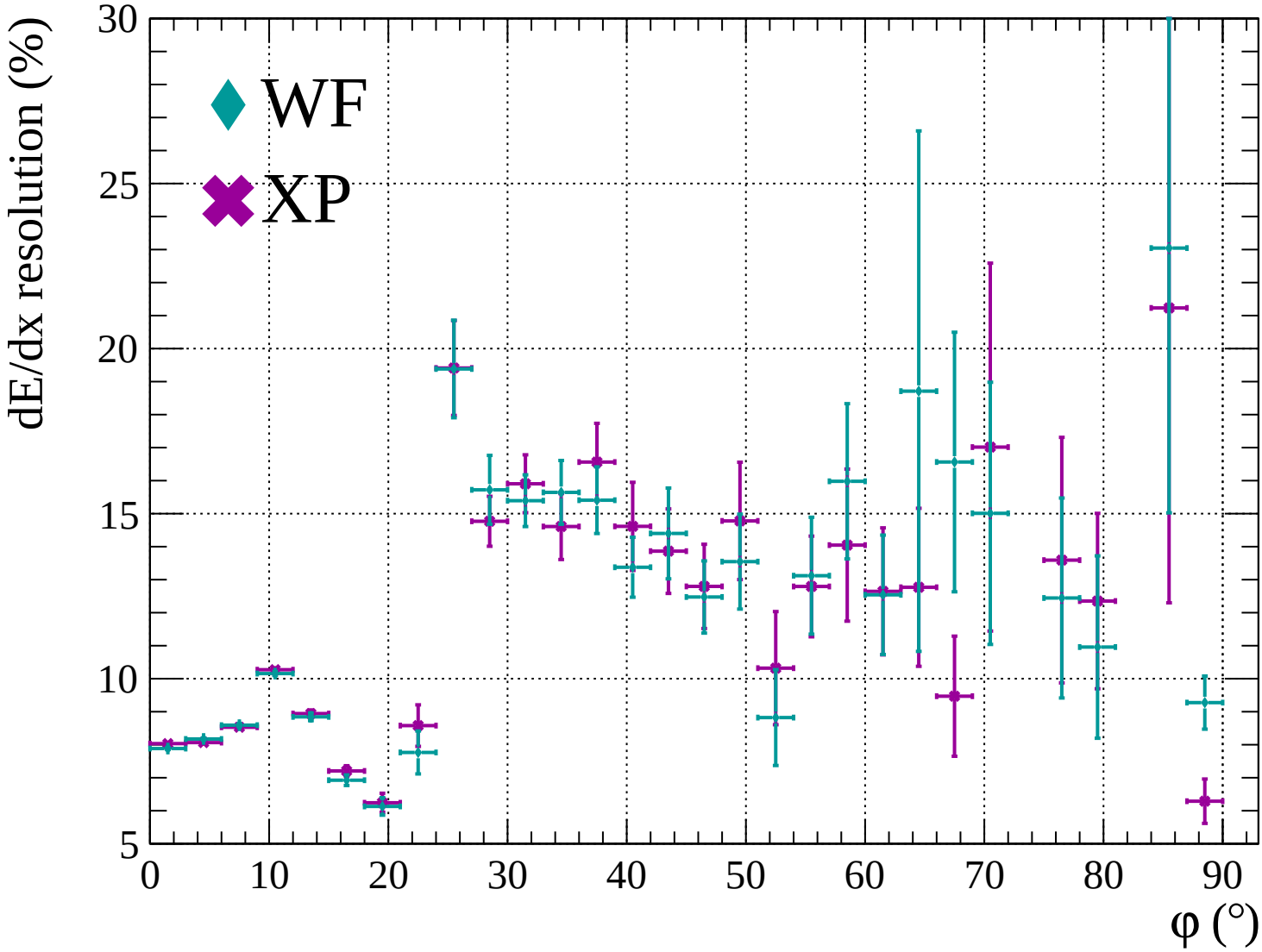


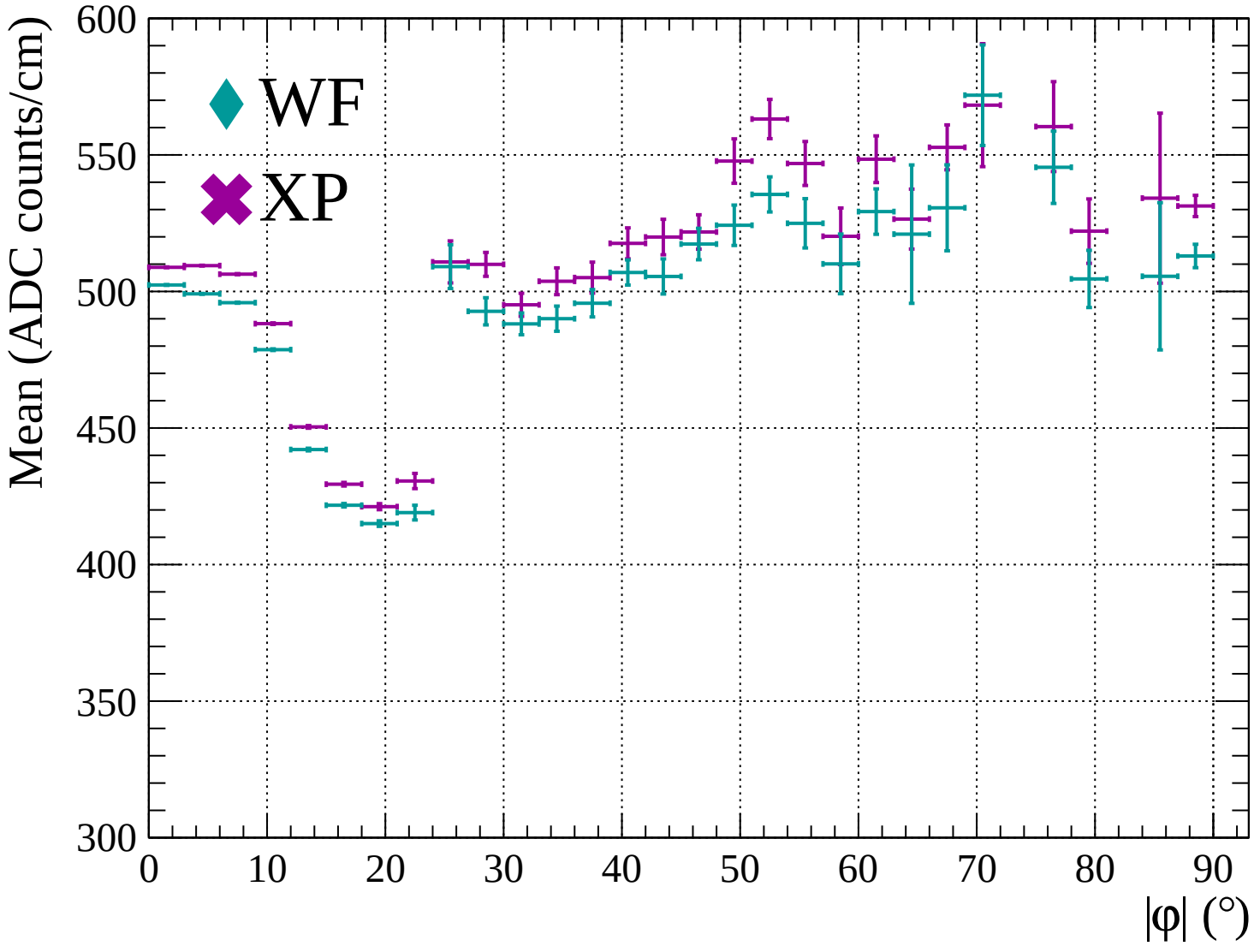


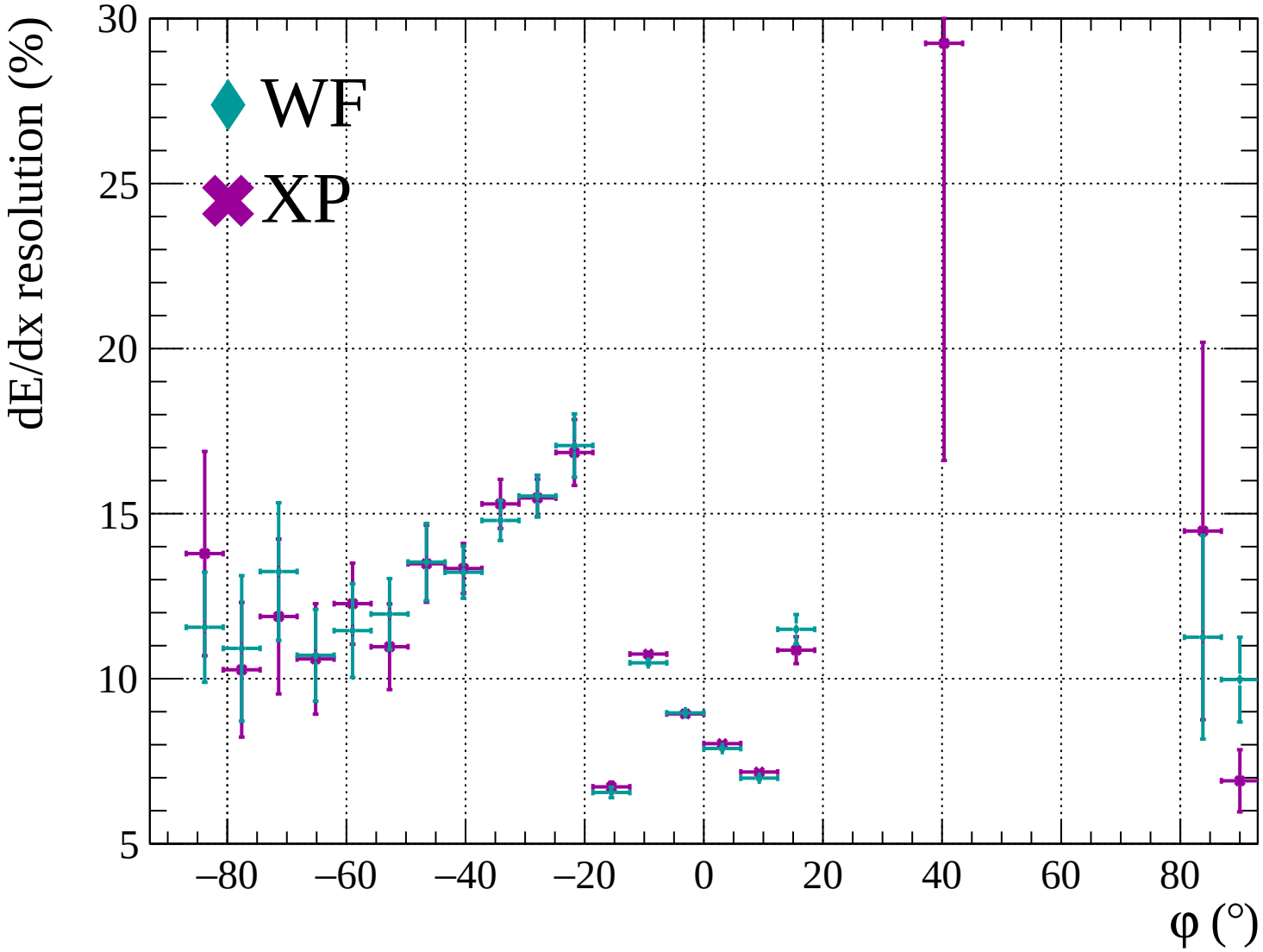


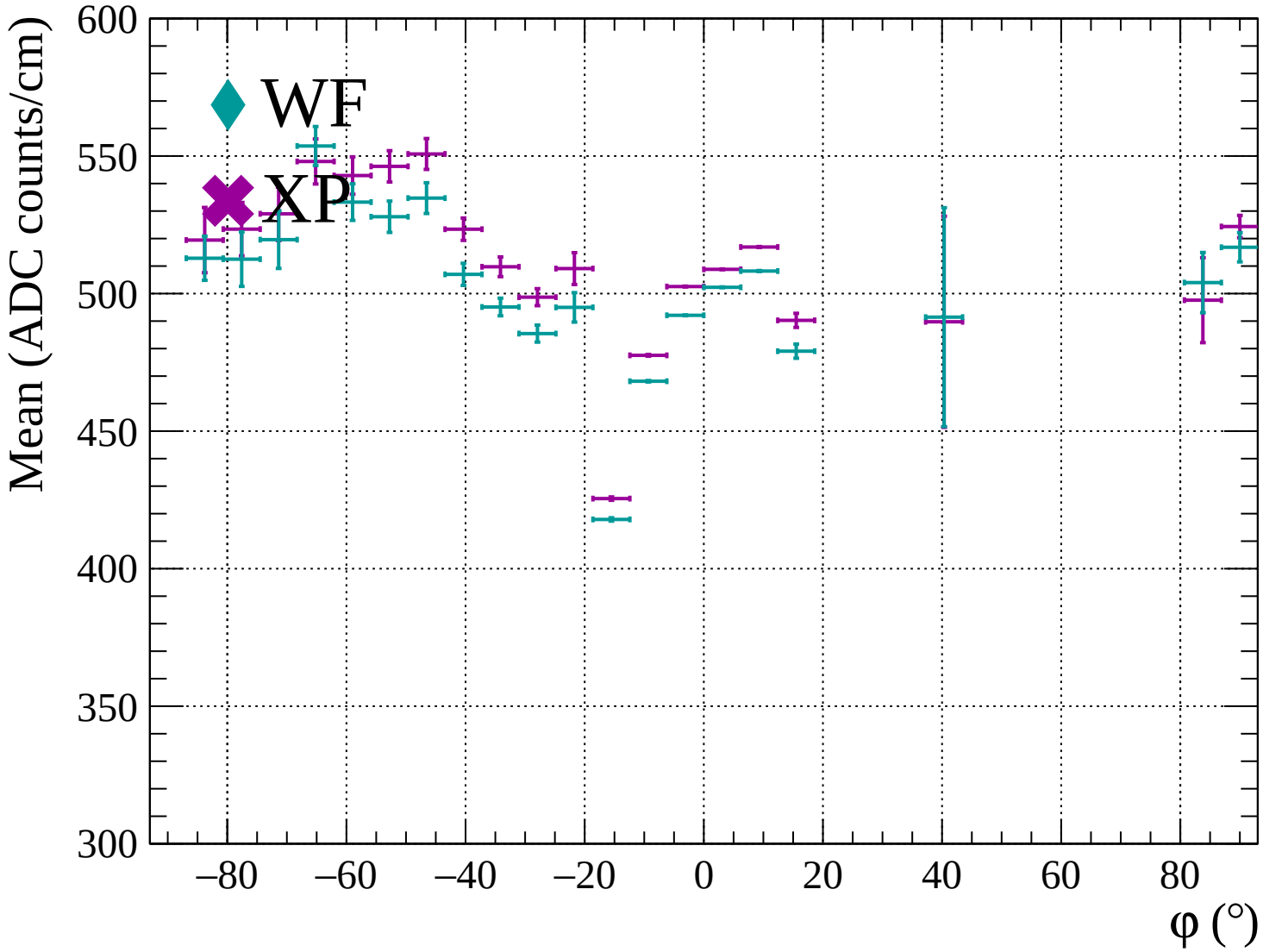


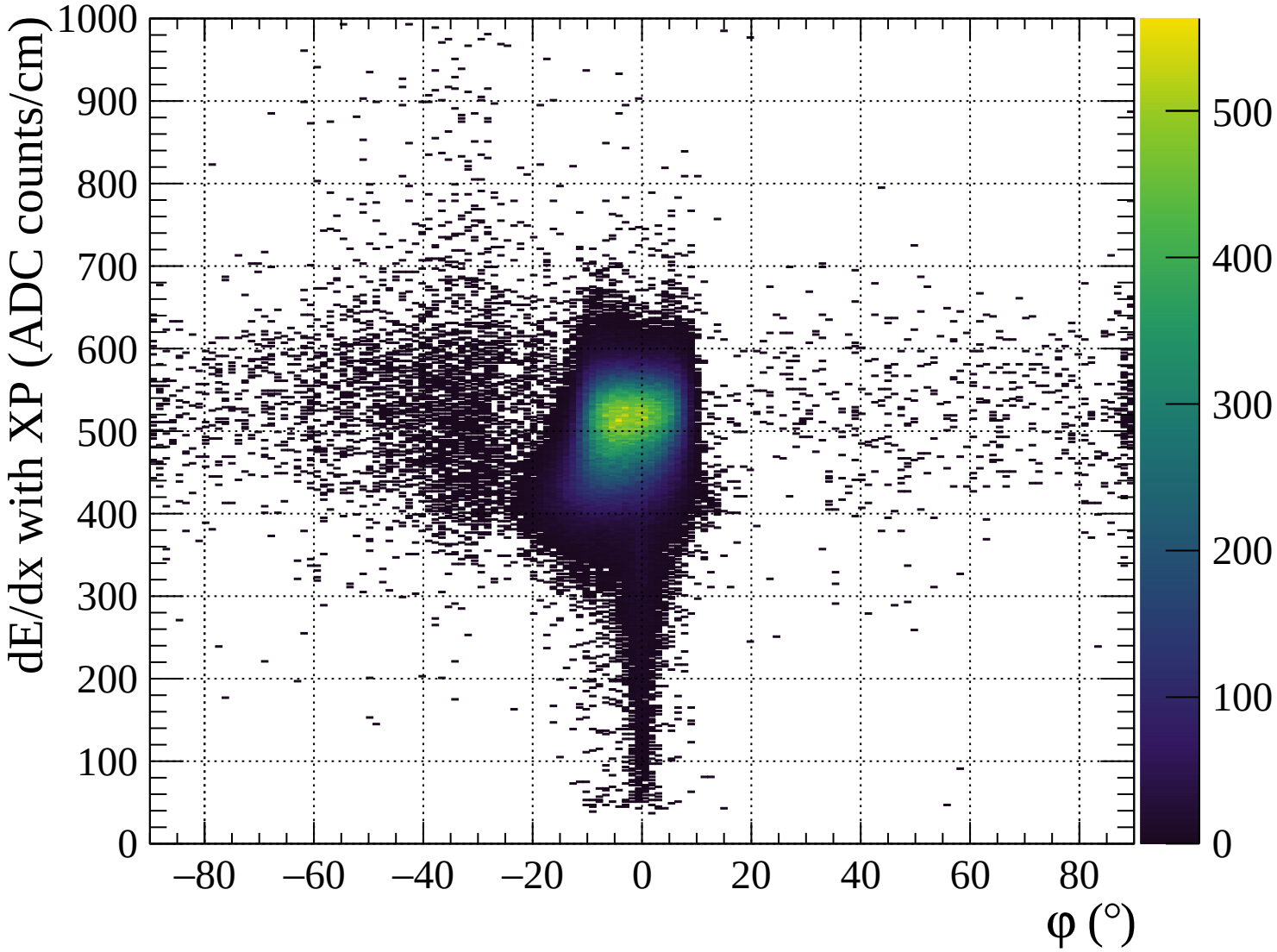


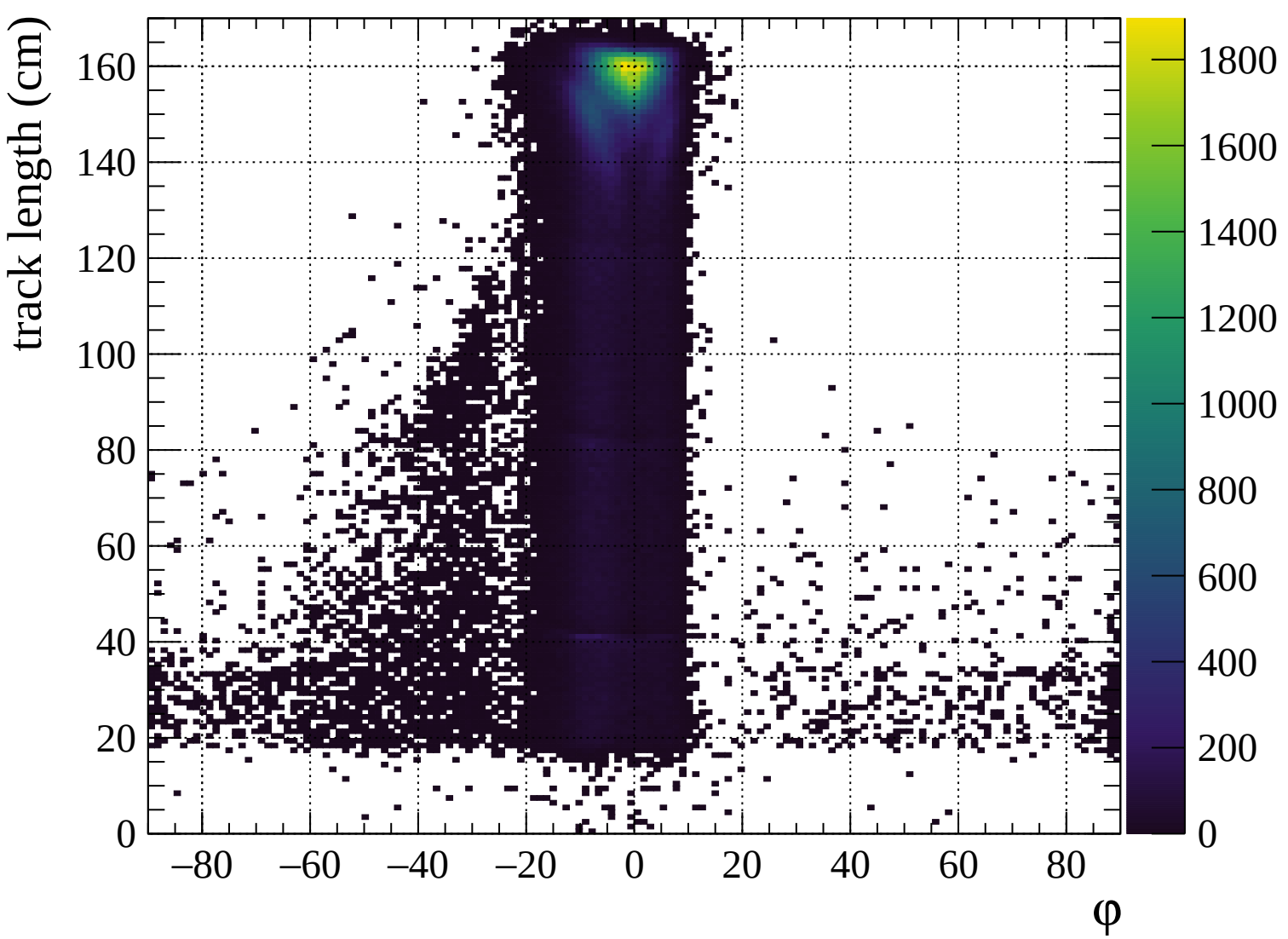


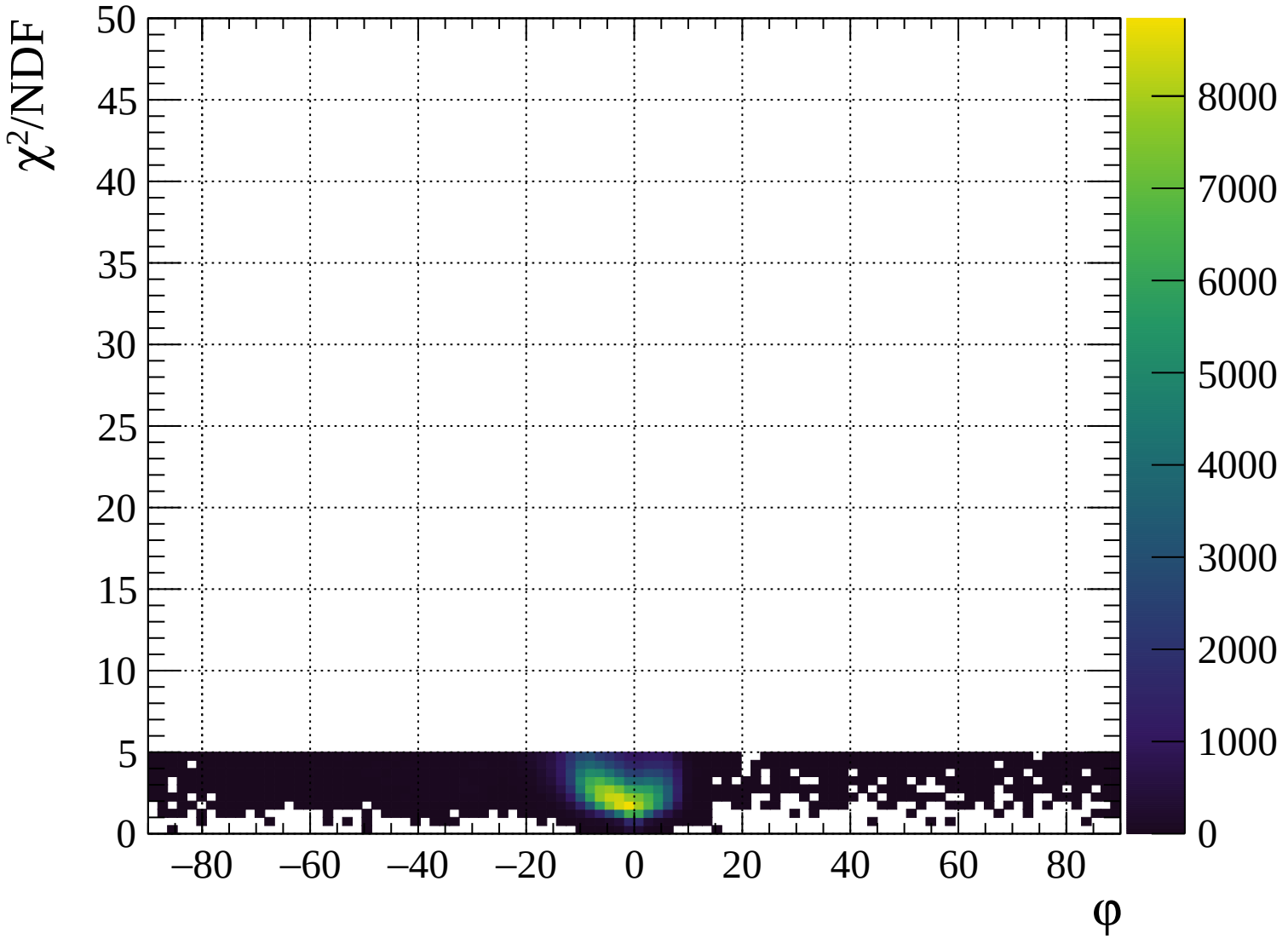


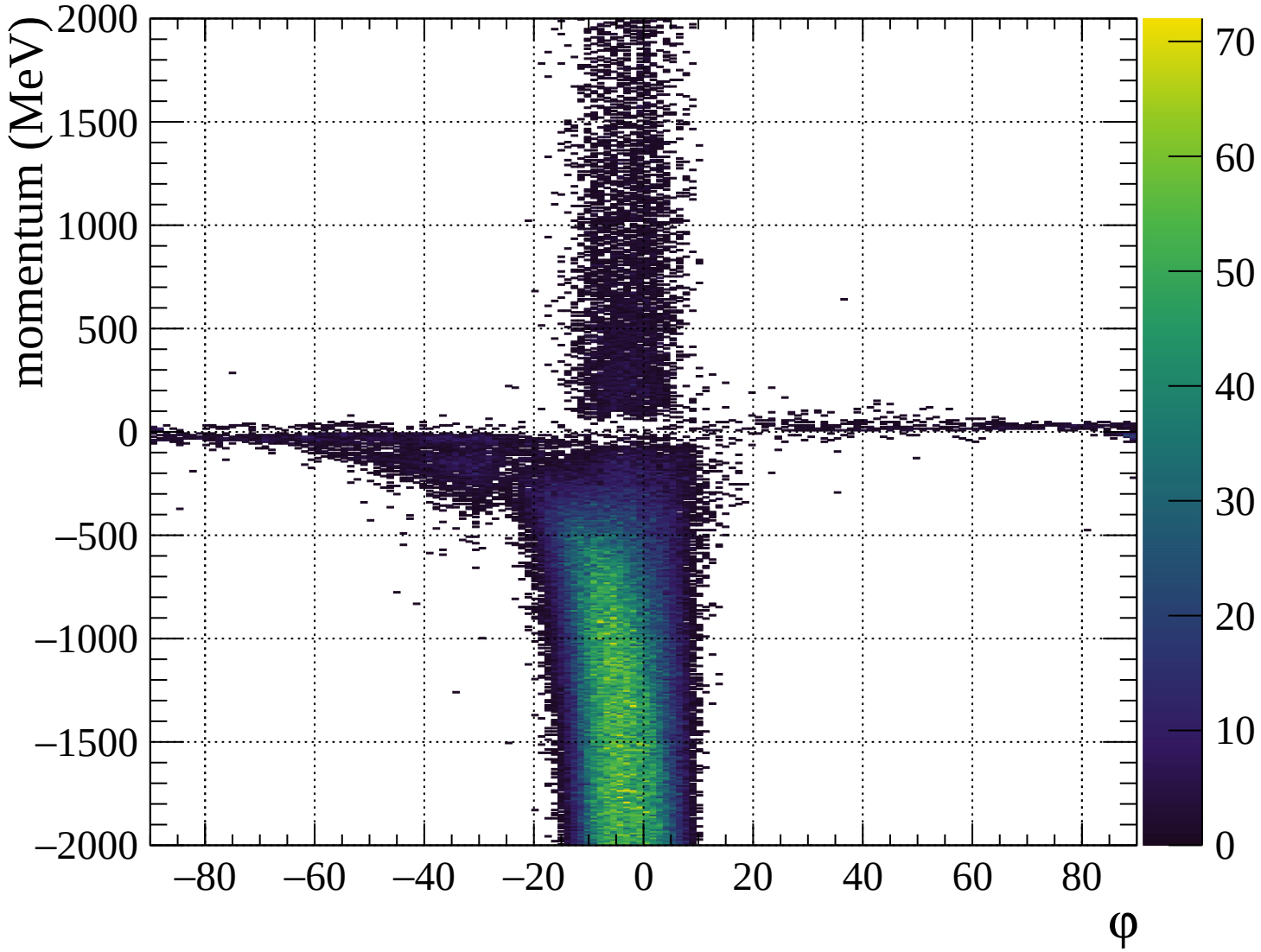


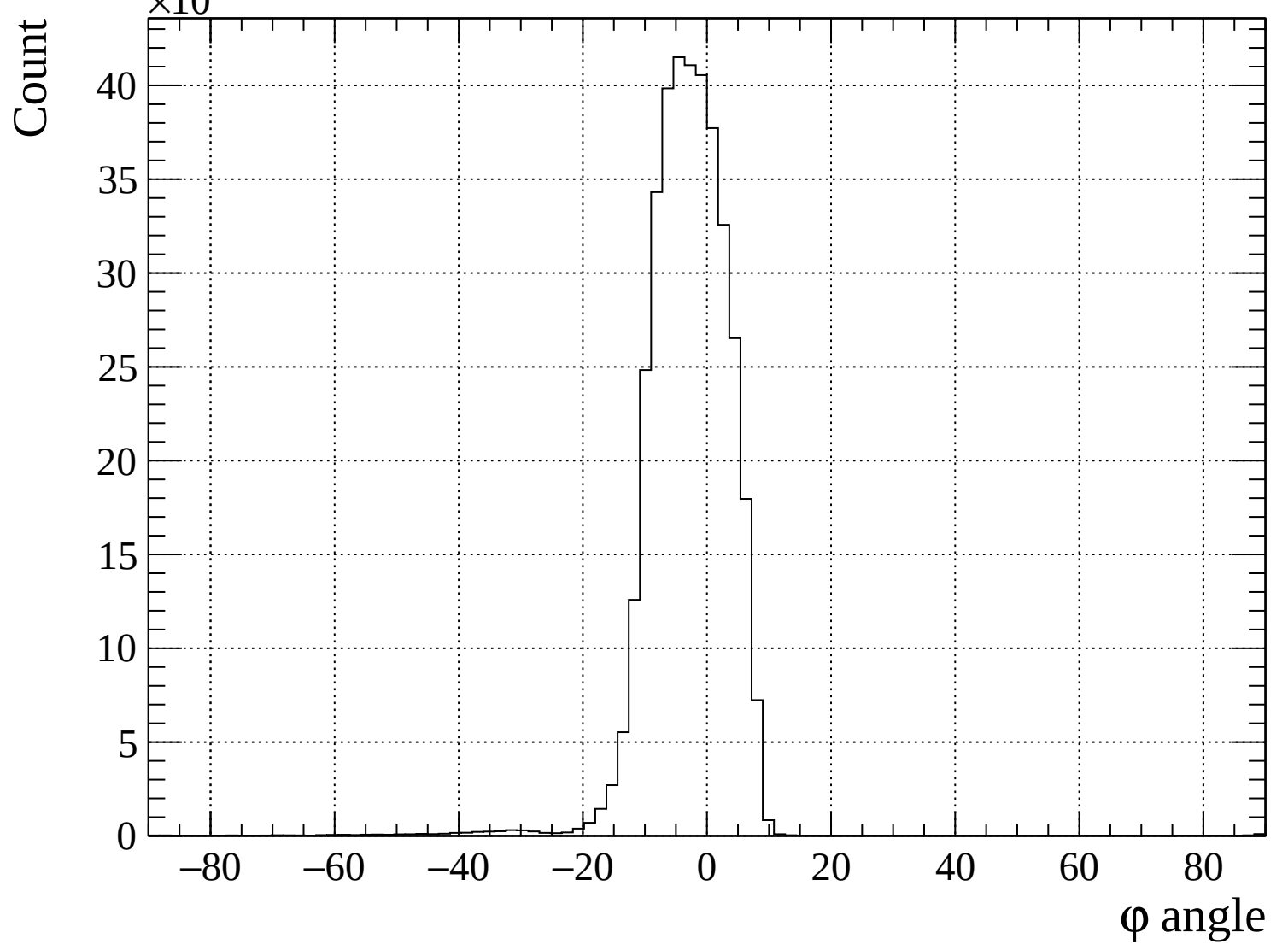


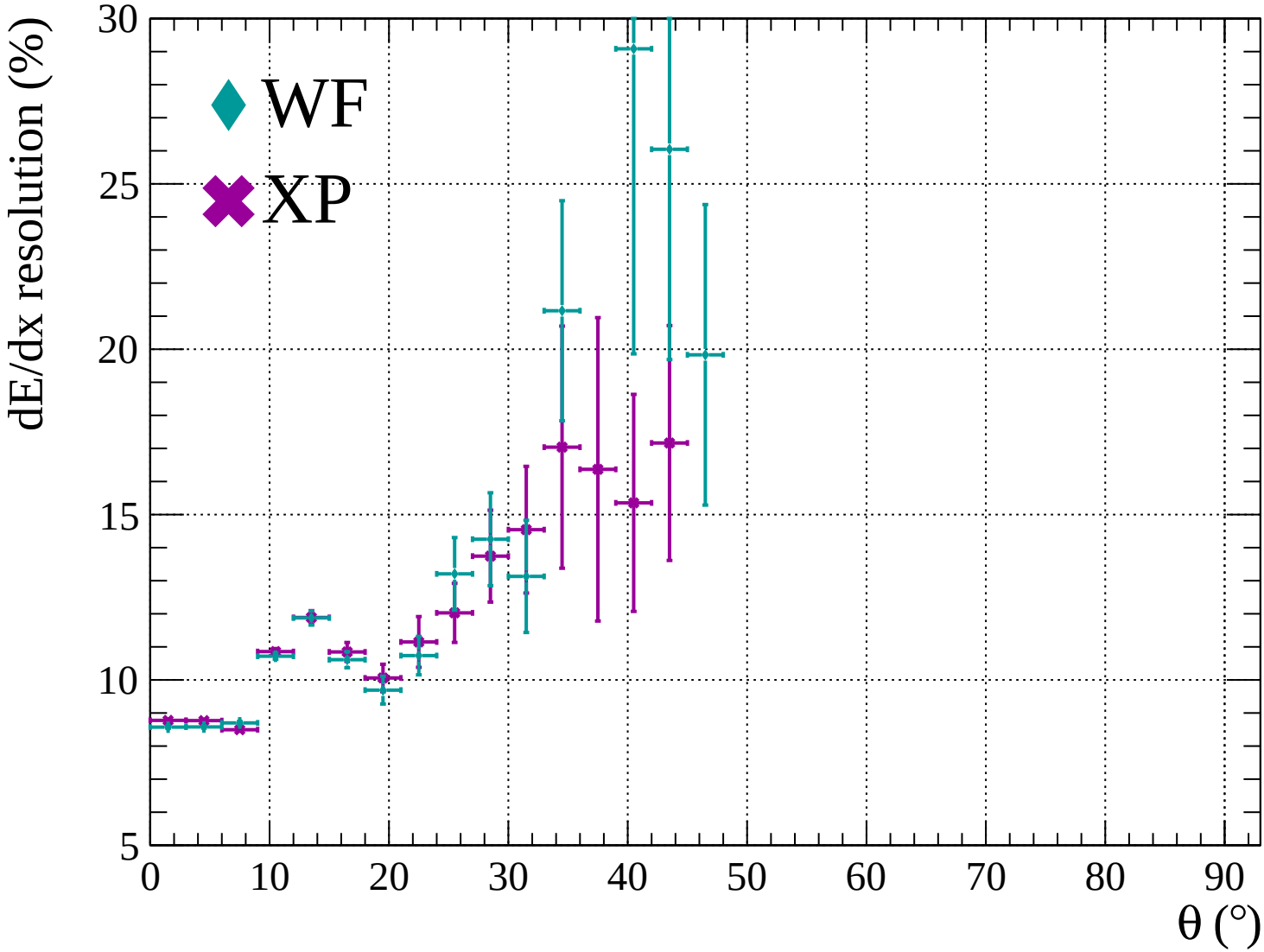


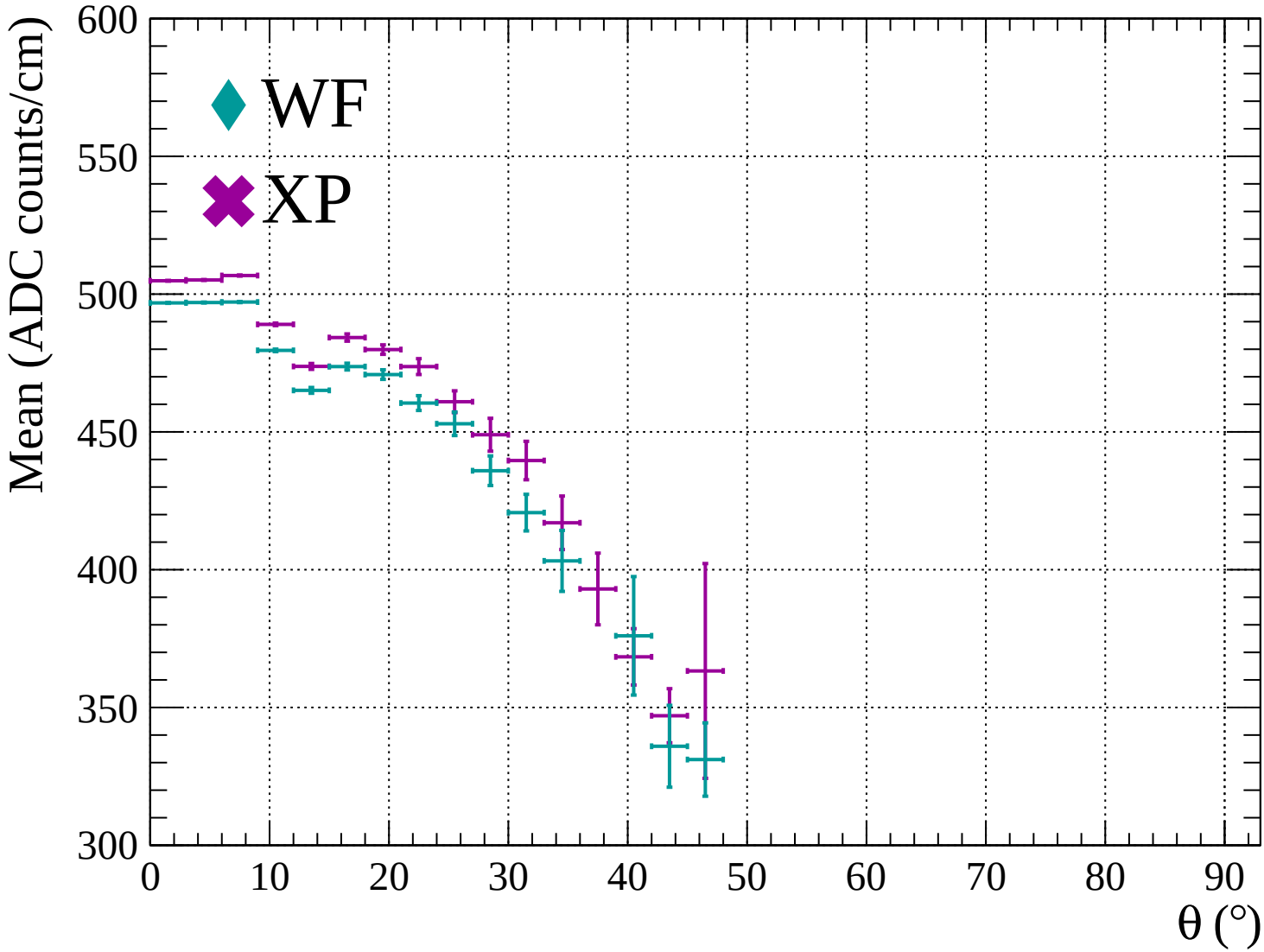


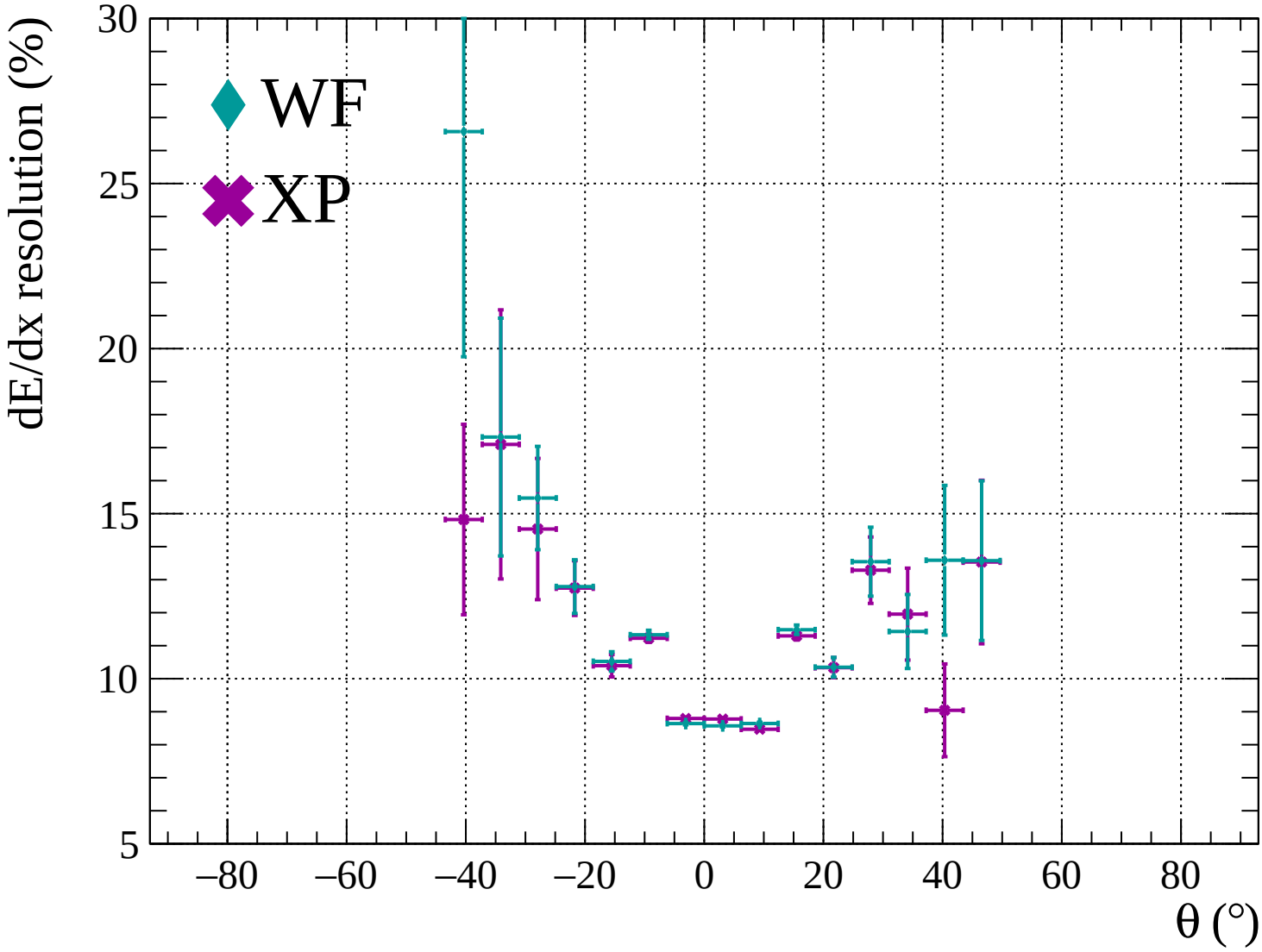


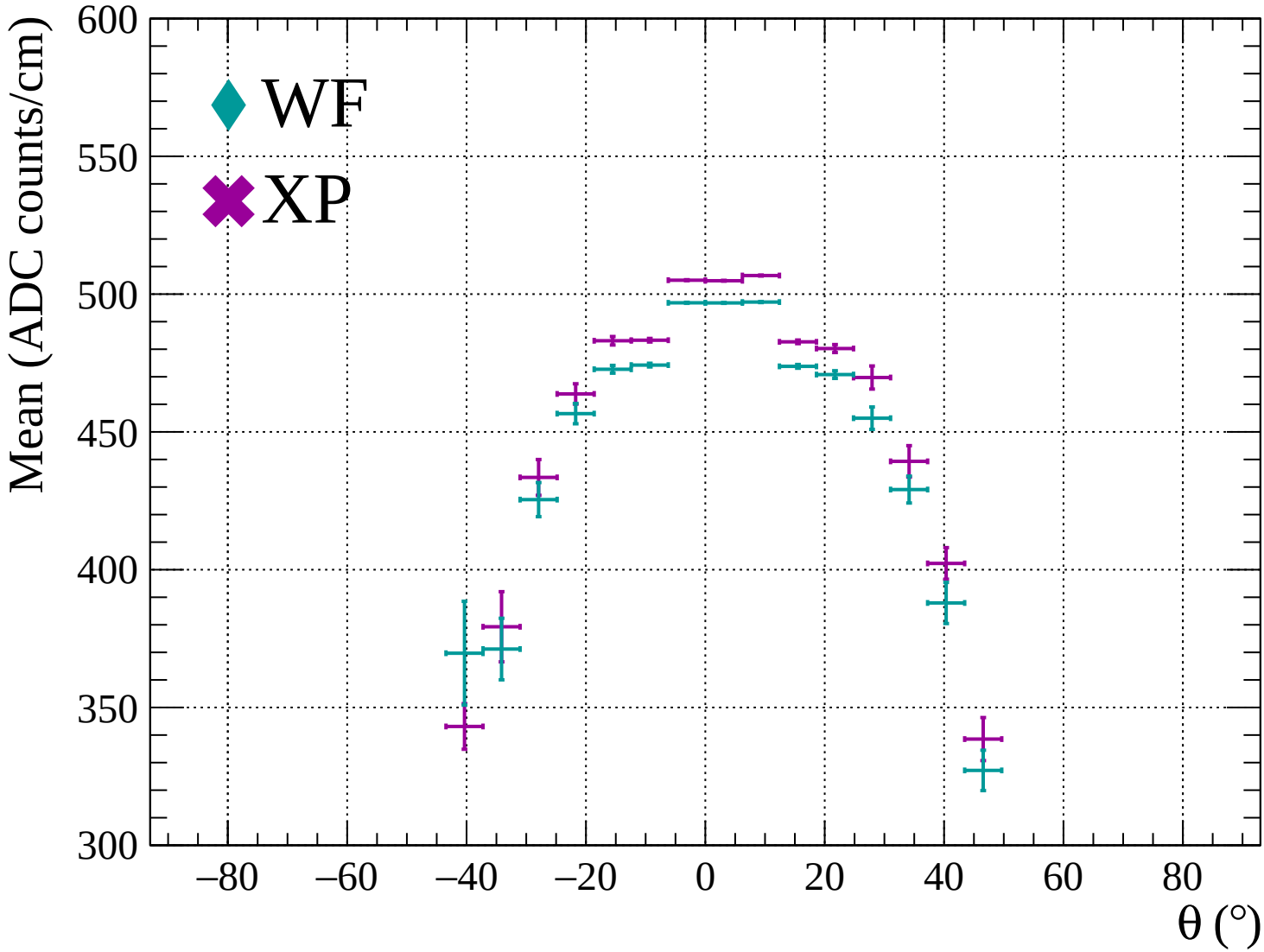


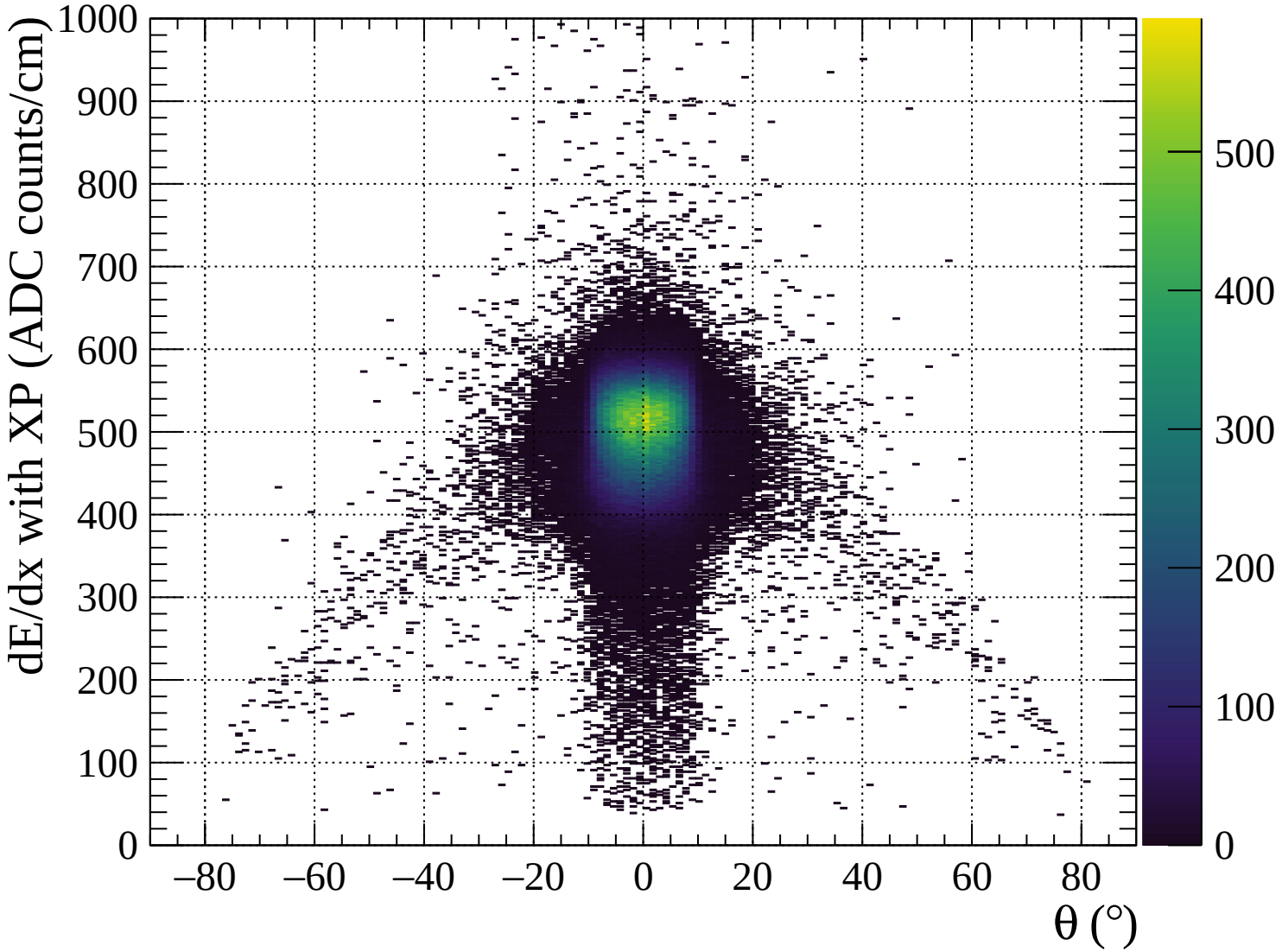


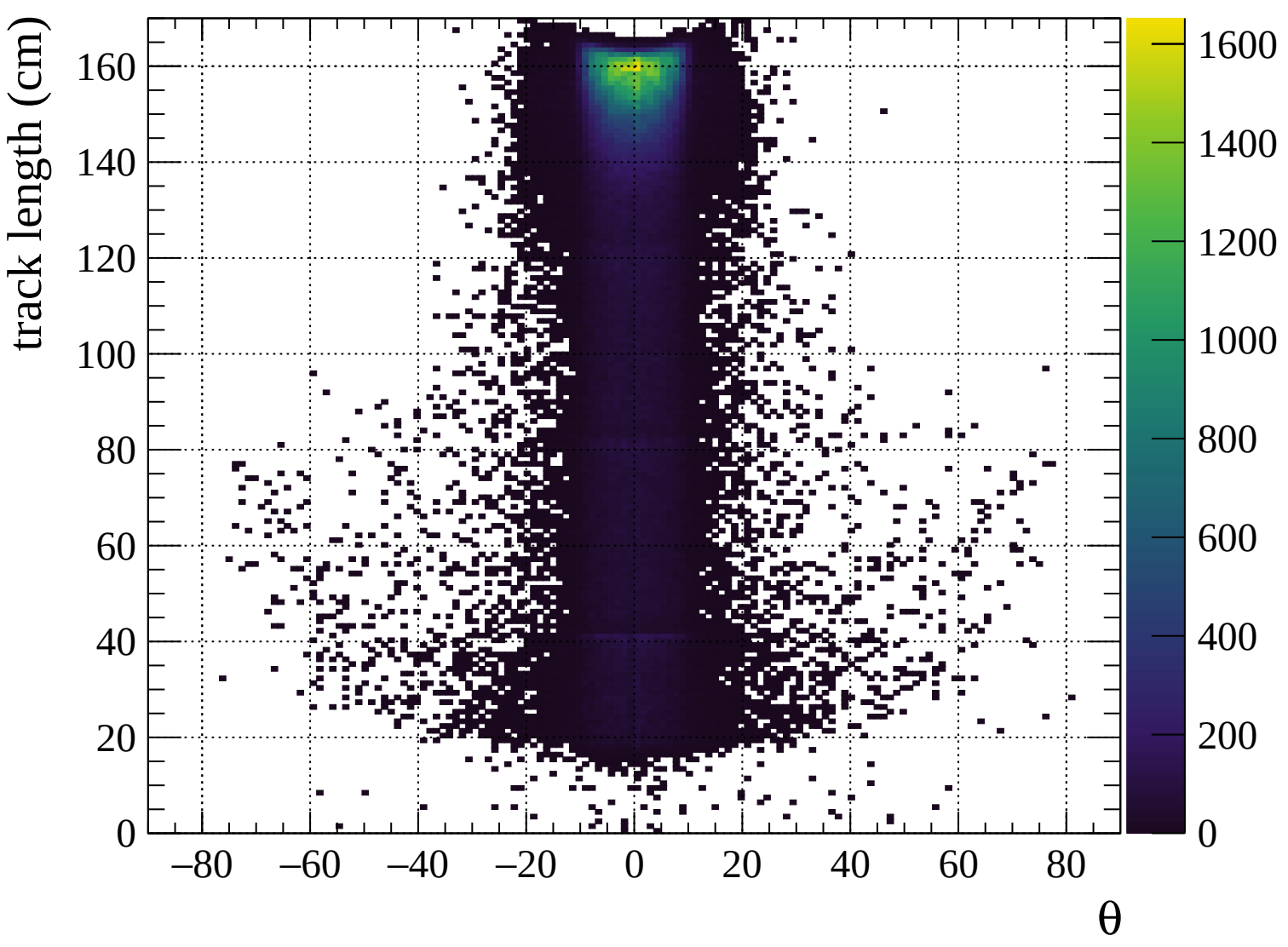


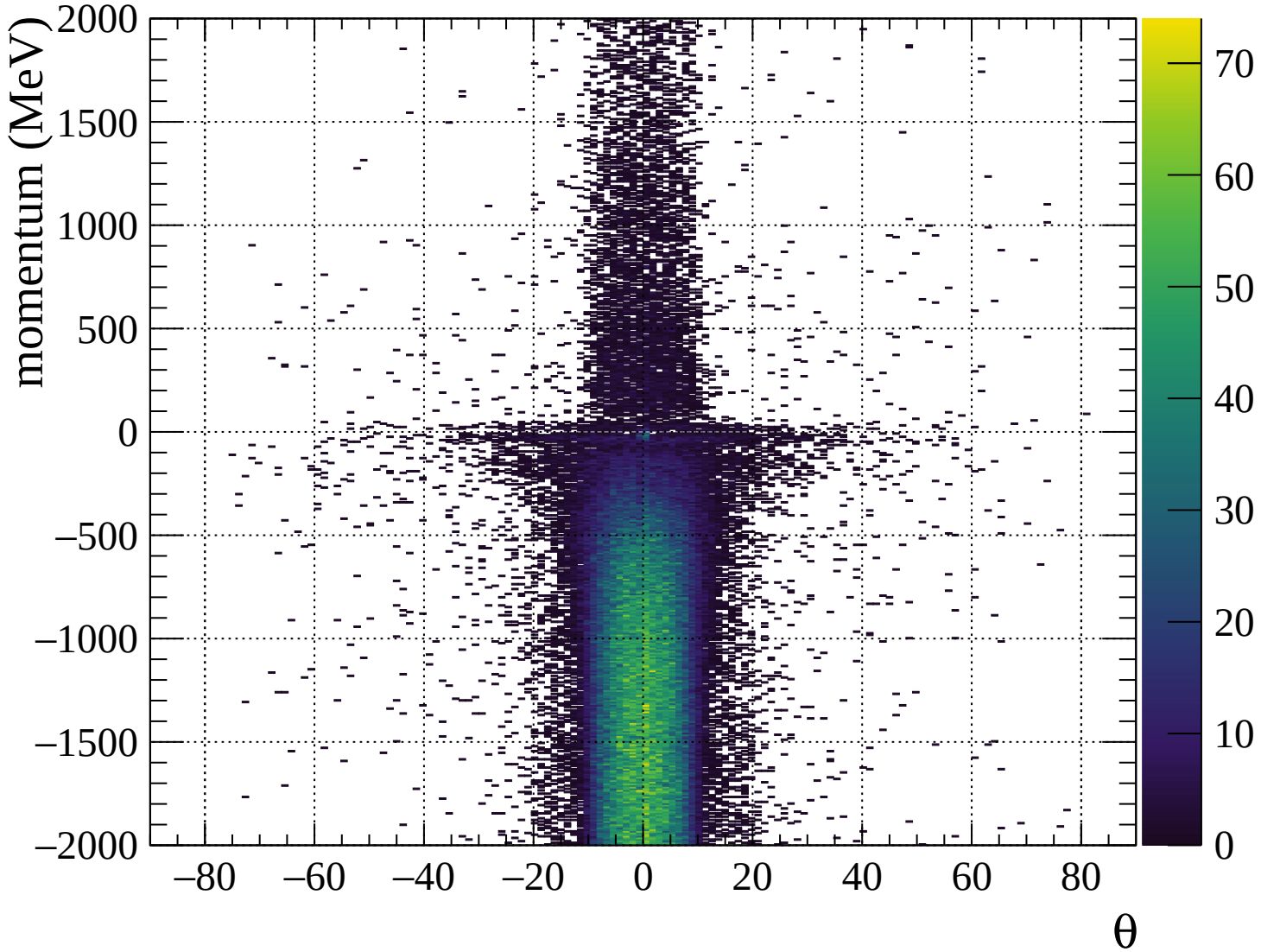


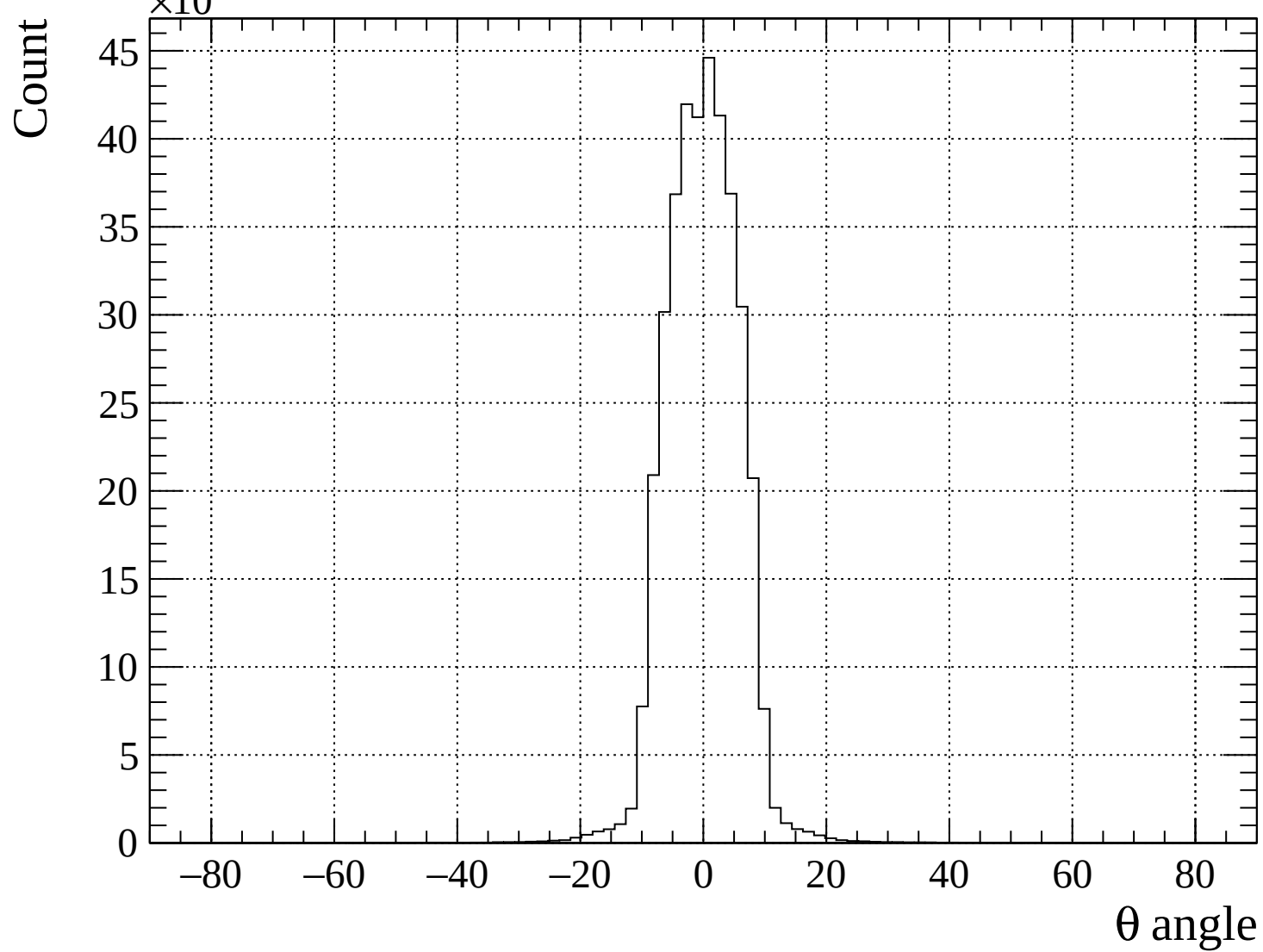


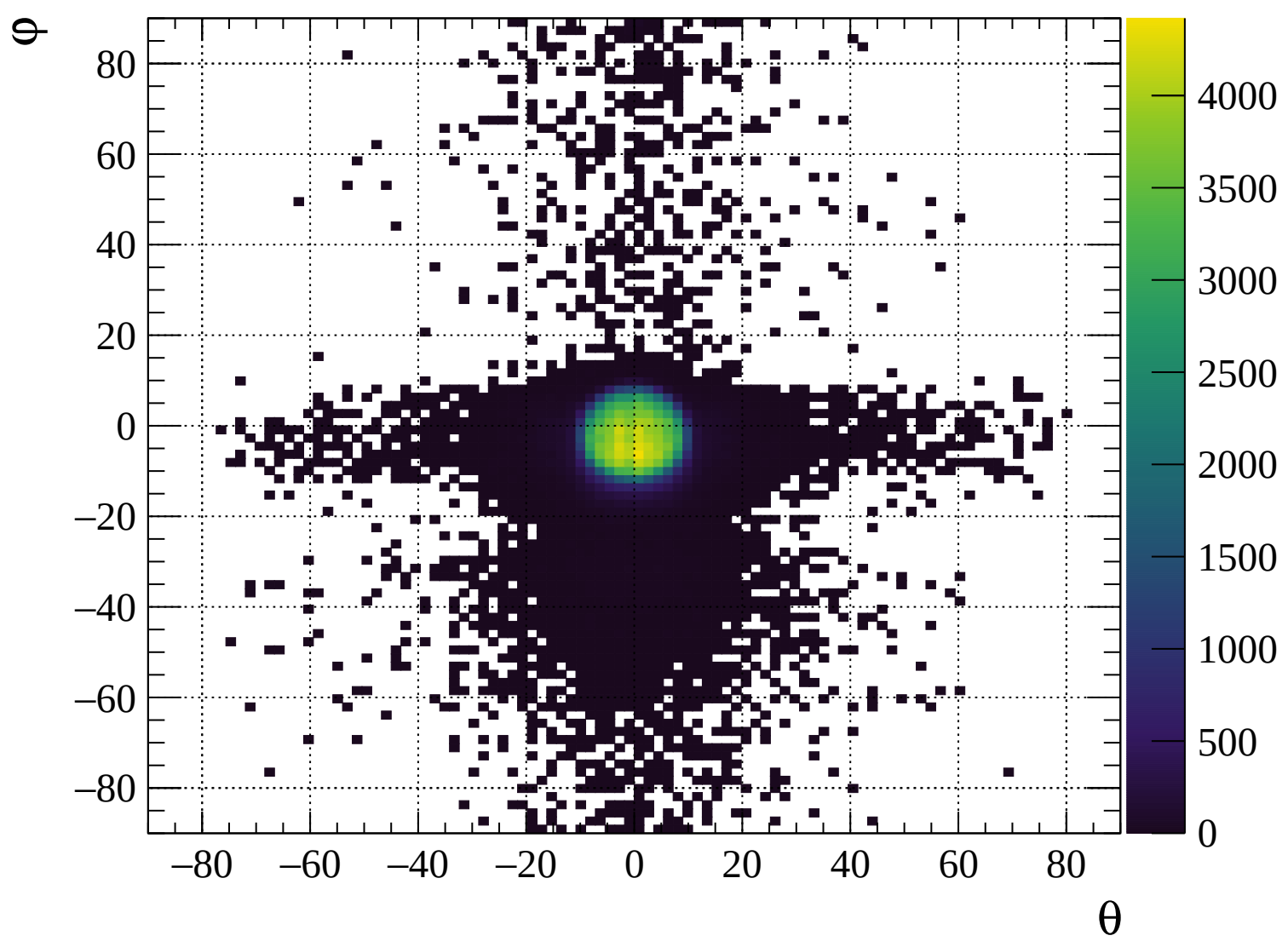


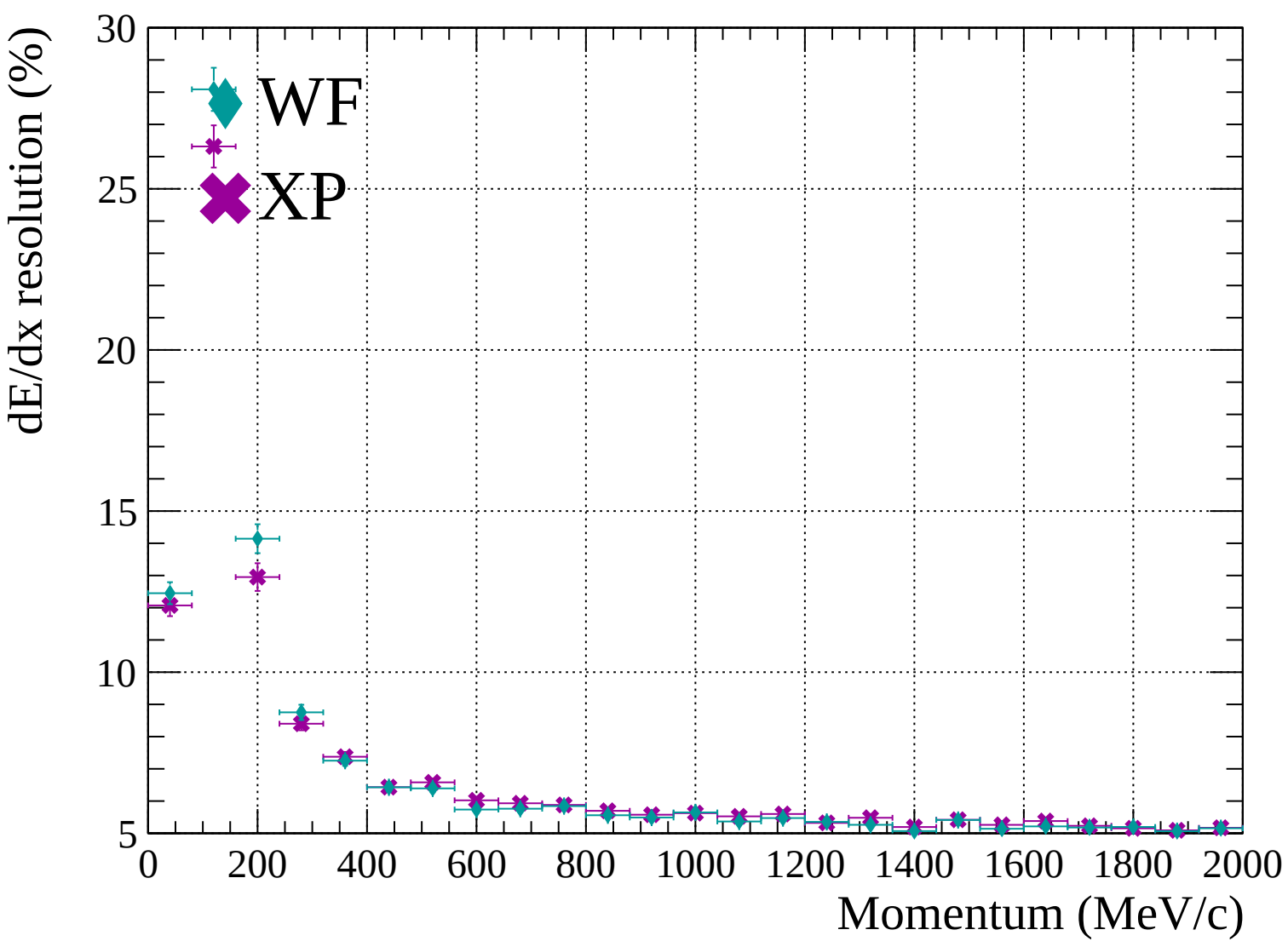


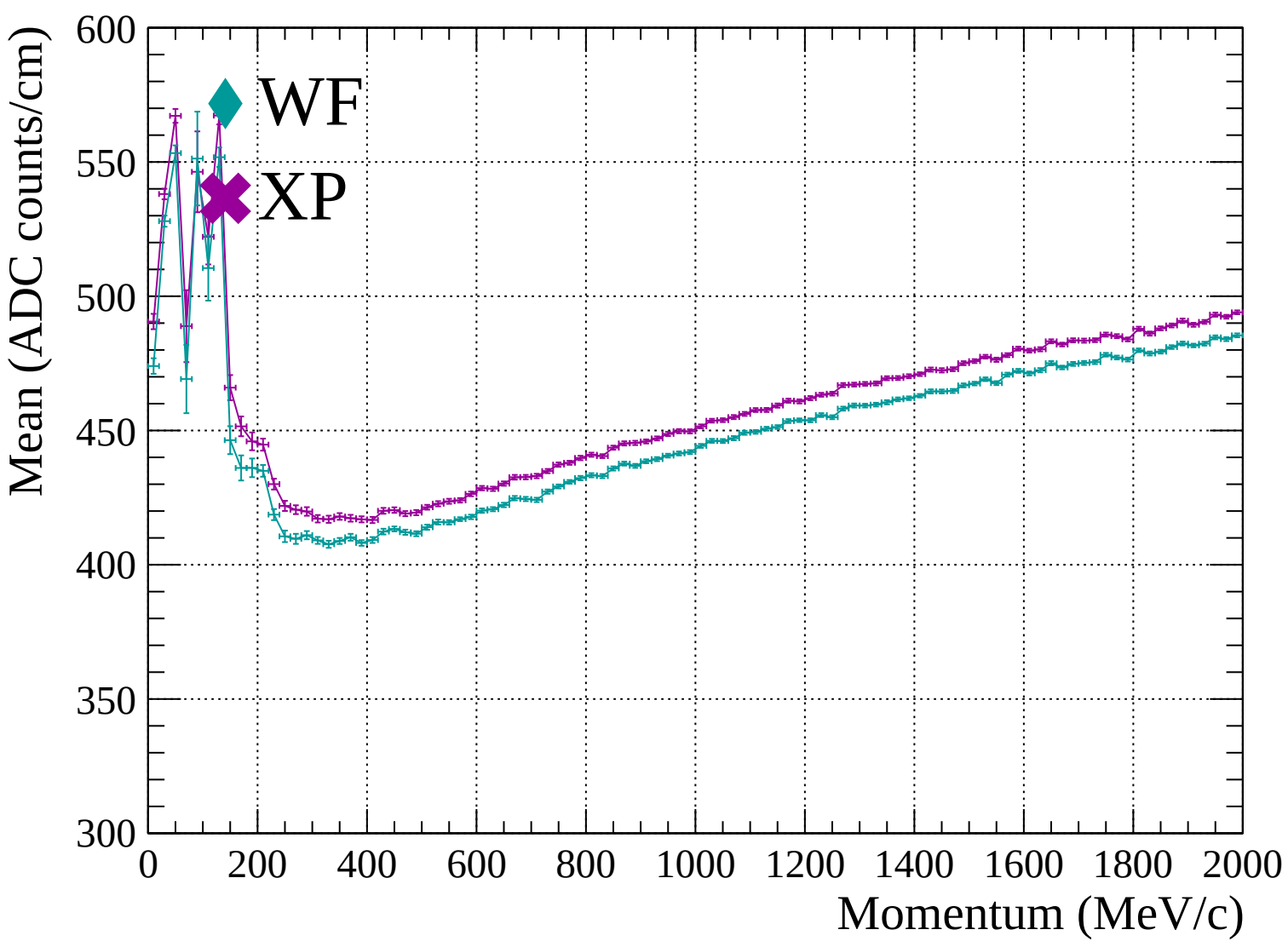


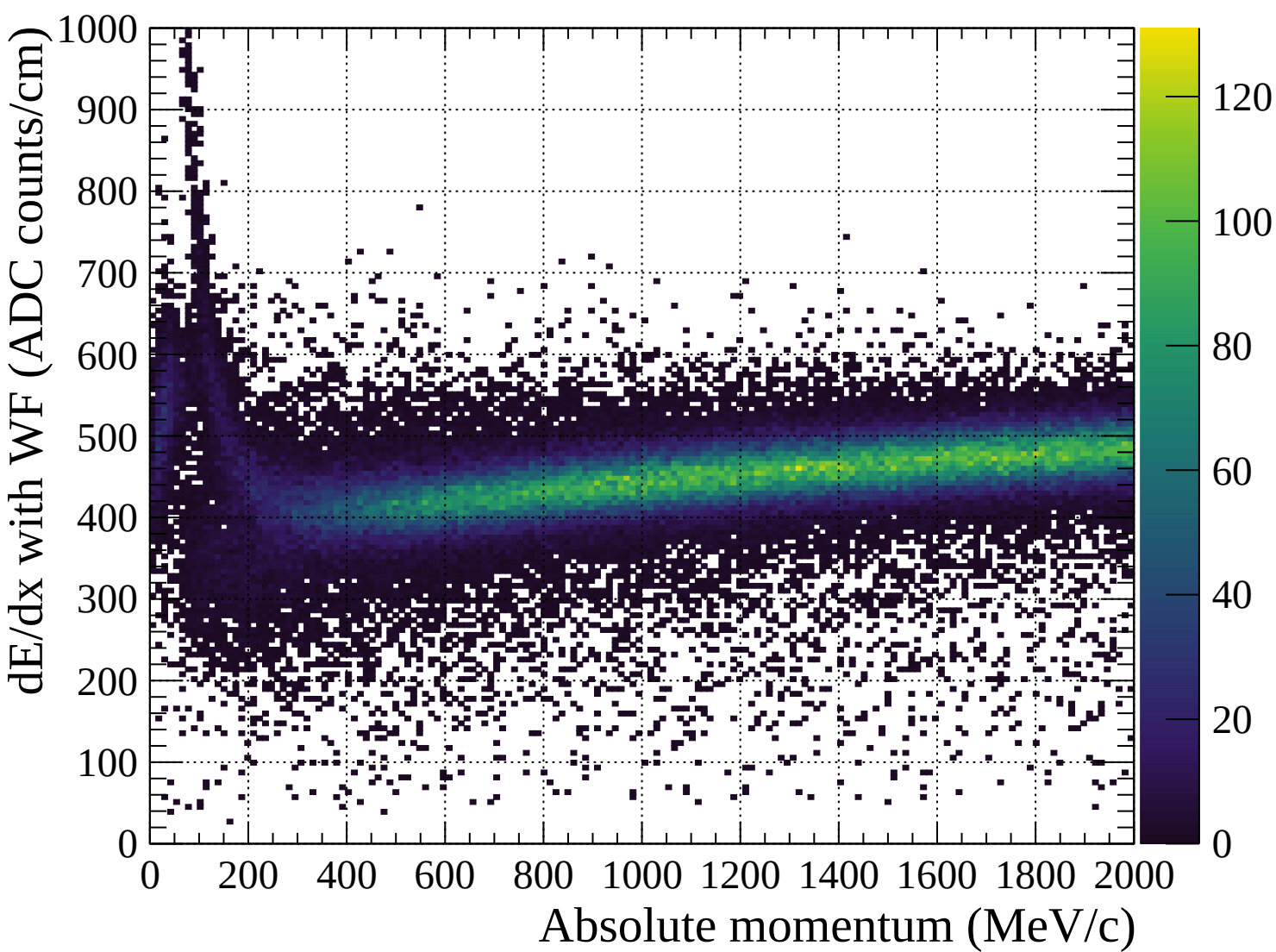


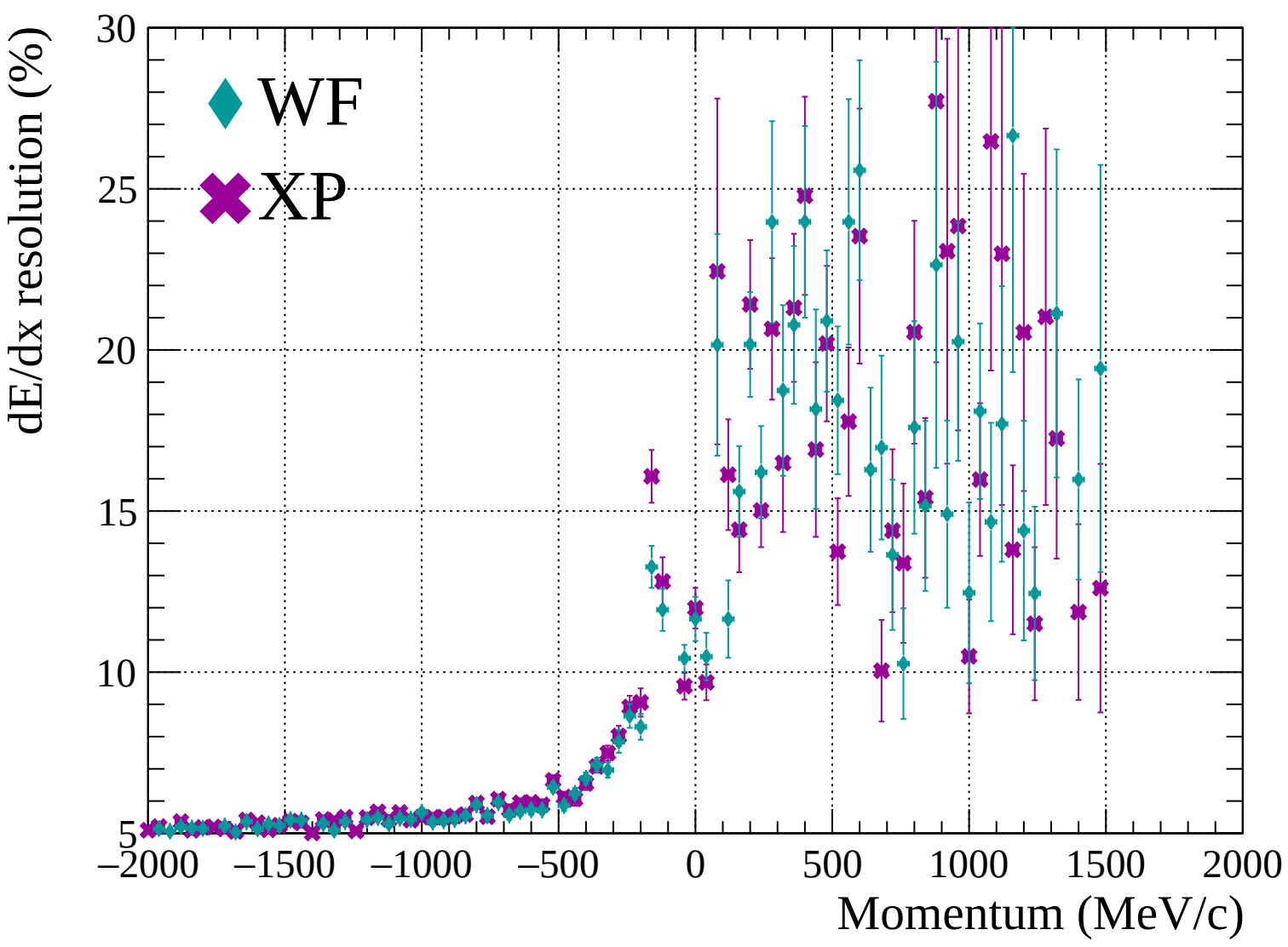


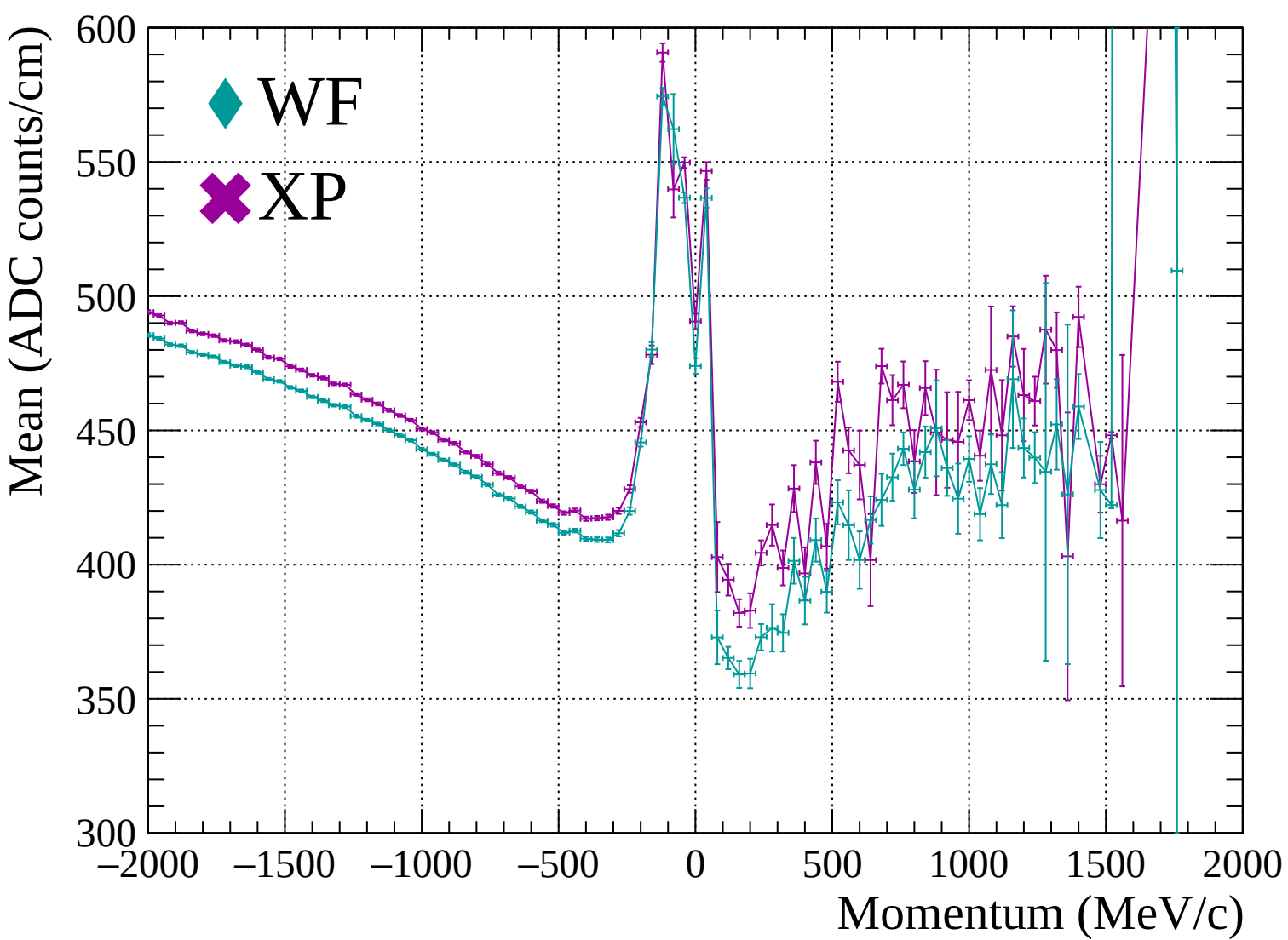


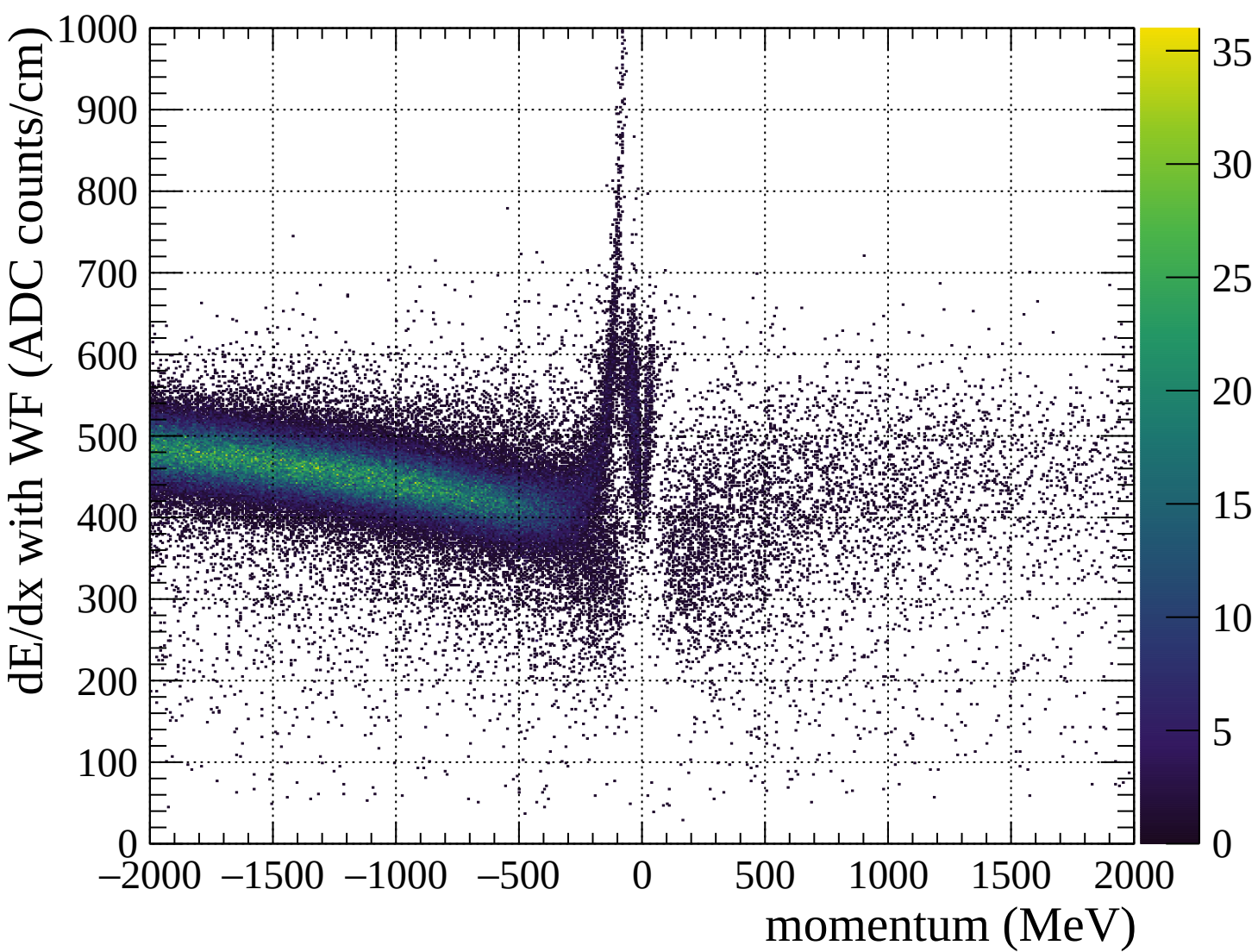


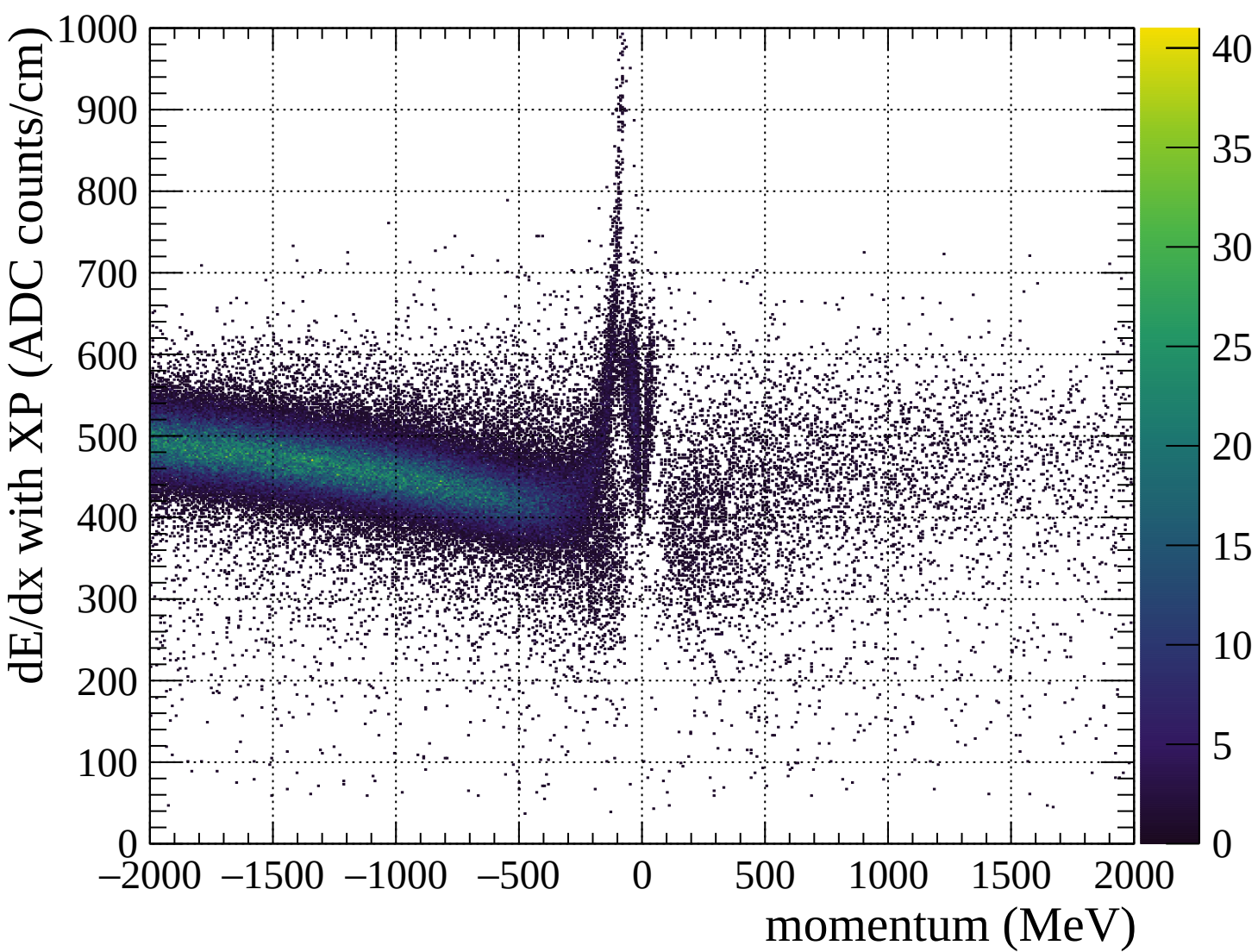


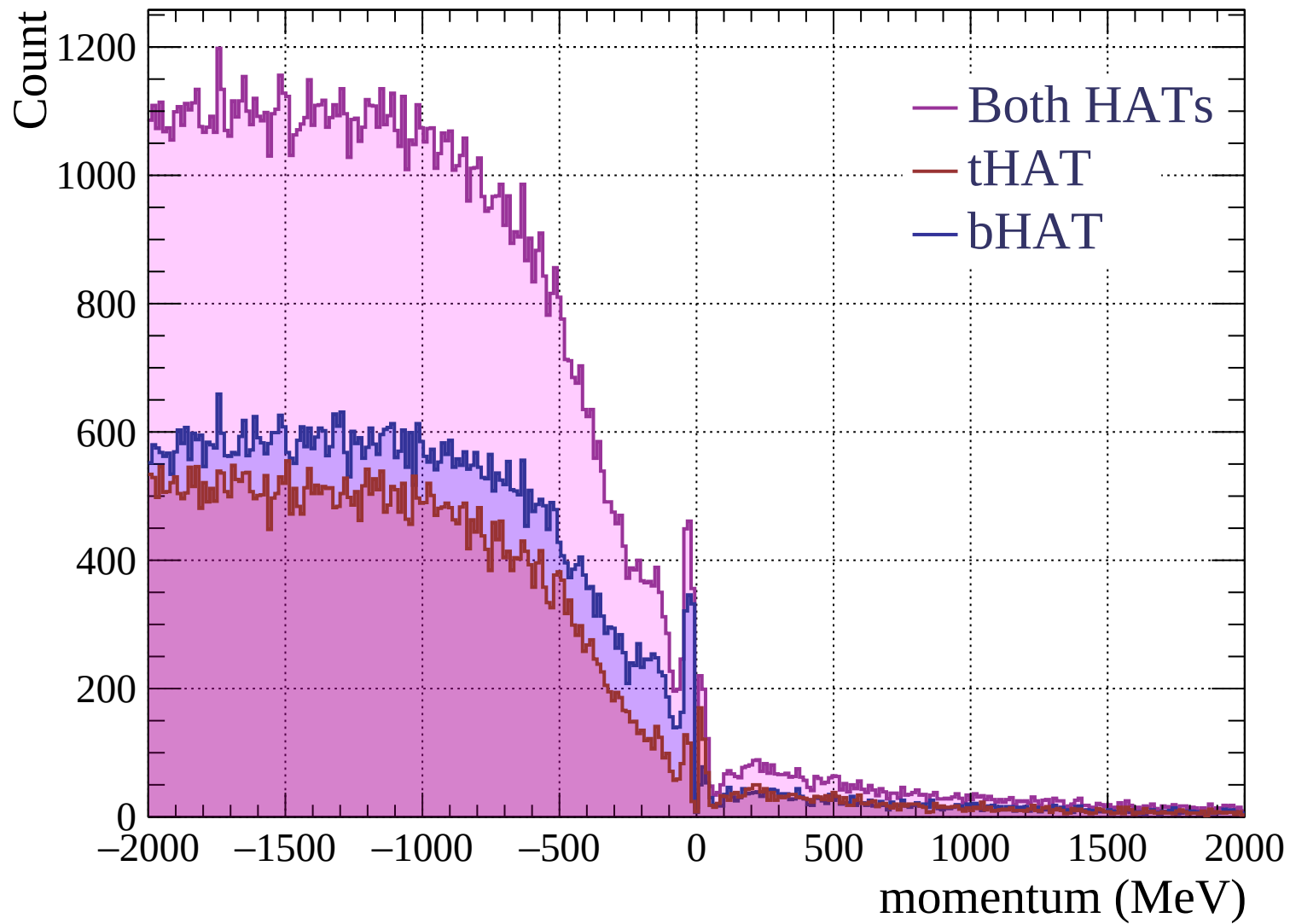


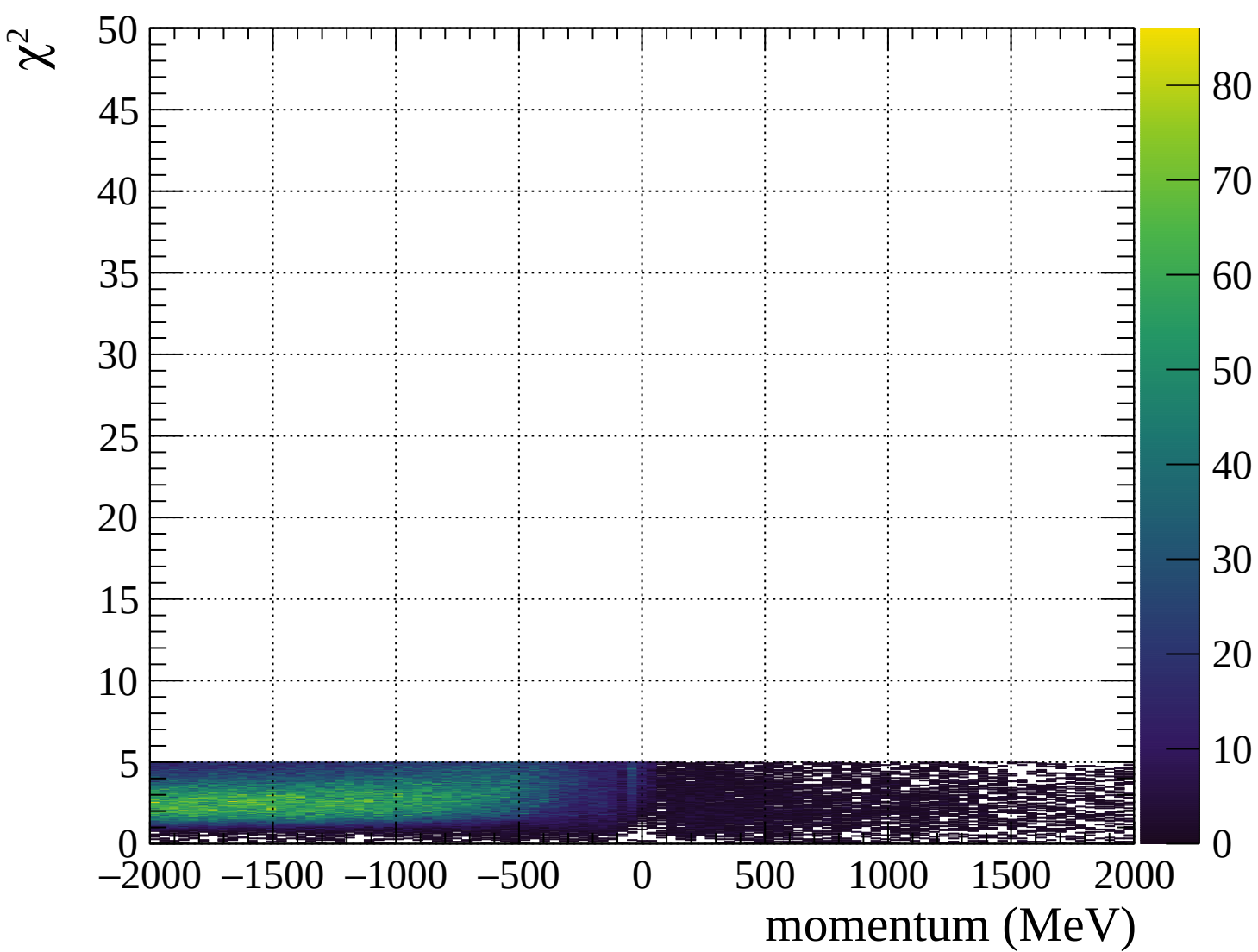




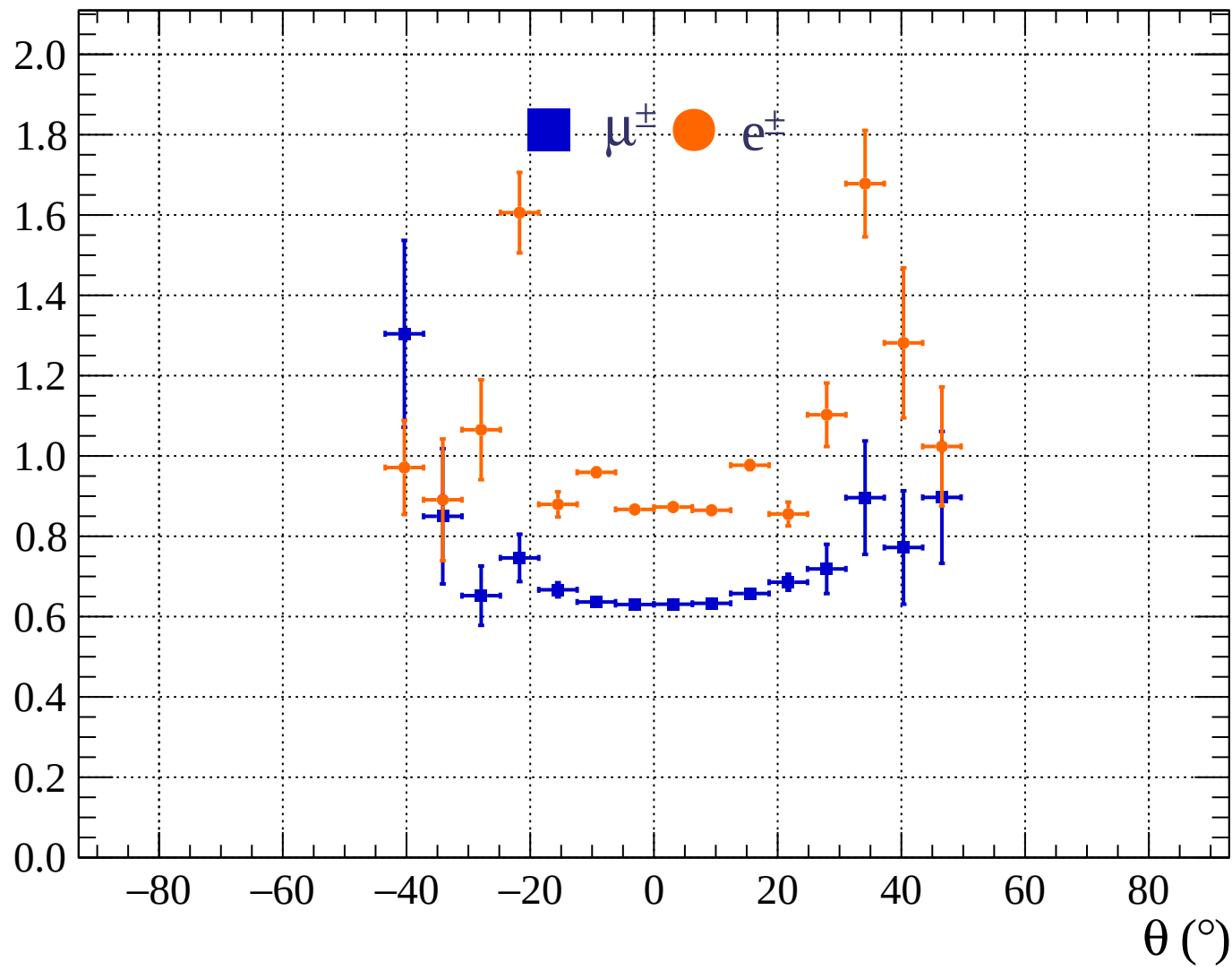


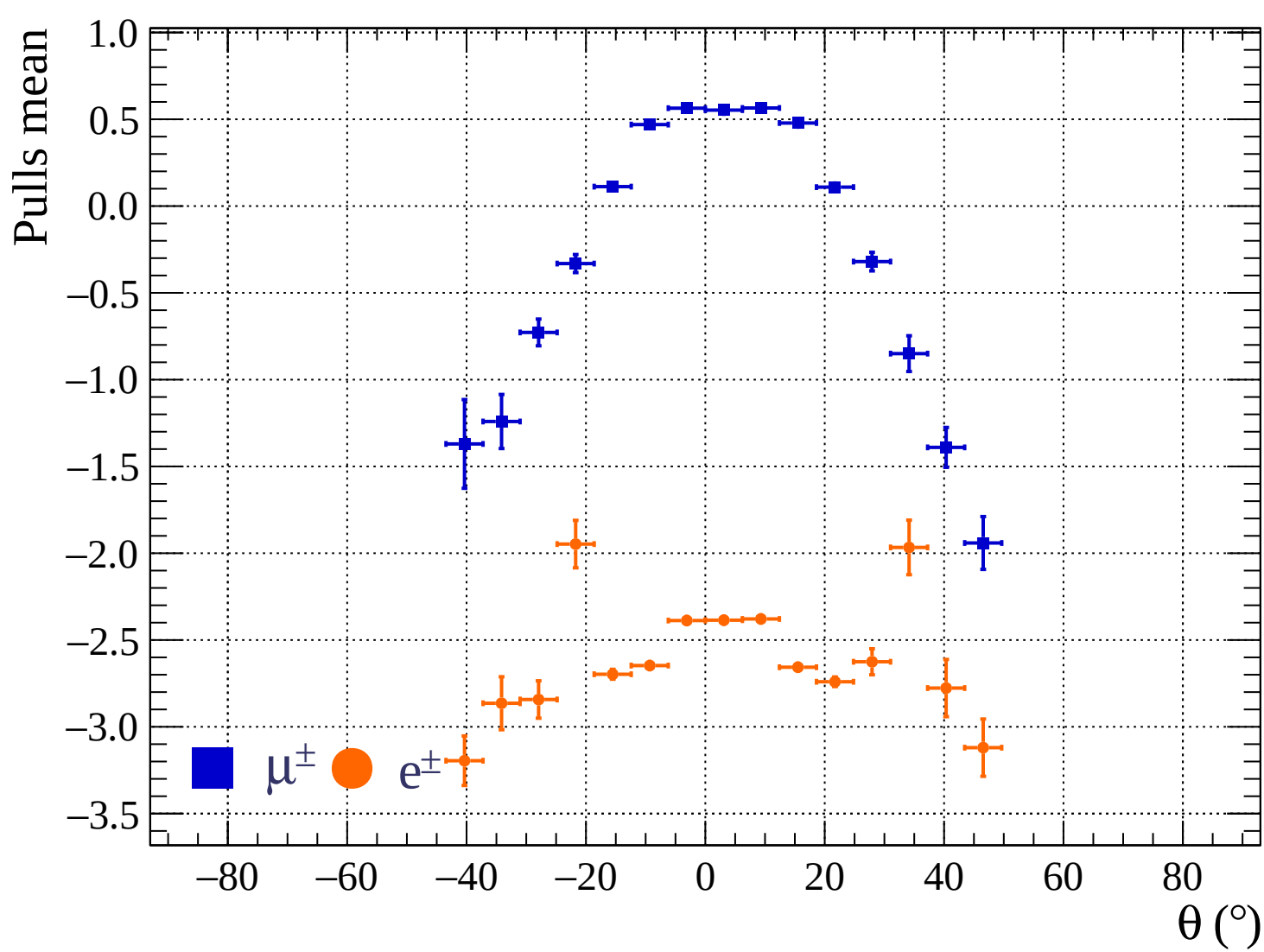




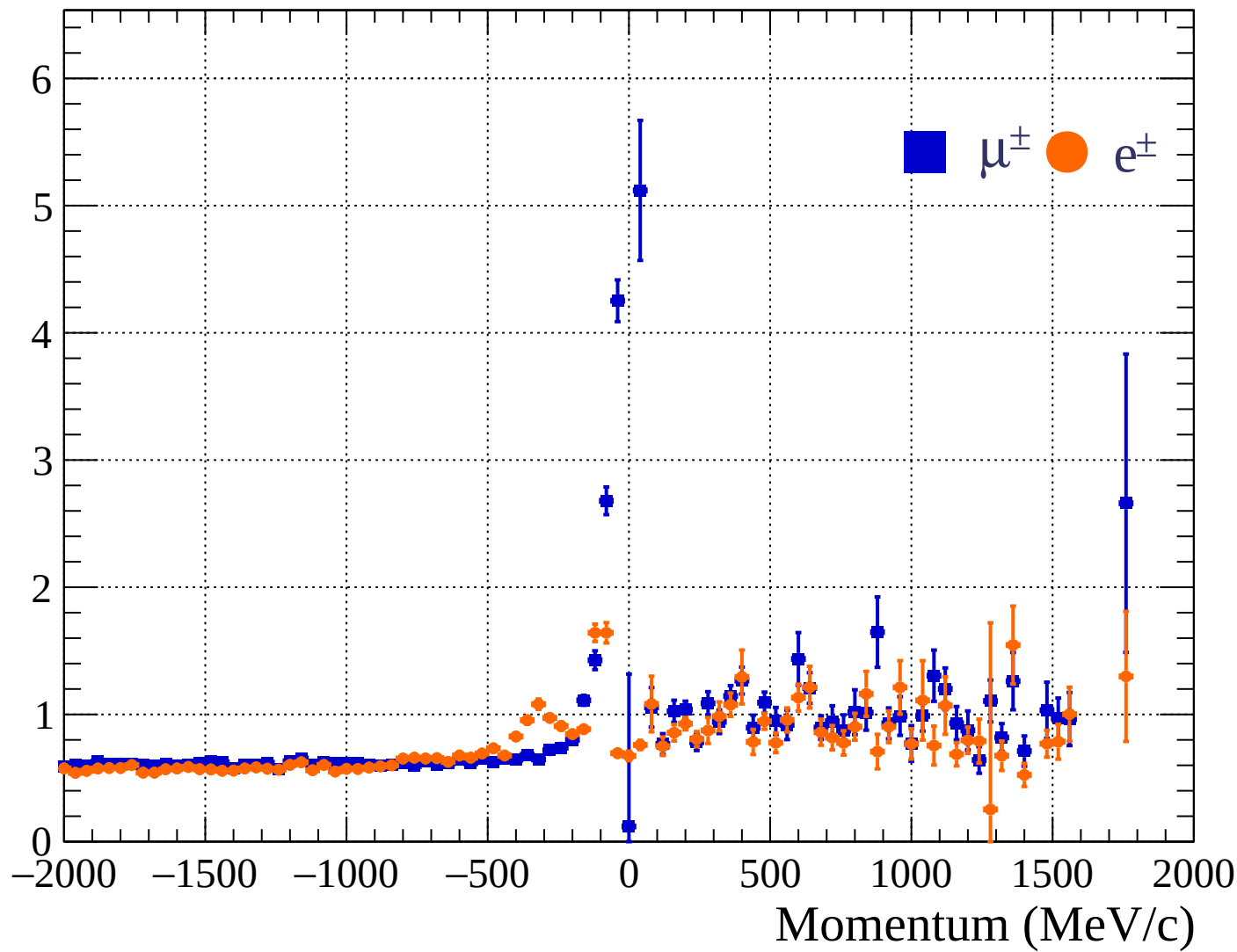


Pulls standard deviation

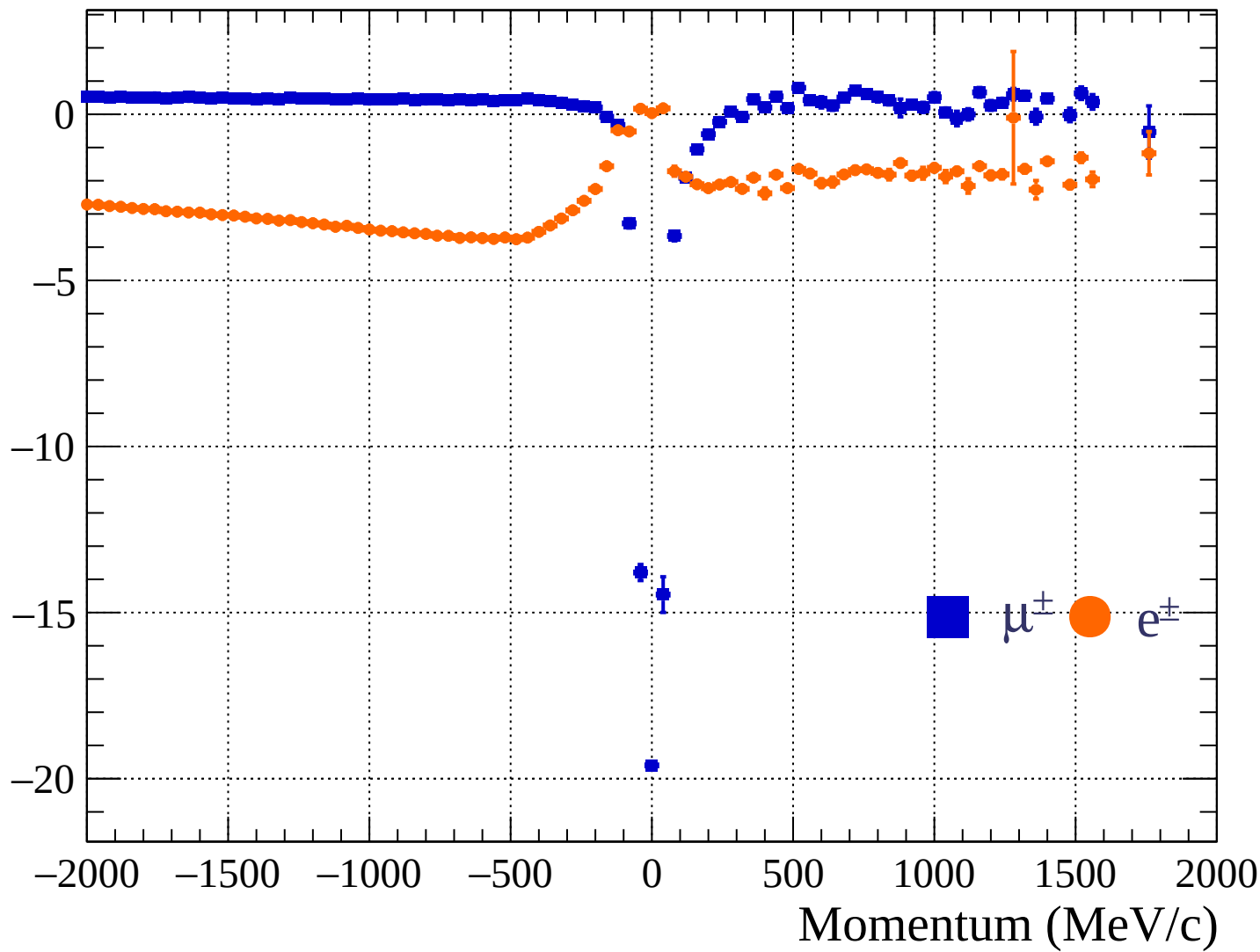


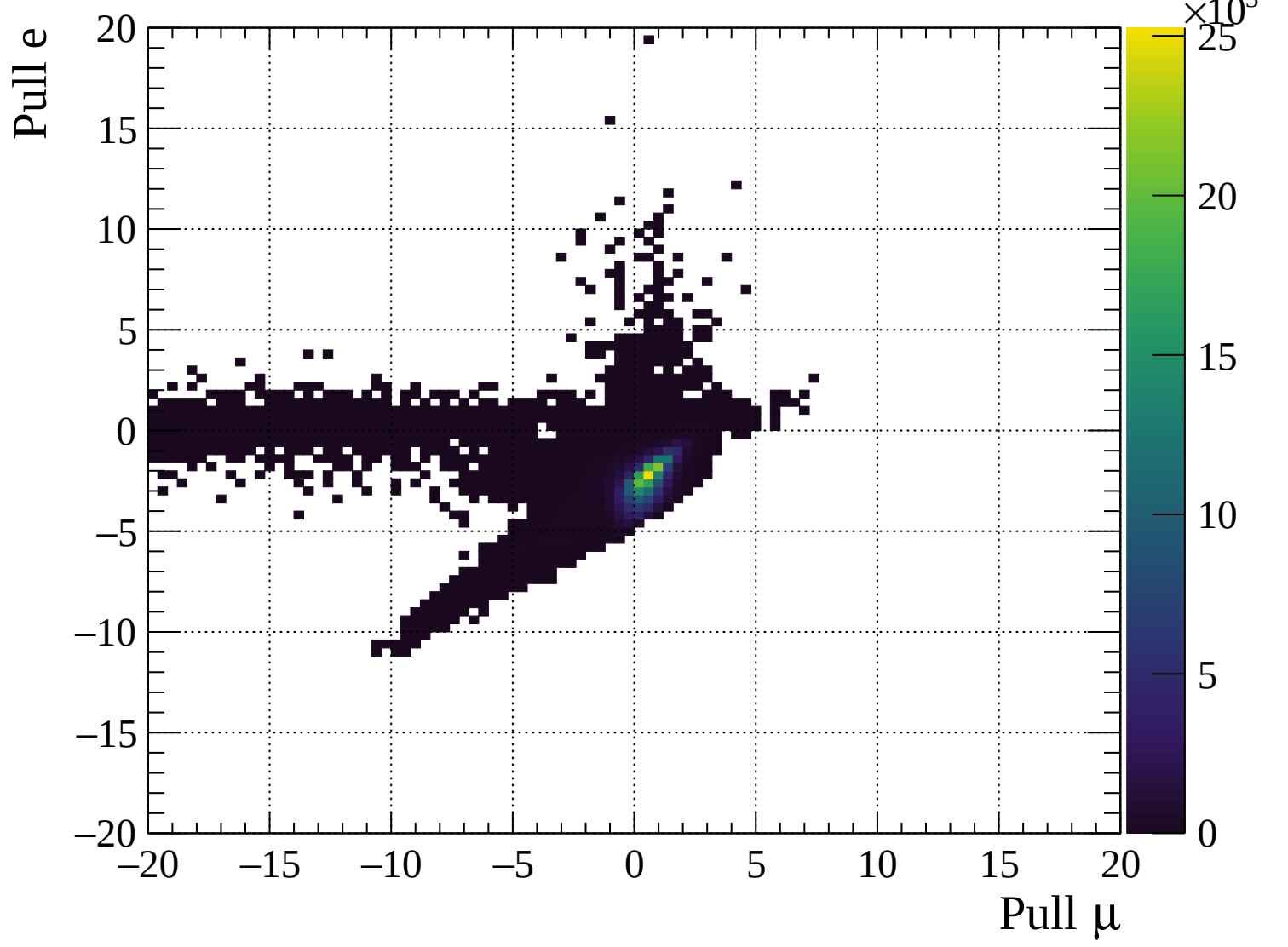


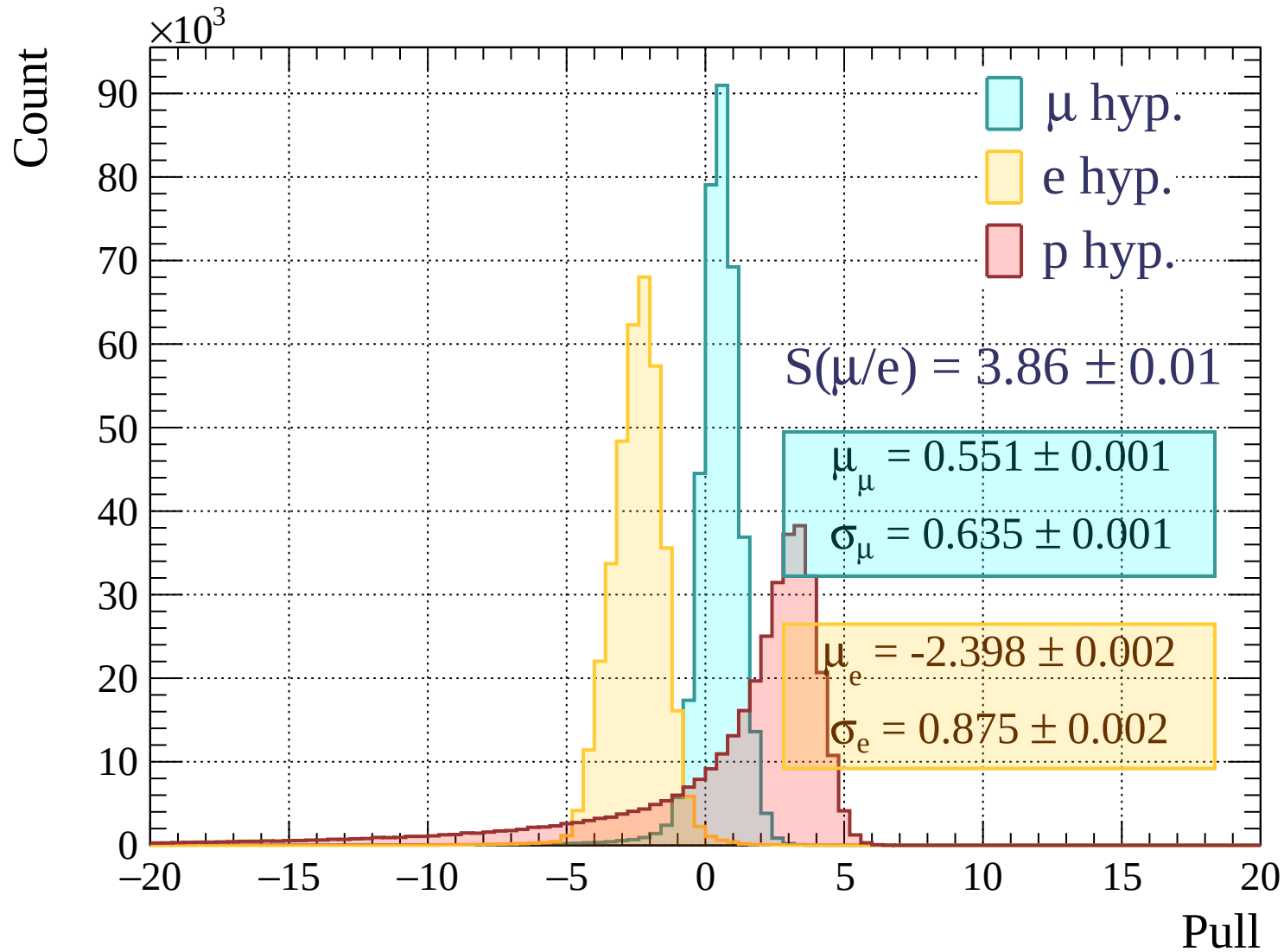
Pulls standard deviation

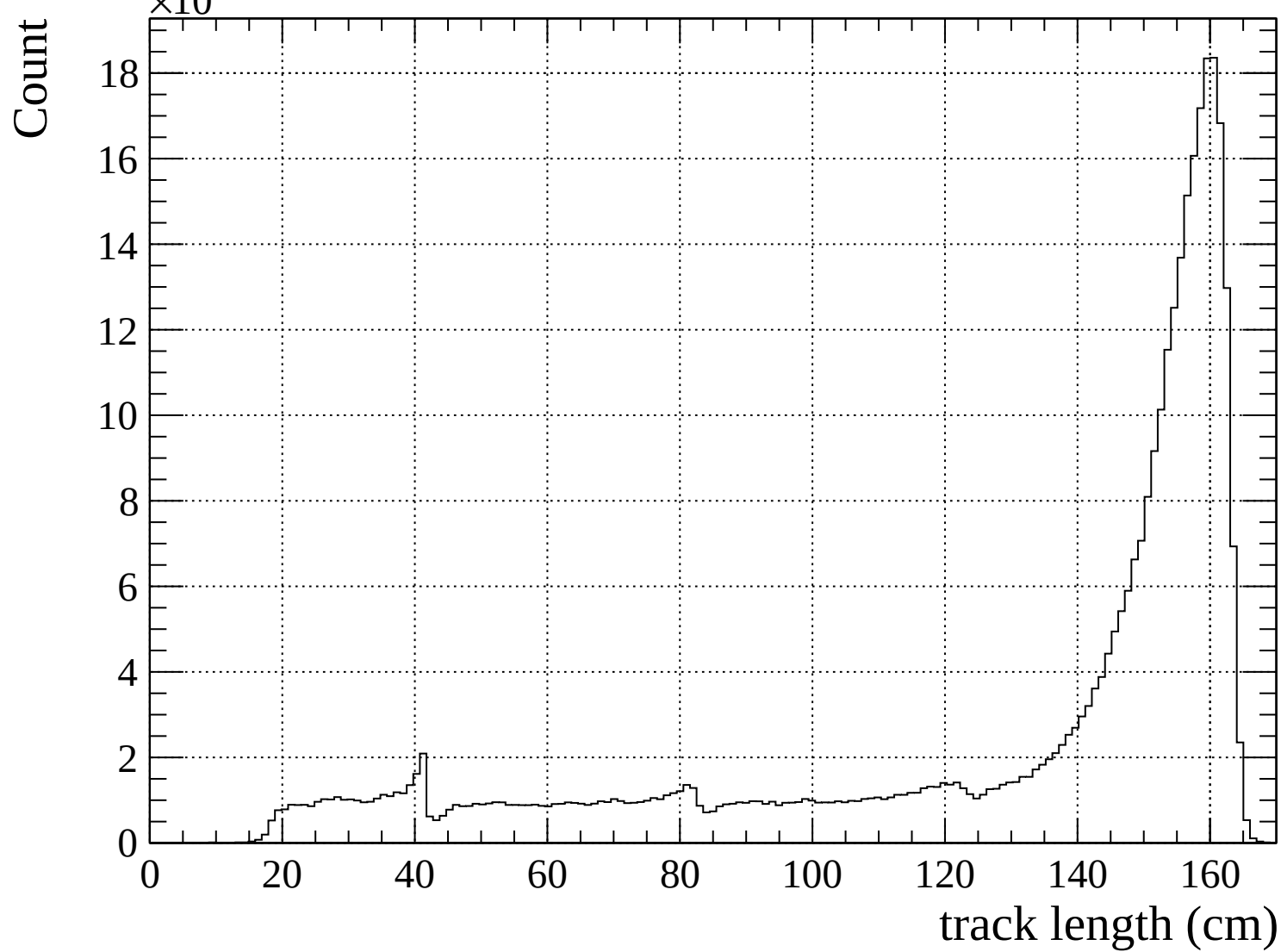


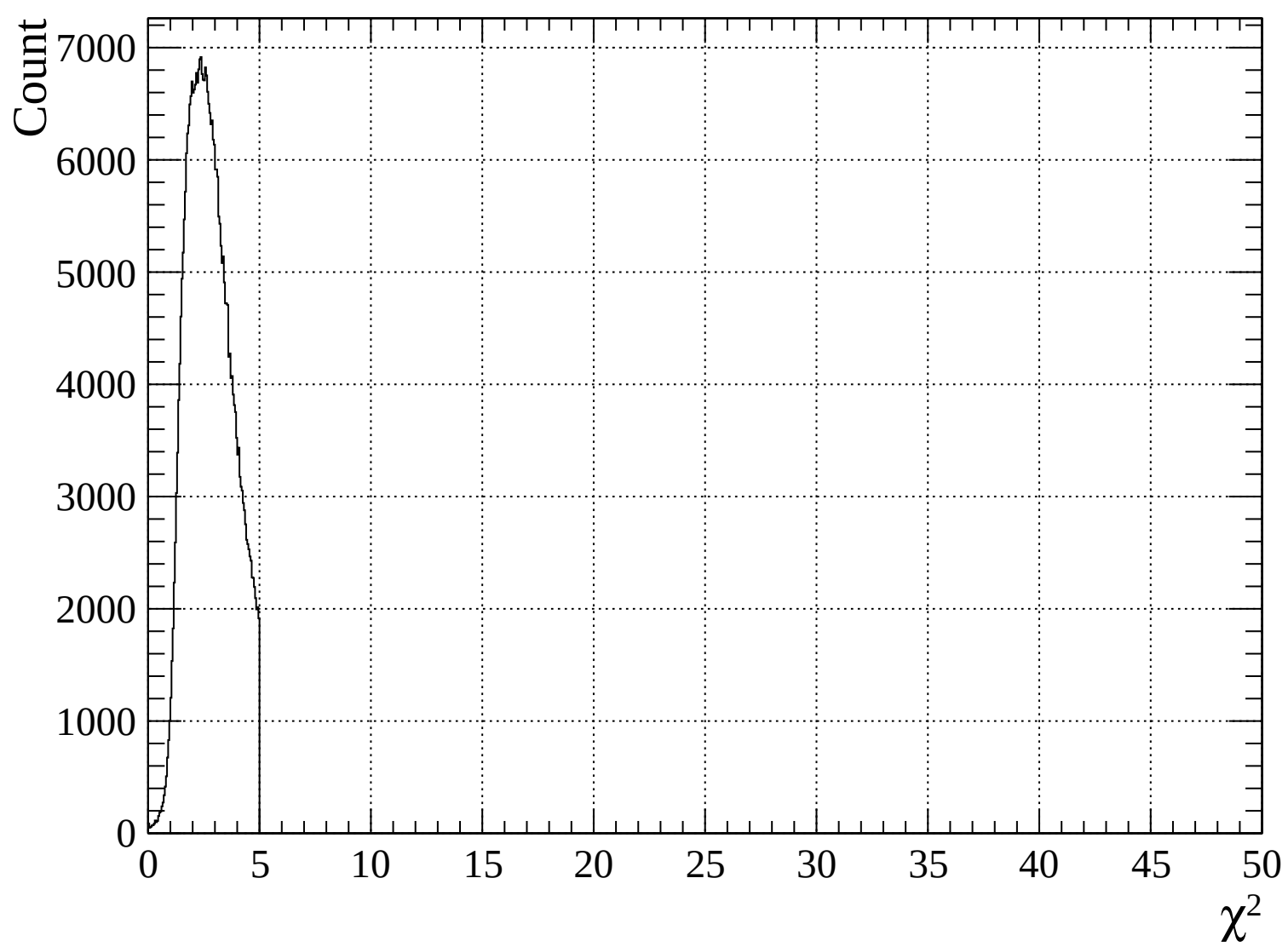
Pulls mean

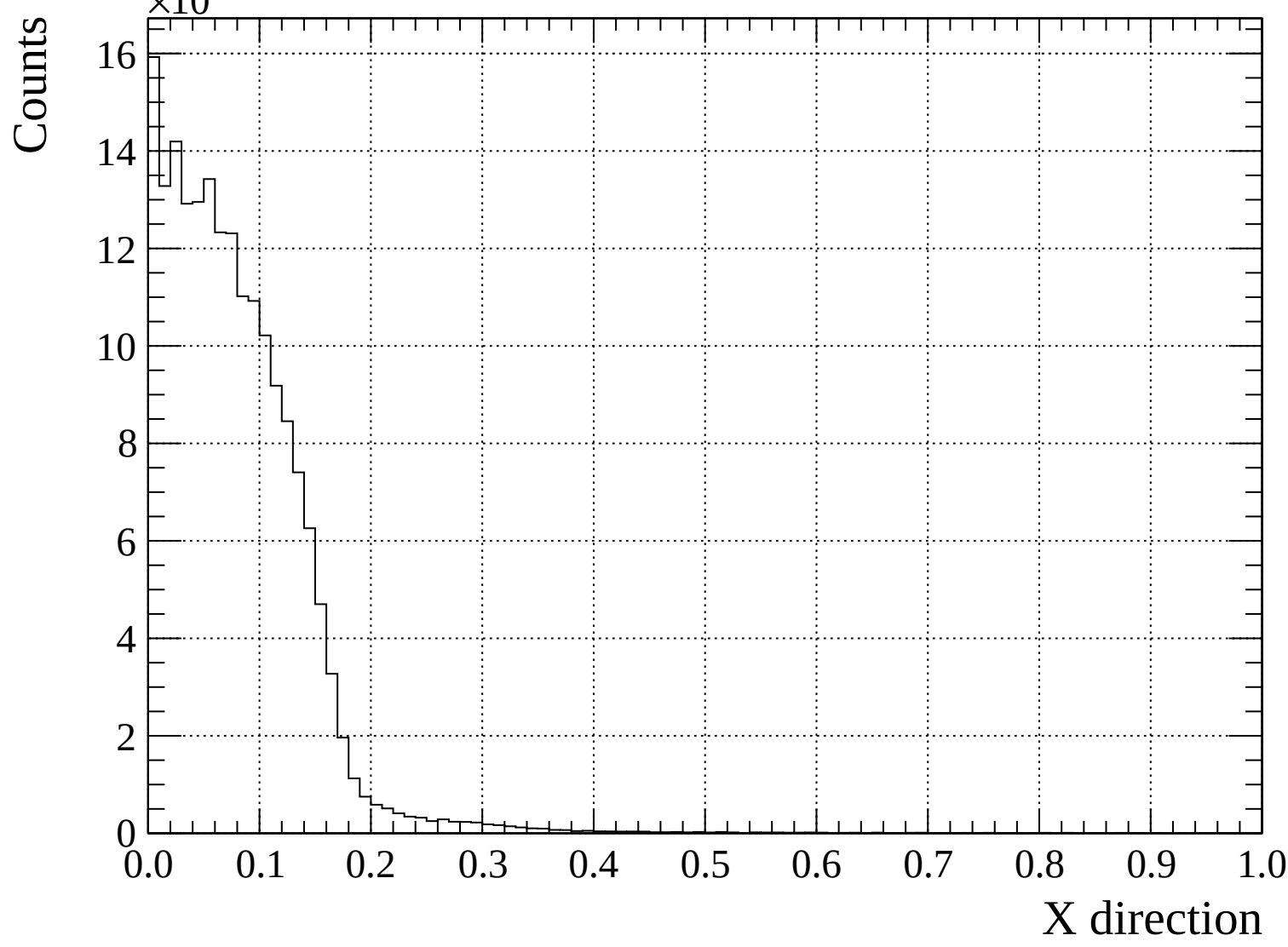


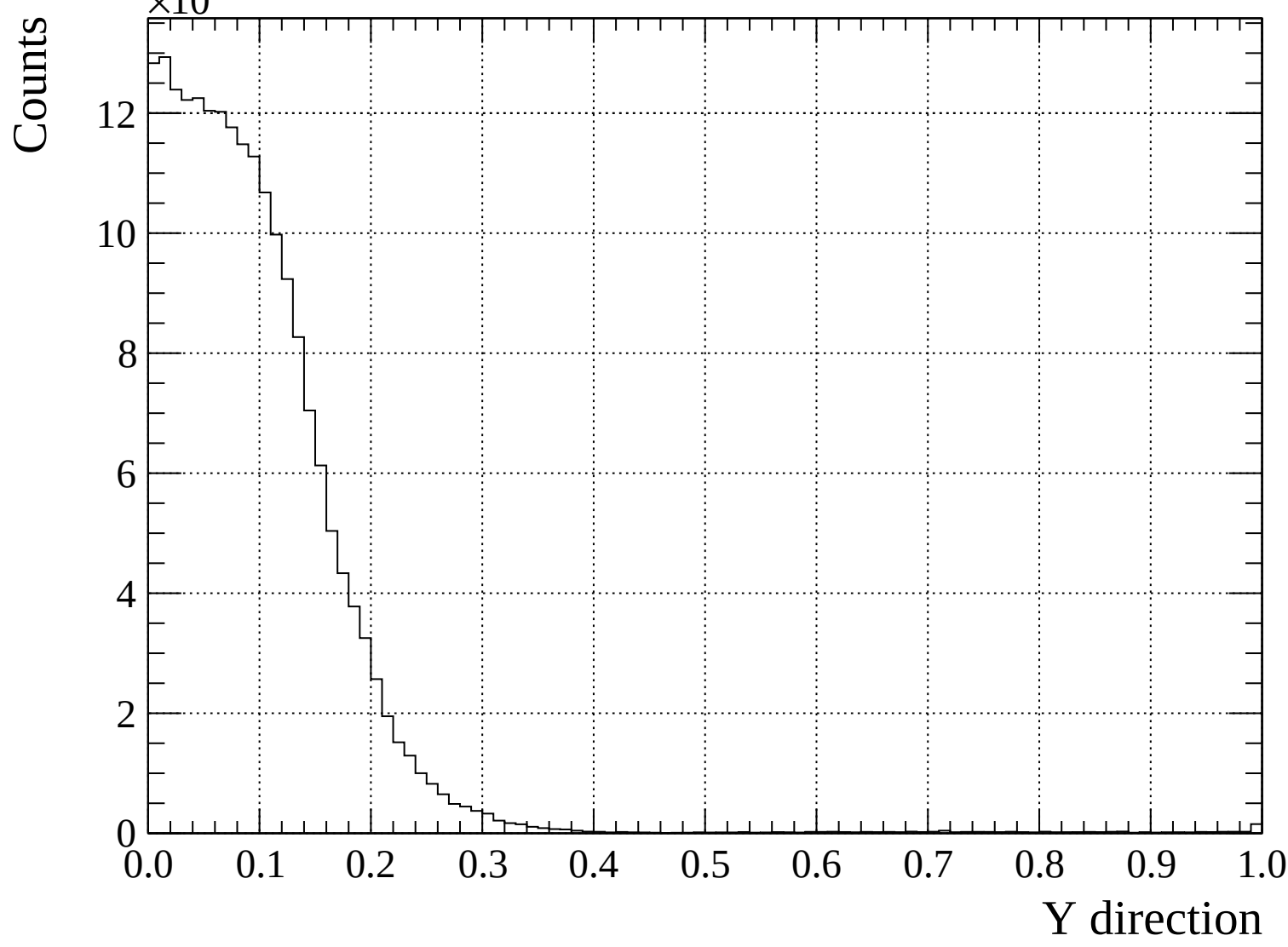


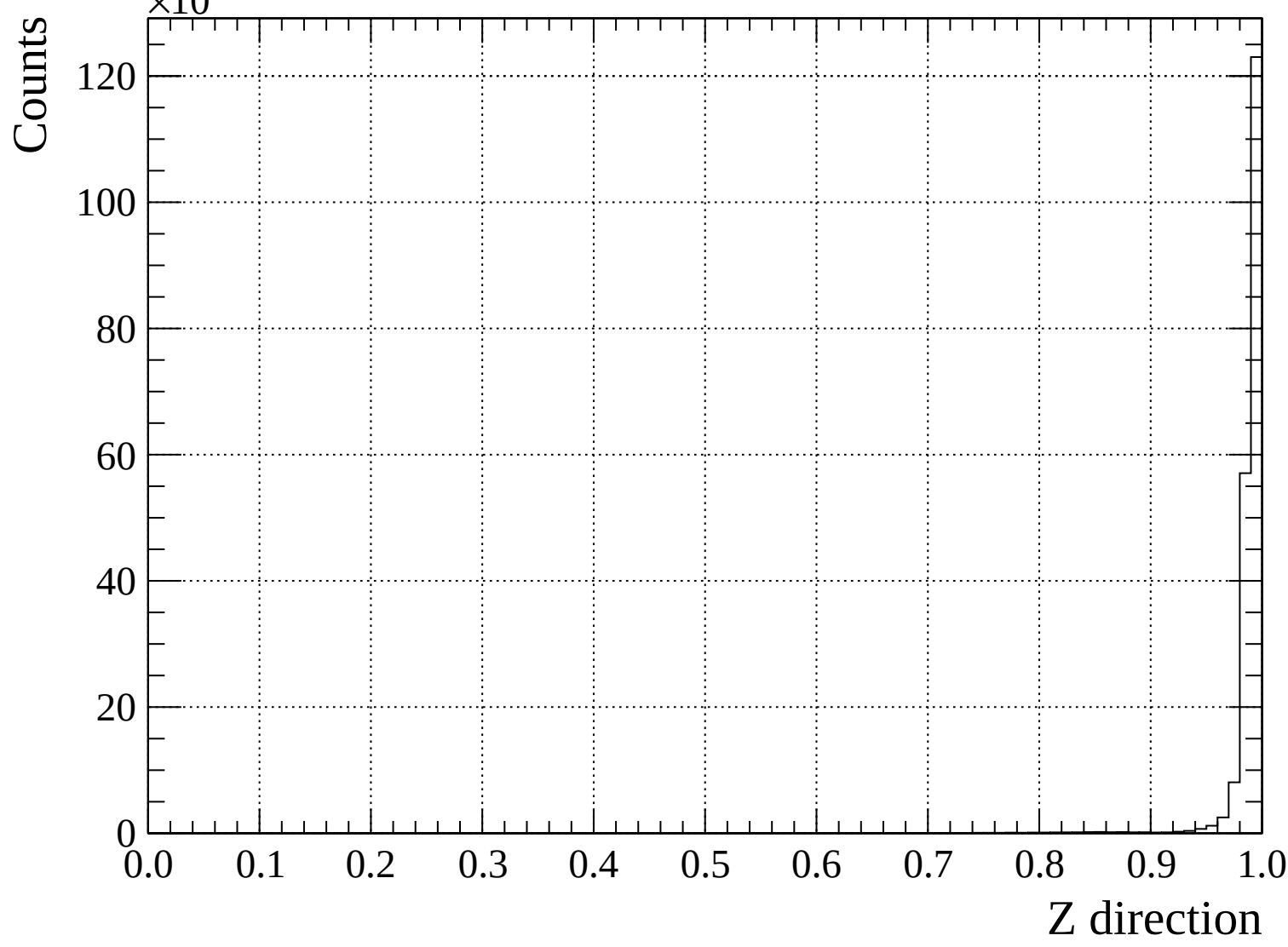


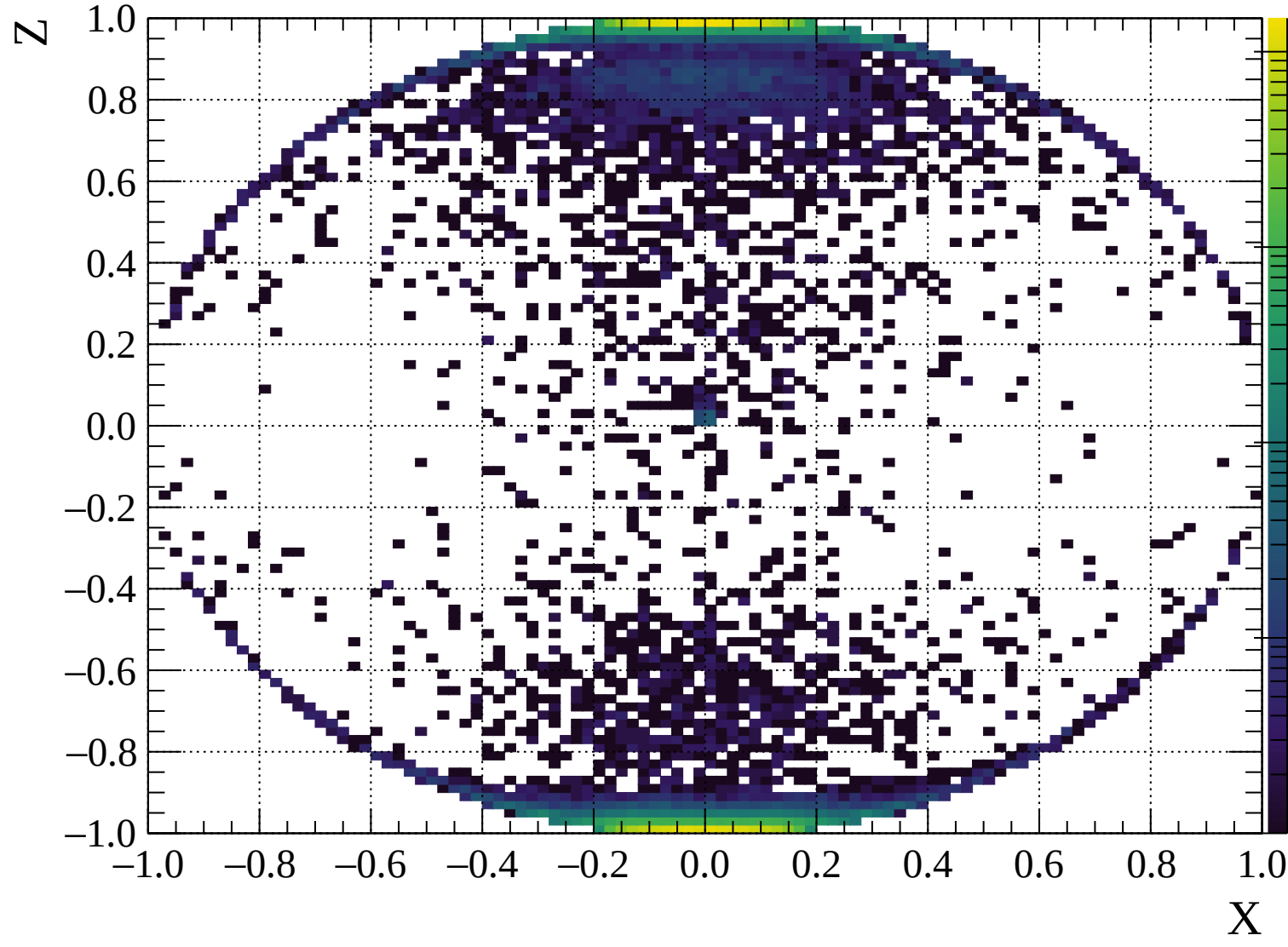


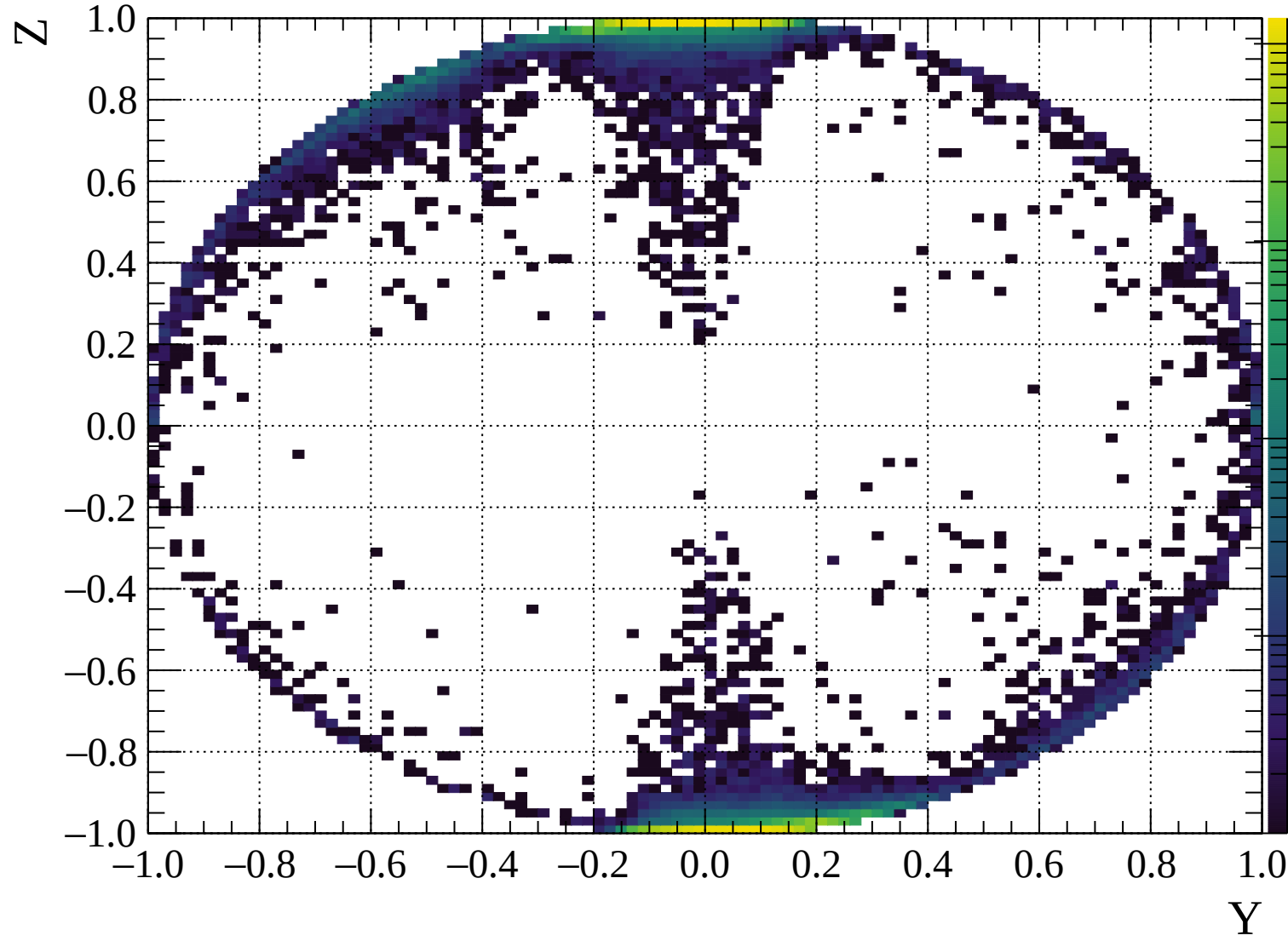


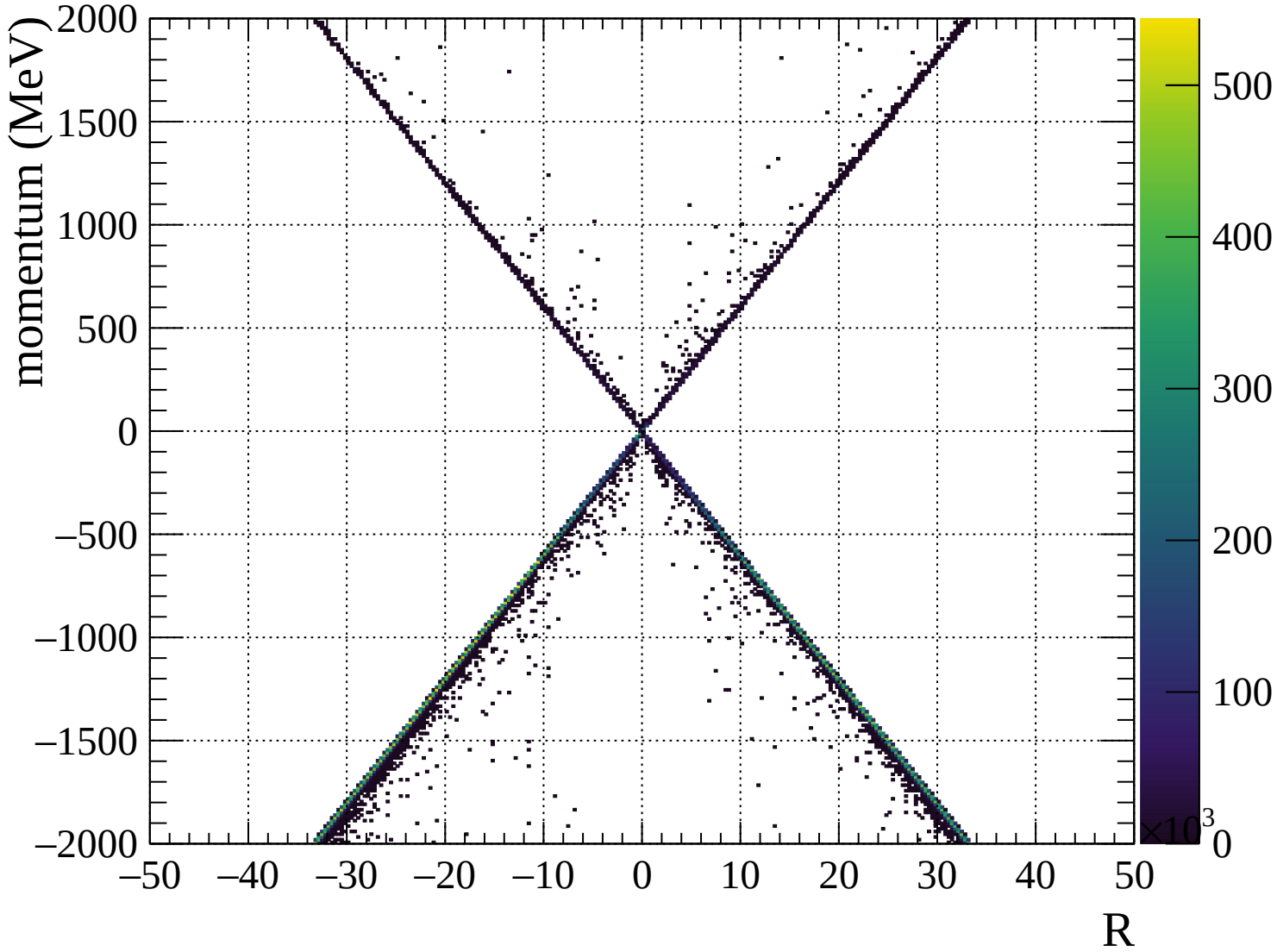


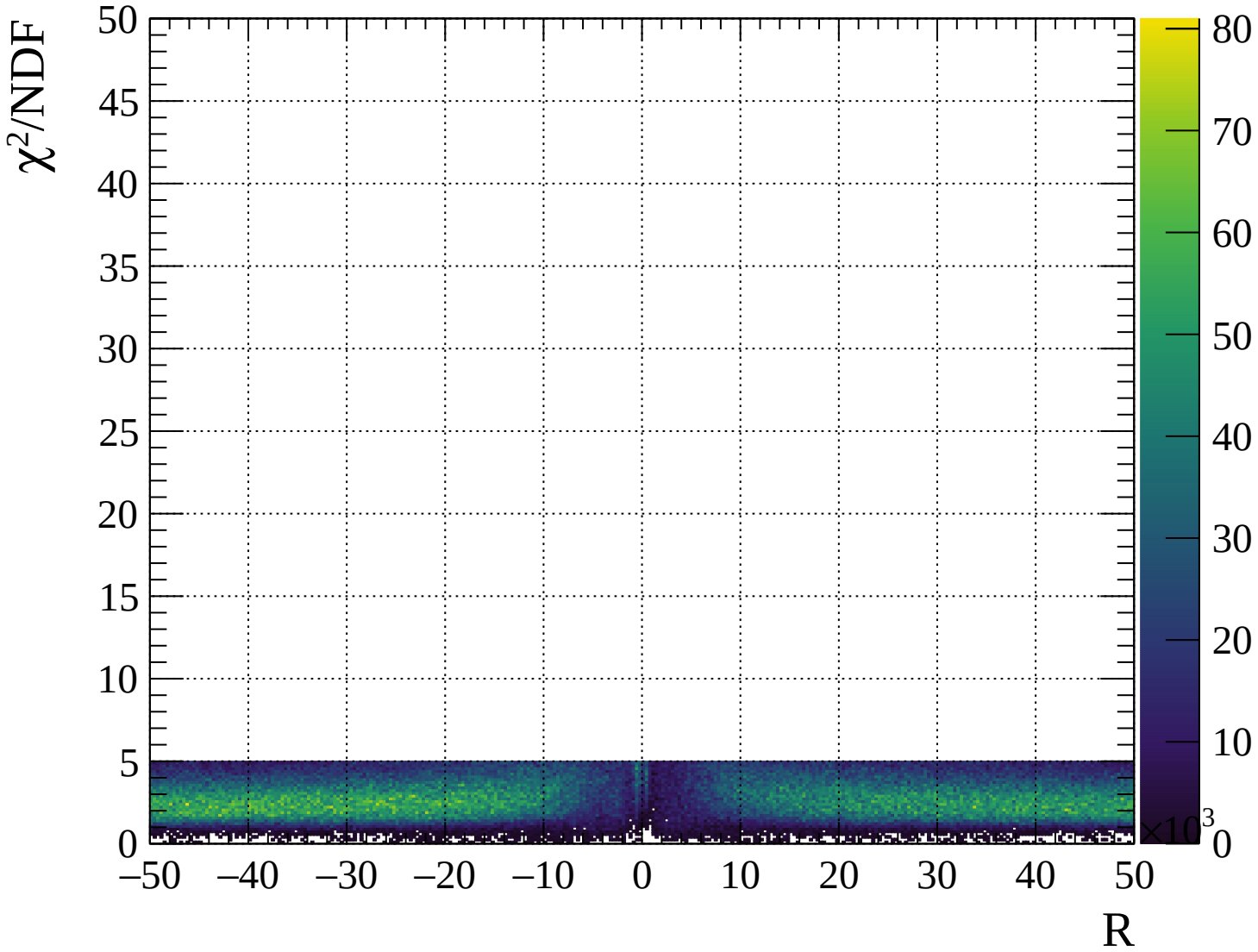




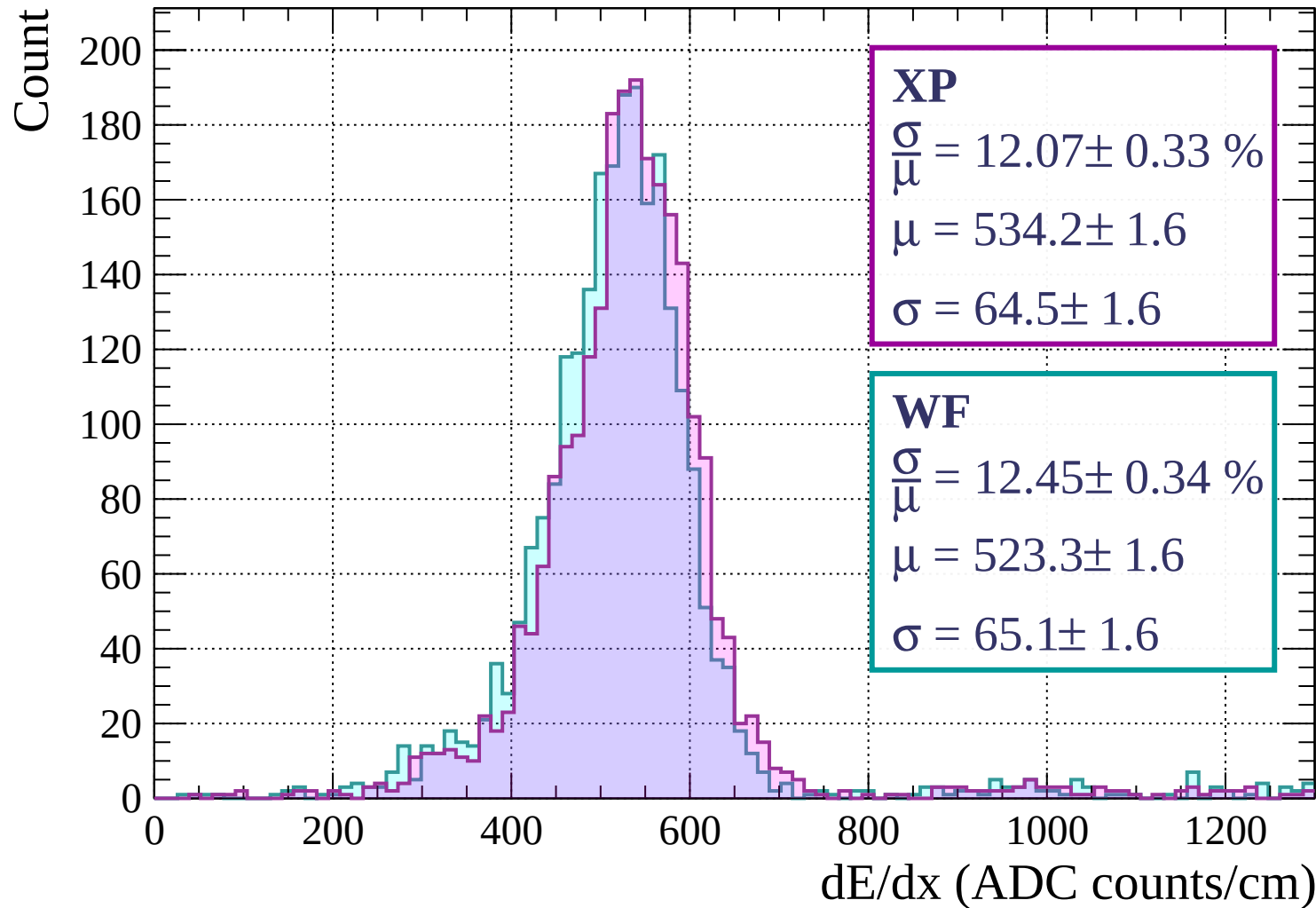






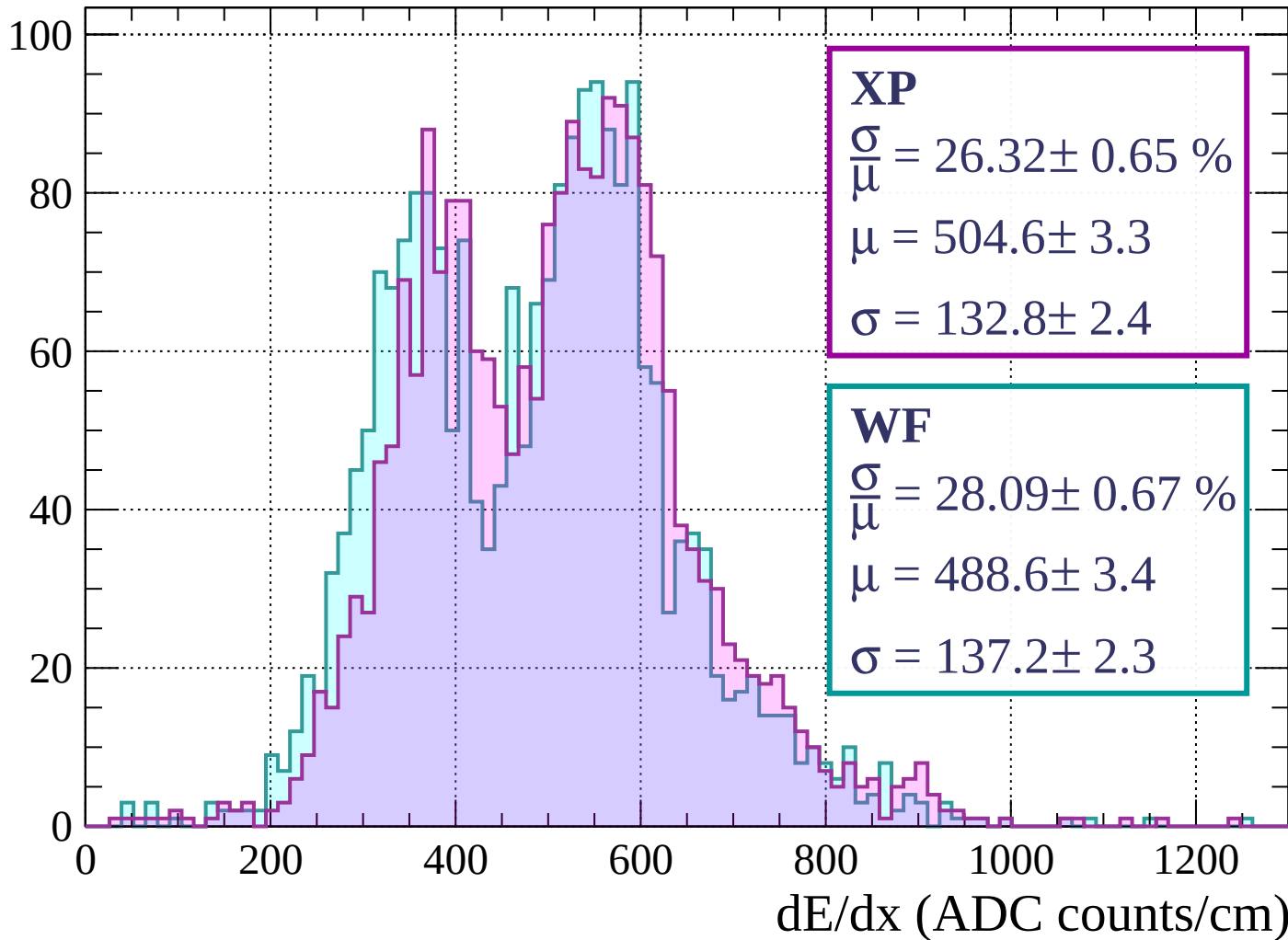


Energy loss | $0 < p < 80$



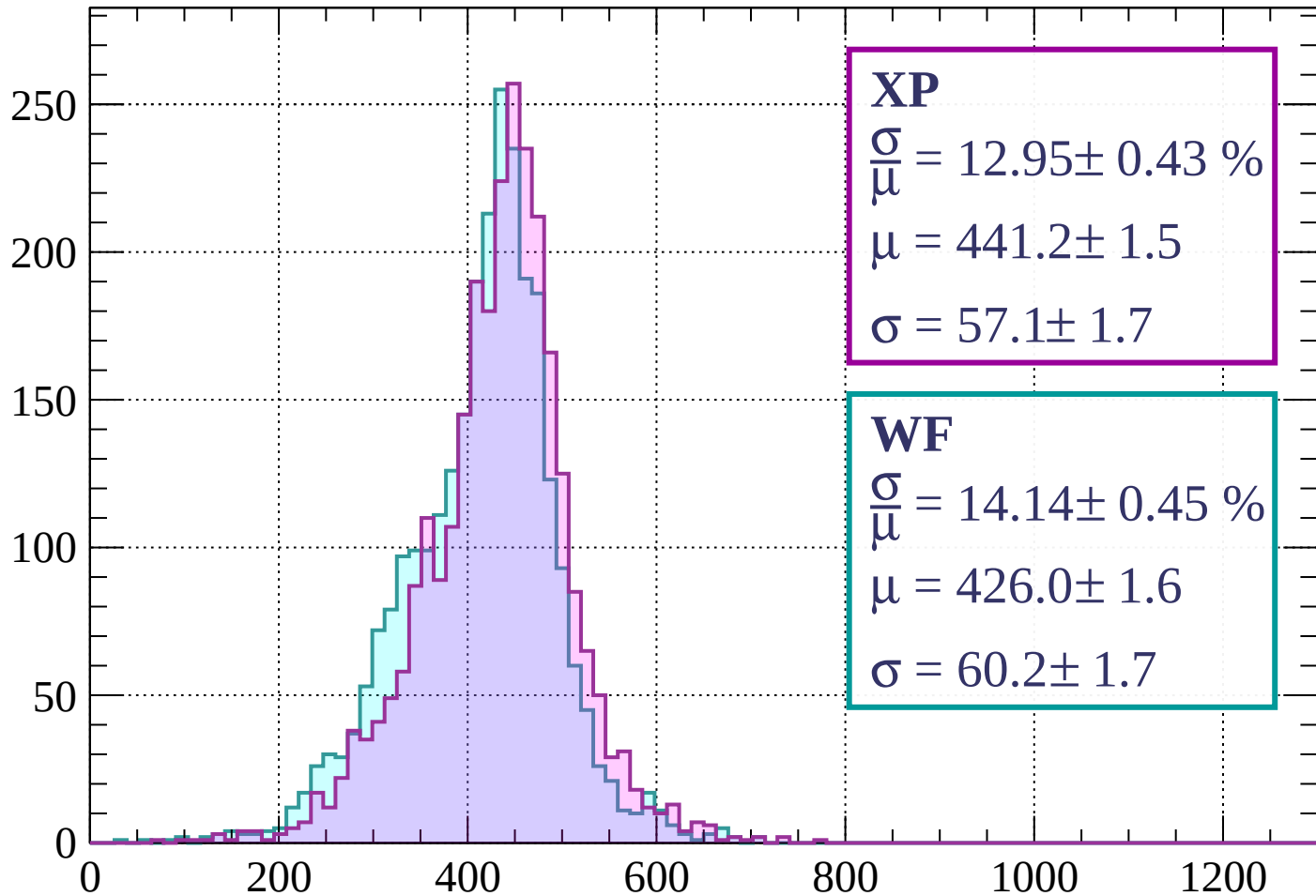
Energy loss | $80 < p < 160$

Count



Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 12.95 \pm 0.43 \%$$

$$\mu = 441.2 \pm 1.5$$

$$\sigma = 57.1 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 14.14 \pm 0.45 \%$$

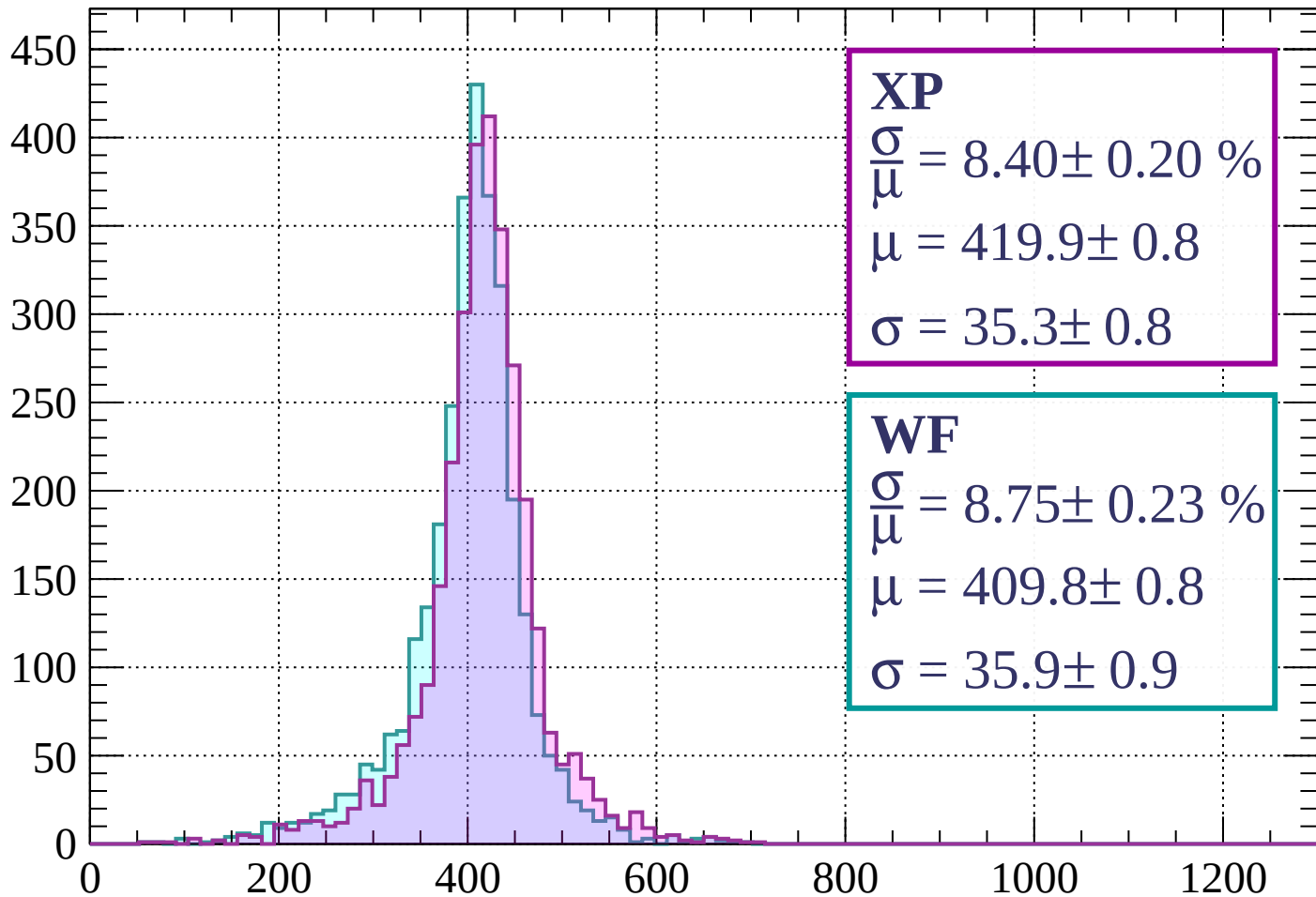
$$\mu = 426.0 \pm 1.6$$

$$\sigma = 60.2 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 8.40 \pm 0.20 \%$$

$$\mu = 419.9 \pm 0.8$$

$$\sigma = 35.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.75 \pm 0.23 \%$$

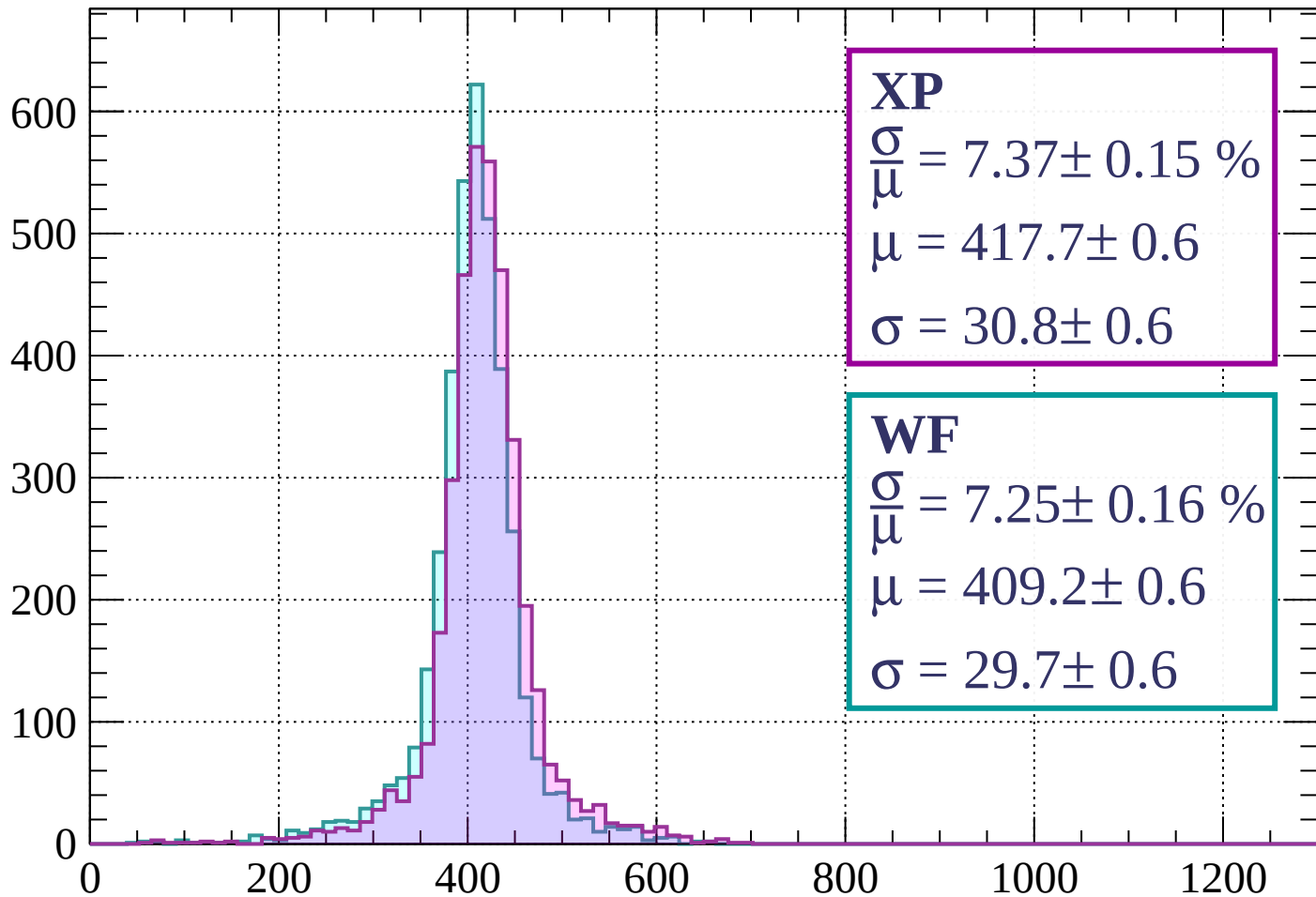
$$\mu = 409.8 \pm 0.8$$

$$\sigma = 35.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$

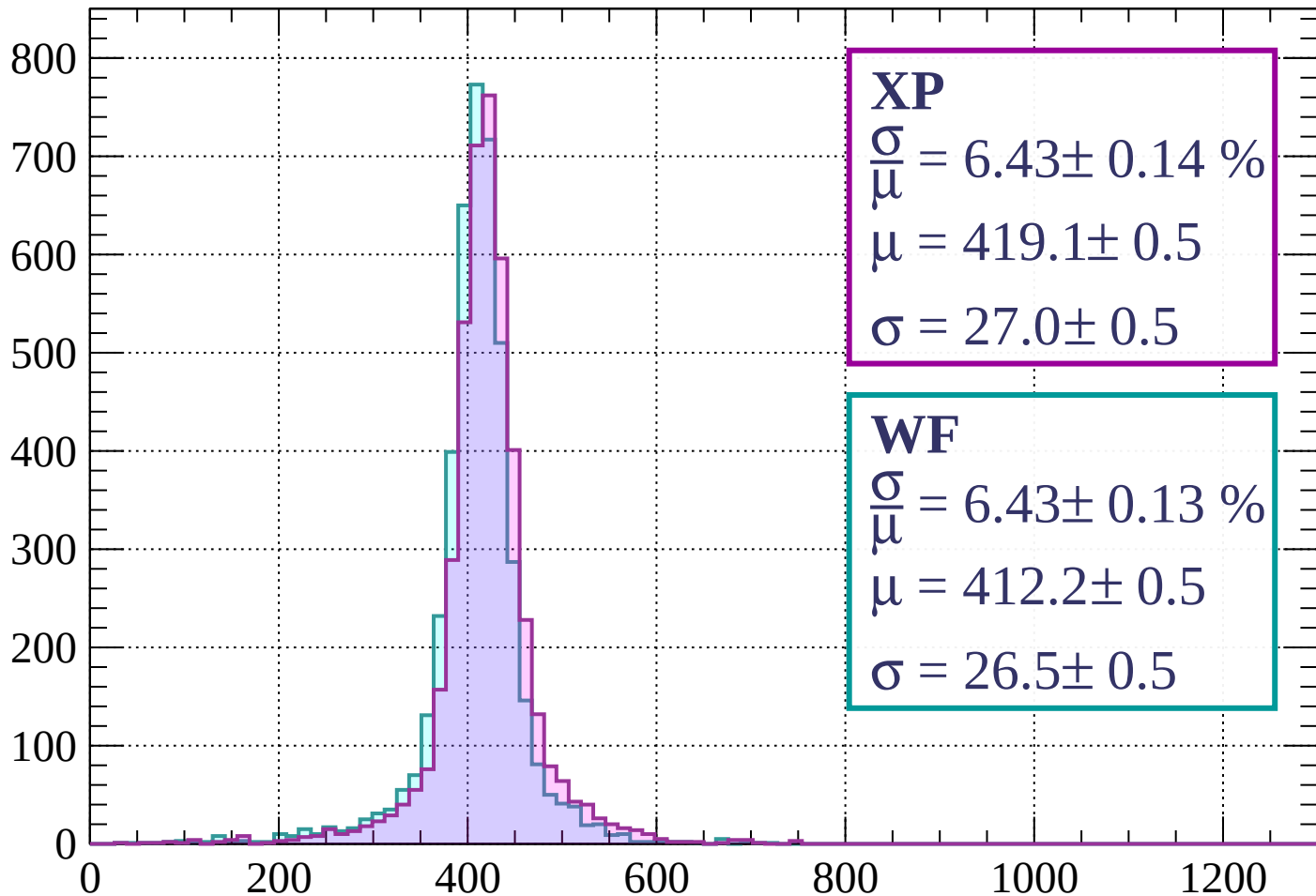
Count



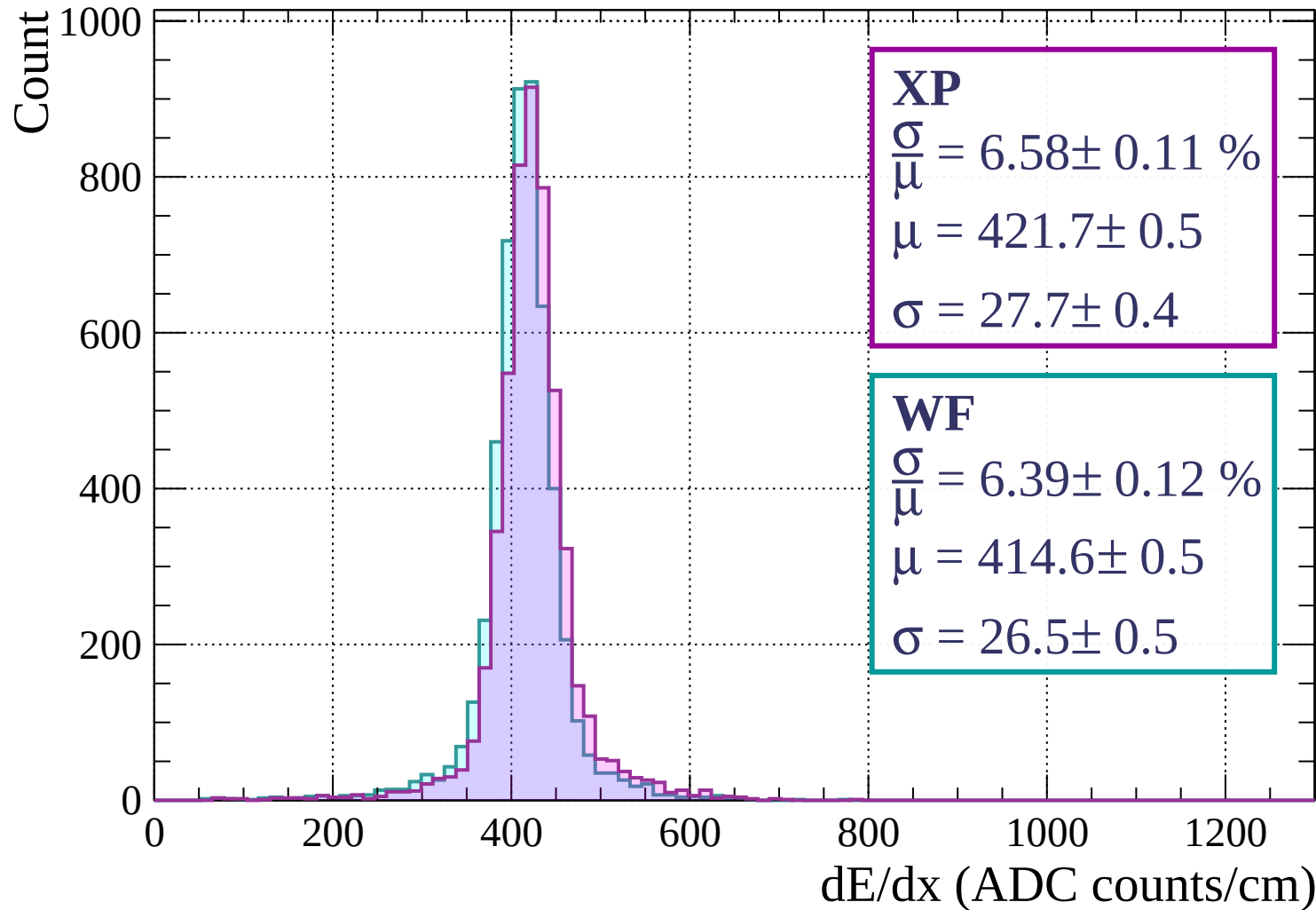
dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$

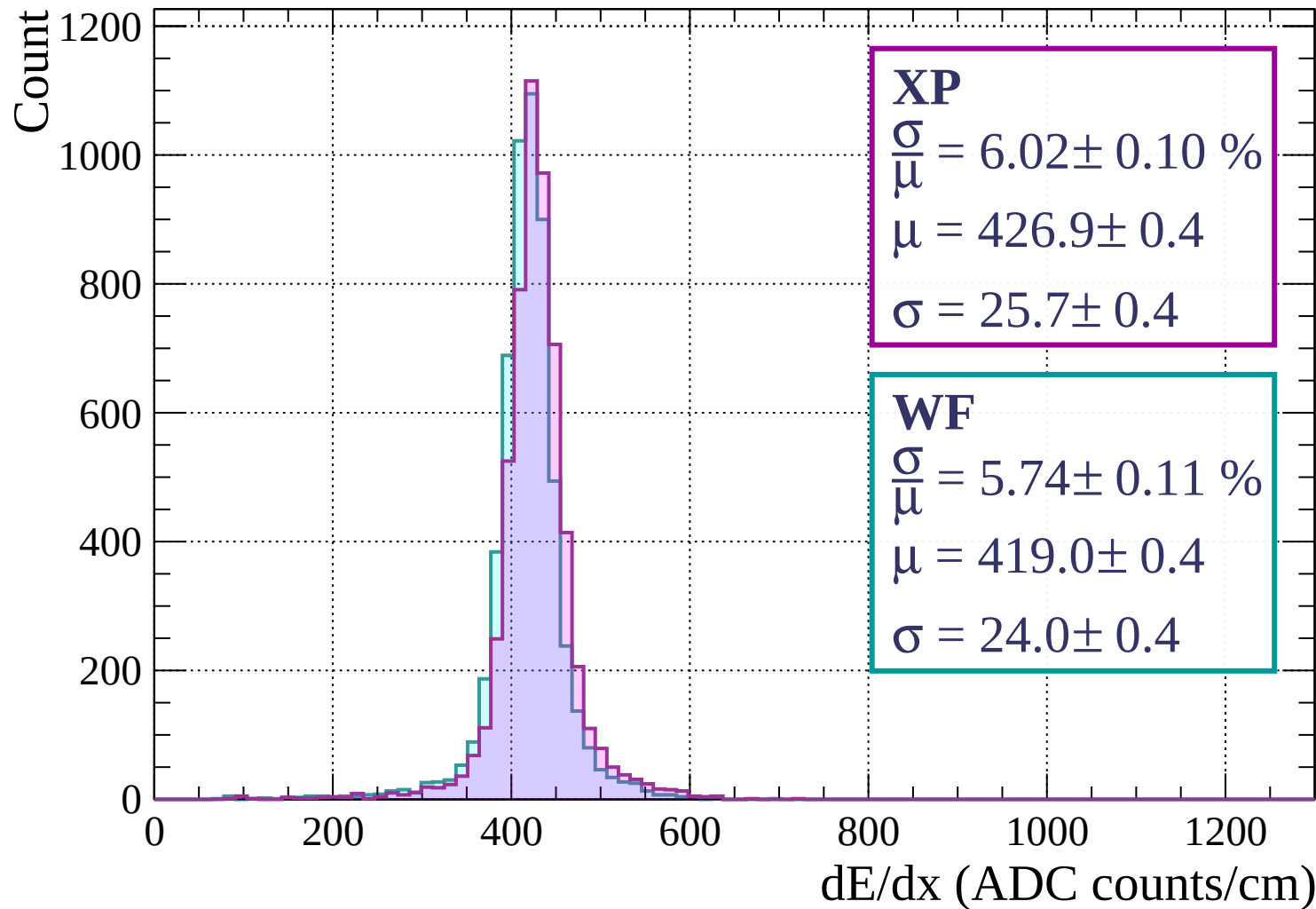
Count



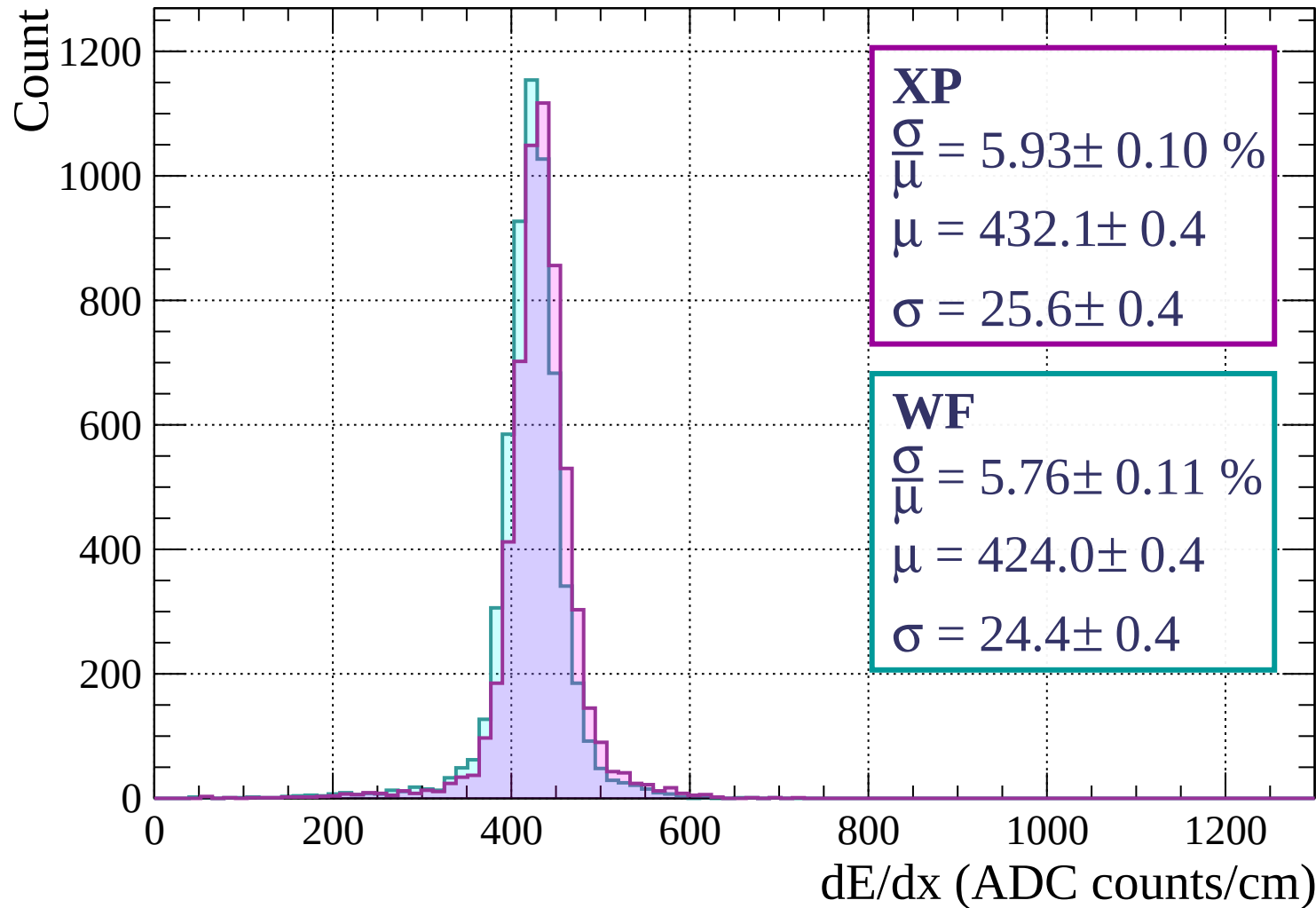
Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$



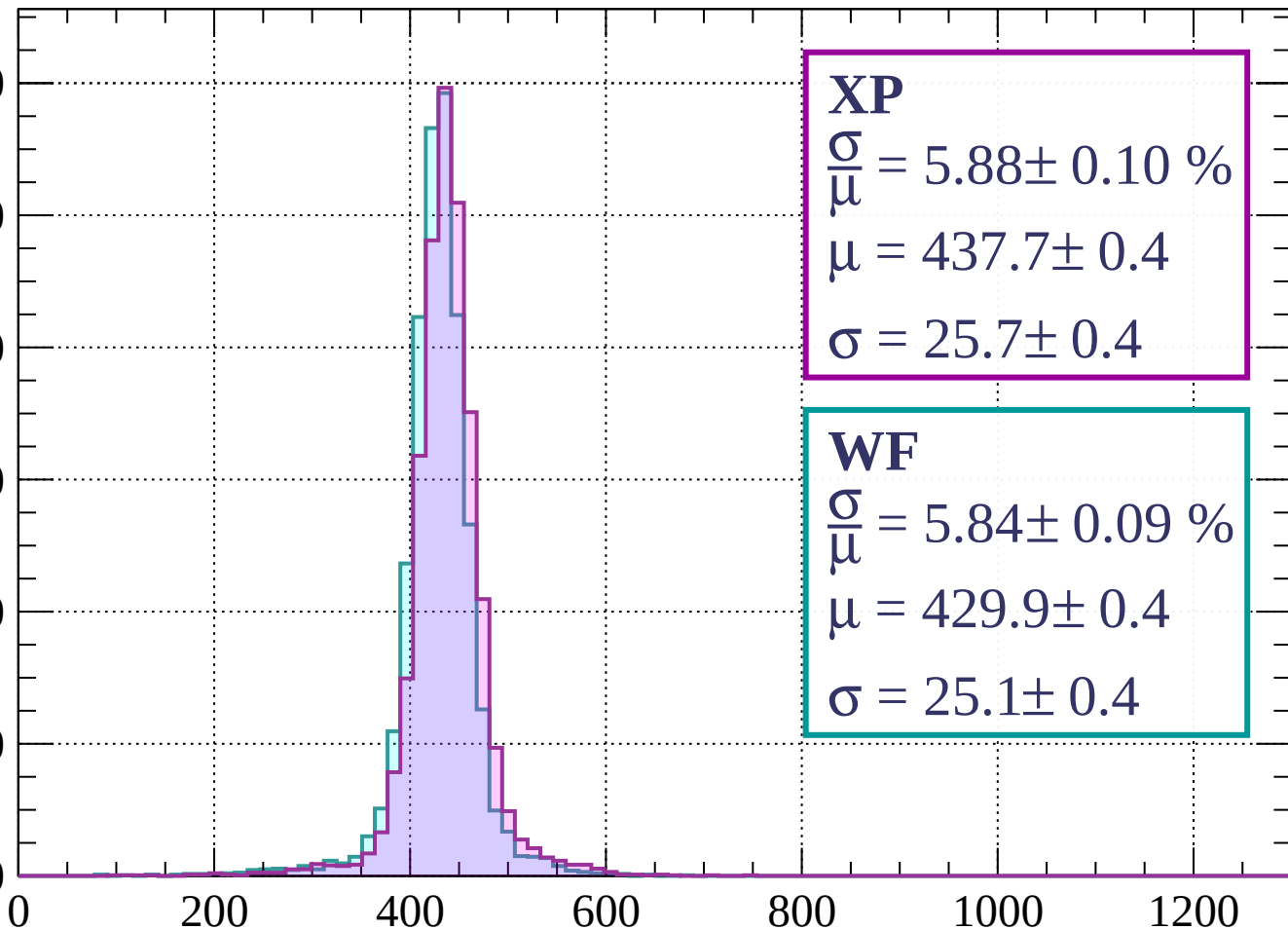
Energy loss | $640 < p < 720$



Energy loss | $720 < p < 800$

Count

1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.88 \pm 0.10 \%$$

$$\mu = 437.7 \pm 0.4$$

$$\sigma = 25.7 \pm 0.4$$

WF

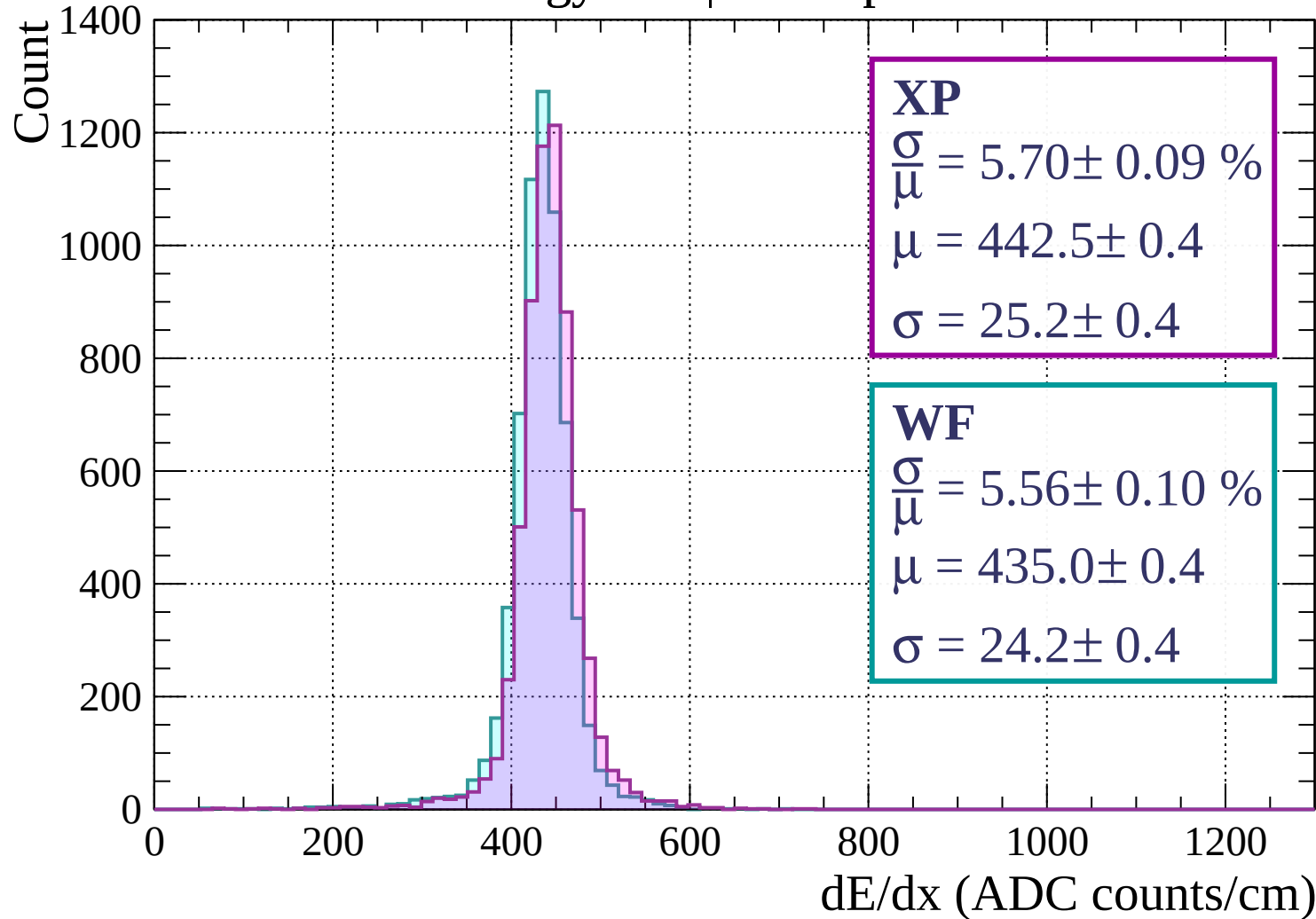
$$\frac{\sigma}{\mu} = 5.84 \pm 0.09 \%$$

$$\mu = 429.9 \pm 0.4$$

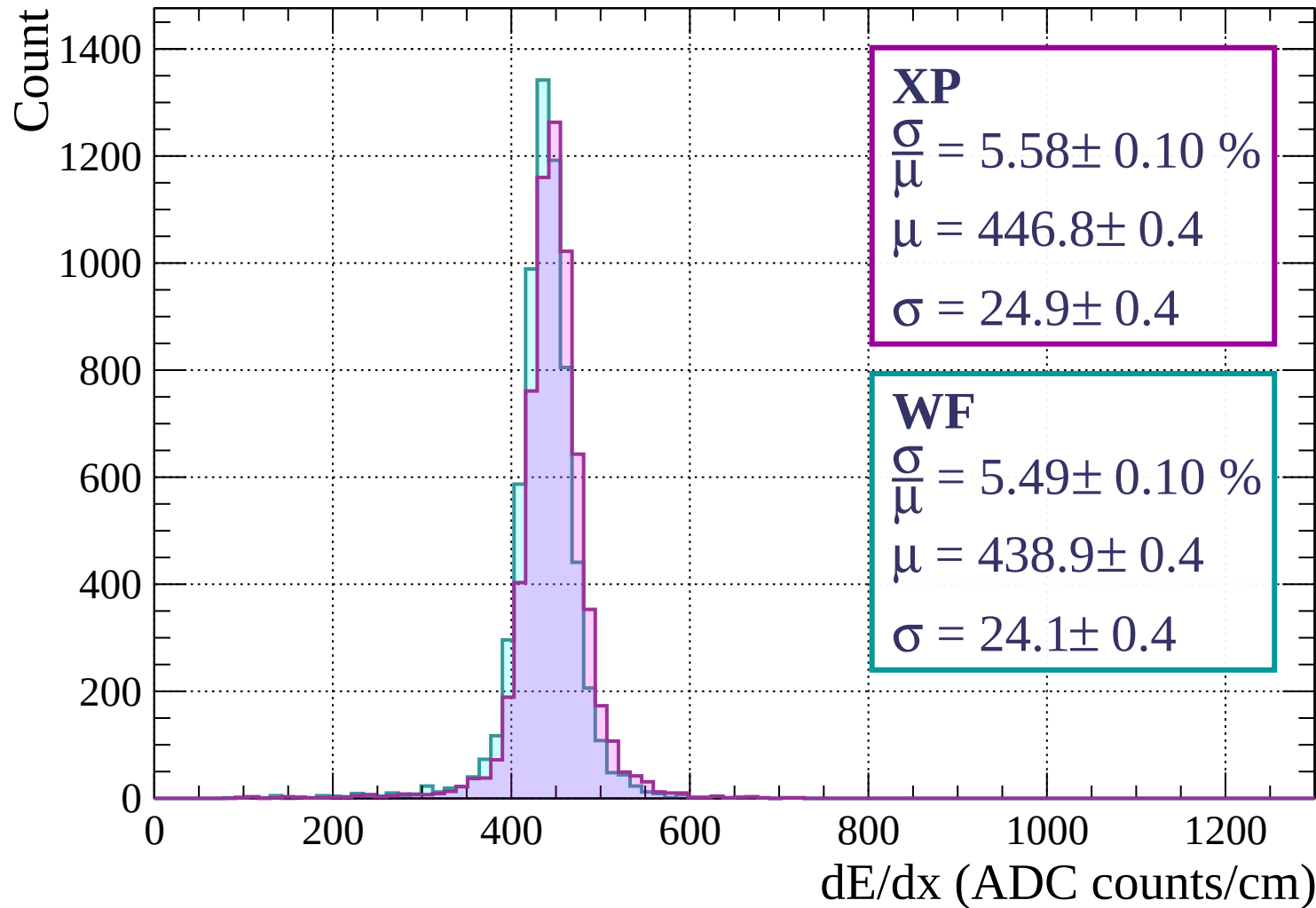
$$\sigma = 25.1 \pm 0.4$$

dE/dx (ADC counts/cm)

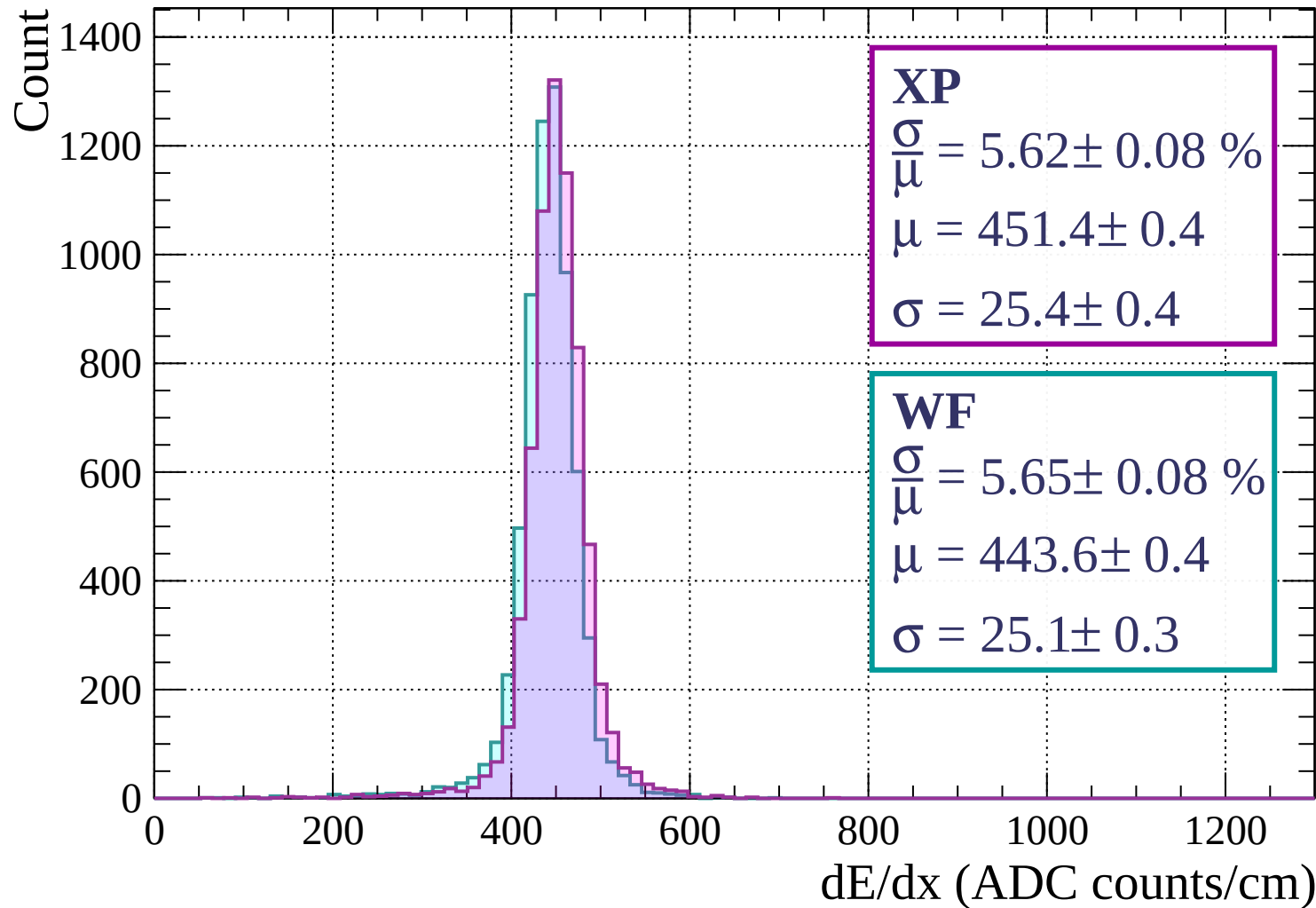
Energy loss | $800 < p < 880$



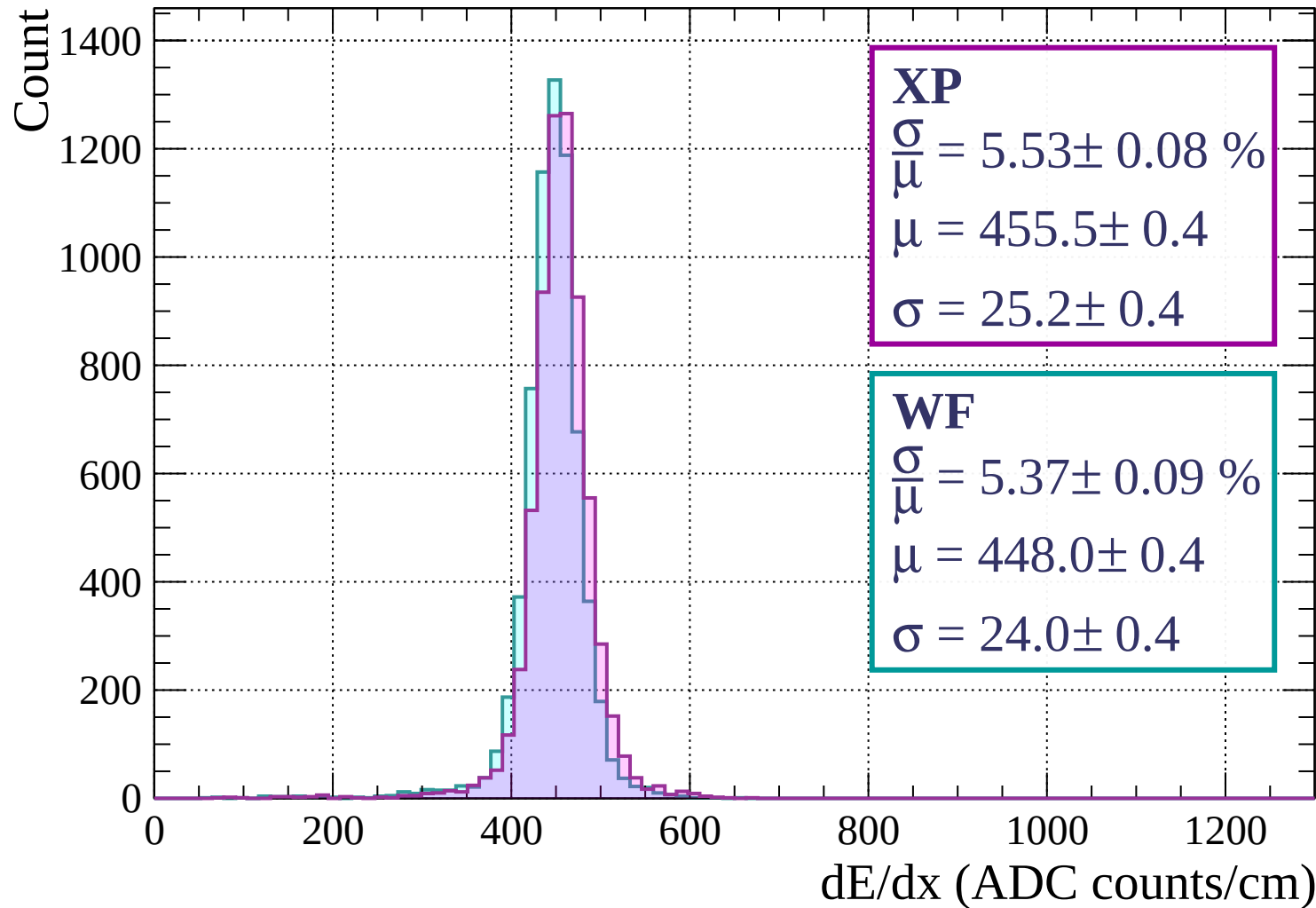
Energy loss | $880 < p < 960$



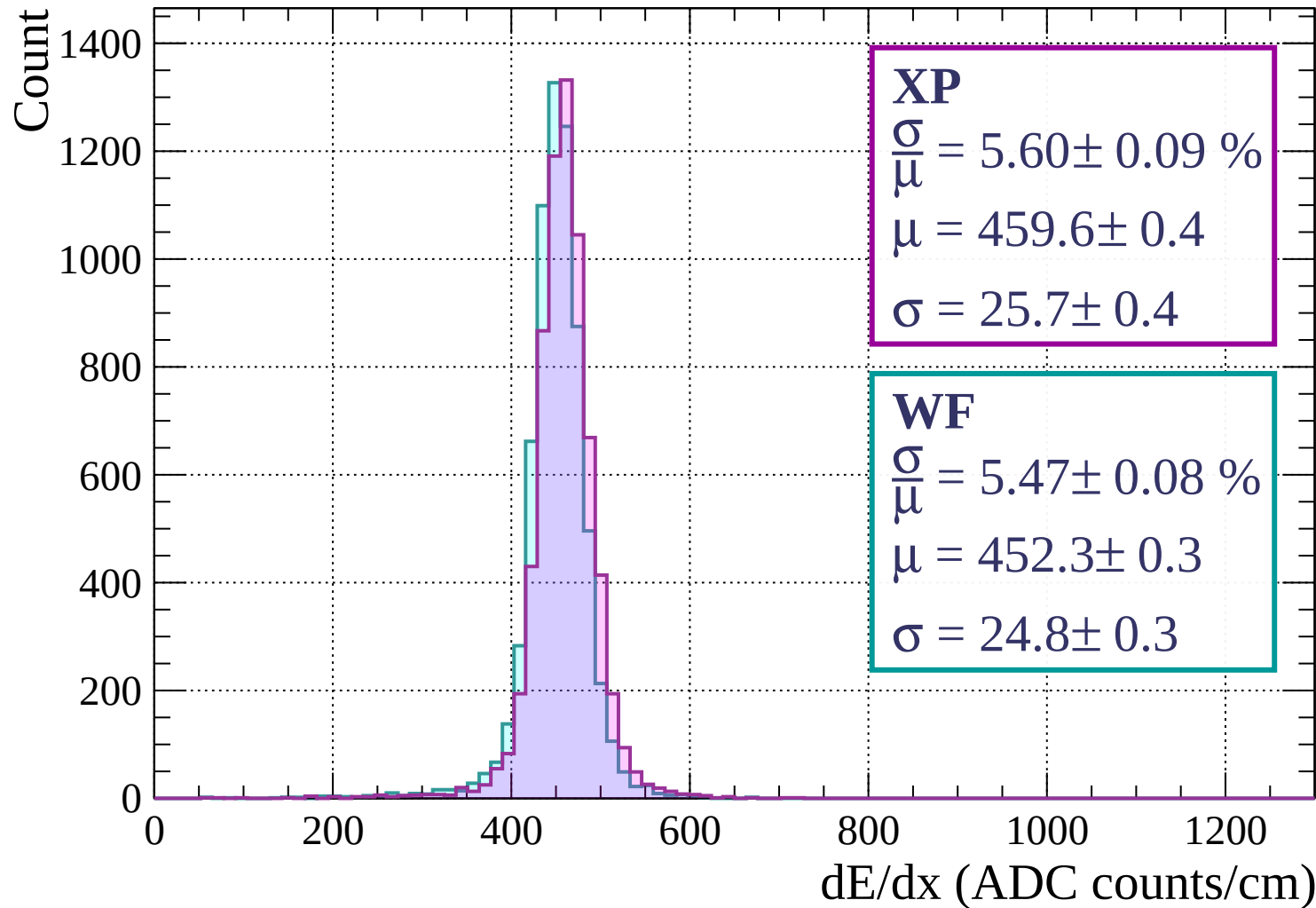
Energy loss | $960 < p < 1040$



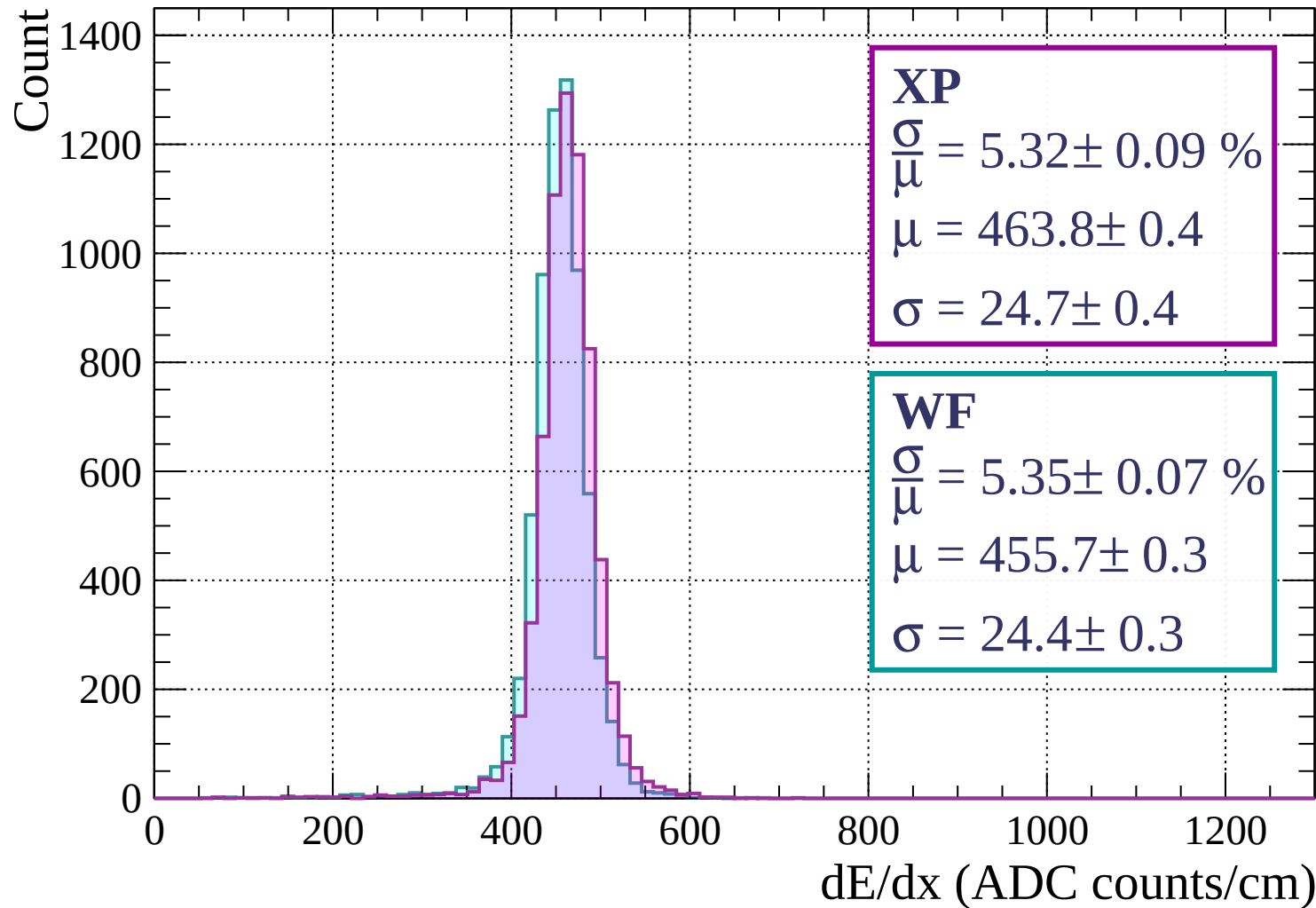
Energy loss | $1040 < p < 1120$



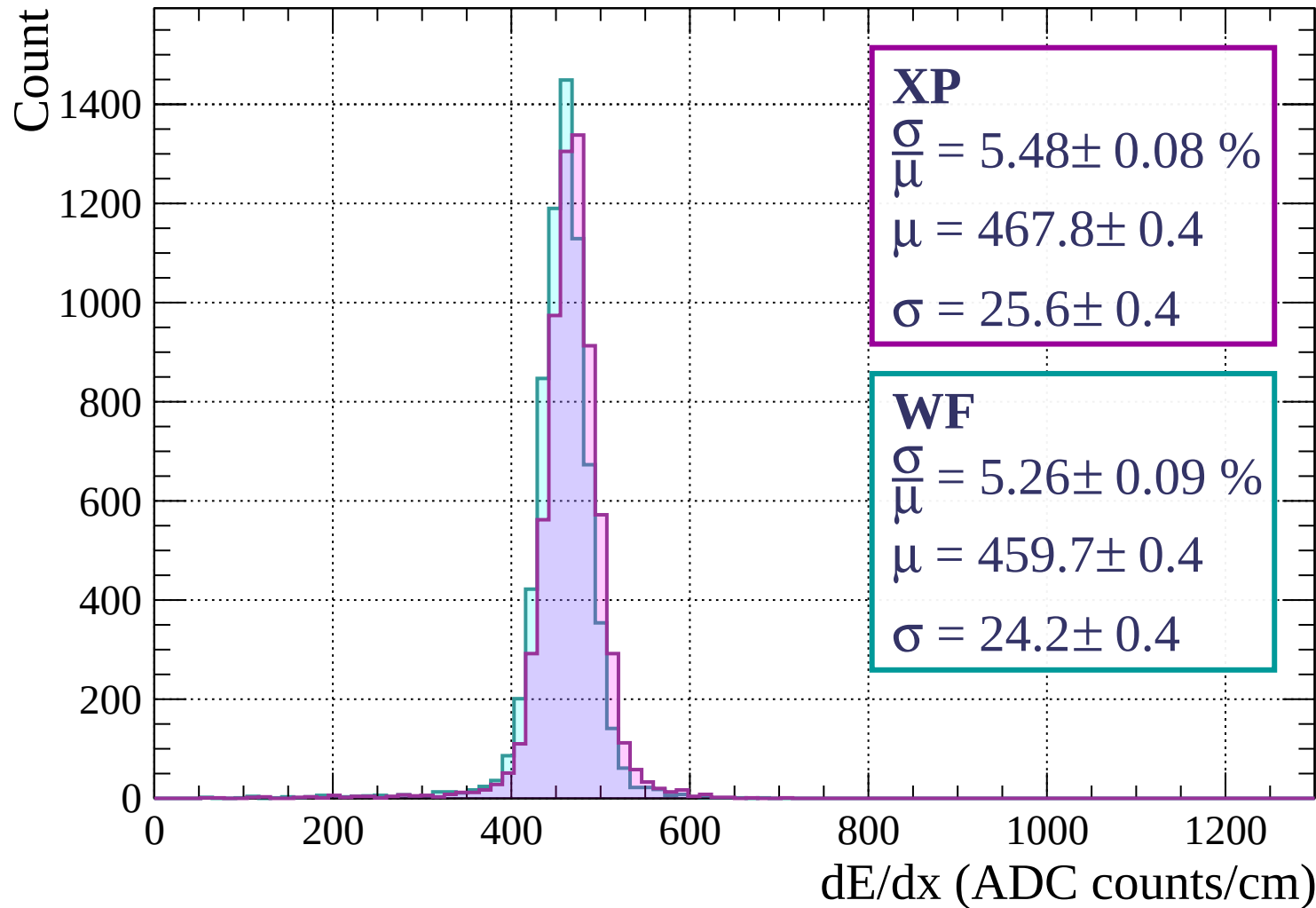
Energy loss | $1120 < p < 1200$



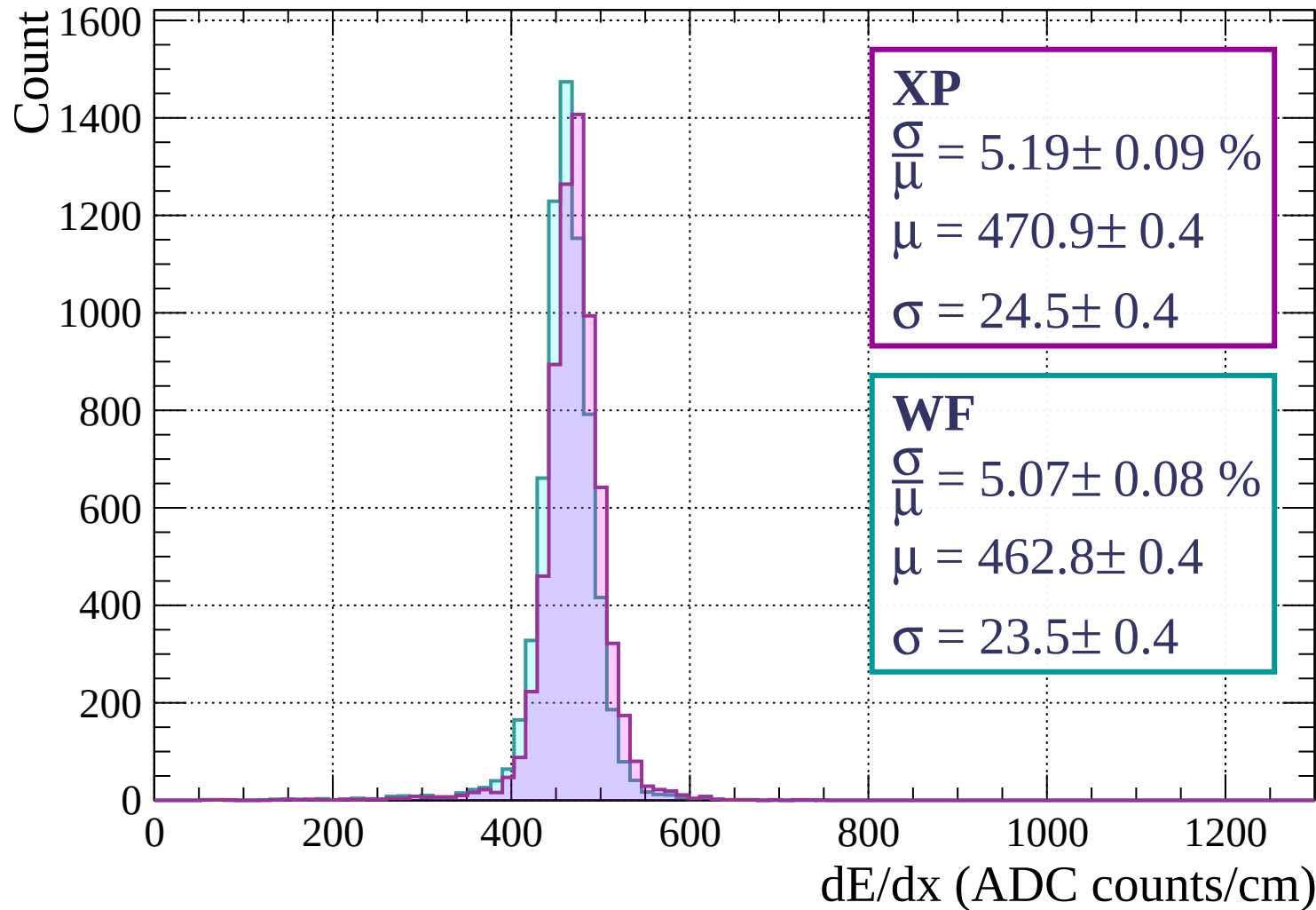
Energy loss | $1200 < p < 1280$



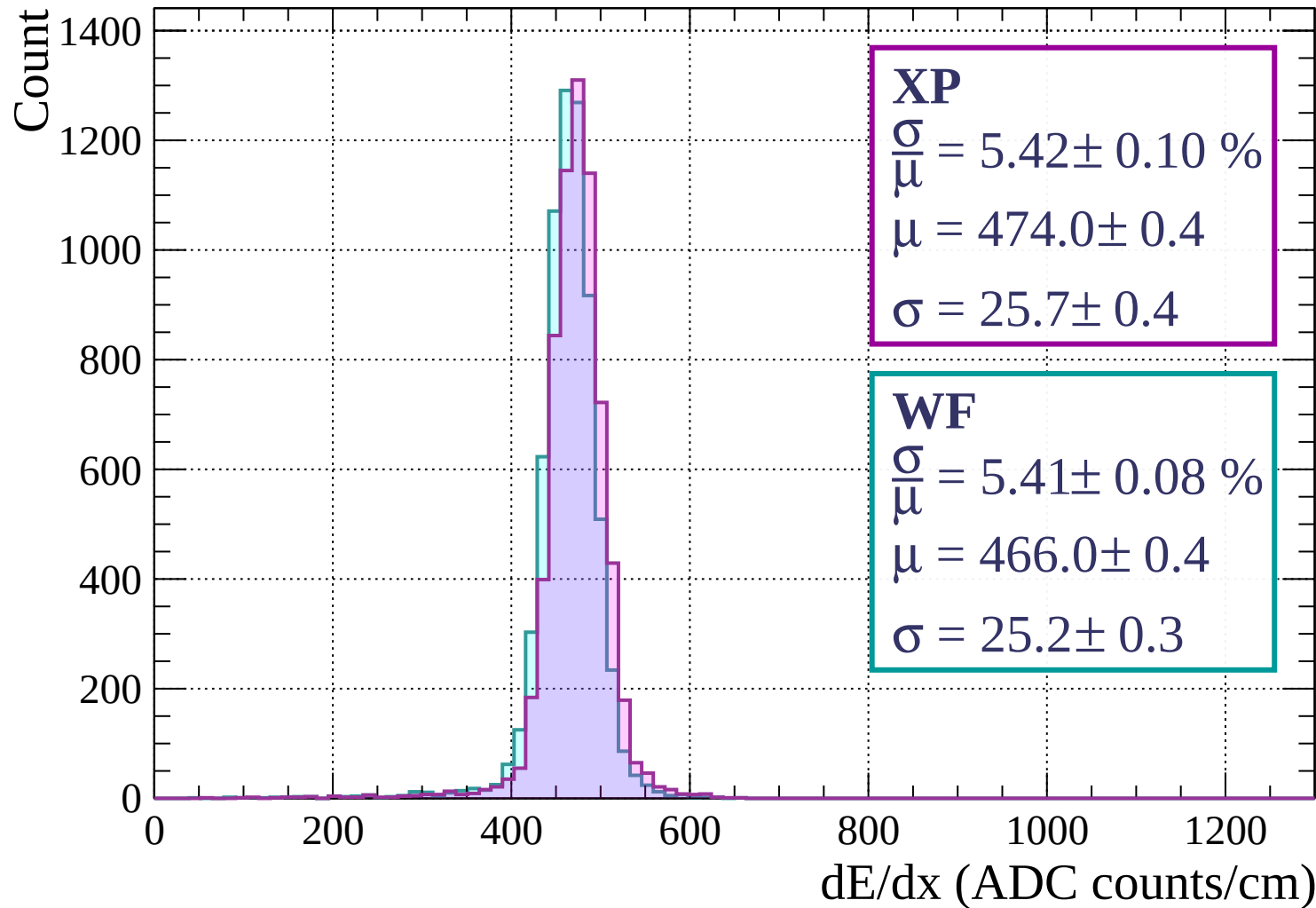
Energy loss | $1280 < p < 1360$



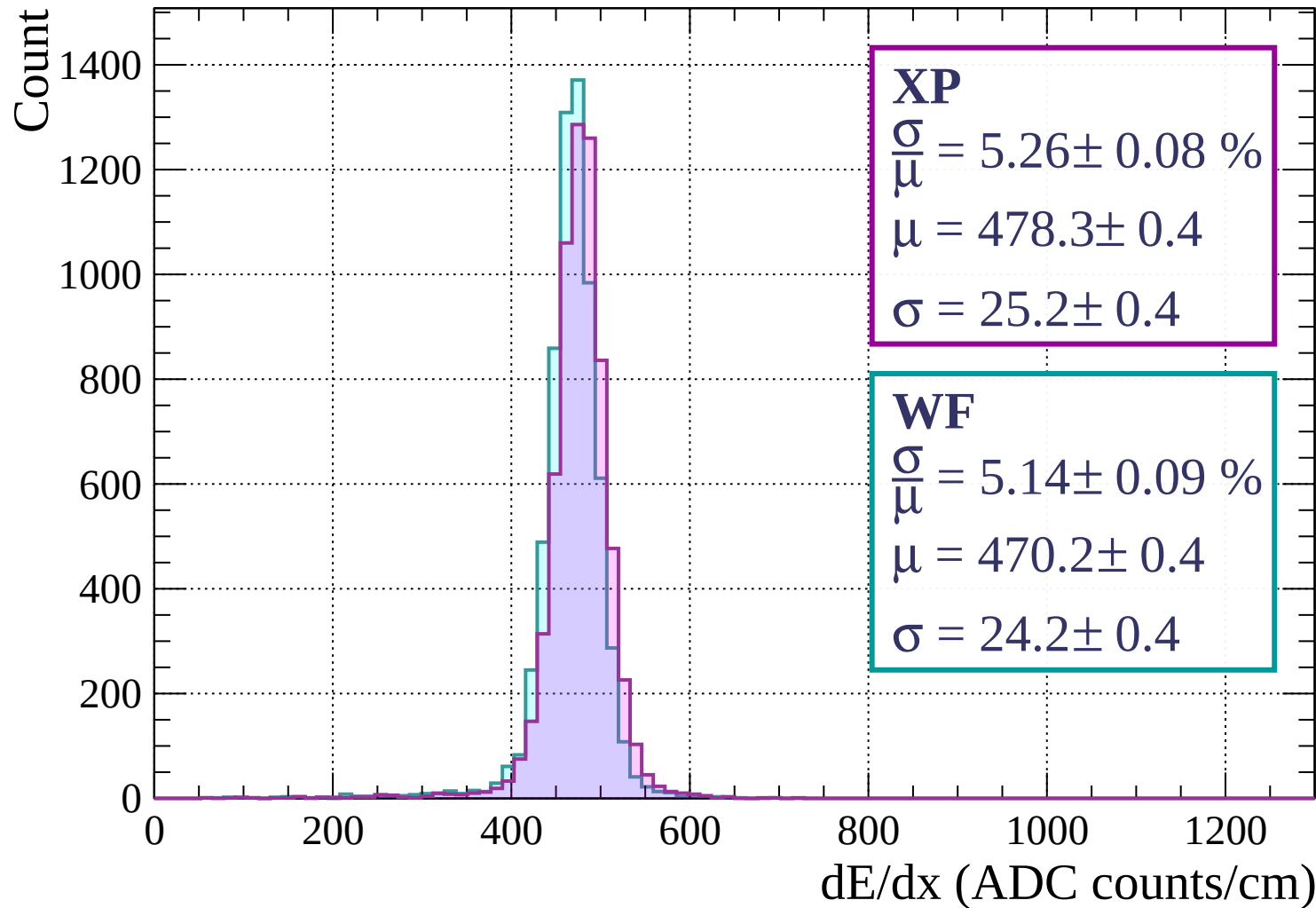
Energy loss | $1360 < p < 1440$



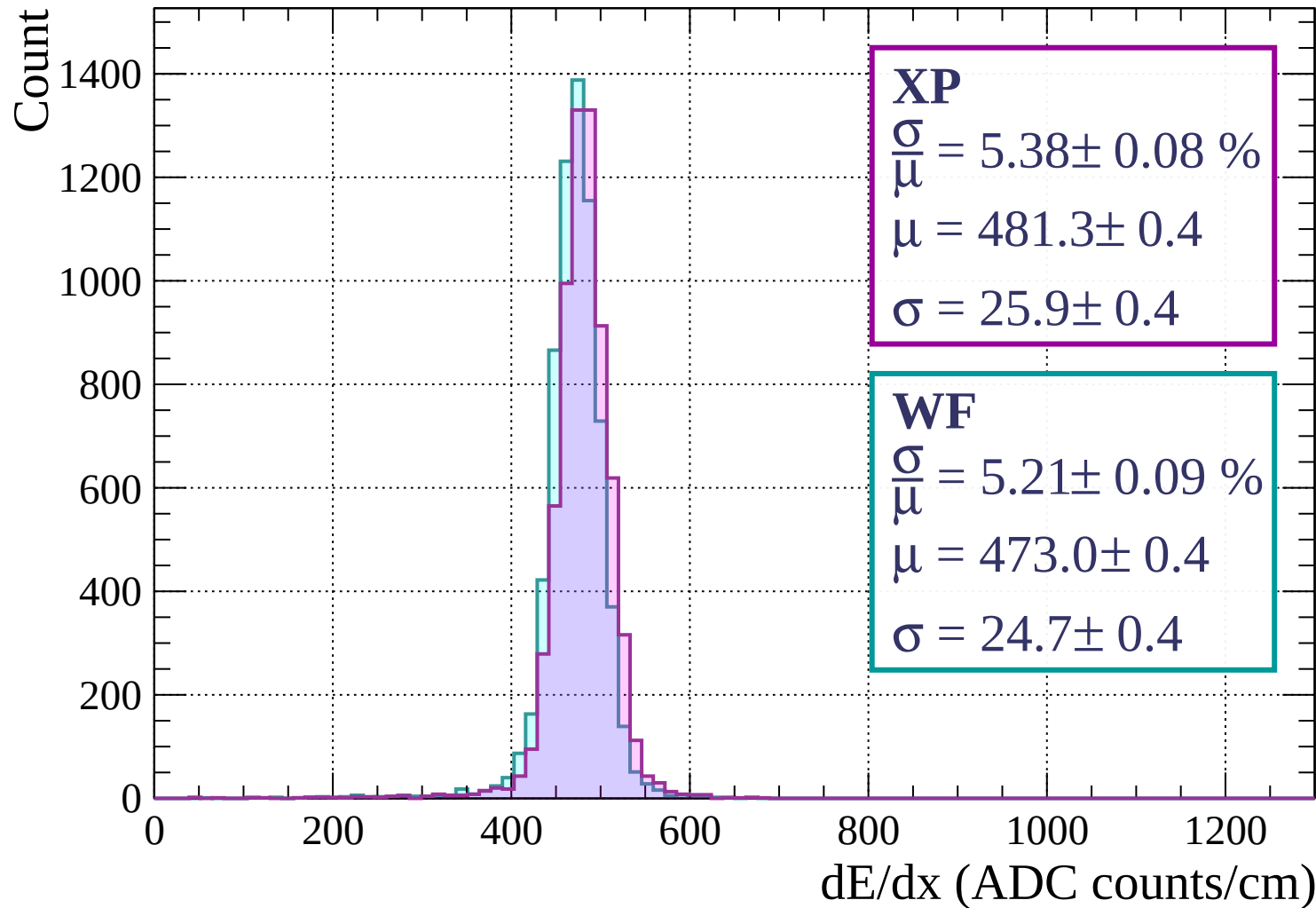
Energy loss | $1440 < p < 1520$



Energy loss | $1520 < p < 1600$

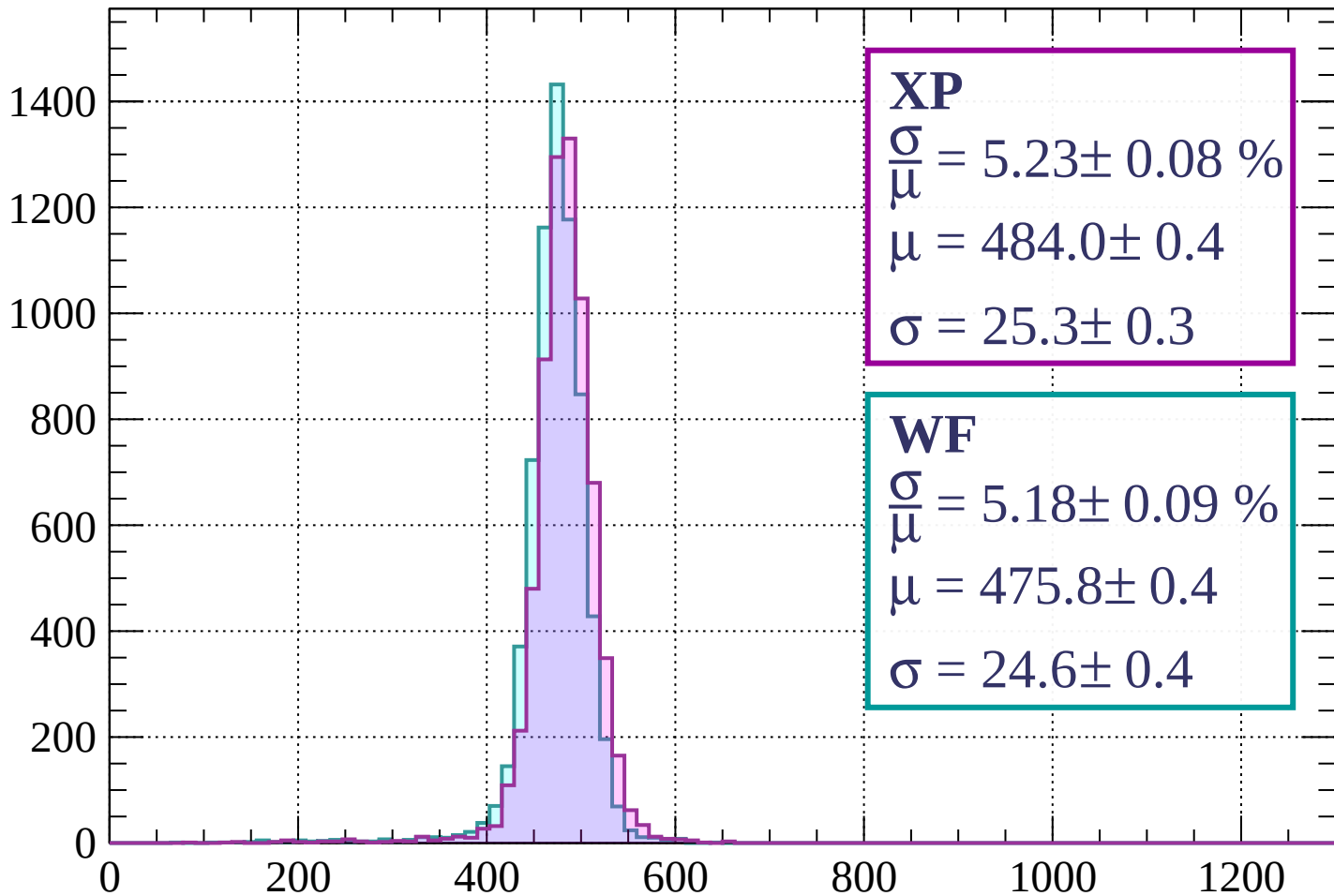


Energy loss | $1600 < p < 1680$



Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 5.23 \pm 0.08 \%$$

$$\mu = 484.0 \pm 0.4$$

$$\sigma = 25.3 \pm 0.3$$

WF

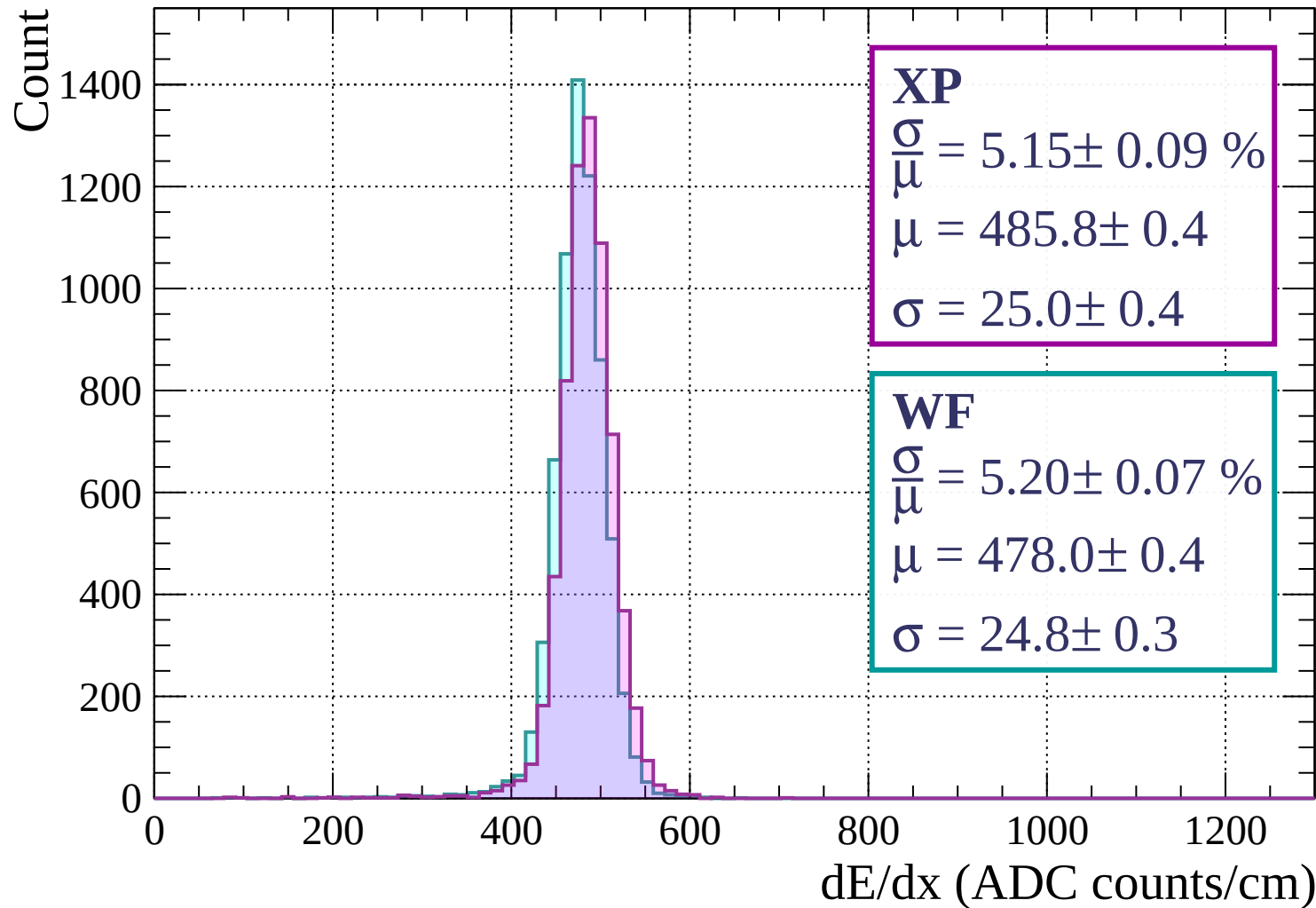
$$\frac{\sigma}{\mu} = 5.18 \pm 0.09 \%$$

$$\mu = 475.8 \pm 0.4$$

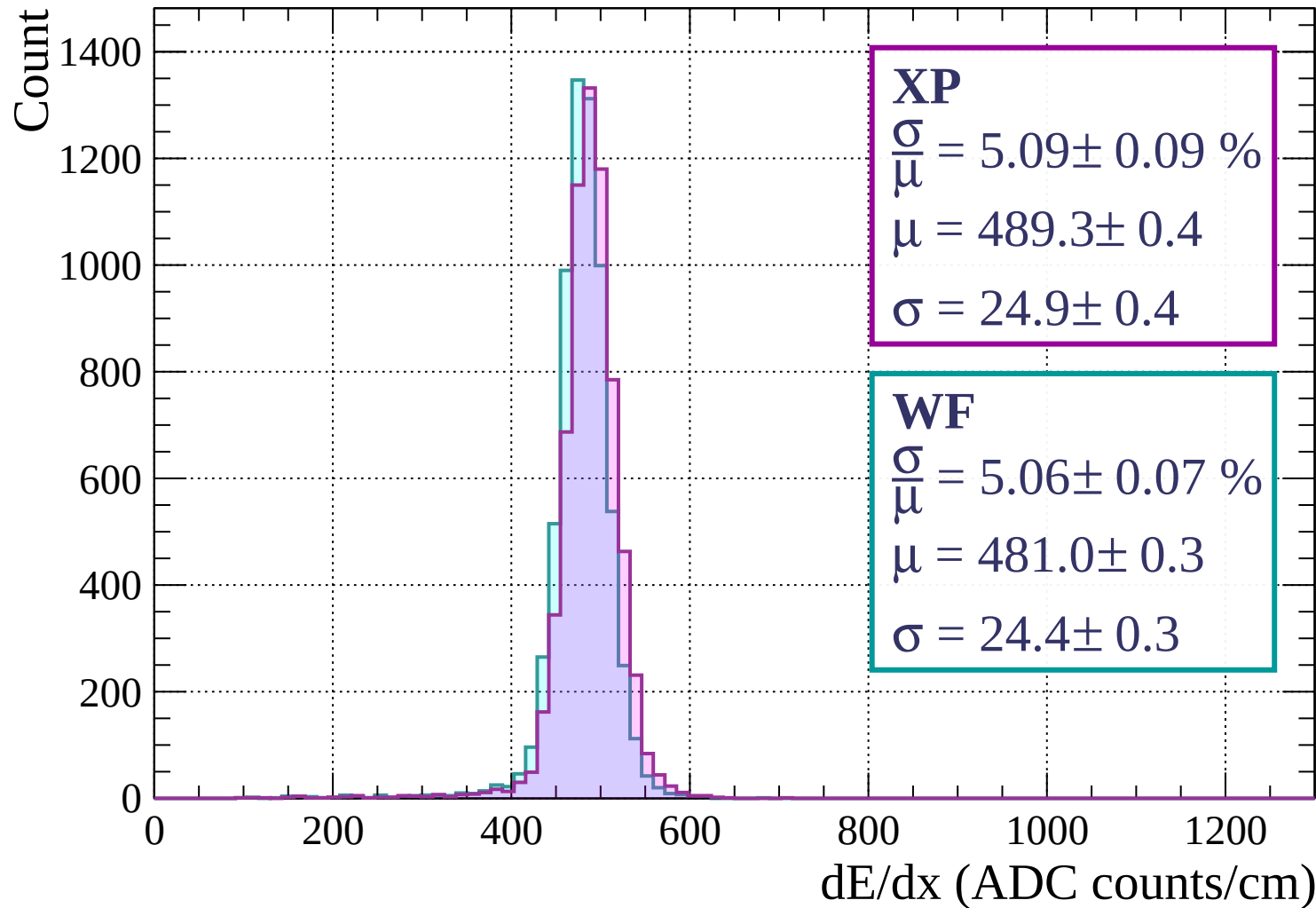
$$\sigma = 24.6 \pm 0.4$$

dE/dx (ADC counts/cm)

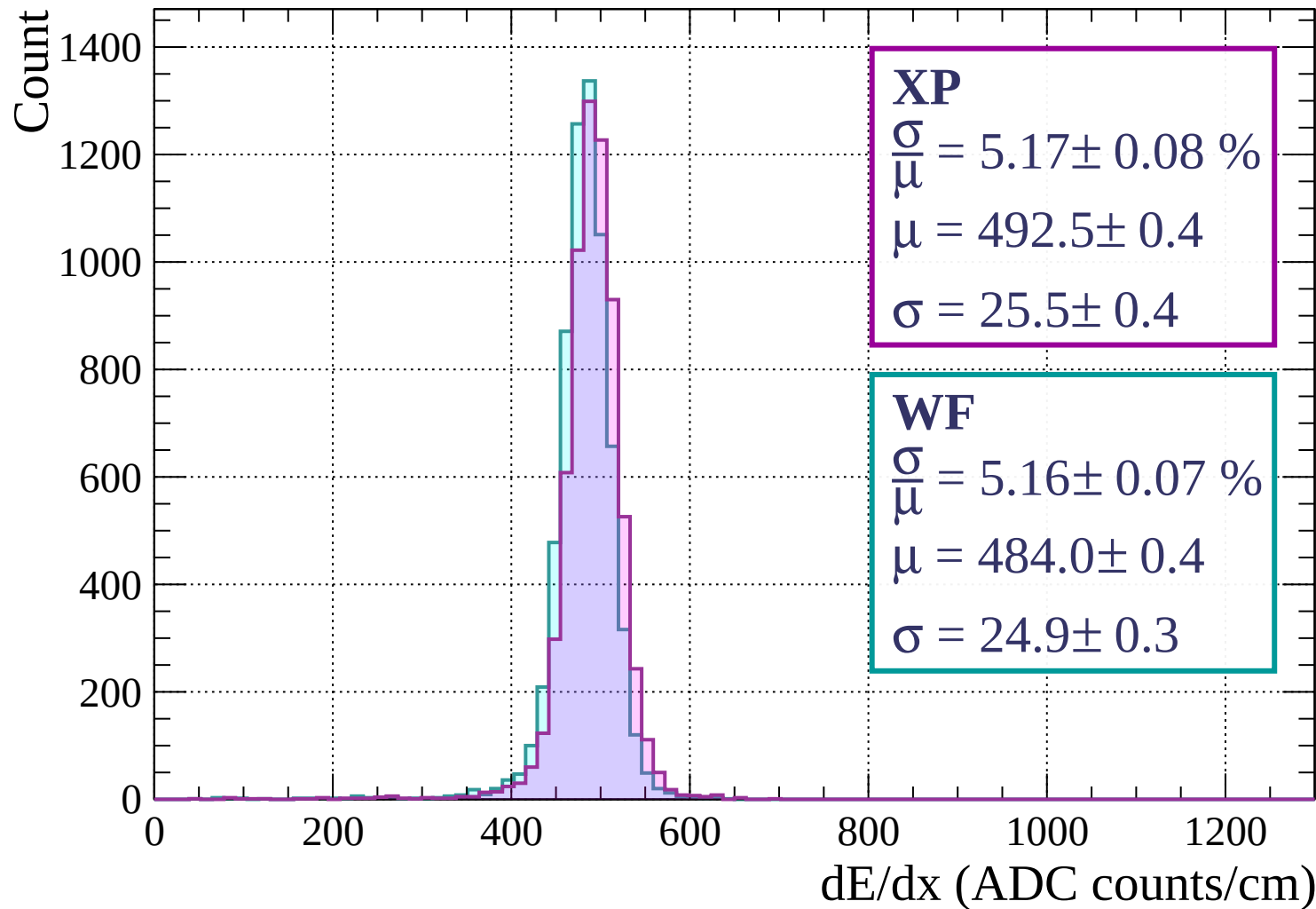
Energy loss | $1760 < p < 1840$



Energy loss | $1840 < p < 1920$



Energy loss | $1920 < p < 2000$



Energy loss | $0 < \phi < 3$

Count

$\times 10^3$

16

14

12

10

8

6

4

2

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.03 \pm 0.03 \%$$

$$\mu = 508.8 \pm 0.1$$

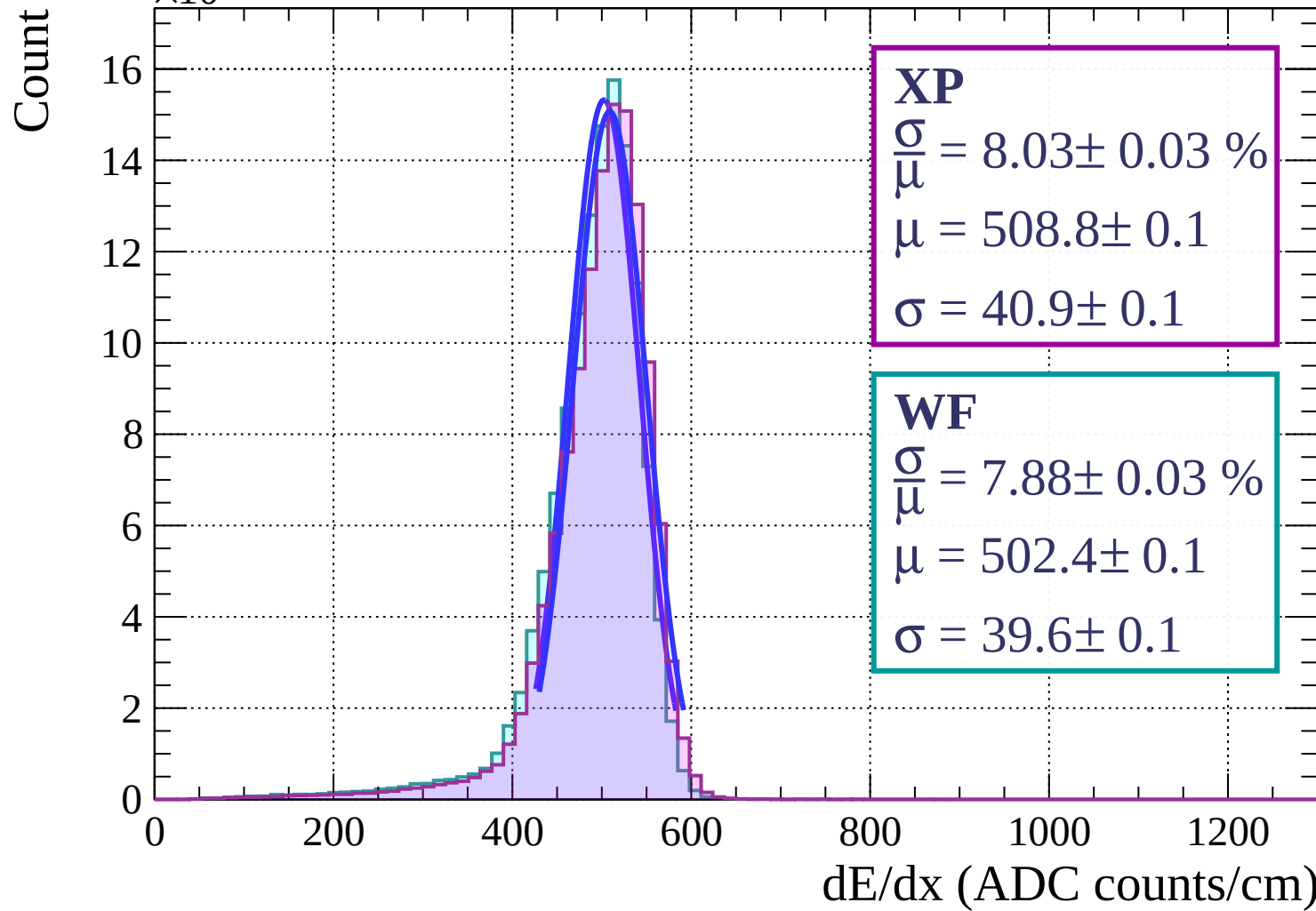
$$\sigma = 40.9 \pm 0.1$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.03 \%$$

$$\mu = 502.4 \pm 0.1$$

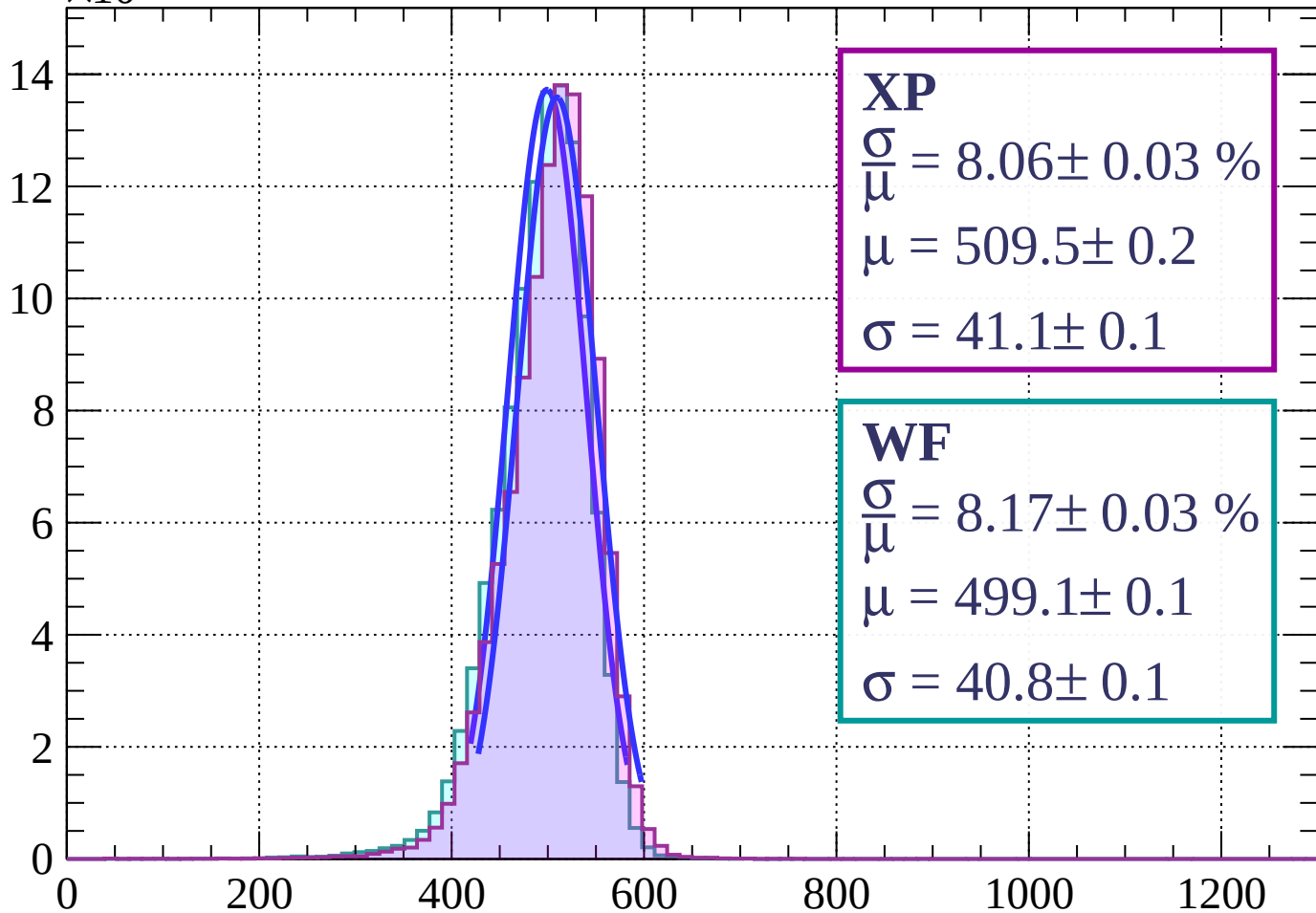
$$\sigma = 39.6 \pm 0.1$$



Energy loss | $3 < \phi < 6$

Count

$\times 10^3$



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.03 \%$$

$$\mu = 509.5 \pm 0.2$$

$$\sigma = 41.1 \pm 0.1$$

WF

$$\frac{\sigma}{\mu} = 8.17 \pm 0.03 \%$$

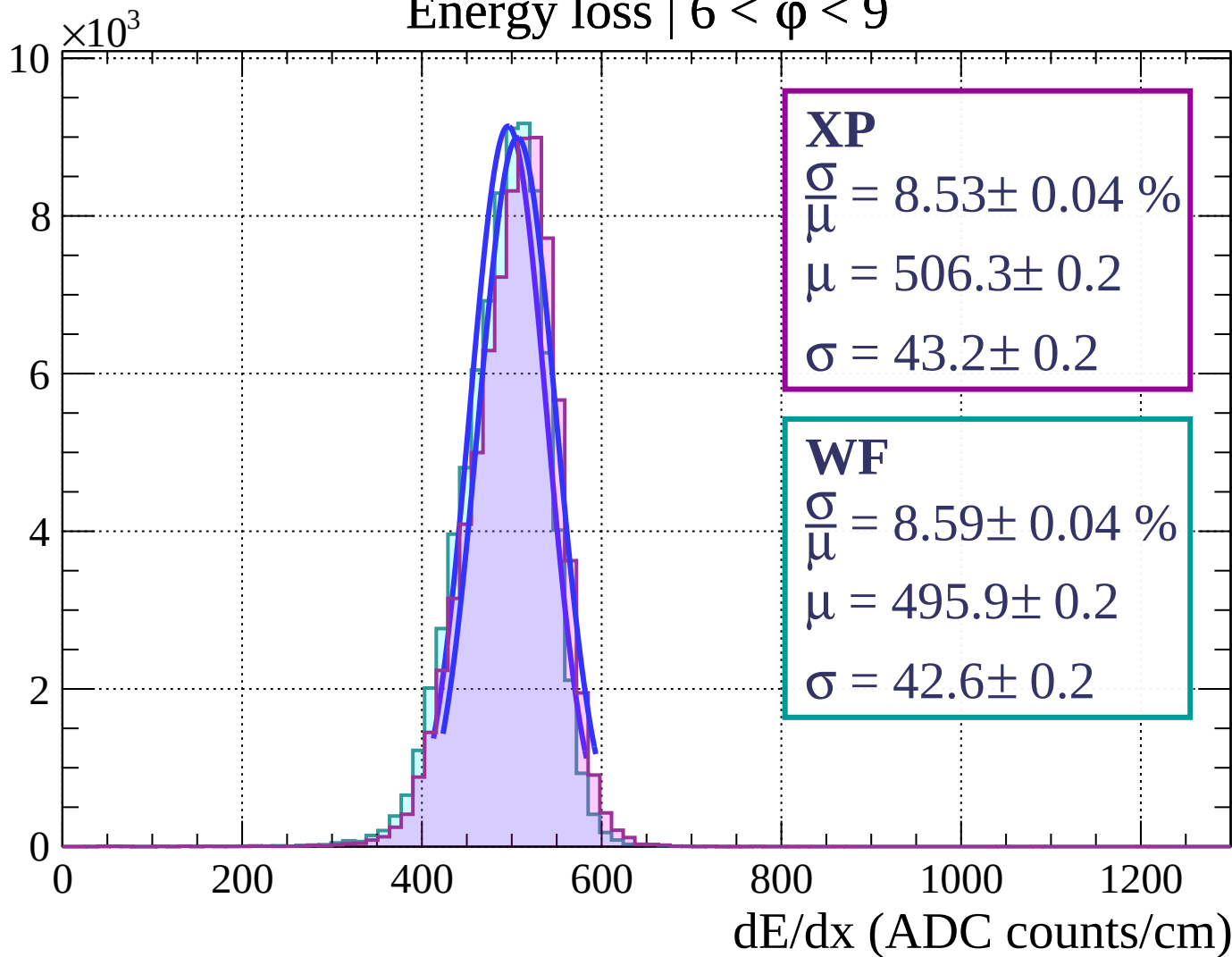
$$\mu = 499.1 \pm 0.1$$

$$\sigma = 40.8 \pm 0.1$$

dE/dx (ADC counts/cm)

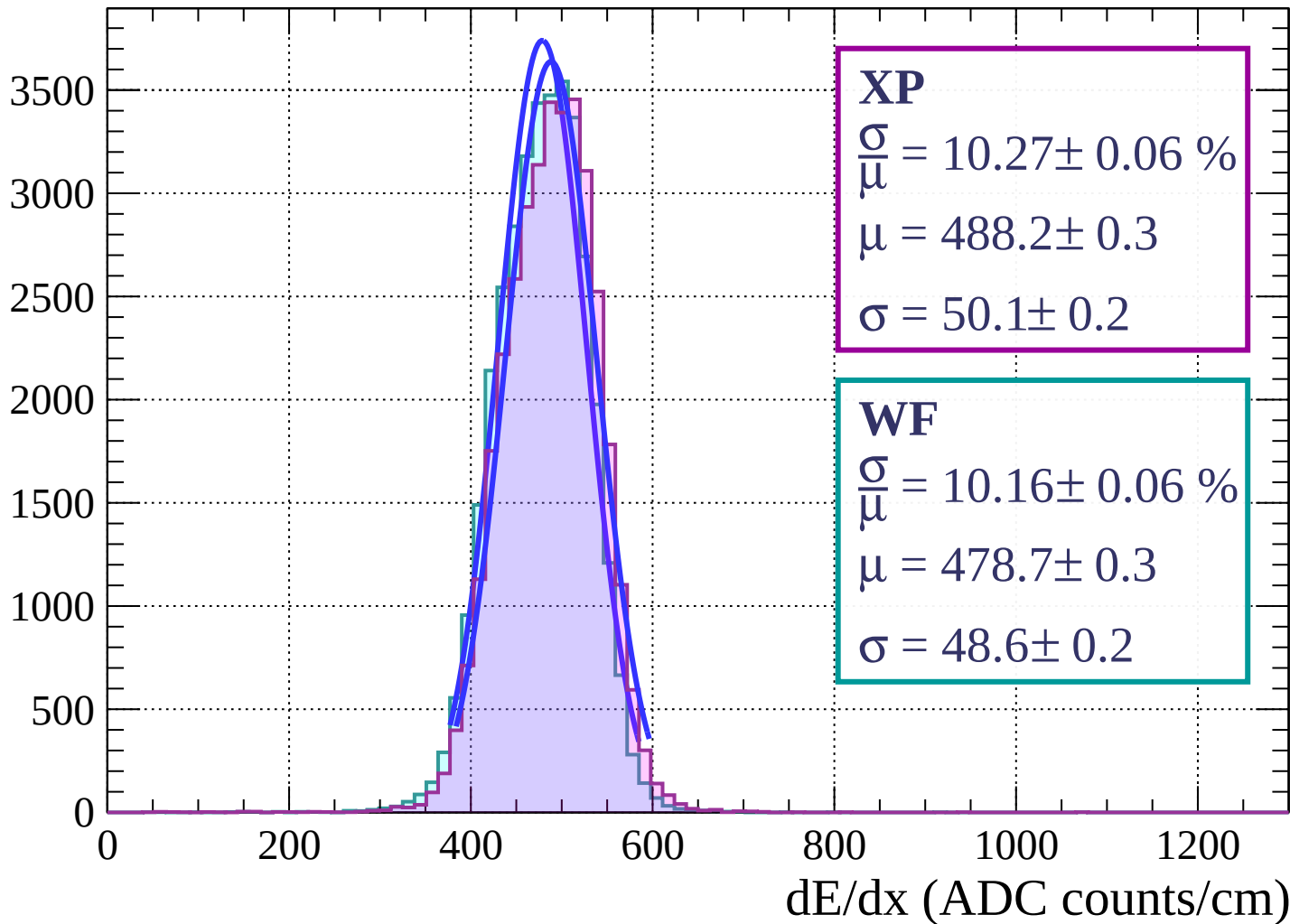
Energy loss | $6 < \phi < 9$

Count

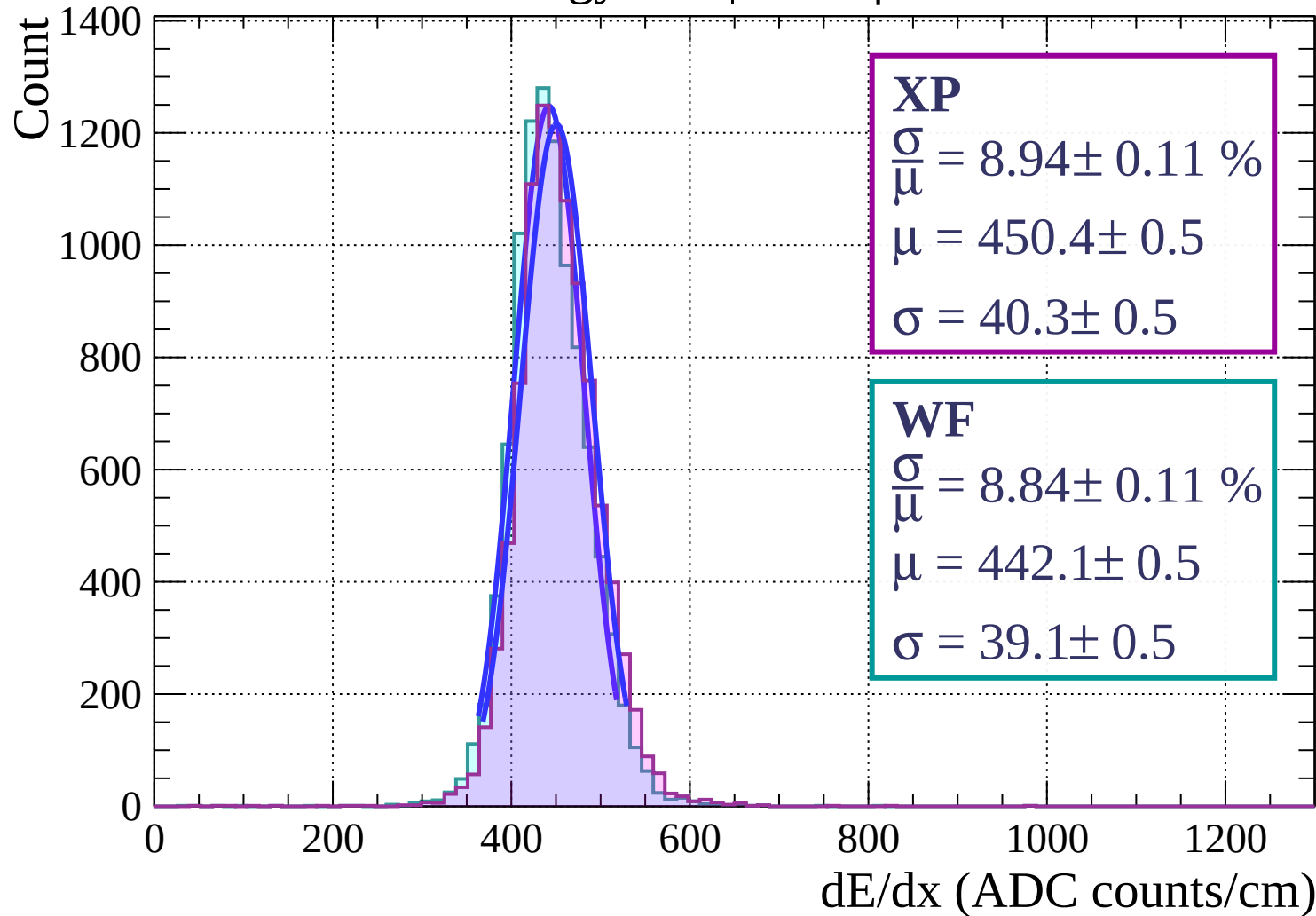


Energy loss | $9 < \phi < 12$

Count



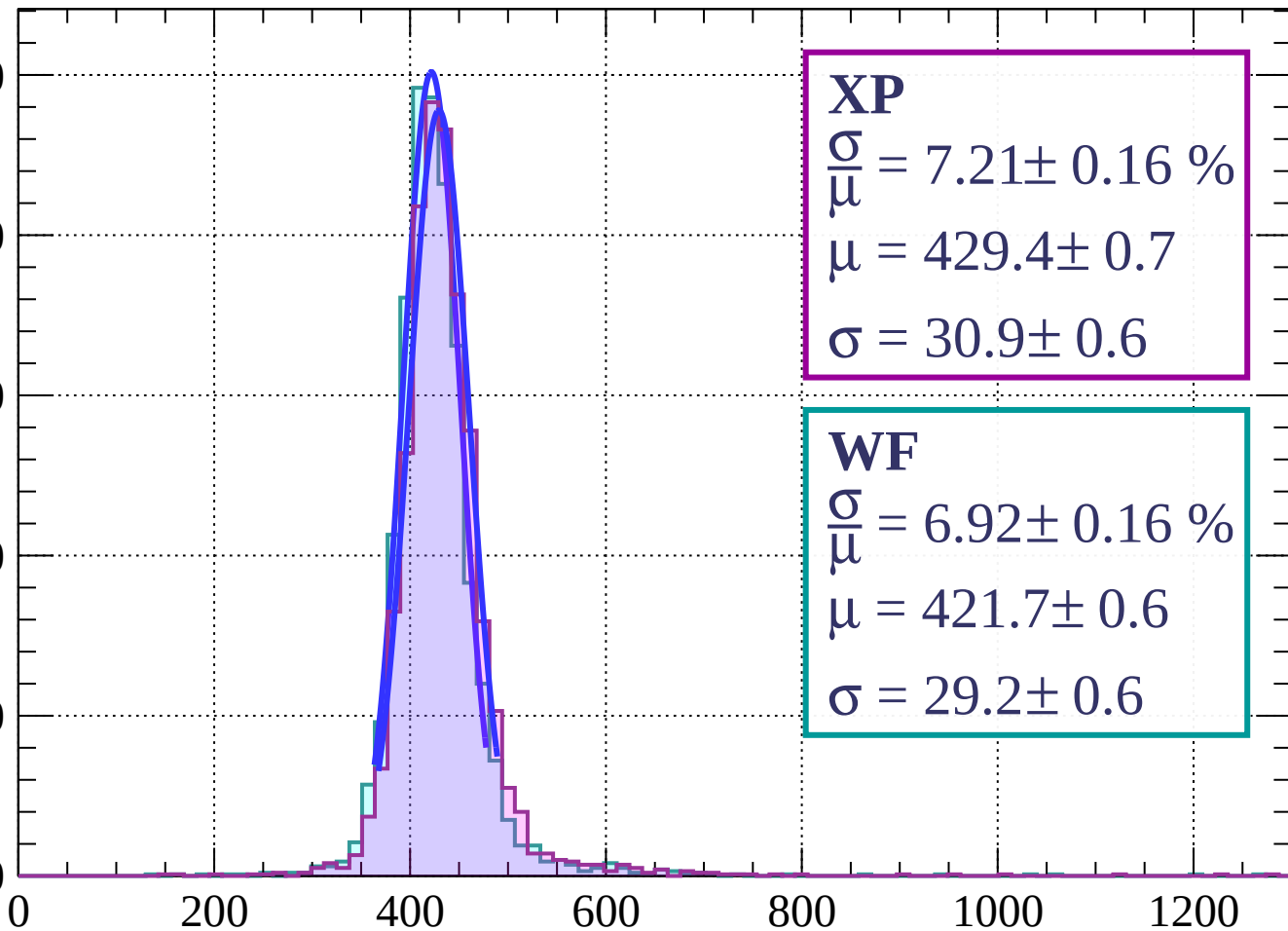
Energy loss | $12 < \phi < 15$



Energy loss | $15 < \phi < 18$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 7.21 \pm 0.16 \%$$

$$\mu = 429.4 \pm 0.7$$

$$\sigma = 30.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.92 \pm 0.16 \%$$

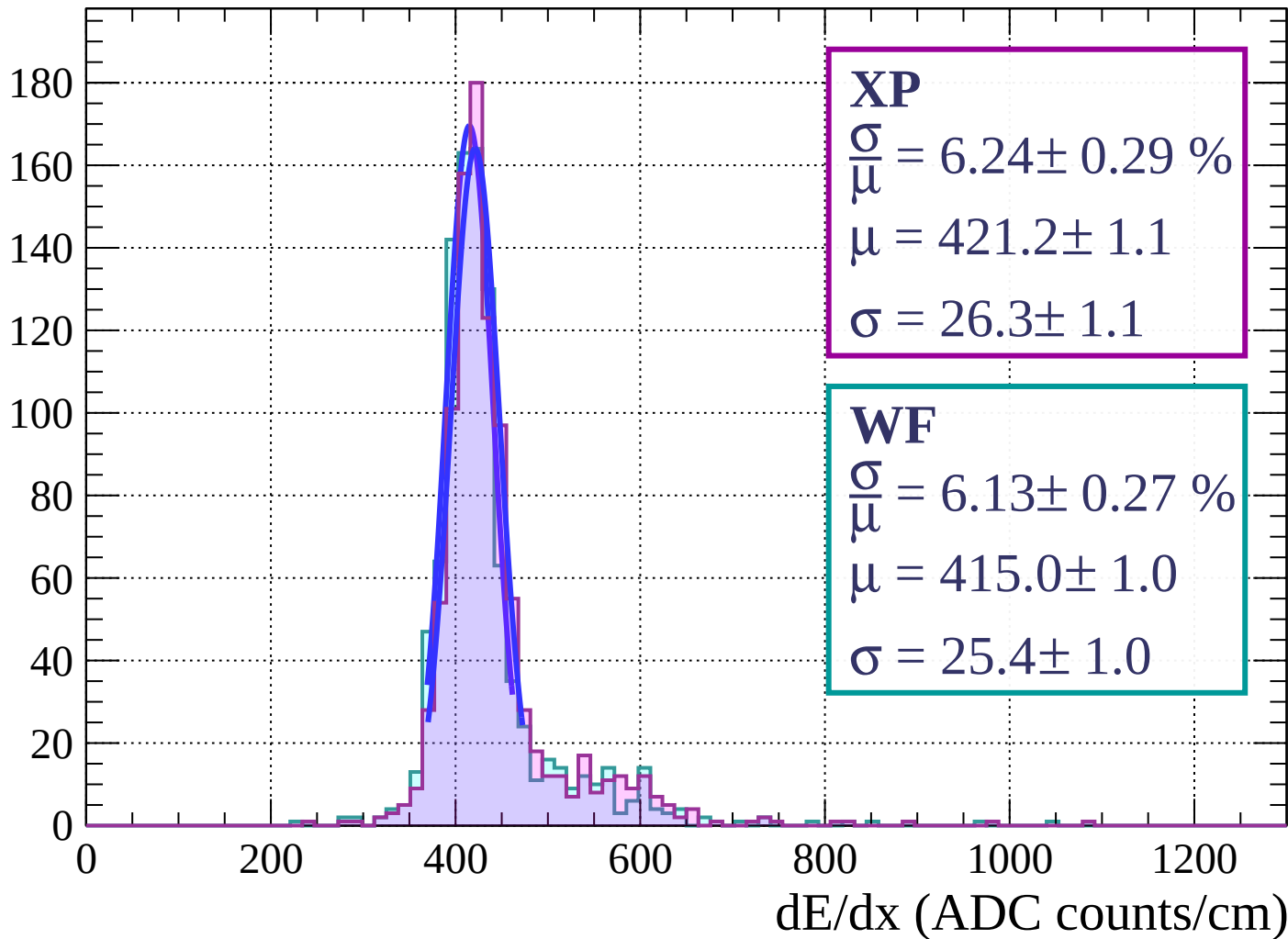
$$\mu = 421.7 \pm 0.6$$

$$\sigma = 29.2 \pm 0.6$$

dE/dx (ADC counts/cm)

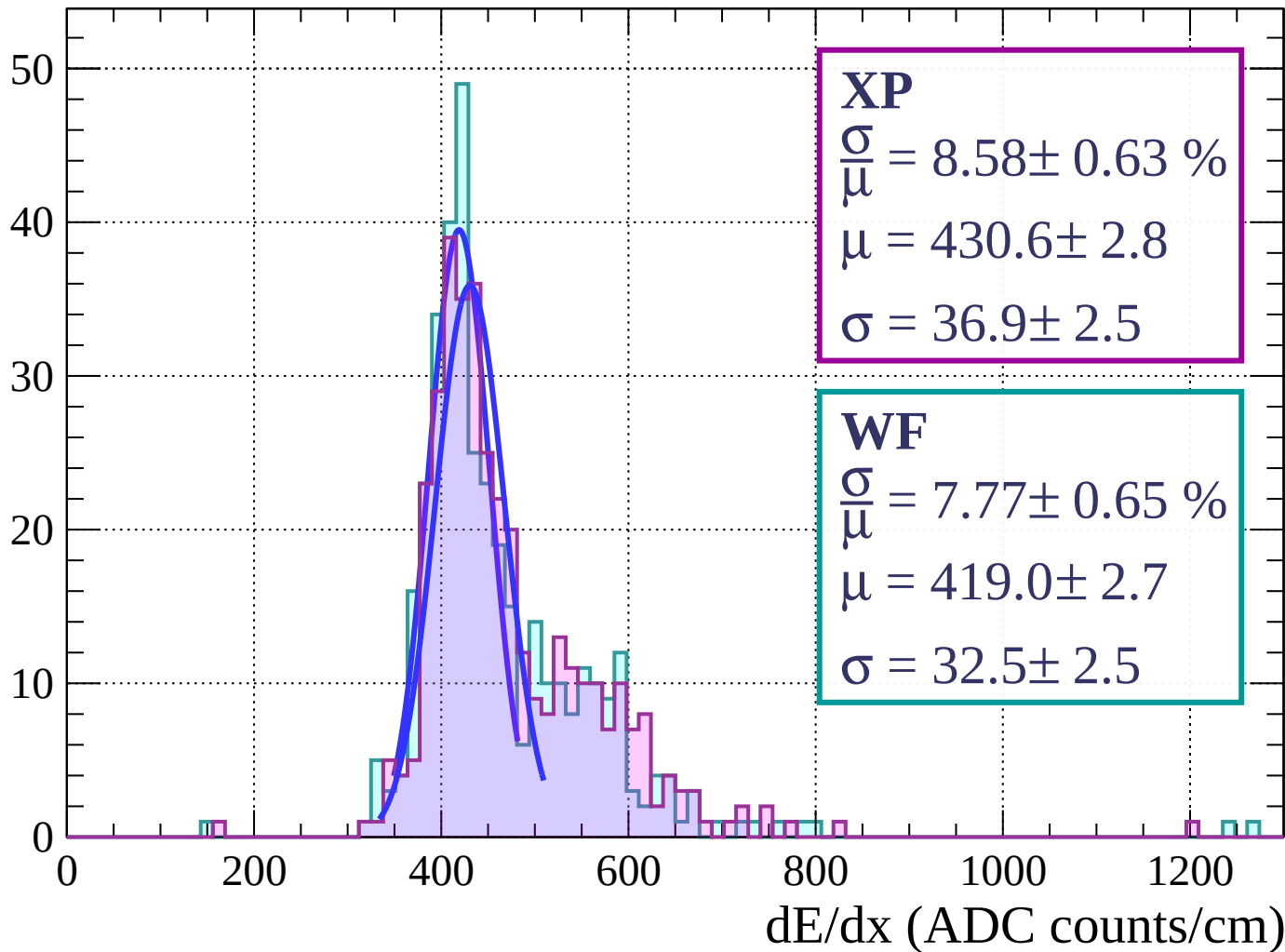
Energy loss | $18 < \phi < 21$

Count



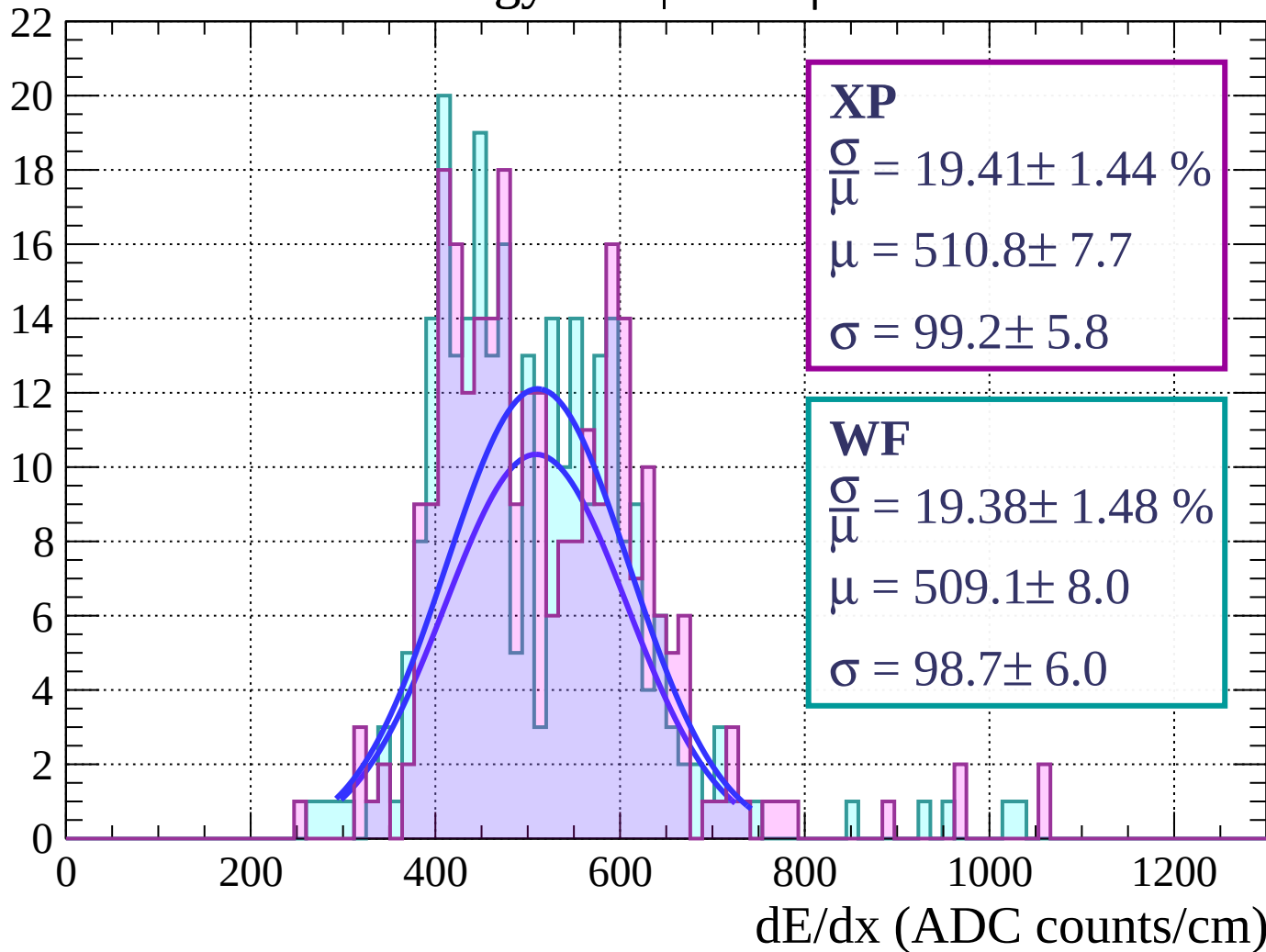
Energy loss | $21 < \phi < 24$

Count



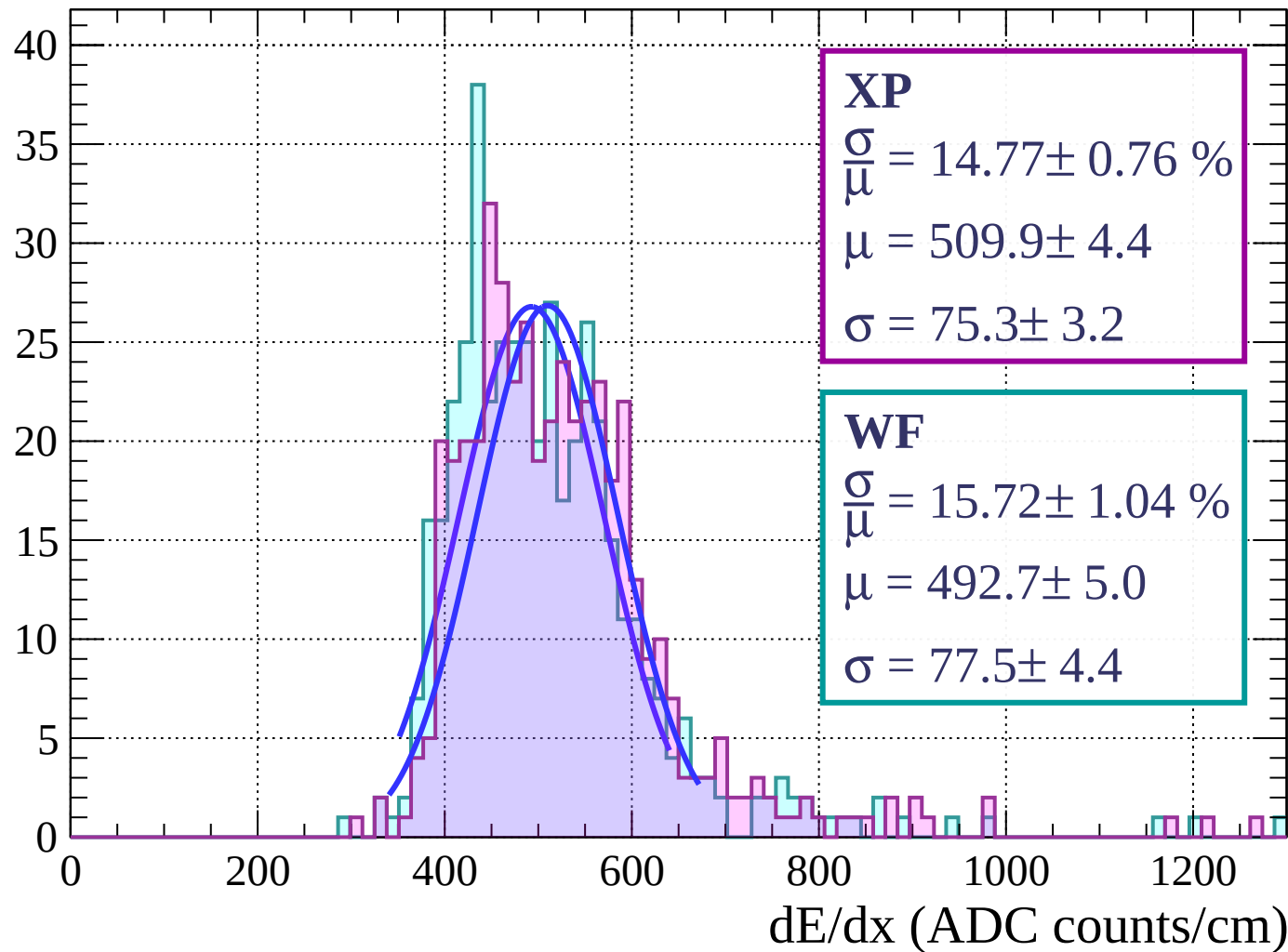
Energy loss | $24 < \phi < 27$

Count



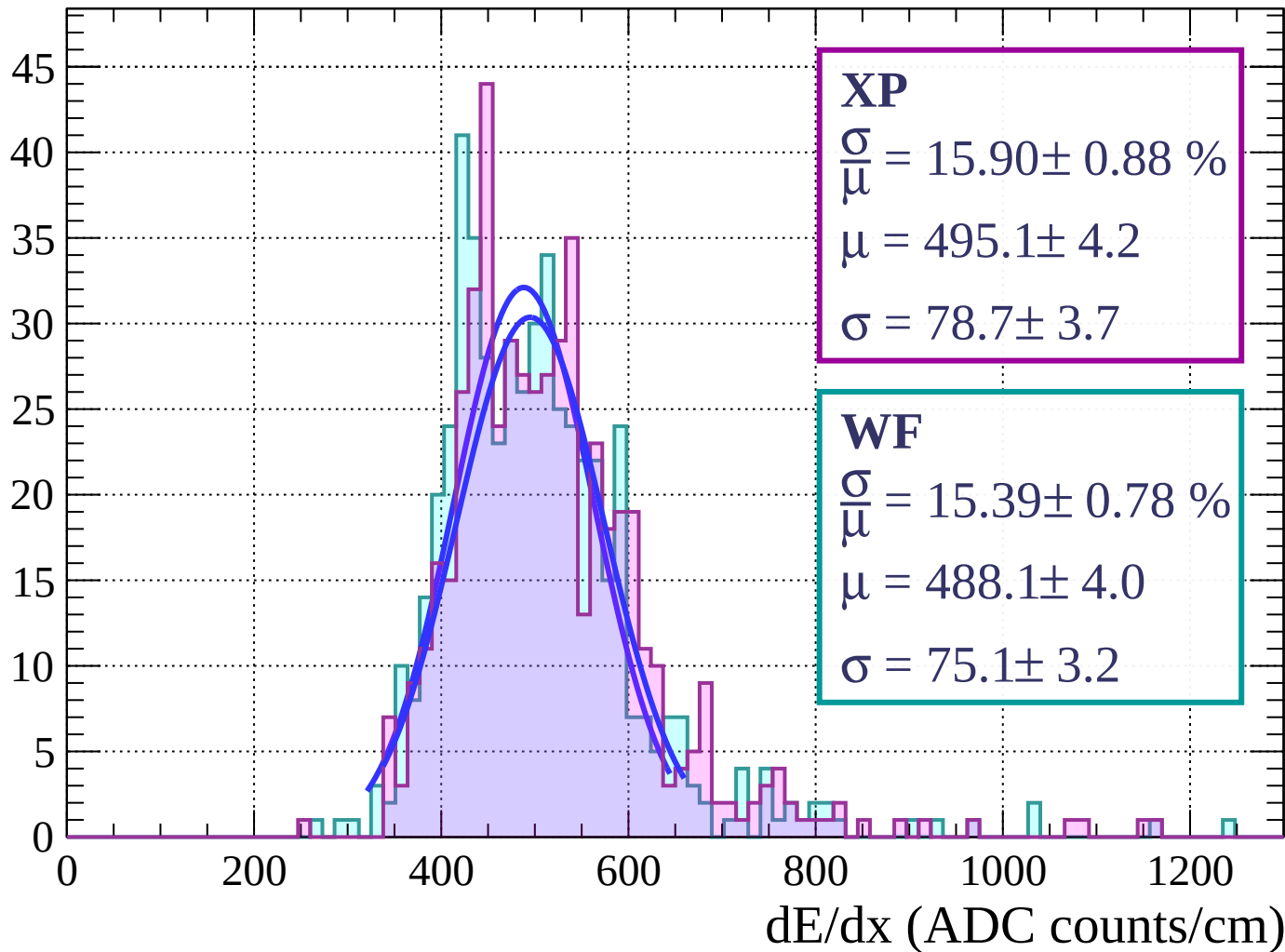
Energy loss | $27 < \phi < 30$

Count



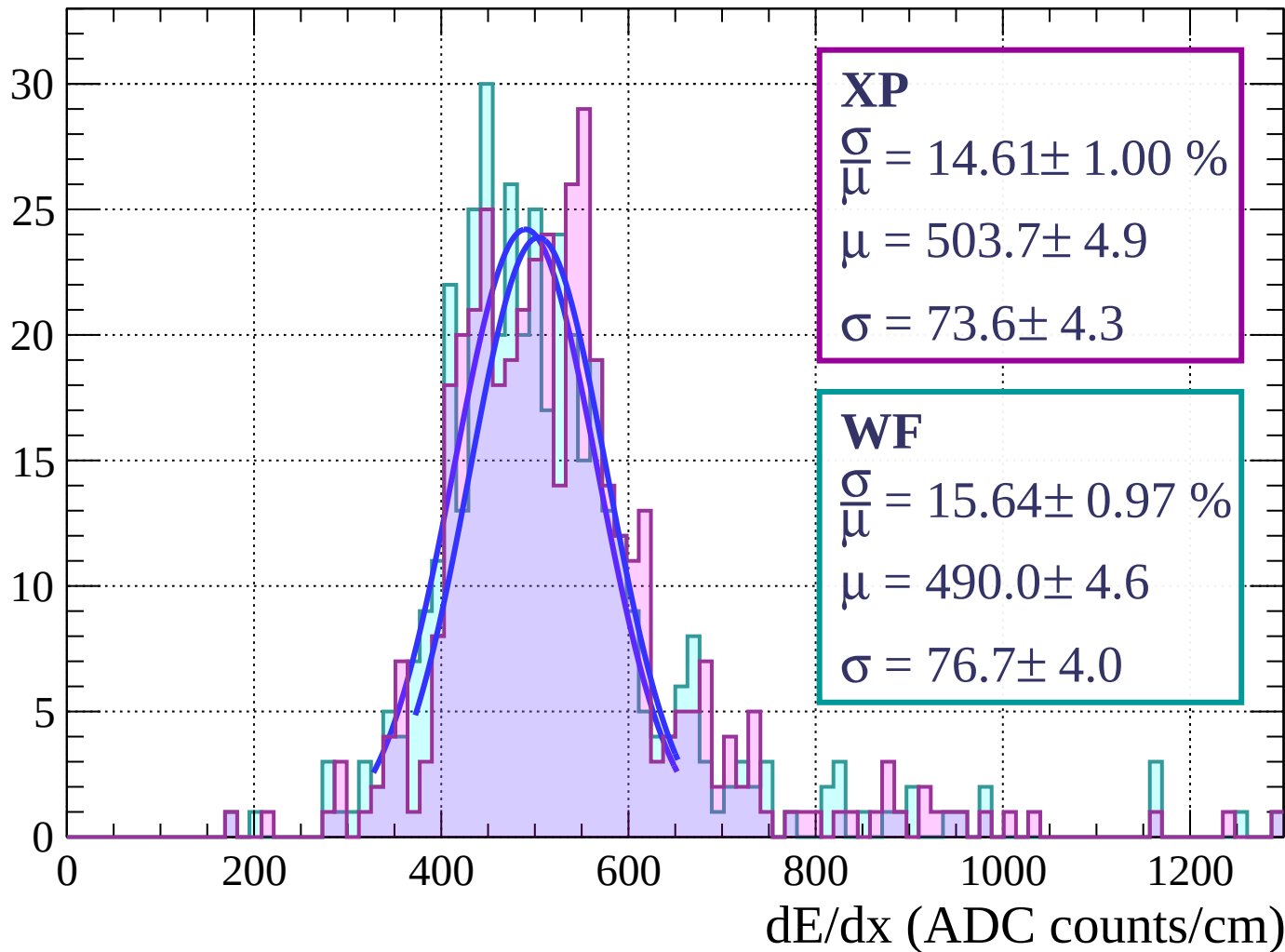
Energy loss | $30 < \phi < 33$

Count



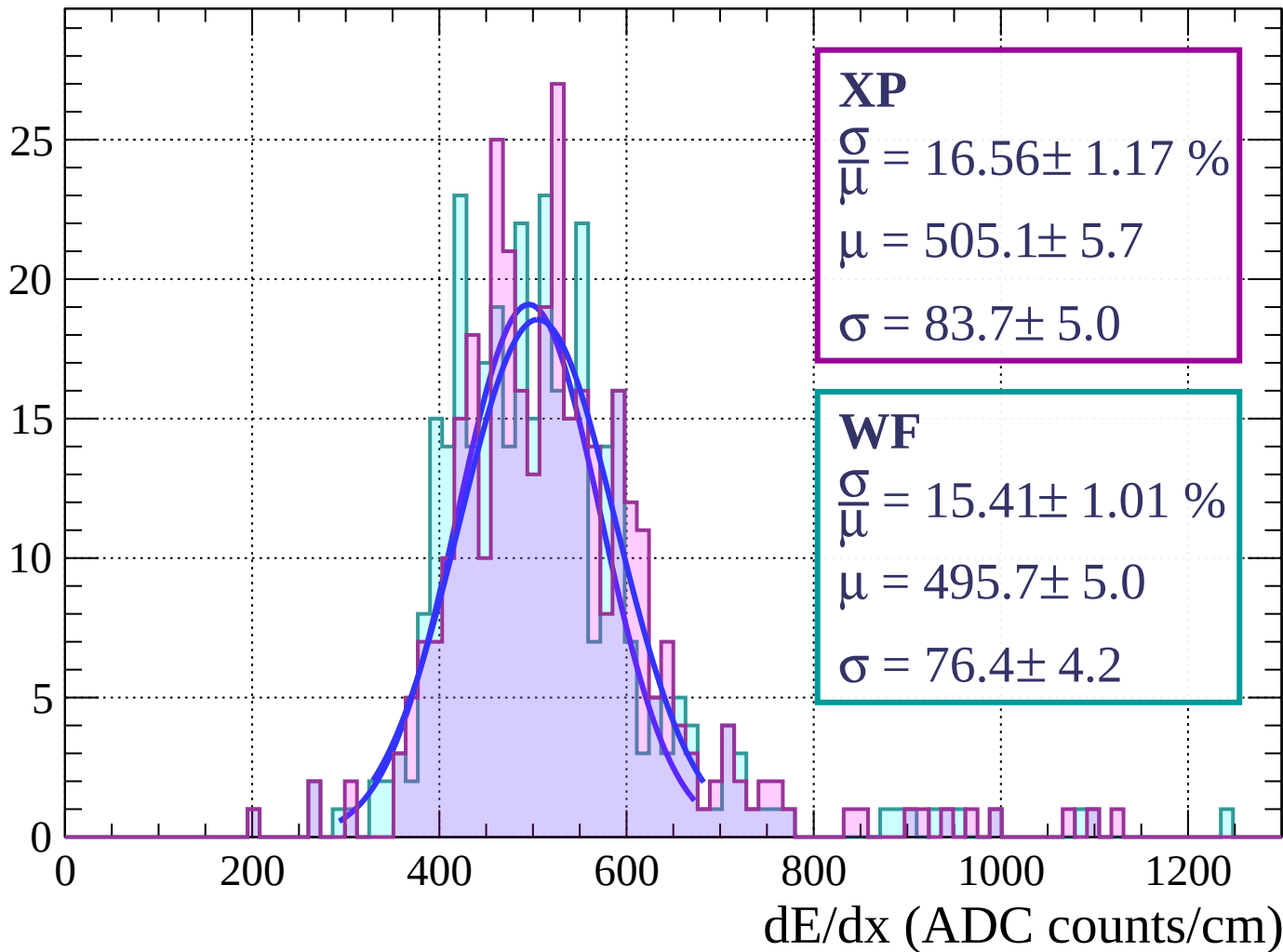
Energy loss | $33 < \phi < 36$

Count



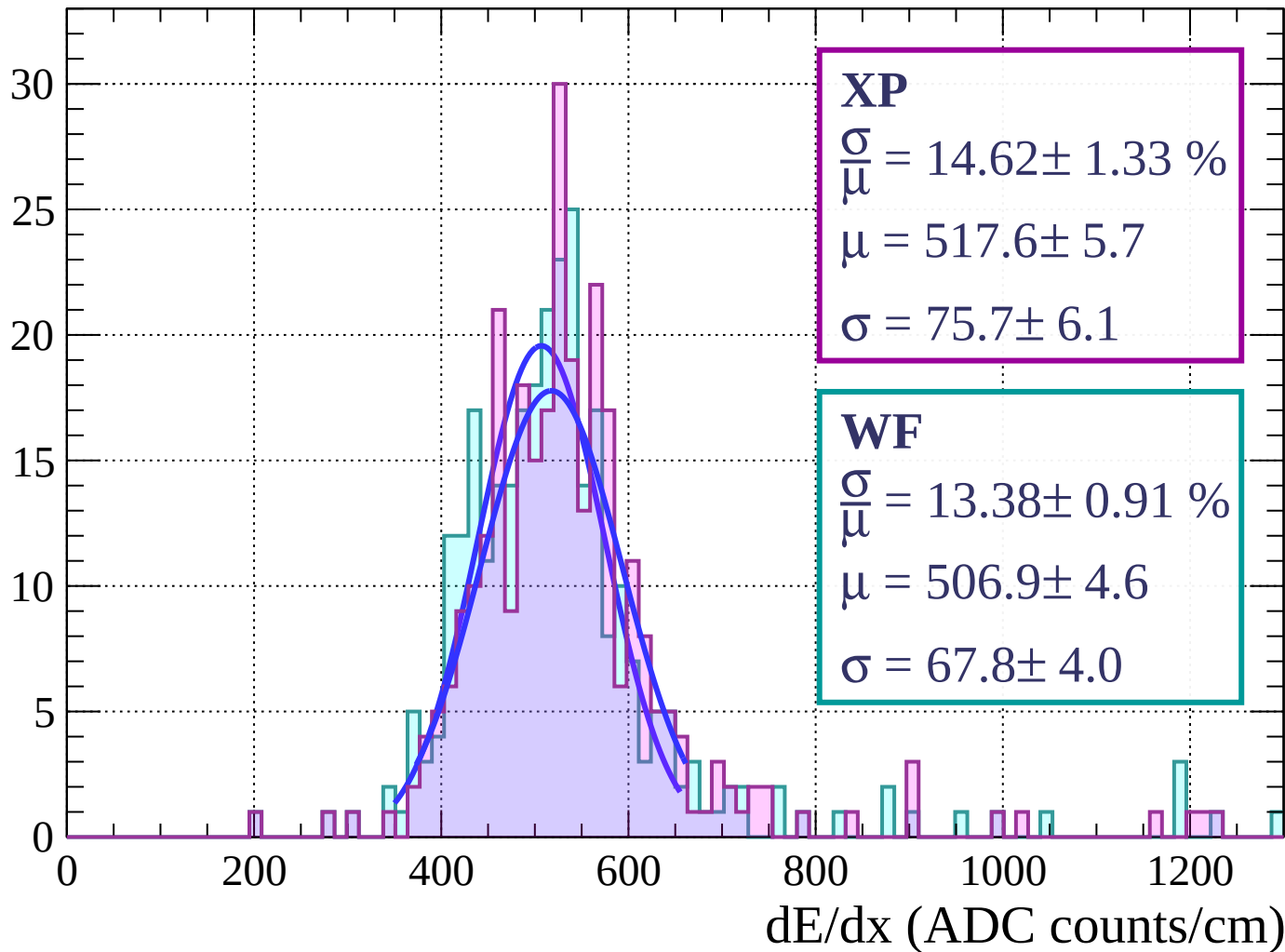
Energy loss | $36 < \phi < 39$

Count



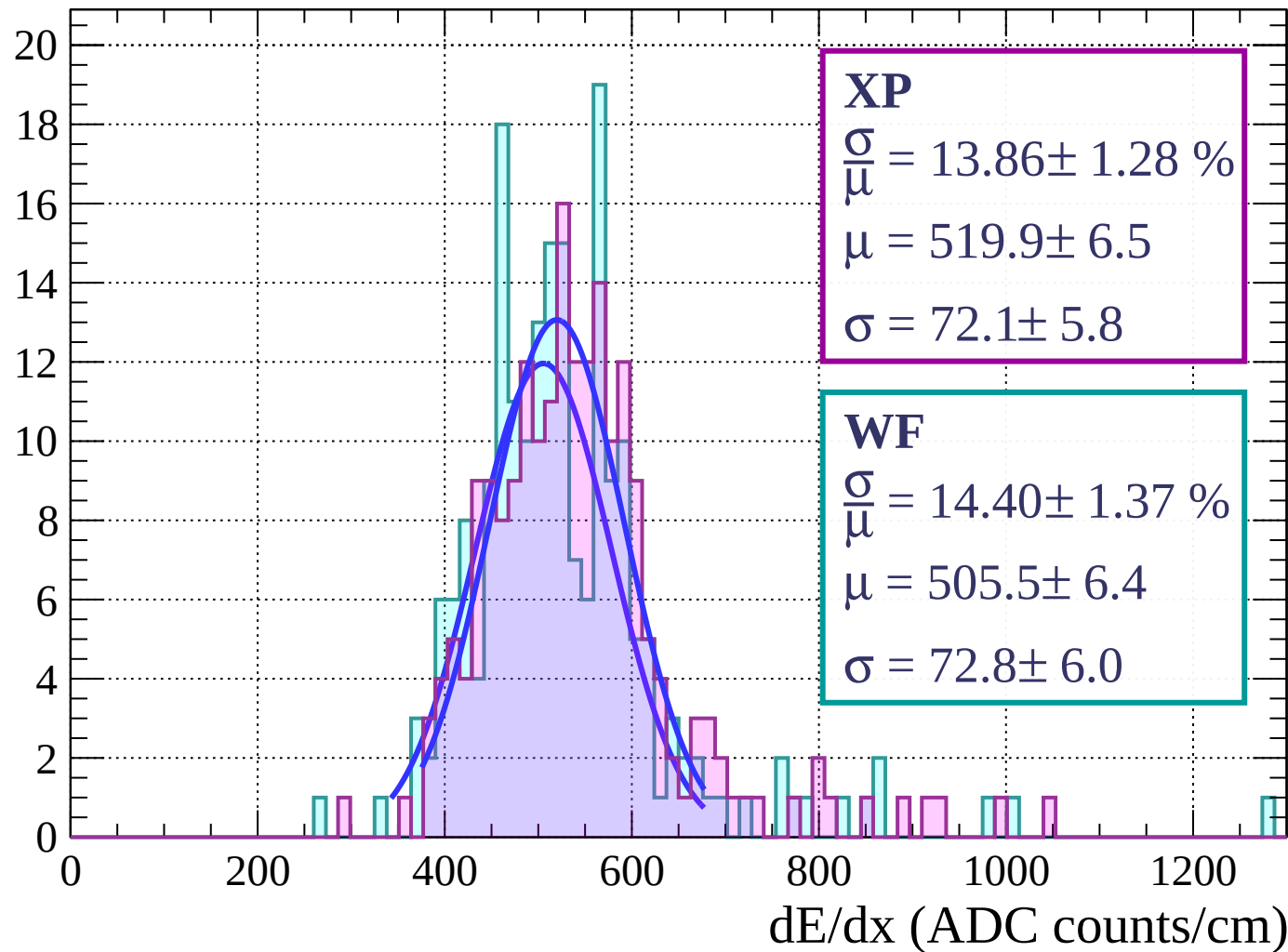
Energy loss | $39 < \phi < 42$

Count



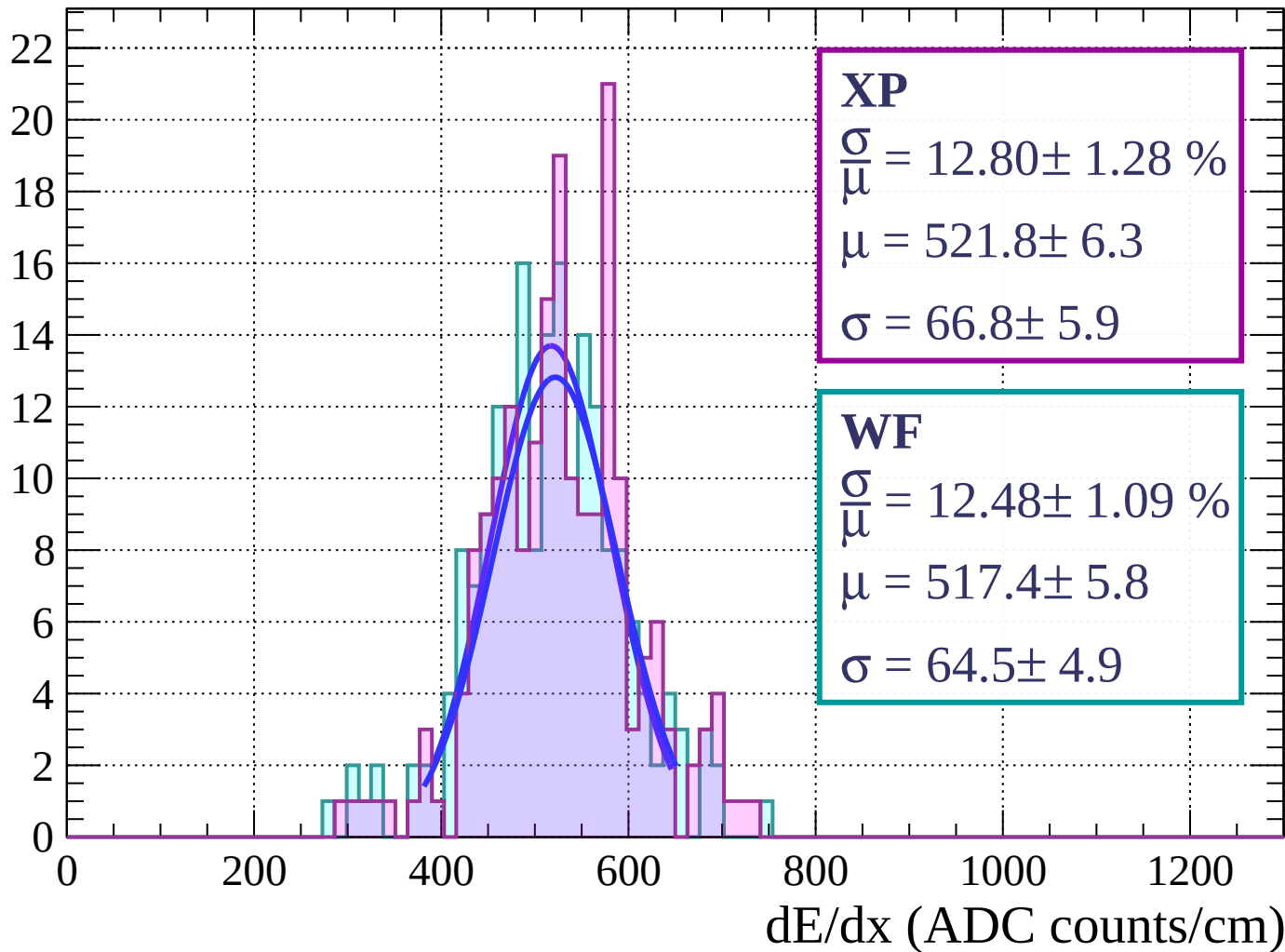
Energy loss | $42 < \phi < 45$

Count



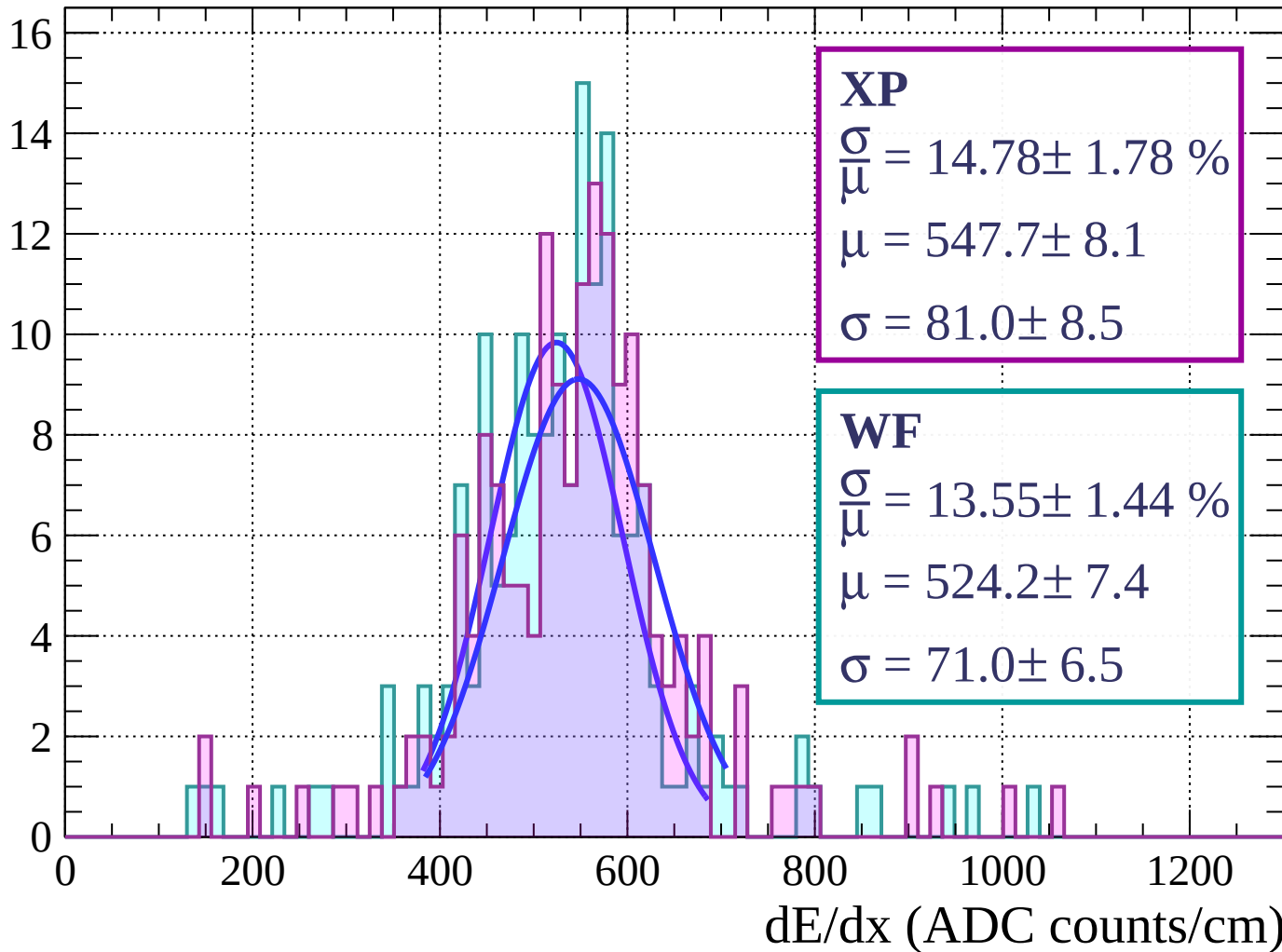
Energy loss | $45 < \phi < 48$

Count



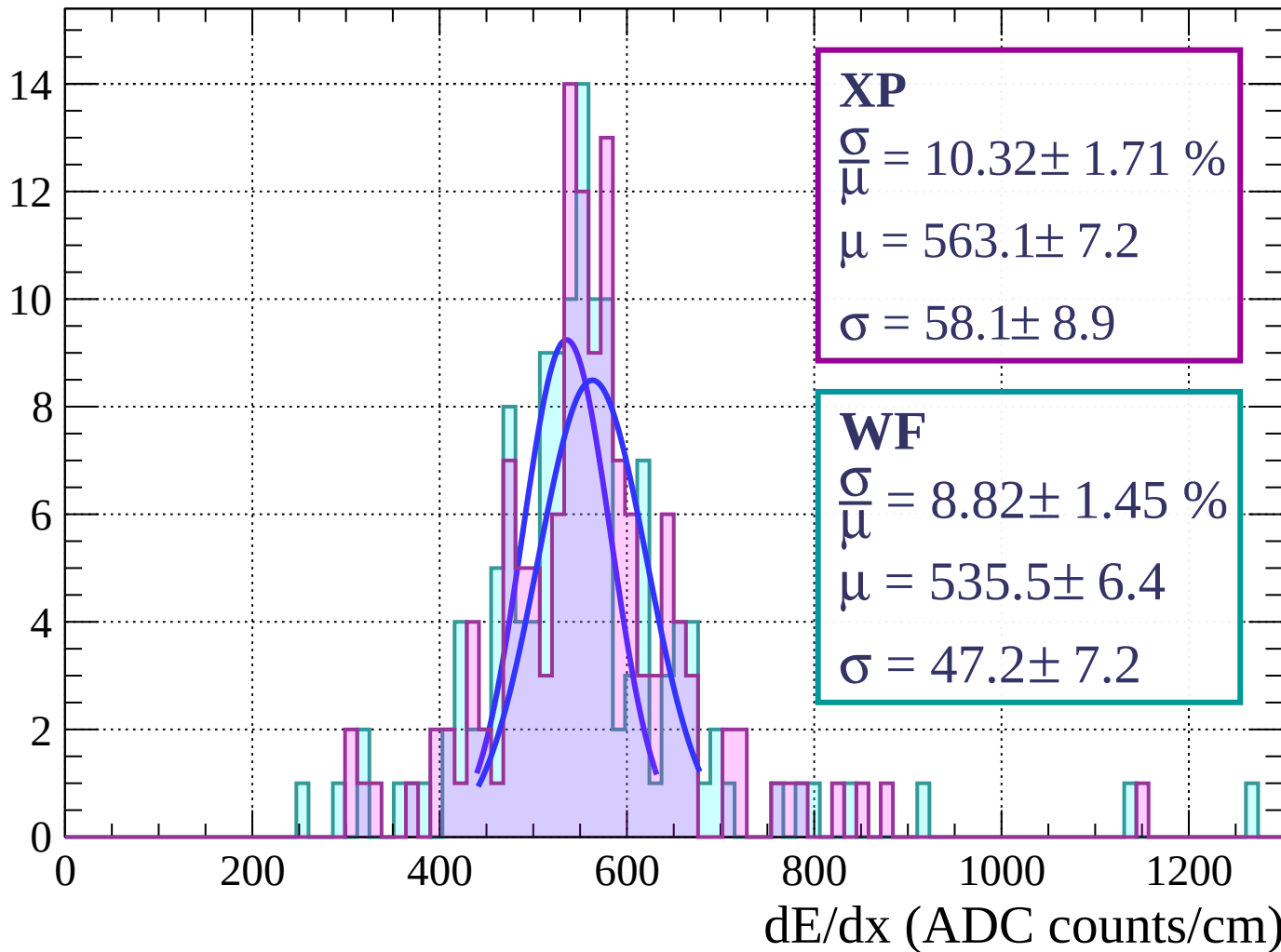
Energy loss | $48 < \phi < 51$

Count



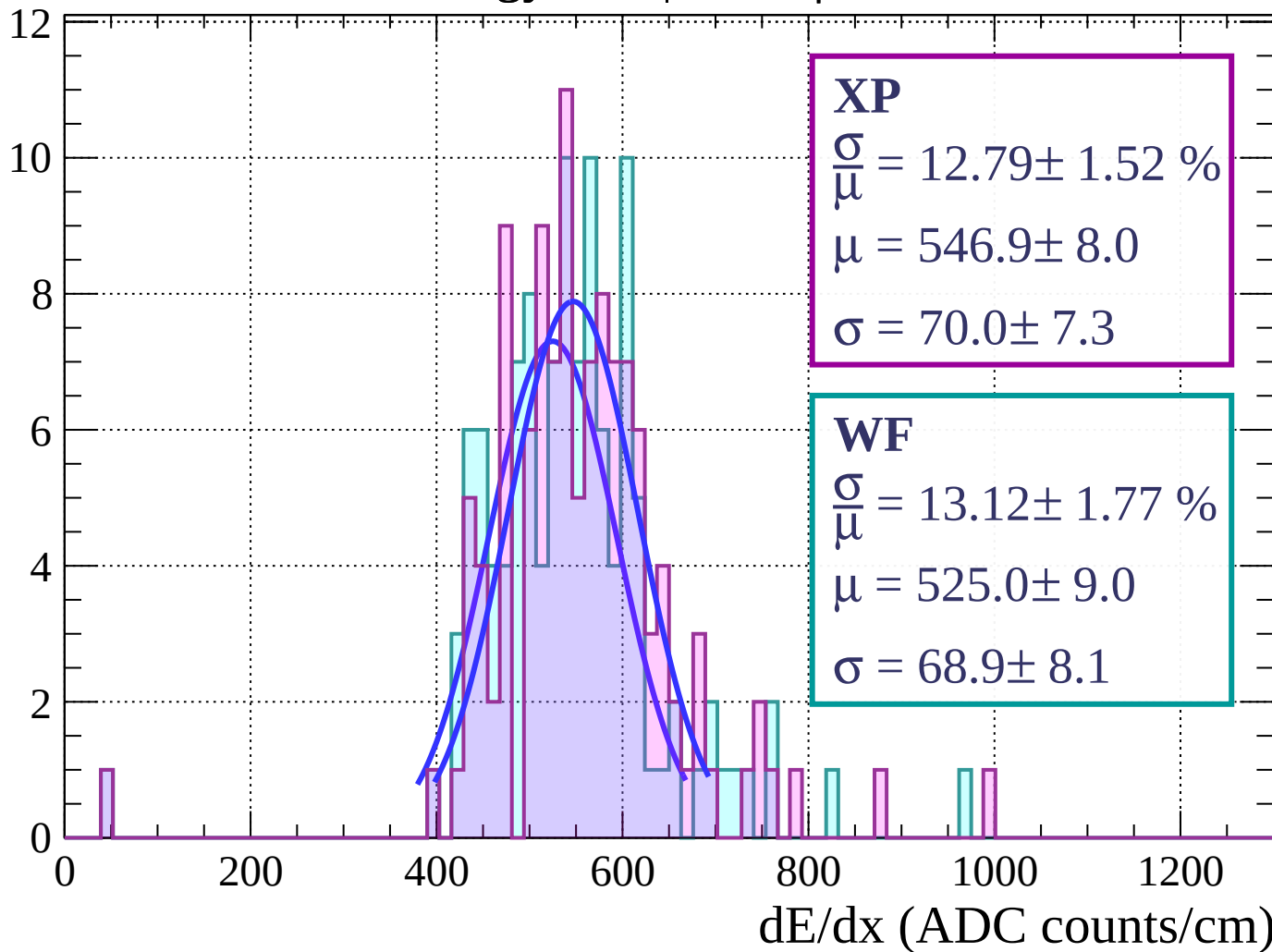
Energy loss | $51 < \phi < 54$

Count



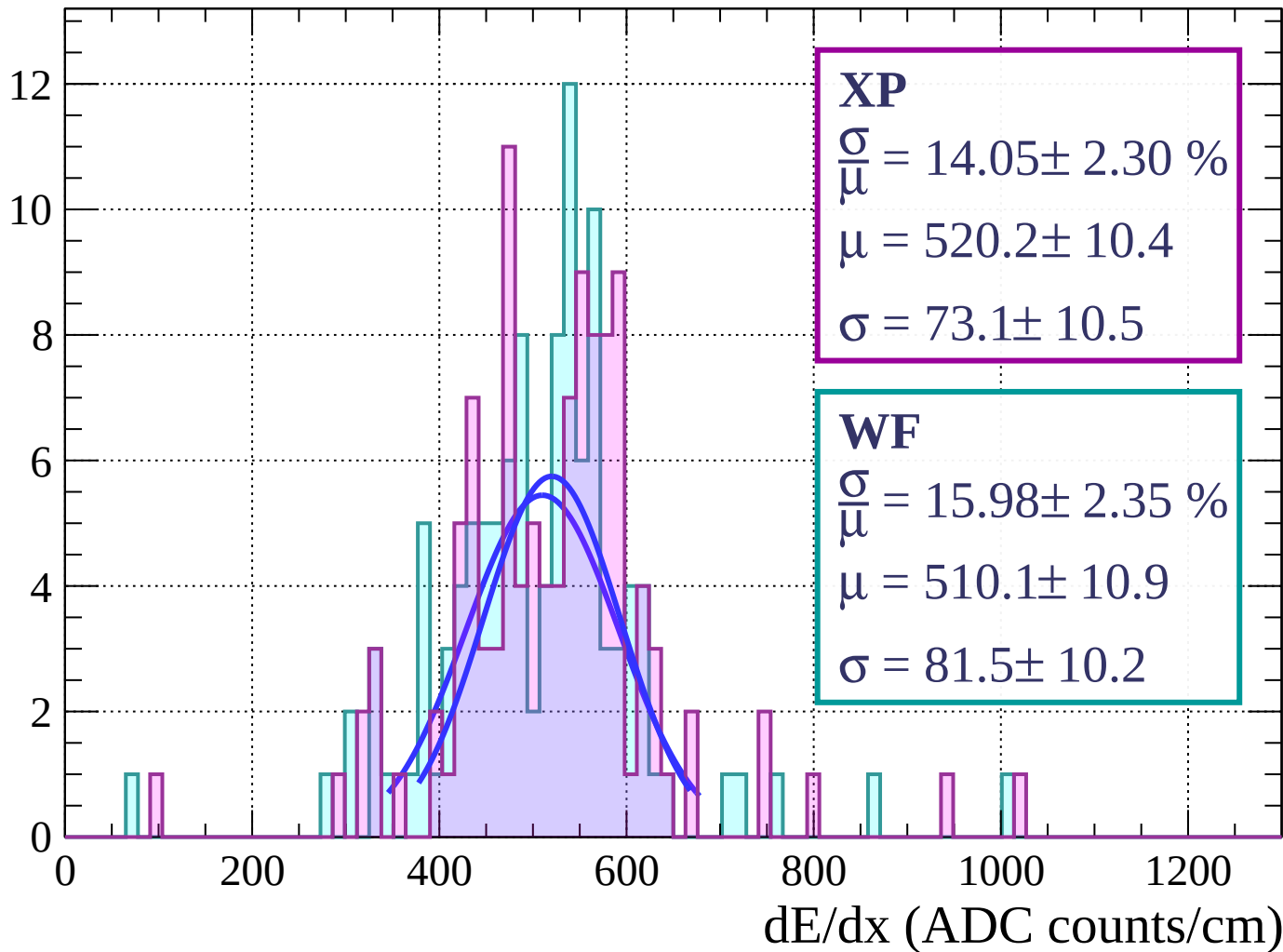
Energy loss | $54 < \phi < 57$

Count



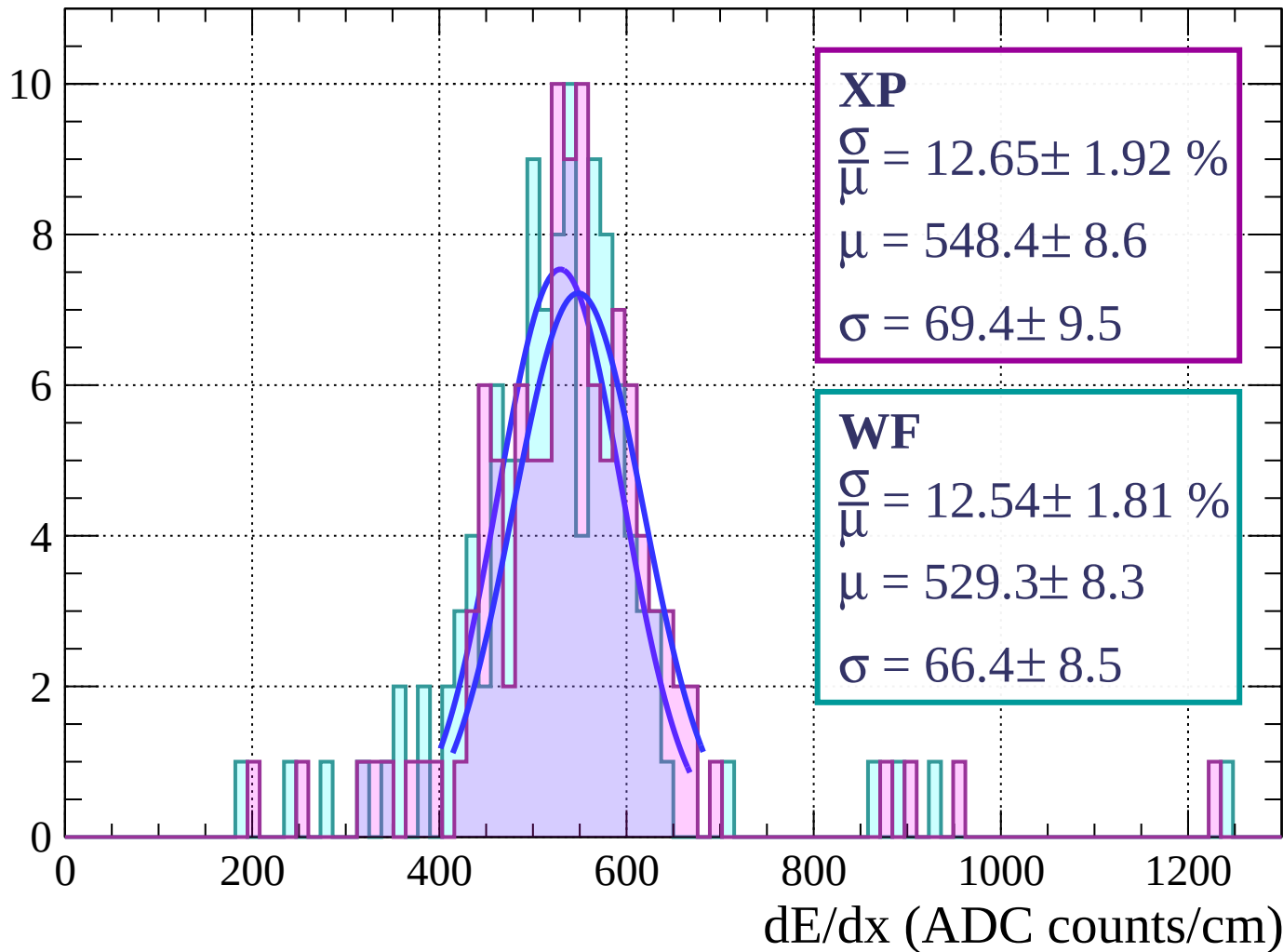
Energy loss | $57 < \phi < 60$

Count



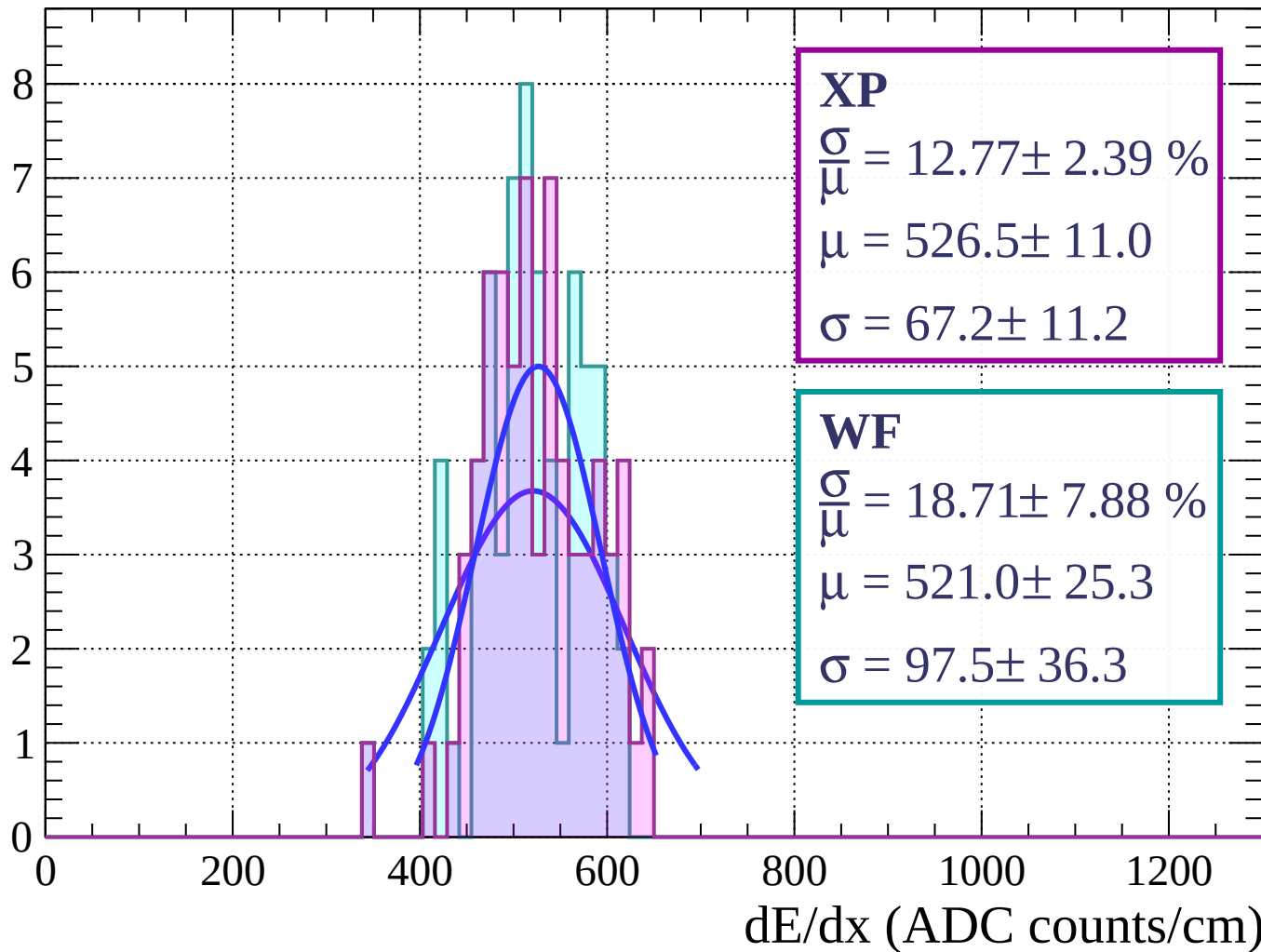
Energy loss | $60 < \phi < 63$

Count



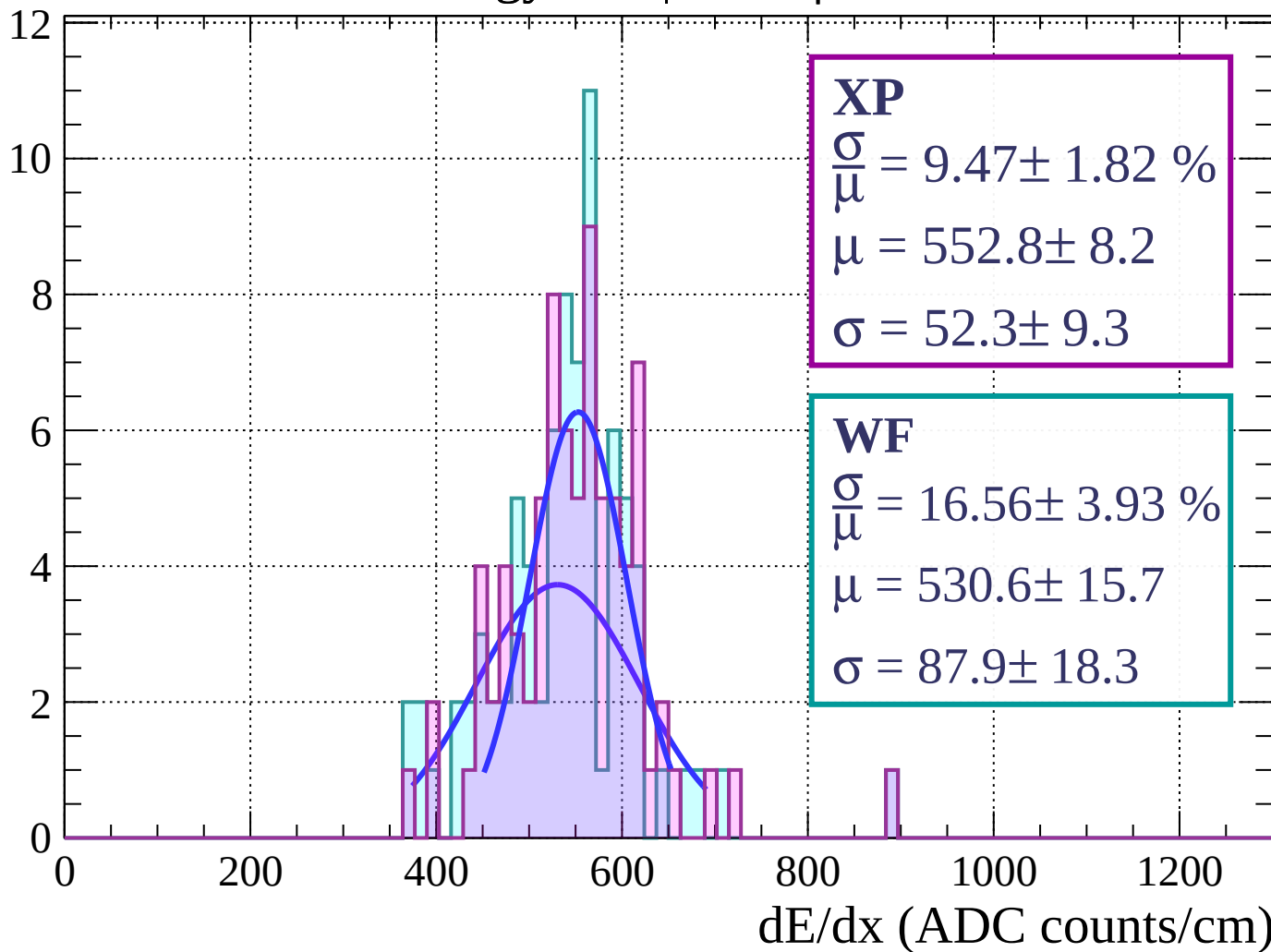
Energy loss | $63 < \phi < 66$

Count



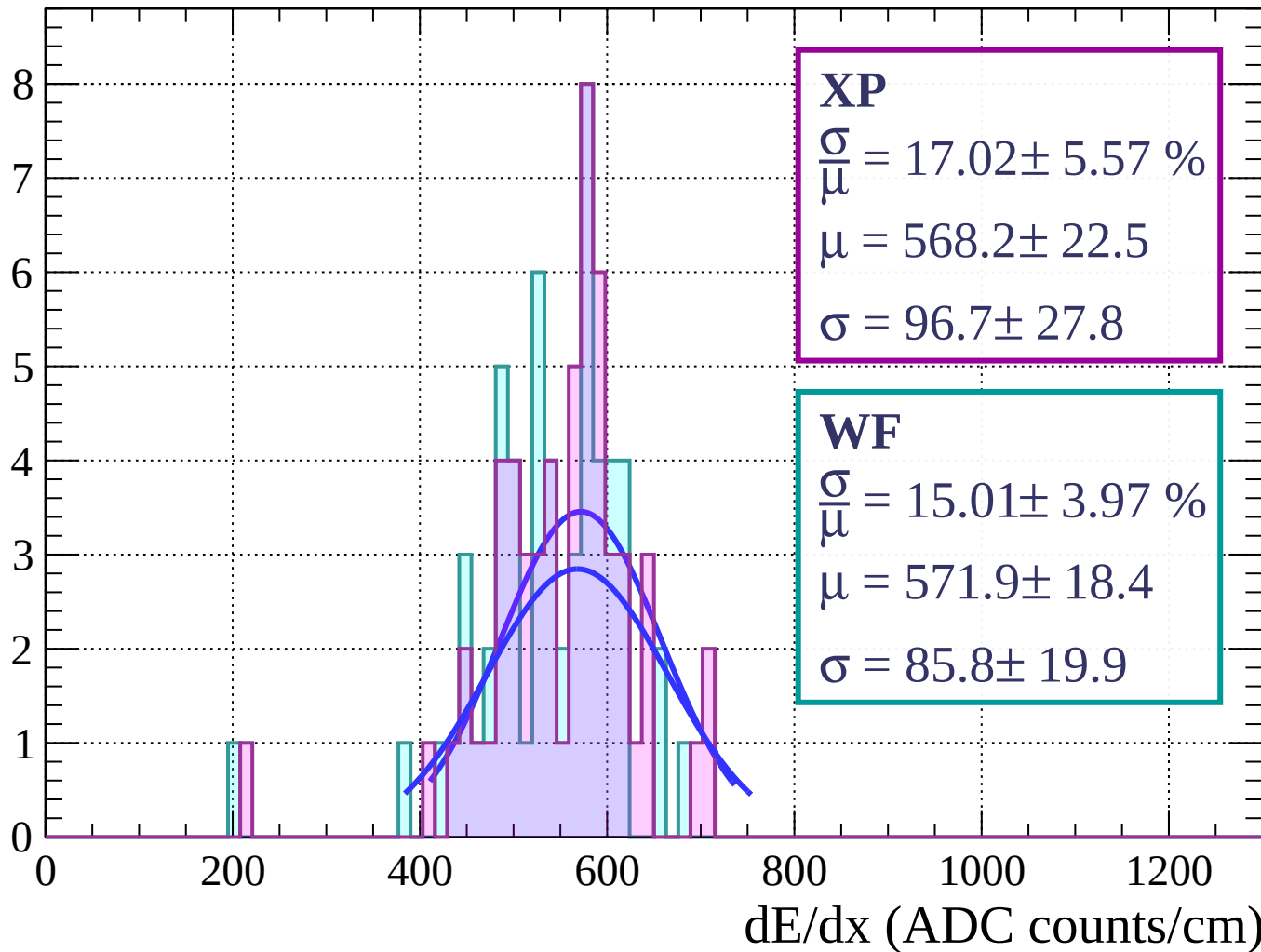
Energy loss | $66 < \phi < 69$

Count



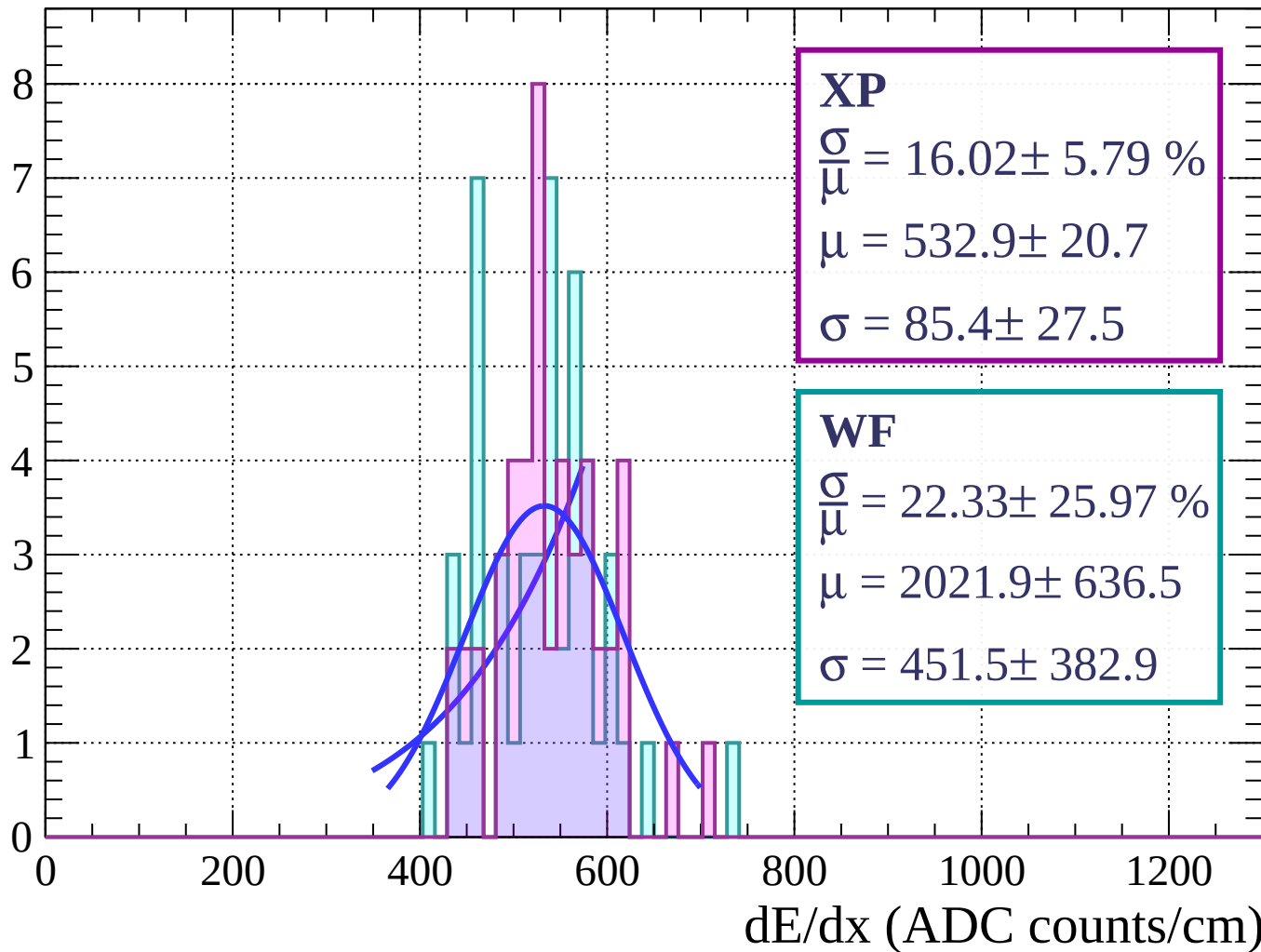
Energy loss | $69 < \phi < 72$

Count



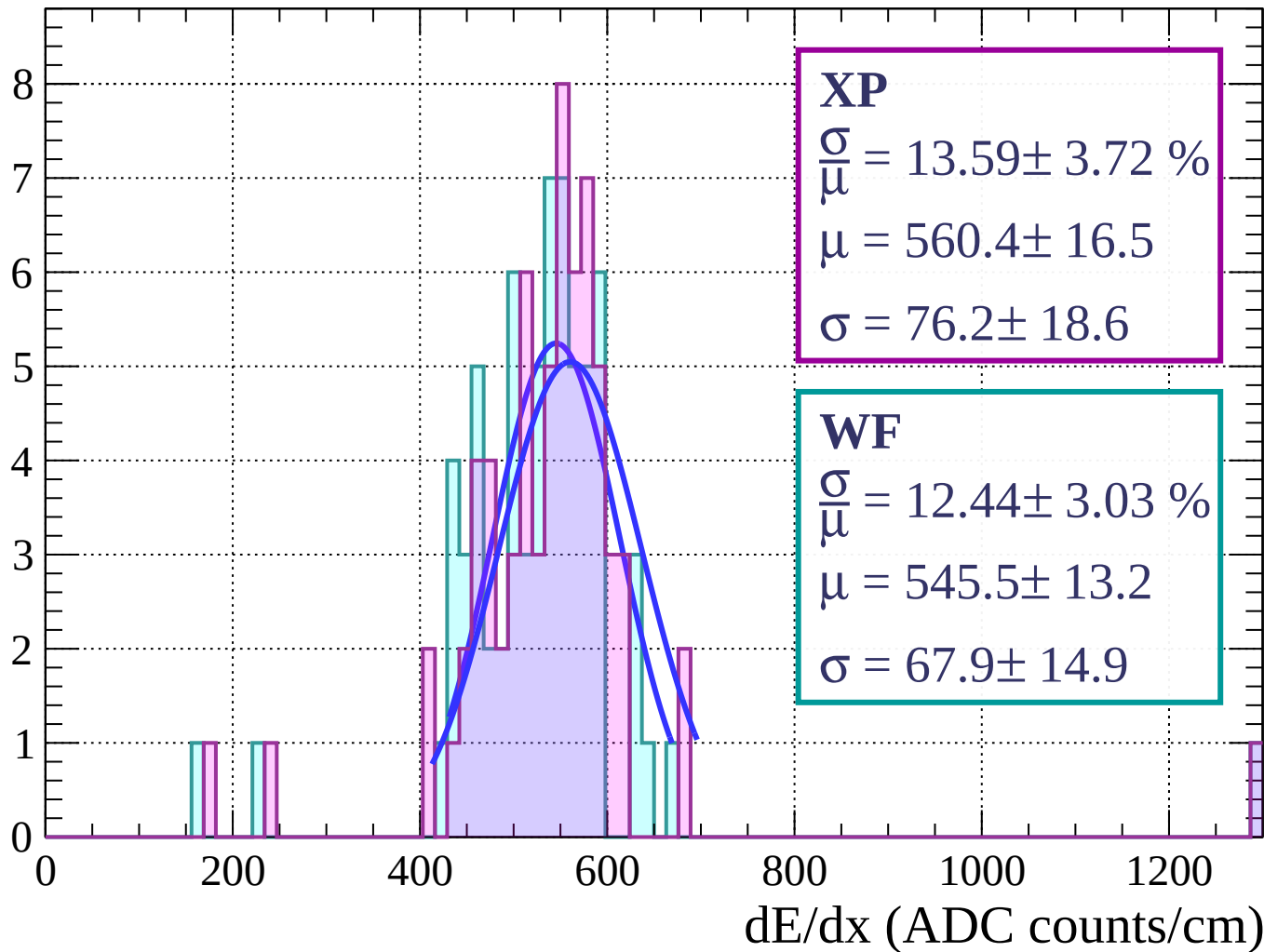
Energy loss | $72 < \phi < 75$

Count



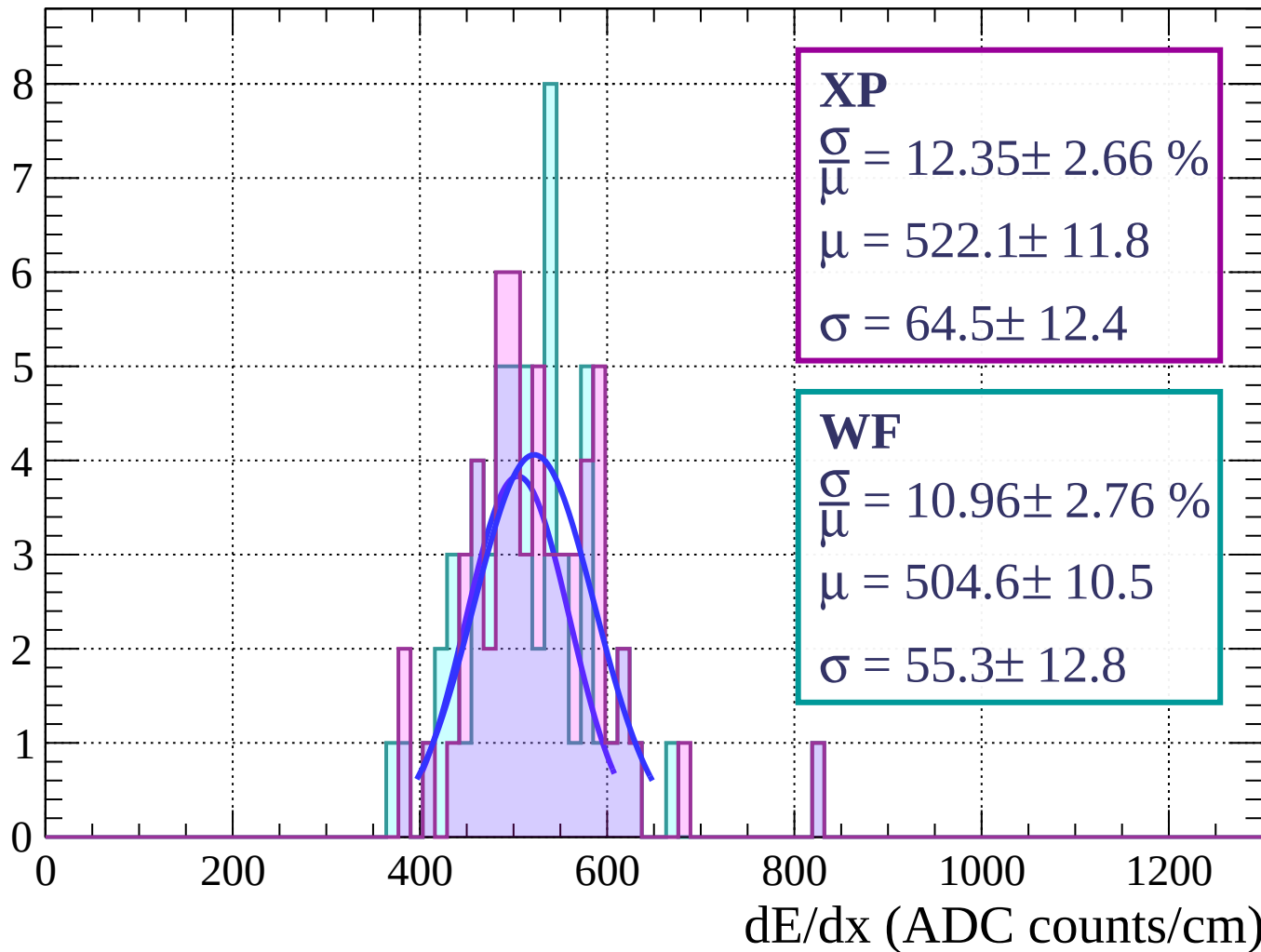
Energy loss | $75 < \phi < 78$

Count



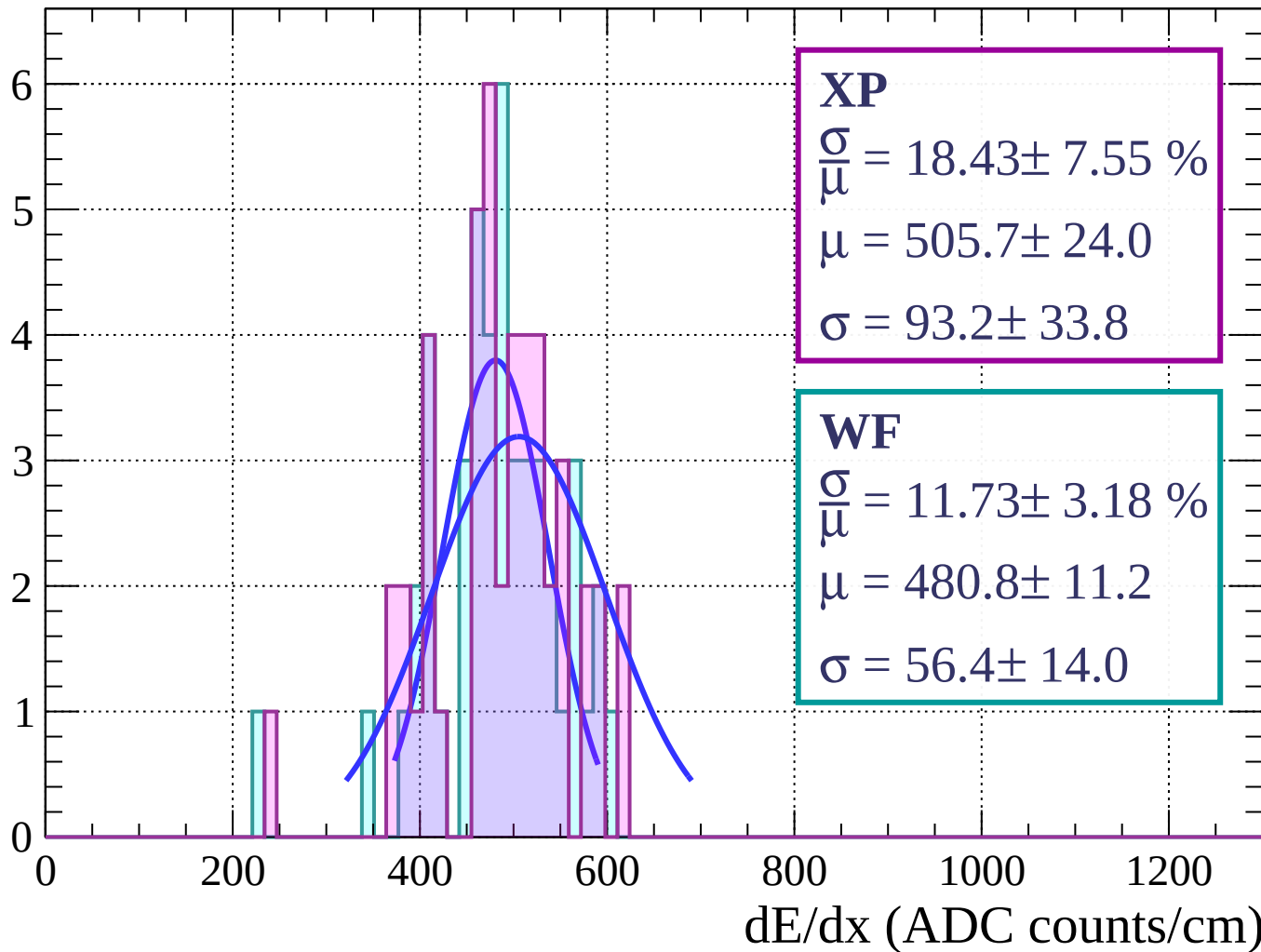
Energy loss | $78 < \phi < 81$

Count



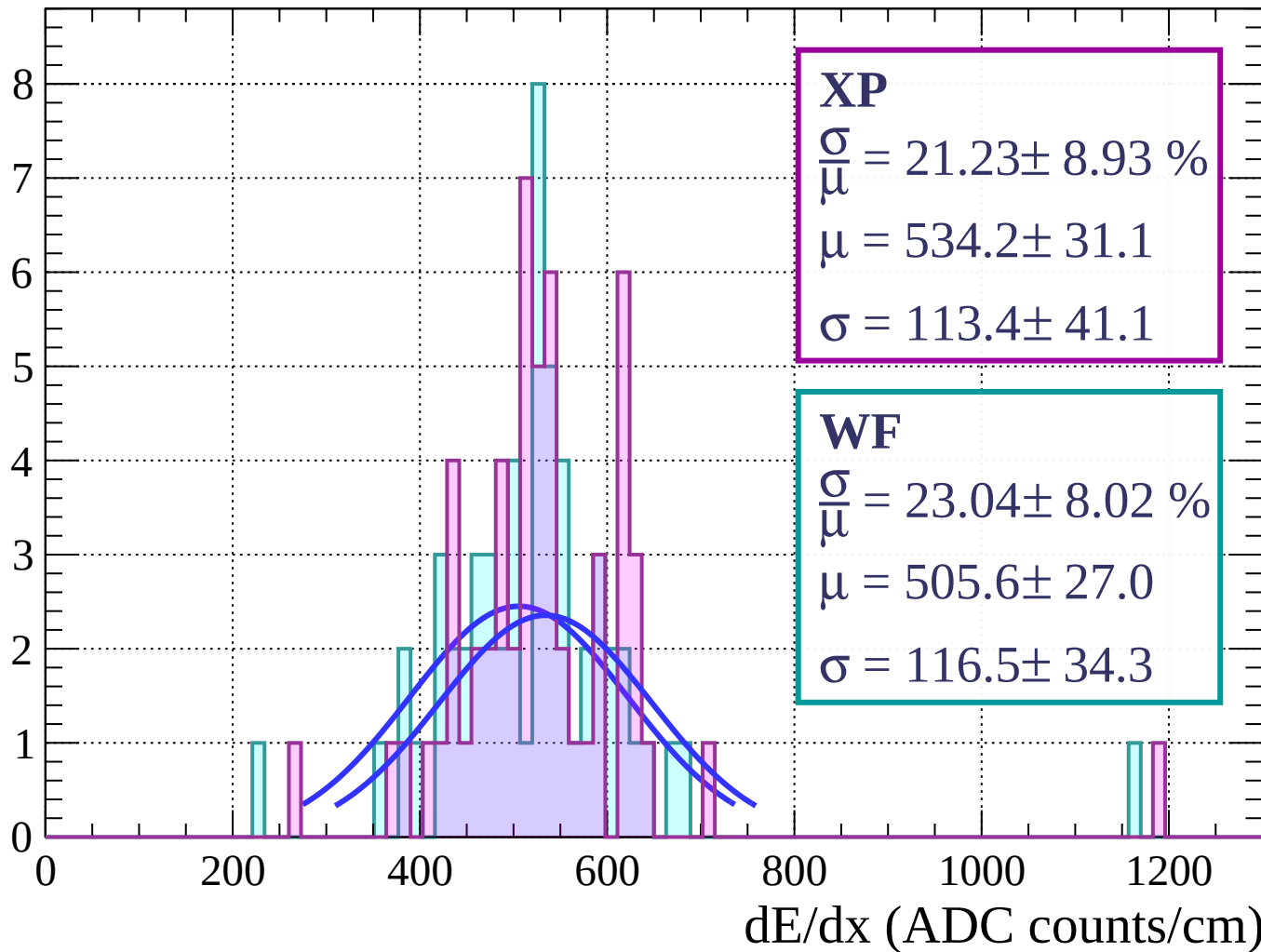
Energy loss | $81 < \phi < 84$

Count



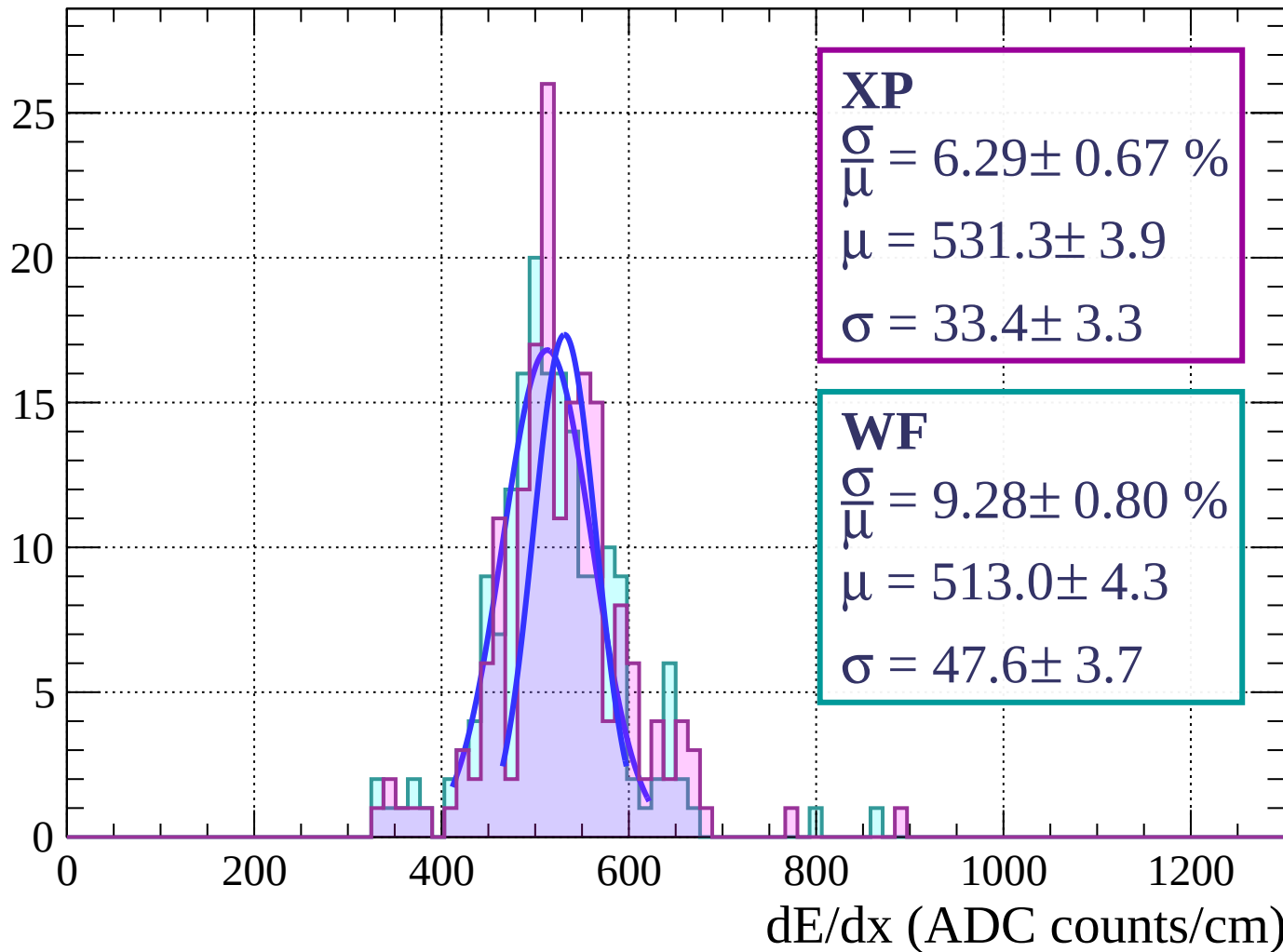
Energy loss | $84 < \phi < 87$

Count



Energy loss | $87 < \phi < 90$

Count



Energy loss | $0 < \theta < 3$

Count

$\times 10^3$

XP

$$\frac{\sigma}{\mu} = 8.78 \pm 0.03 \%$$

$$\mu = 504.8 \pm 0.1$$

$$\sigma = 44.3 \pm 0.1$$

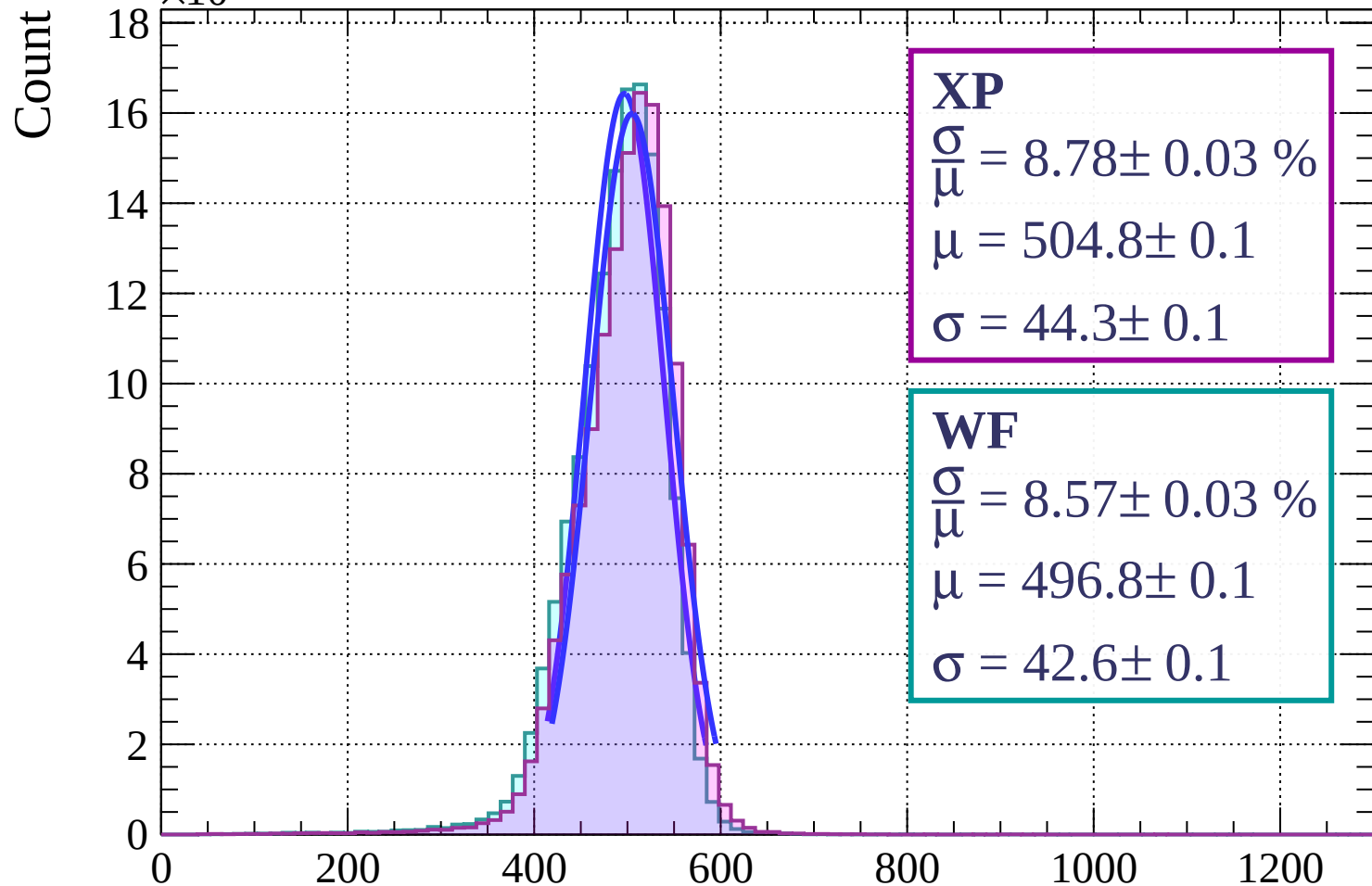
WF

$$\frac{\sigma}{\mu} = 8.57 \pm 0.03 \%$$

$$\mu = 496.8 \pm 0.1$$

$$\sigma = 42.6 \pm 0.1$$

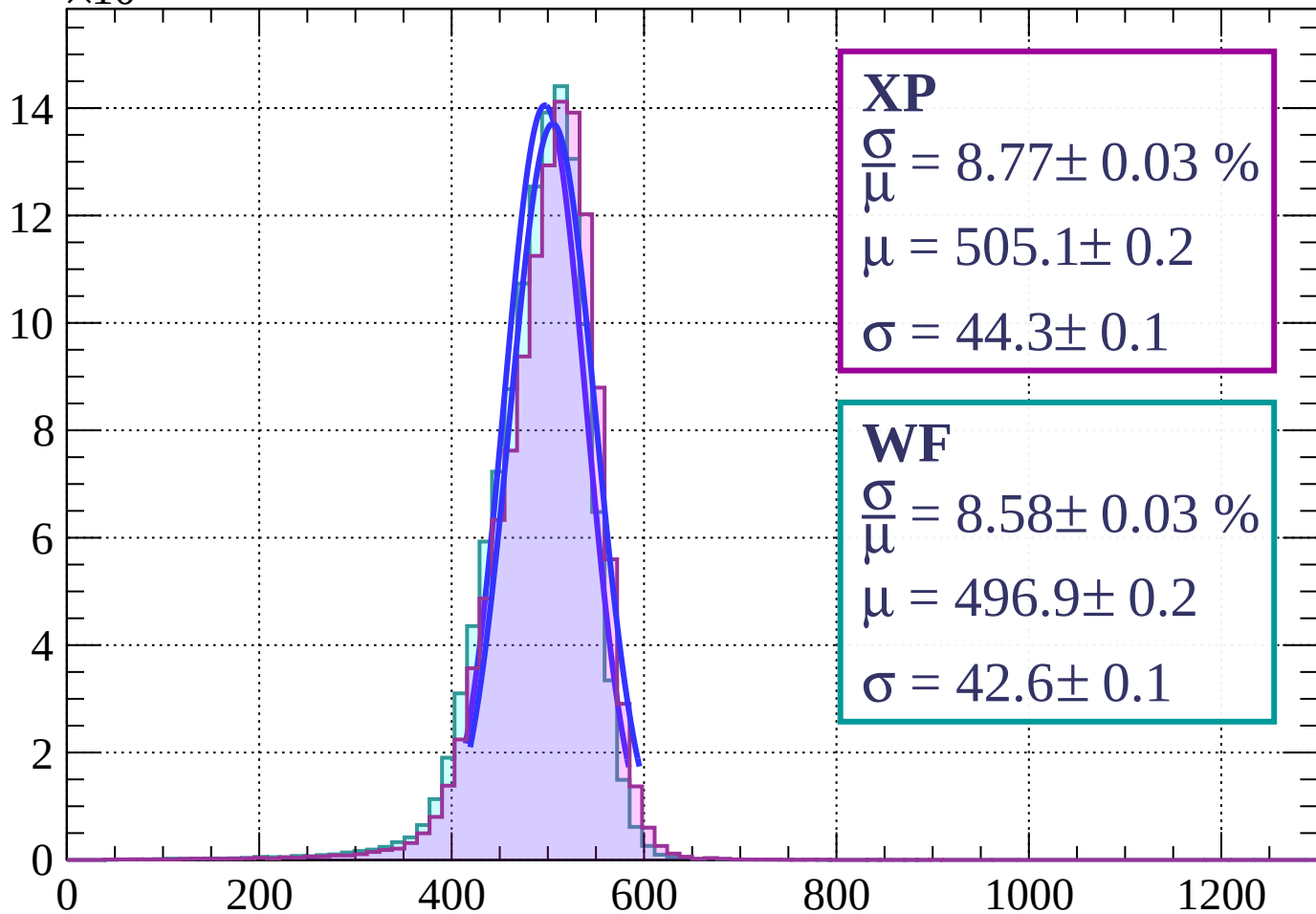
dE/dx (ADC counts/cm)



Energy loss | $3 < \theta < 6$

Count

$\times 10^3$



XP

$$\frac{\sigma}{\mu} = 8.77 \pm 0.03 \%$$

$$\mu = 505.1 \pm 0.2$$

$$\sigma = 44.3 \pm 0.1$$

WF

$$\frac{\sigma}{\mu} = 8.58 \pm 0.03 \%$$

$$\mu = 496.9 \pm 0.2$$

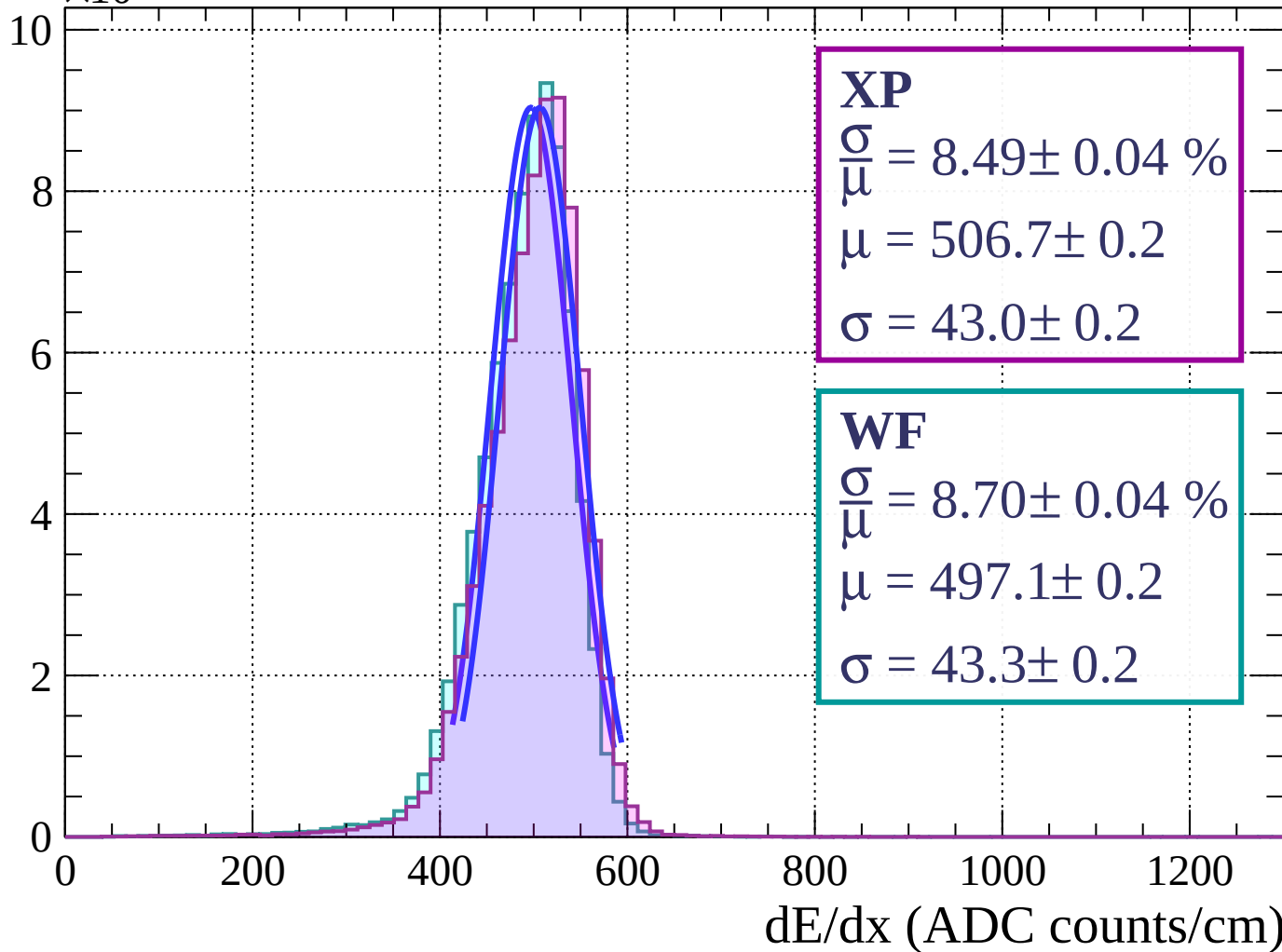
$$\sigma = 42.6 \pm 0.1$$

dE/dx (ADC counts/cm)

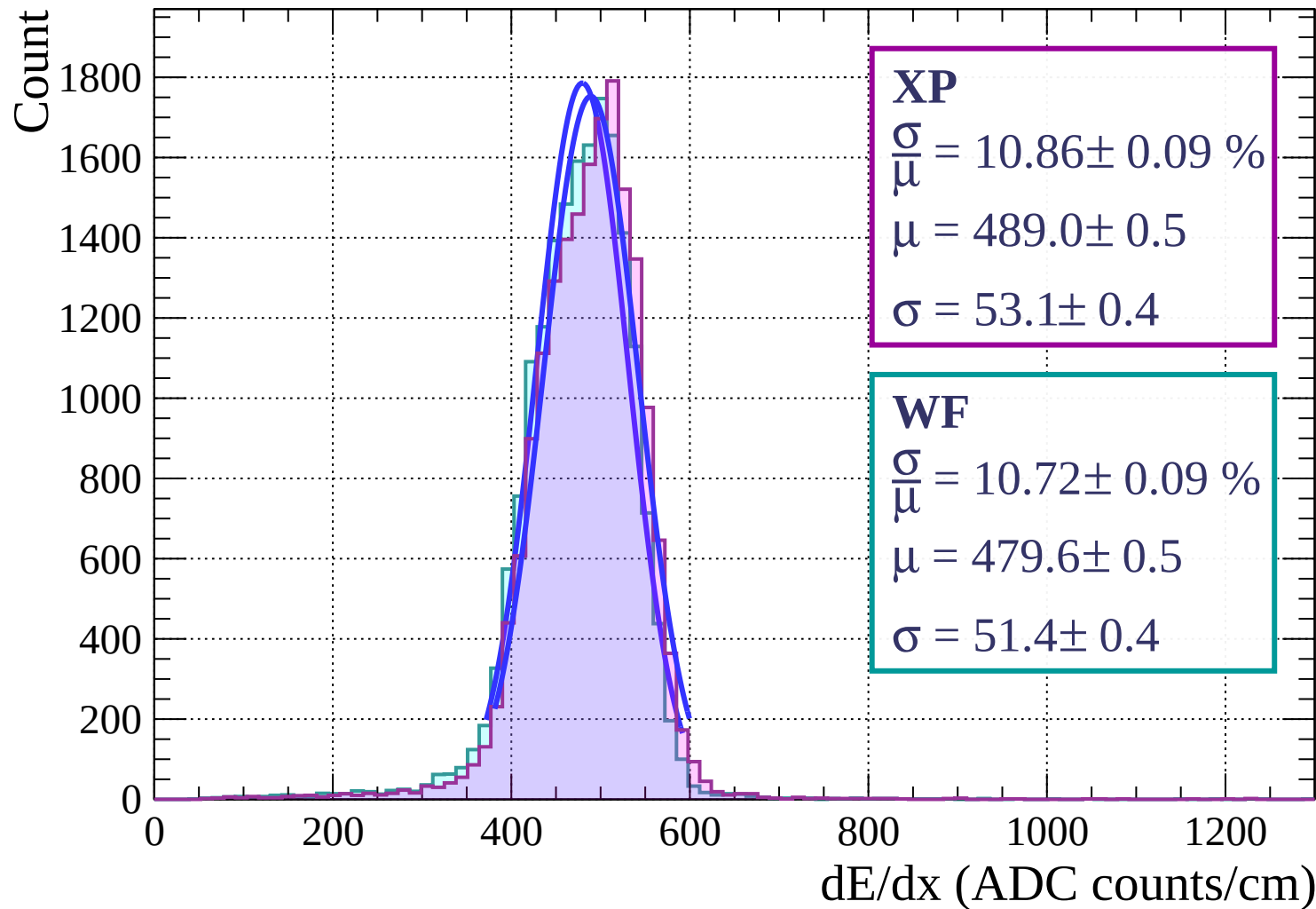
Energy loss | $6 < \theta < 9$

Count

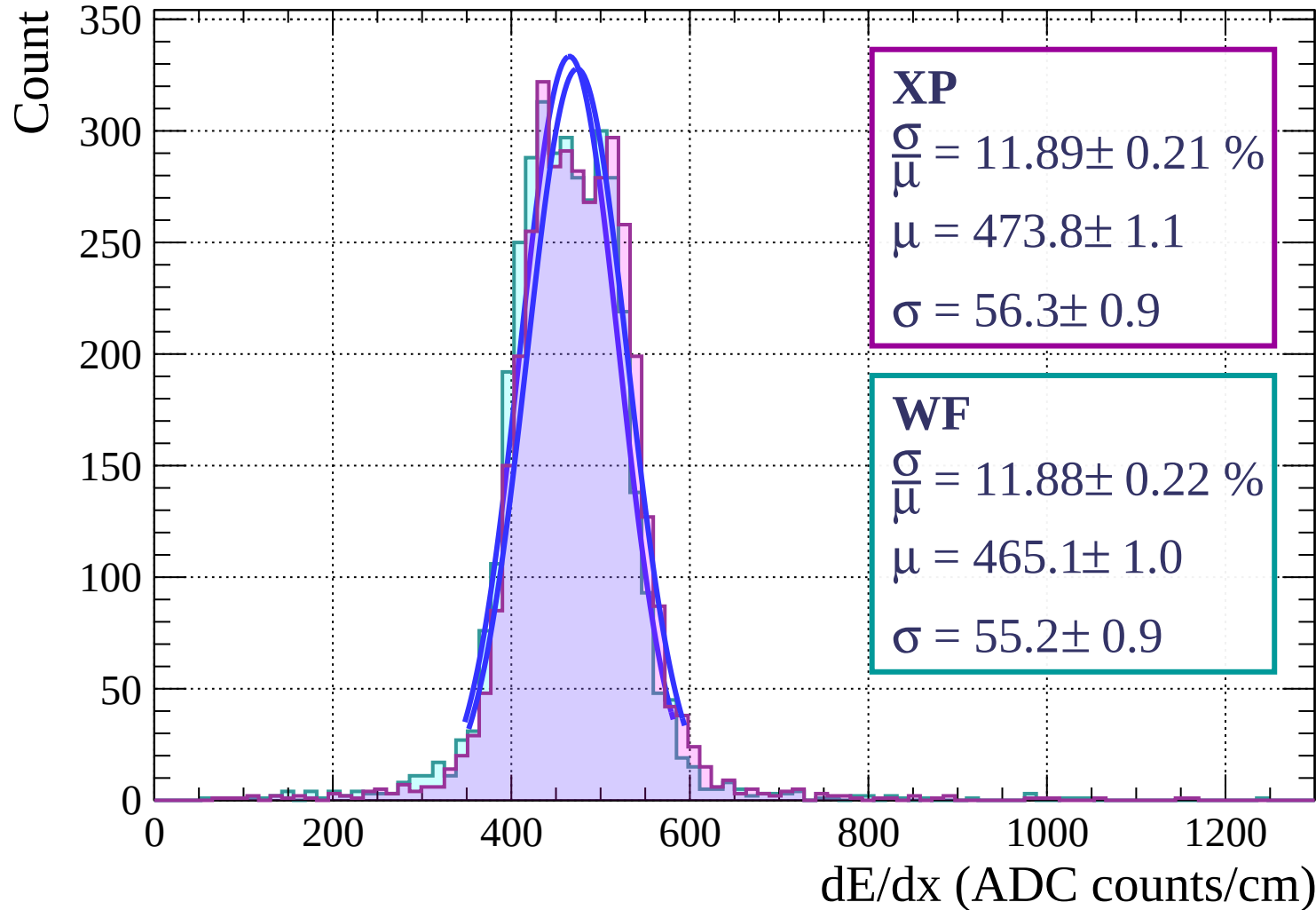
$\times 10^3$



Energy loss | $9 < \theta < 12$

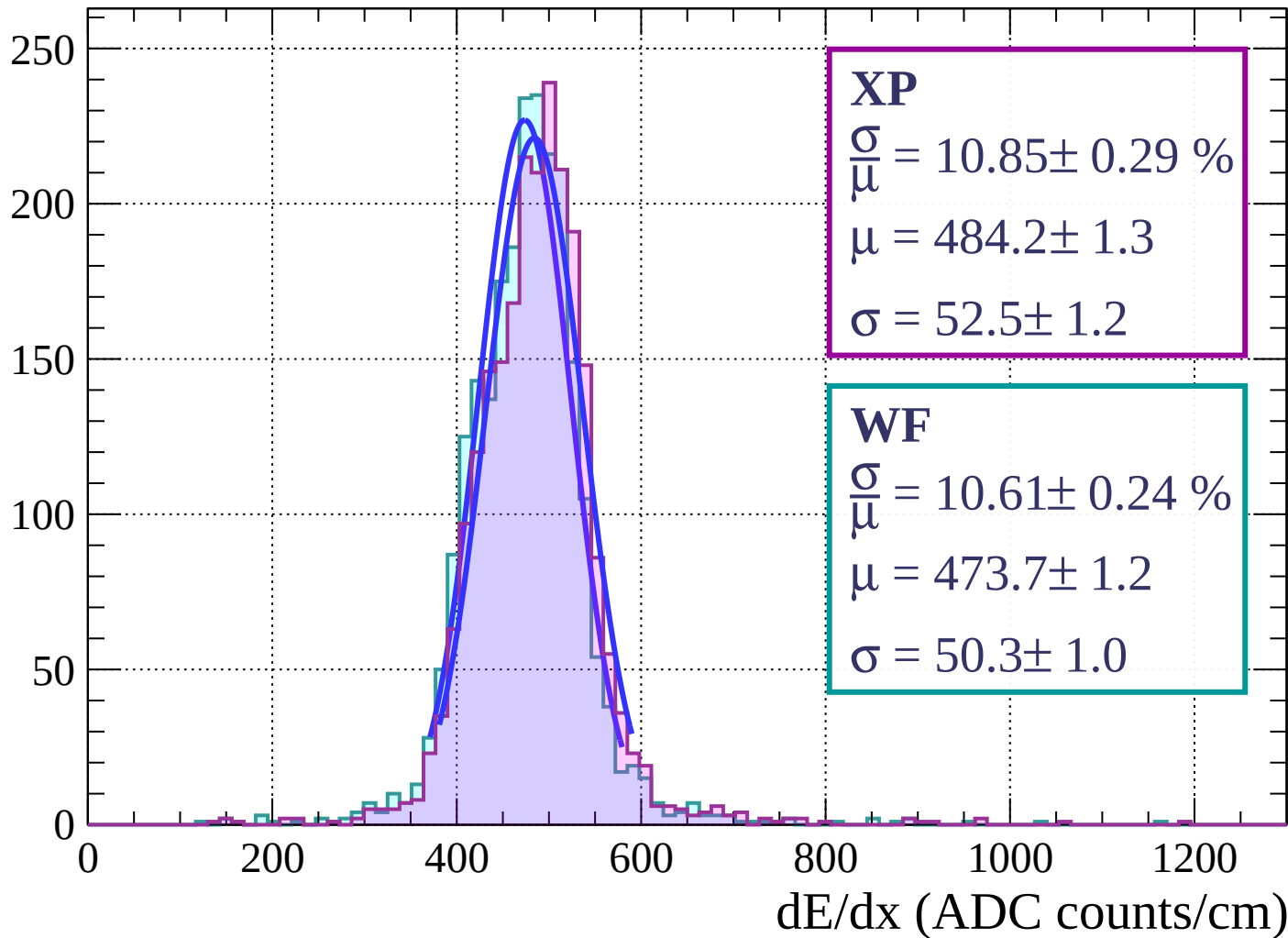


Energy loss | $12 < \theta < 15$



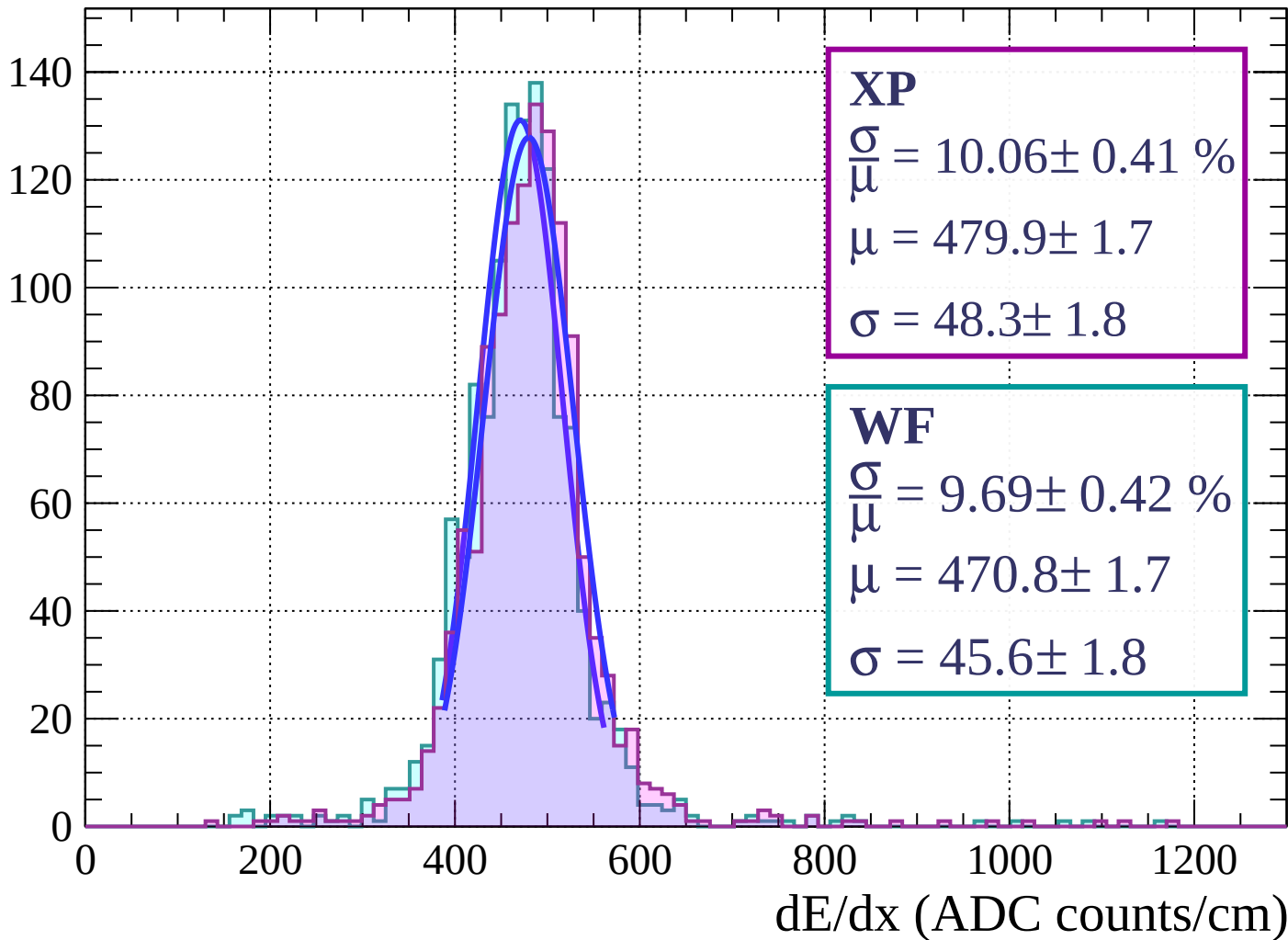
Energy loss | $15 < \theta < 18$

Count



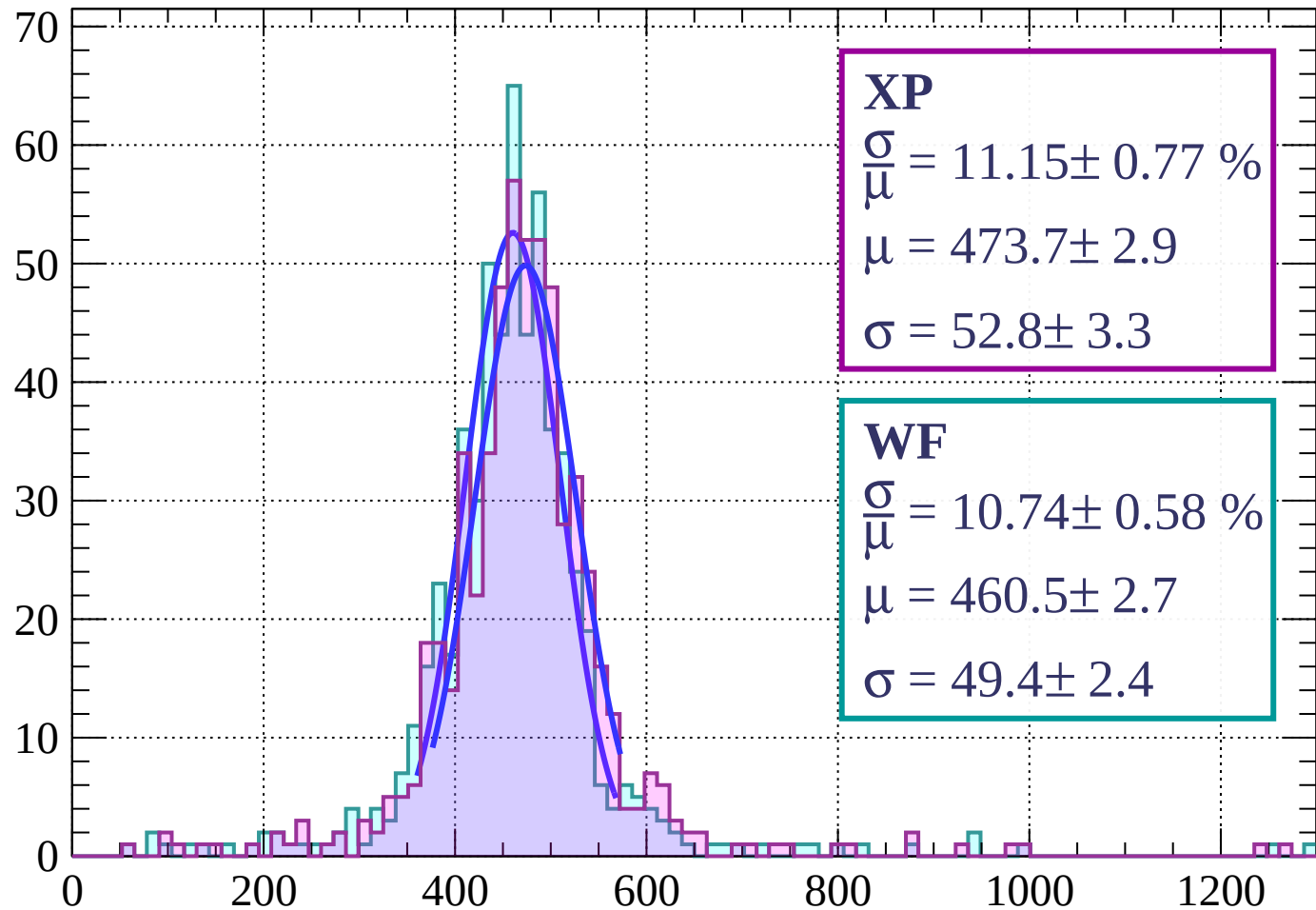
Energy loss | $18 < \theta < 21$

Count



Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 11.15 \pm 0.77 \%$$

$$\mu = 473.7 \pm 2.9$$

$$\sigma = 52.8 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 10.74 \pm 0.58 \%$$

$$\mu = 460.5 \pm 2.7$$

$$\sigma = 49.4 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 27$

Count

XP

$$\frac{\sigma}{\mu} = 12.03 \pm 0.89 \%$$

$$\mu = 460.9 \pm 4.0$$

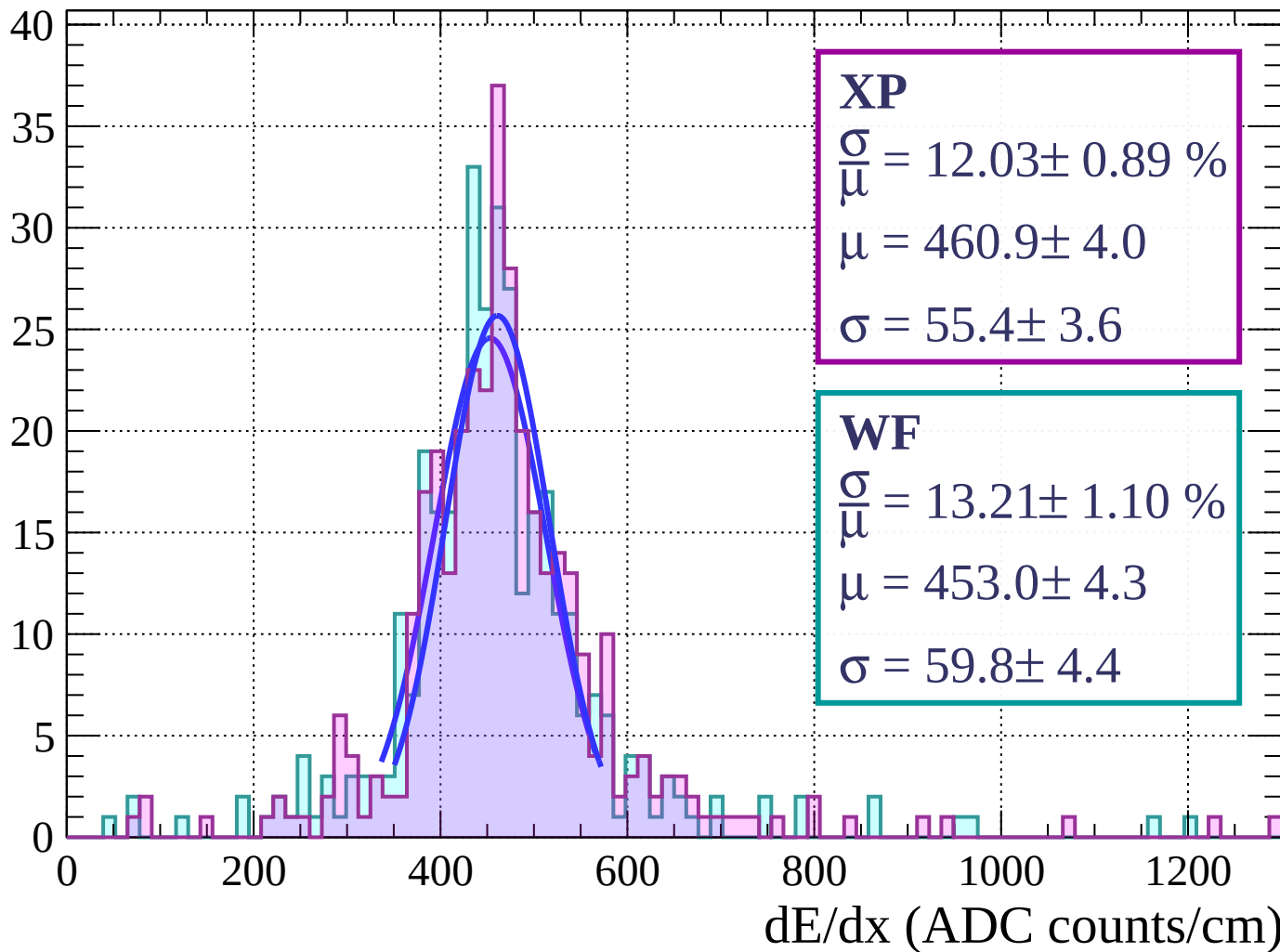
$$\sigma = 55.4 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 13.21 \pm 1.10 \%$$

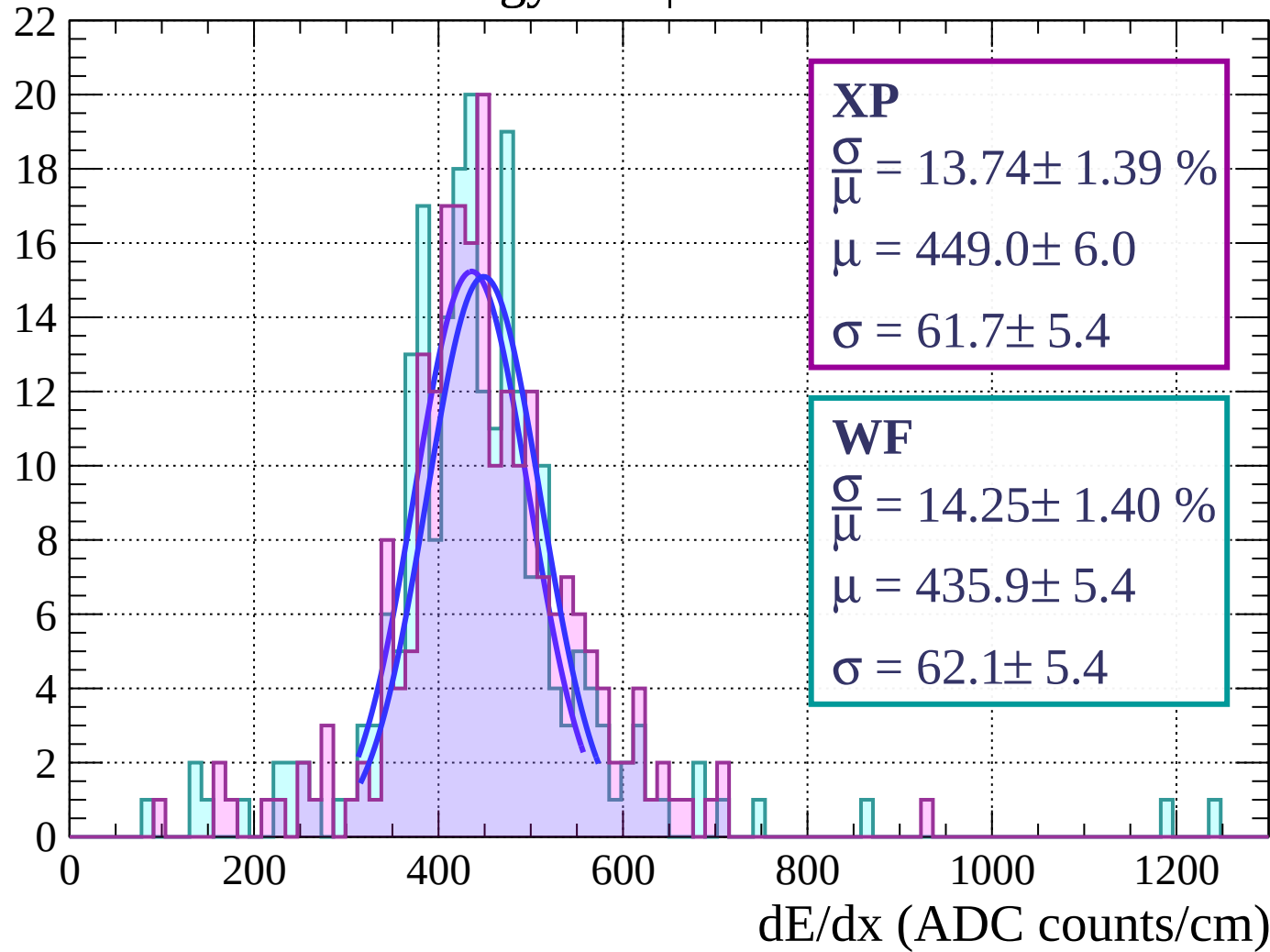
$$\mu = 453.0 \pm 4.3$$

$$\sigma = 59.8 \pm 4.4$$



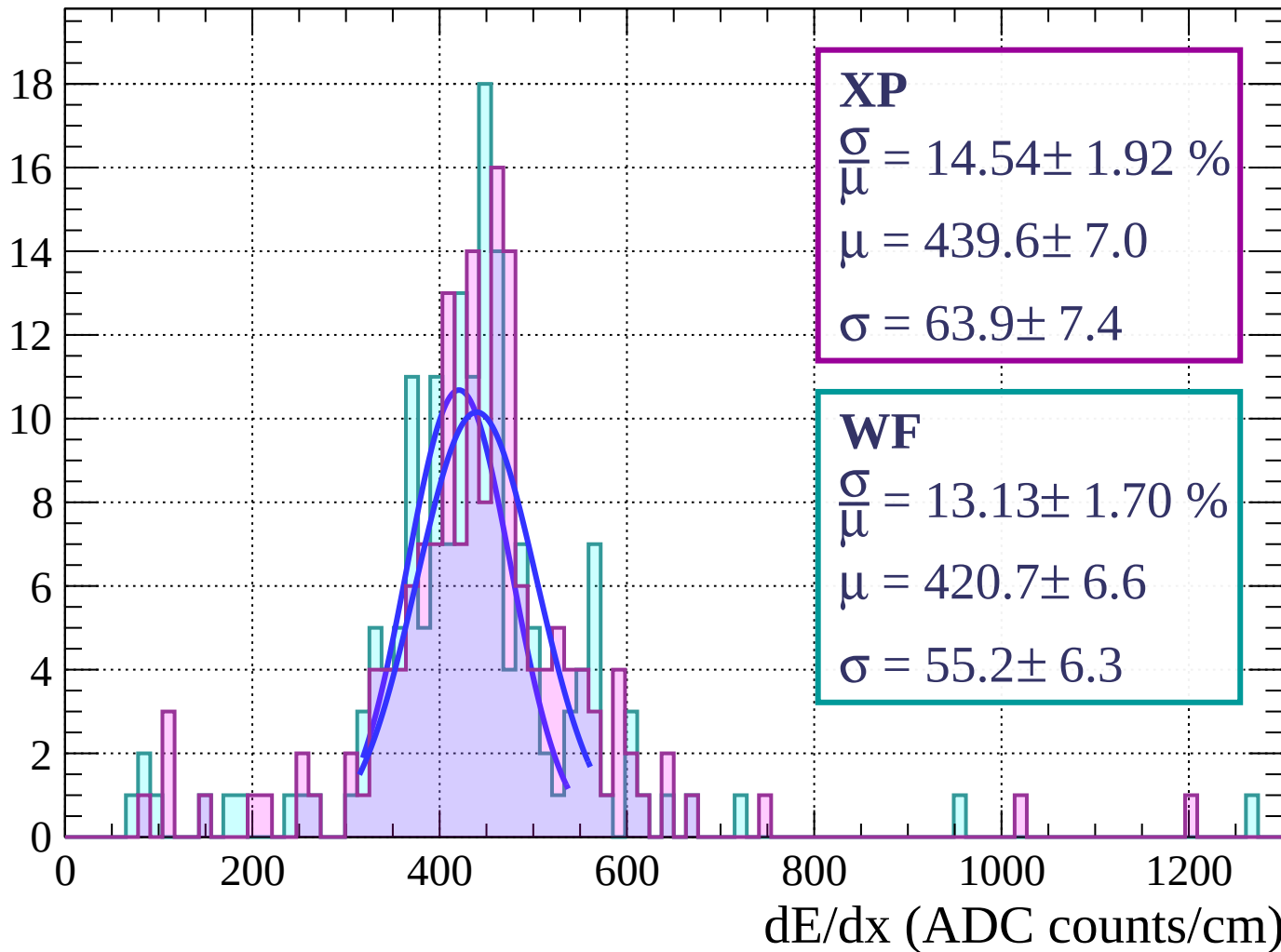
Energy loss | $27 < \theta < 30$

Count



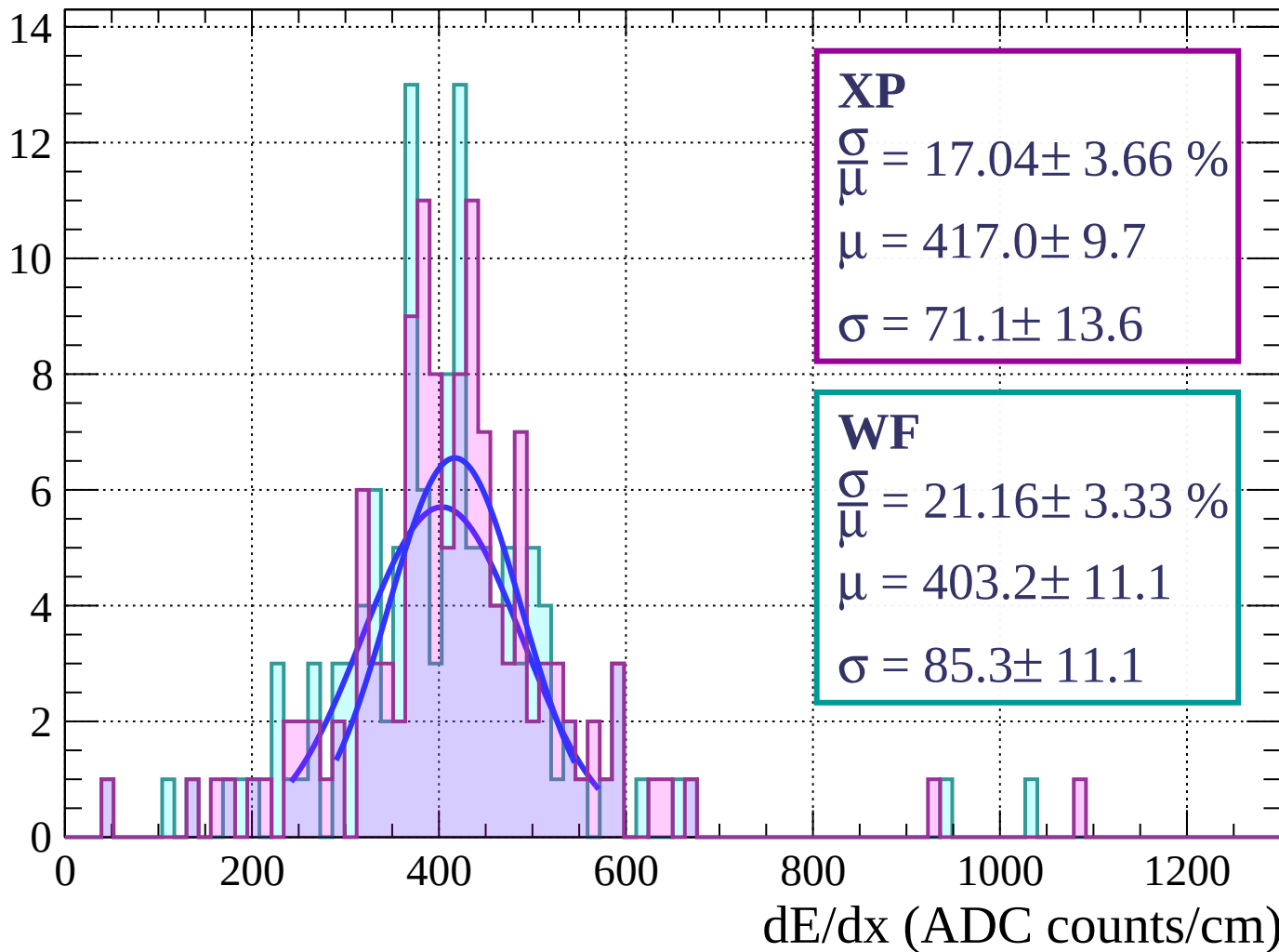
Energy loss | $30 < \theta < 33$

Count



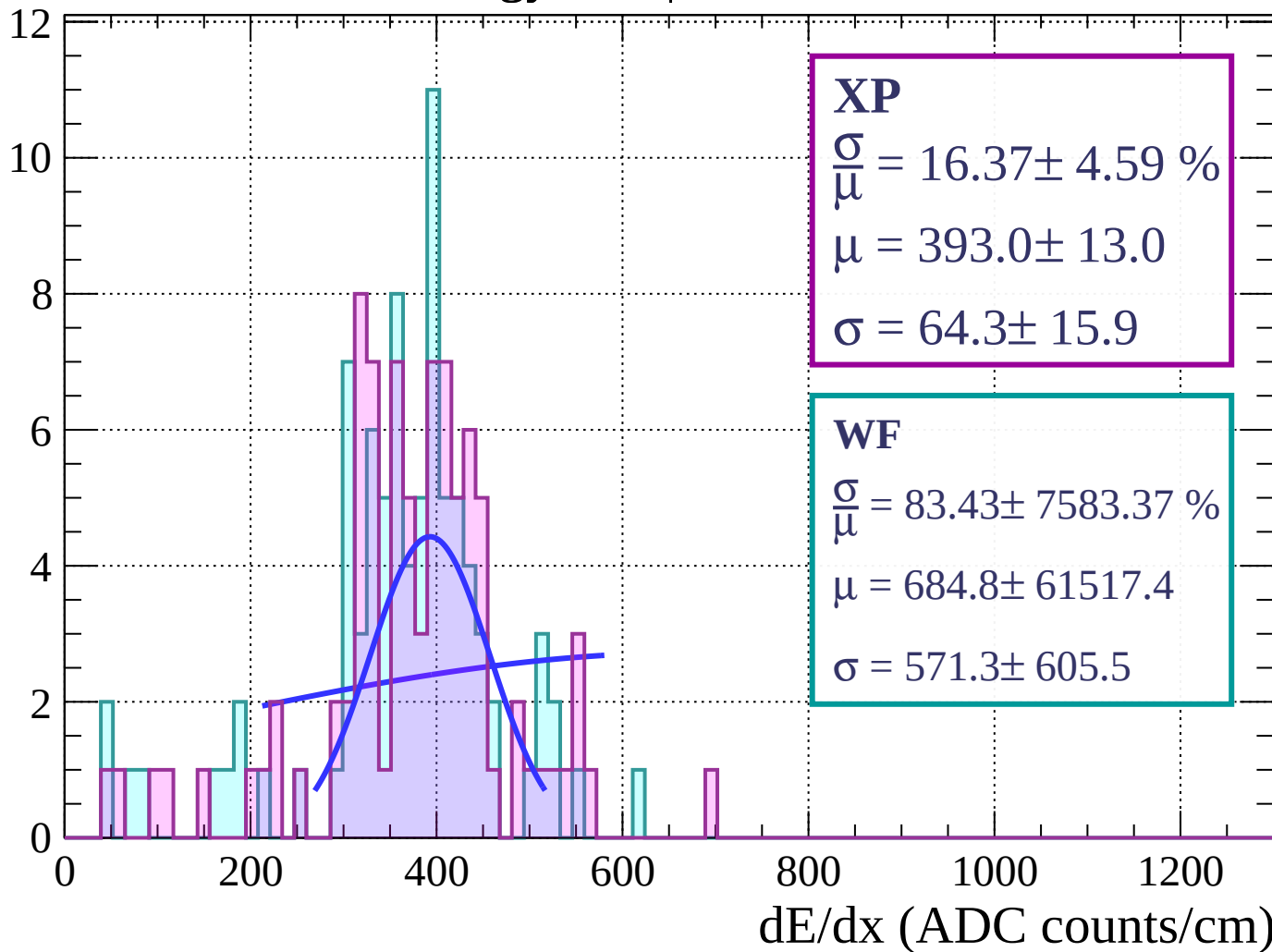
Energy loss | $33 < \theta < 36$

Count



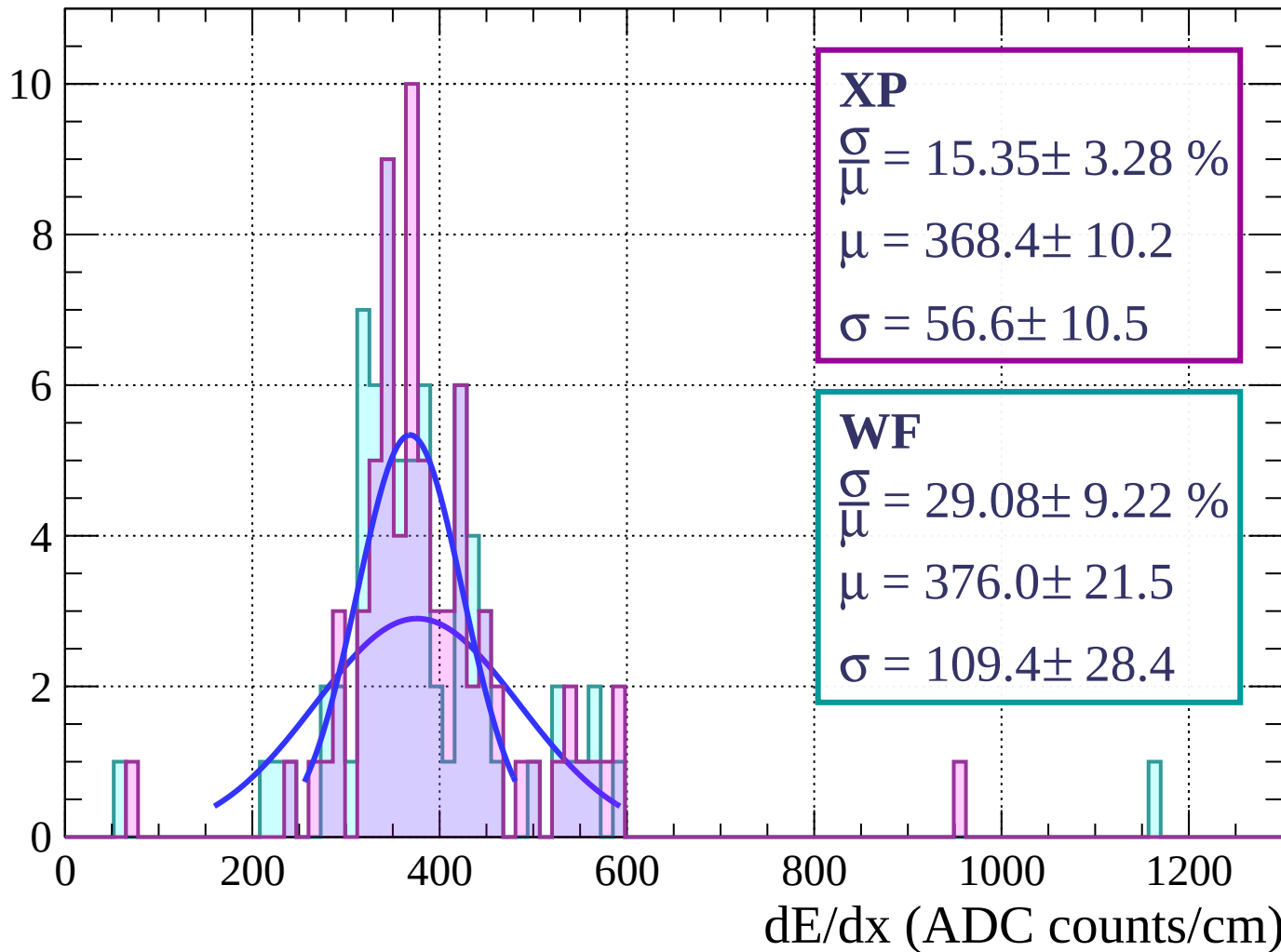
Energy loss | $36 < \theta < 39$

Count



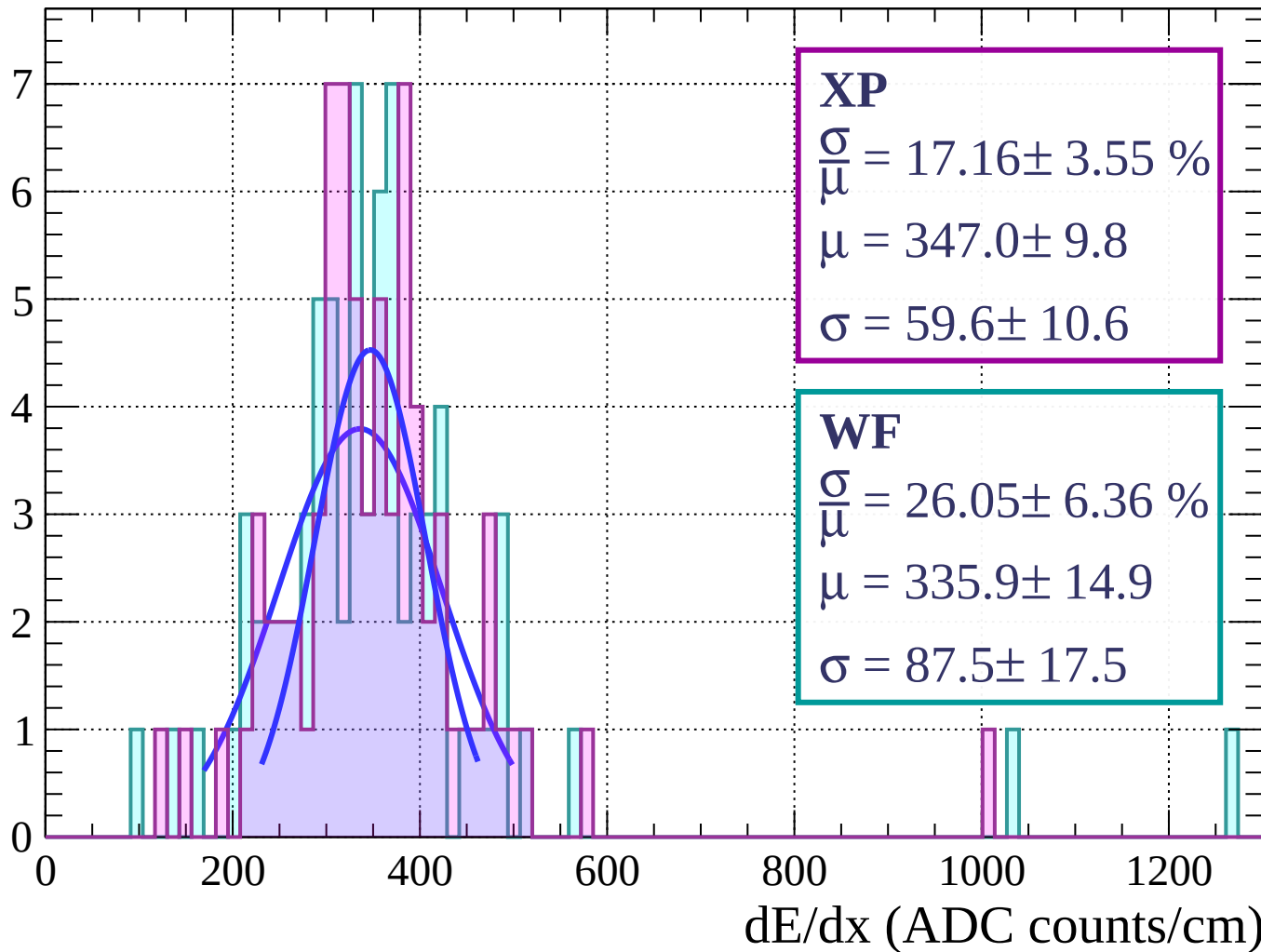
Energy loss | $39 < \theta < 42$

Count



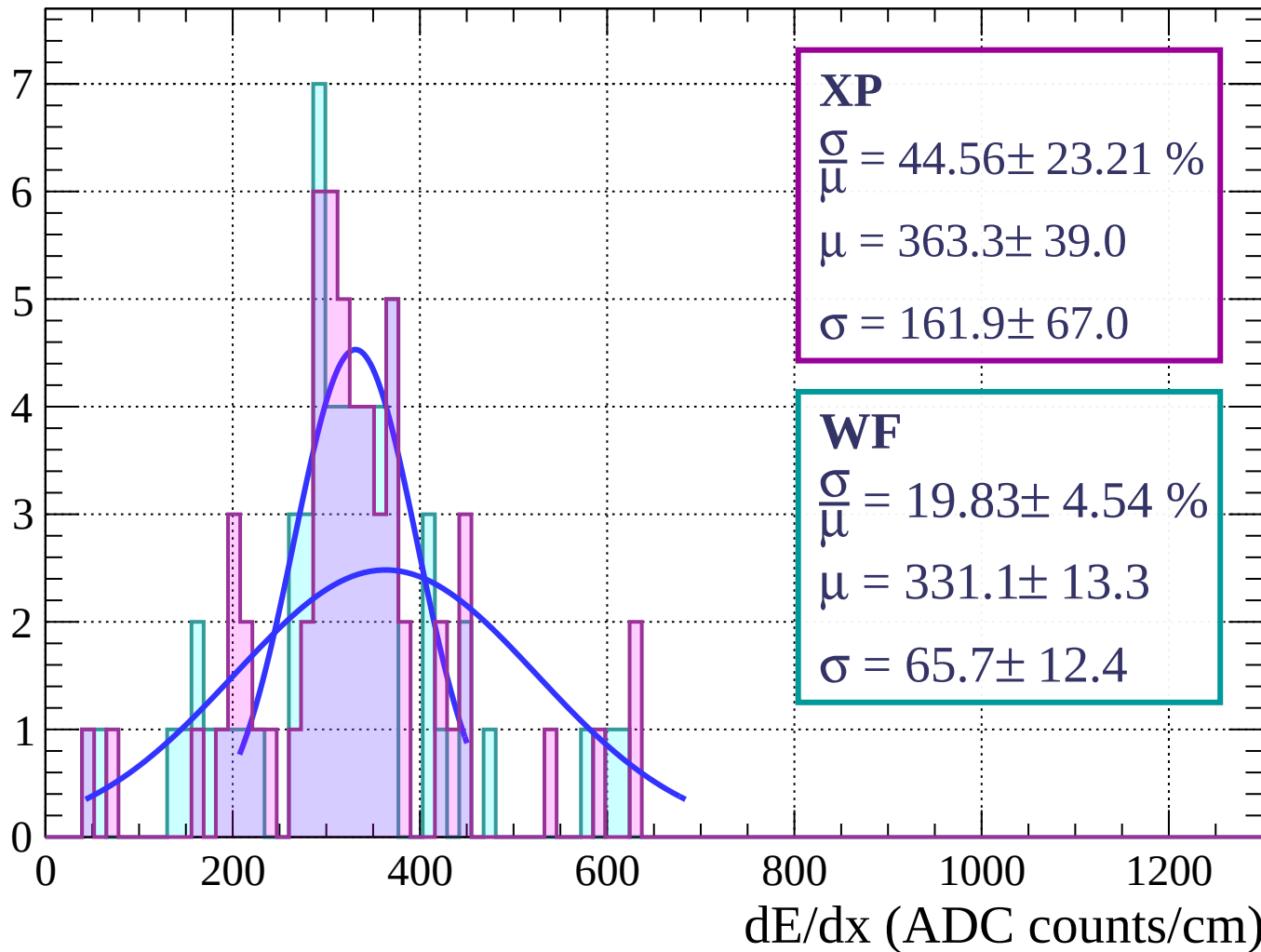
Energy loss | $42 < \theta < 45$

Count



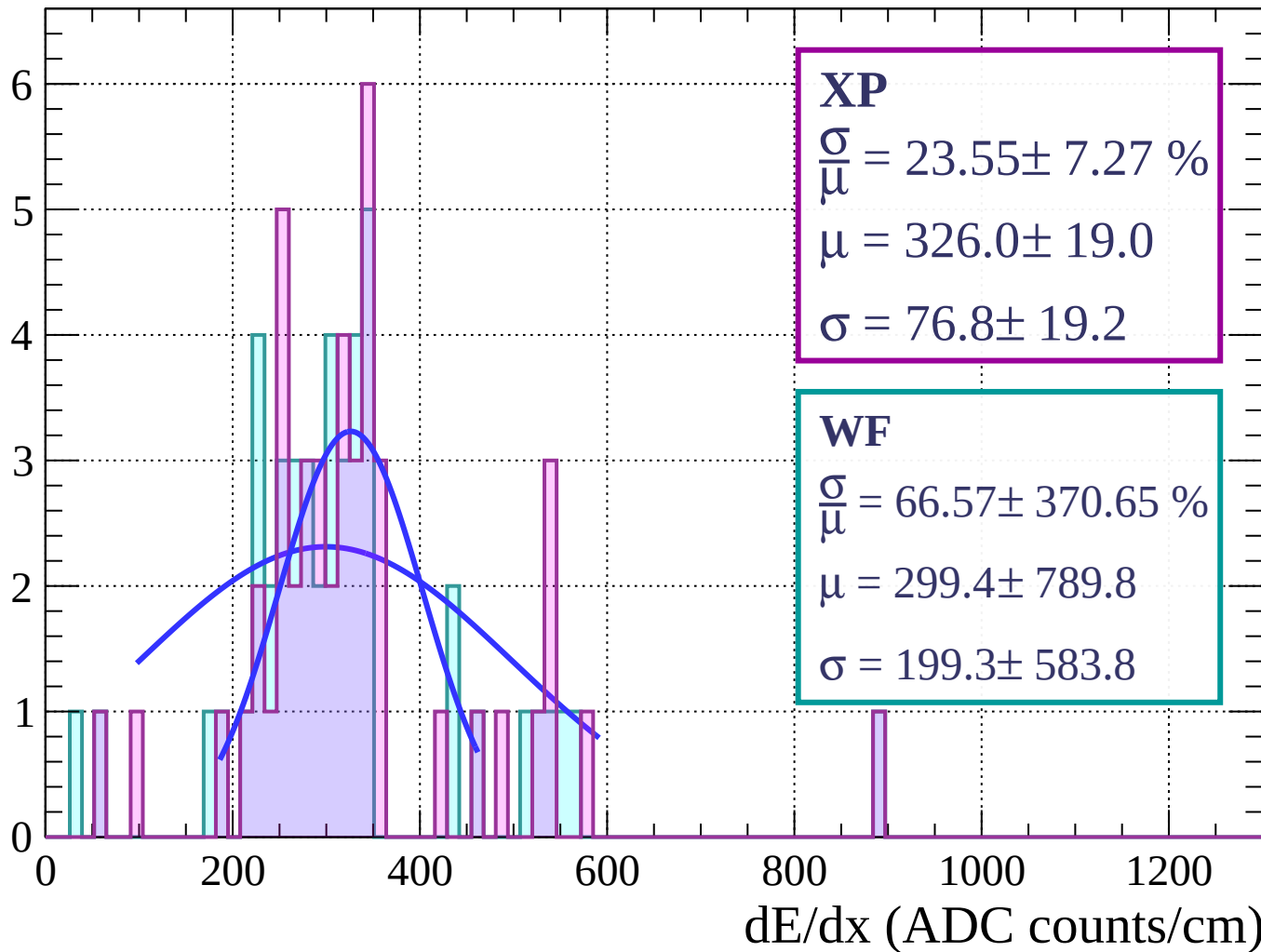
Energy loss | $45 < \theta < 48$

Count



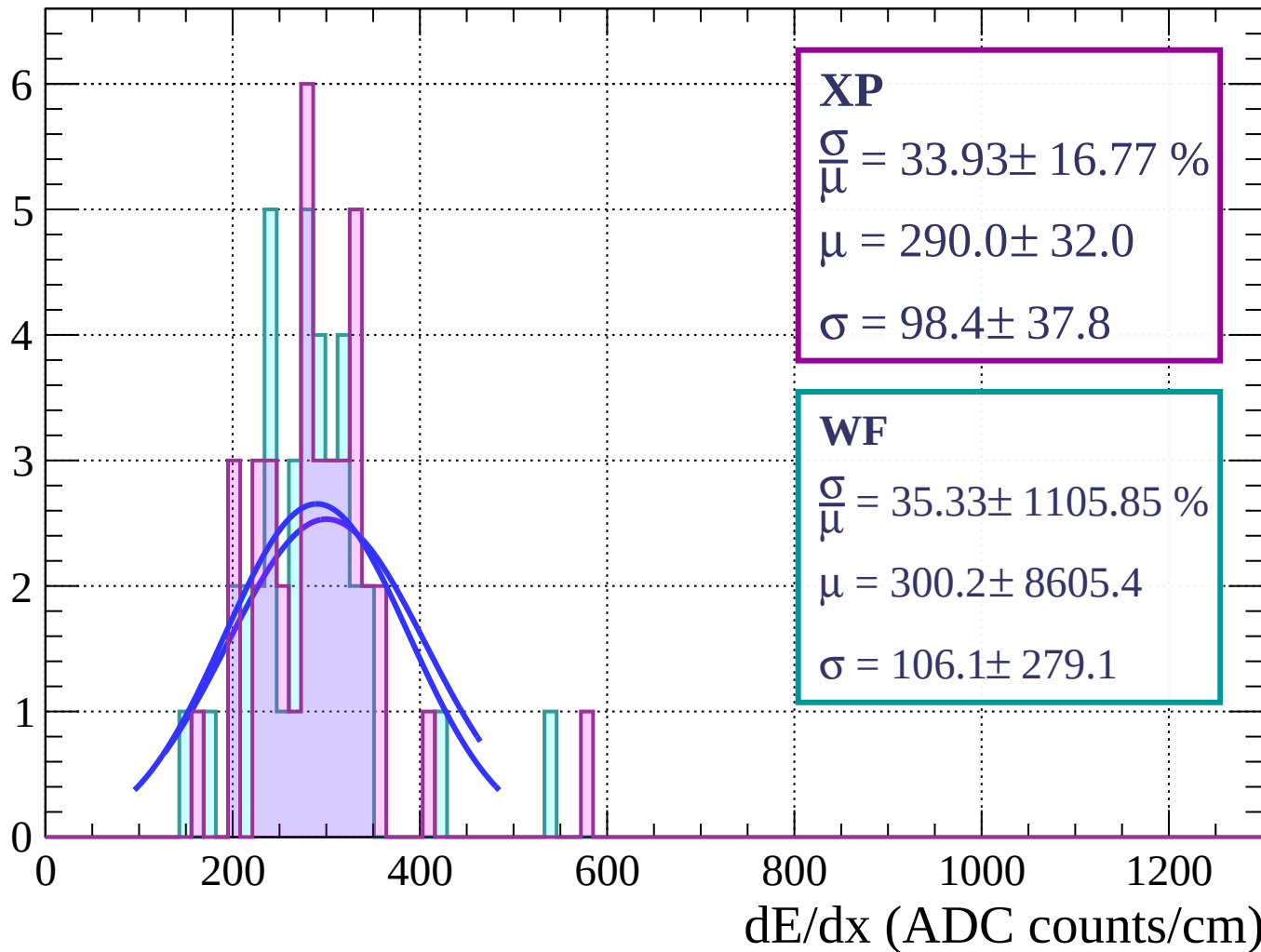
Energy loss | $48 < \theta < 51$

Count



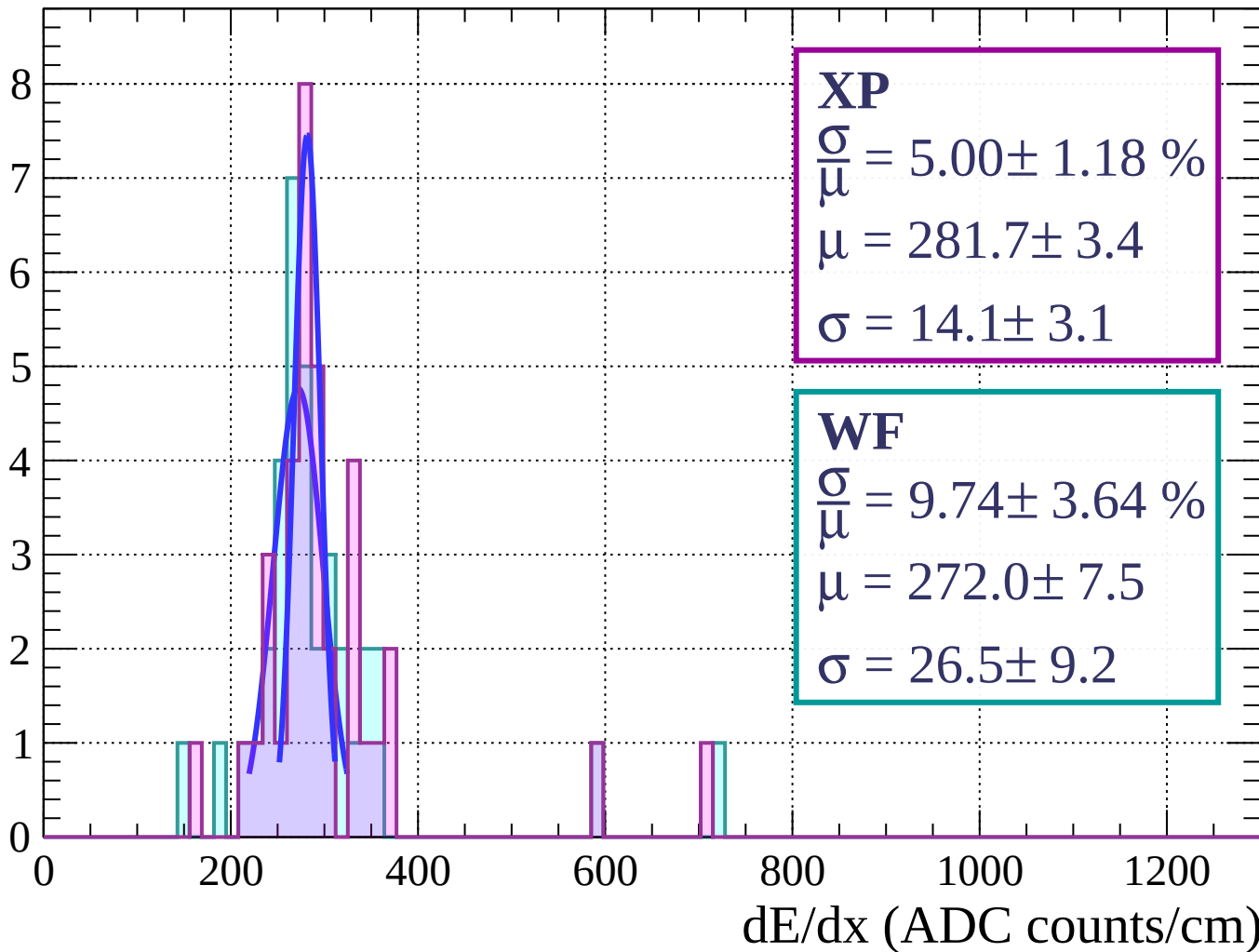
Energy loss | $51 < \theta < 54$

Count



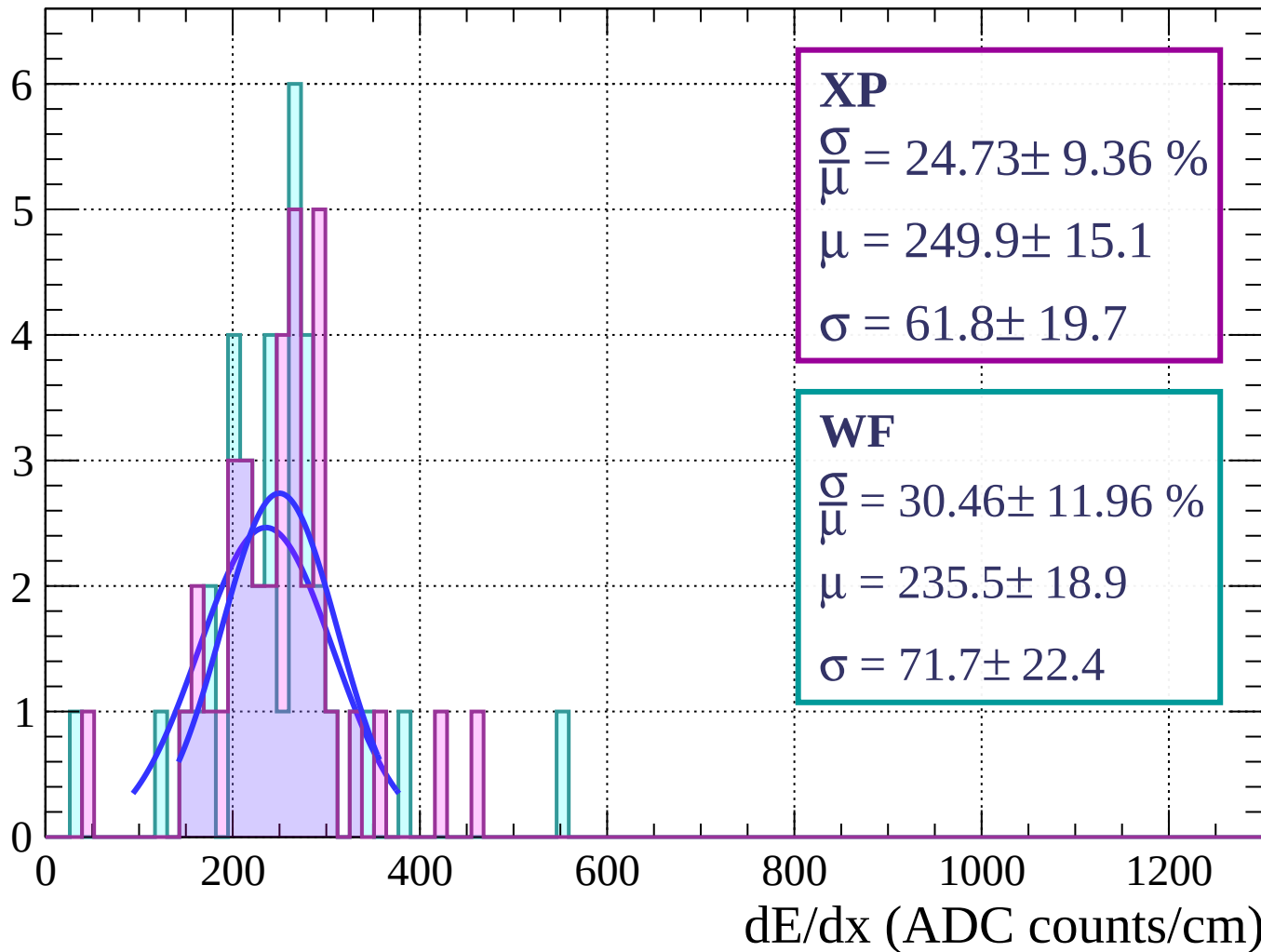
Energy loss | $54 < \theta < 57$

Count



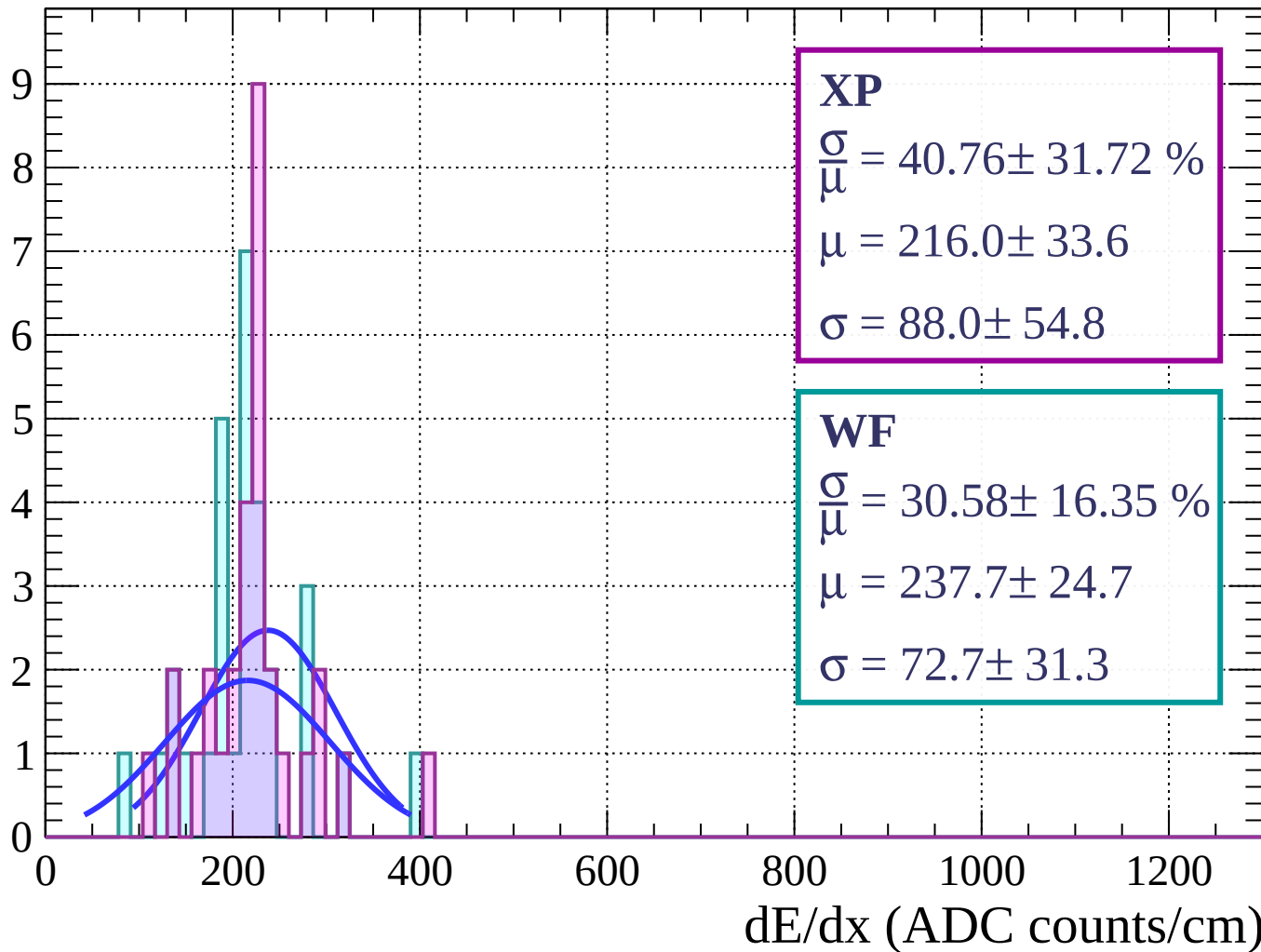
Energy loss | $57 < \theta < 60$

Count



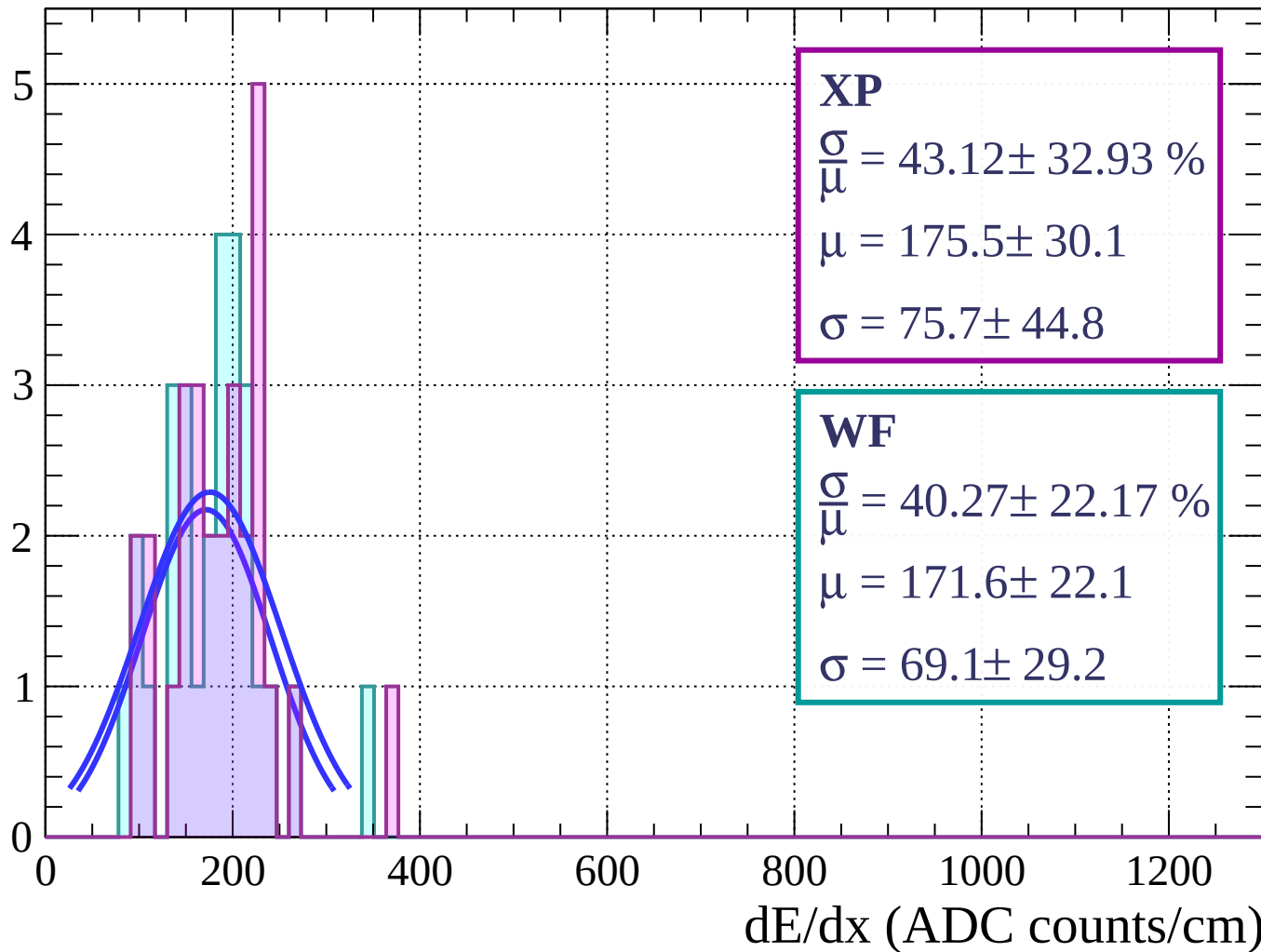
Energy loss | $60 < \theta < 63$

Count



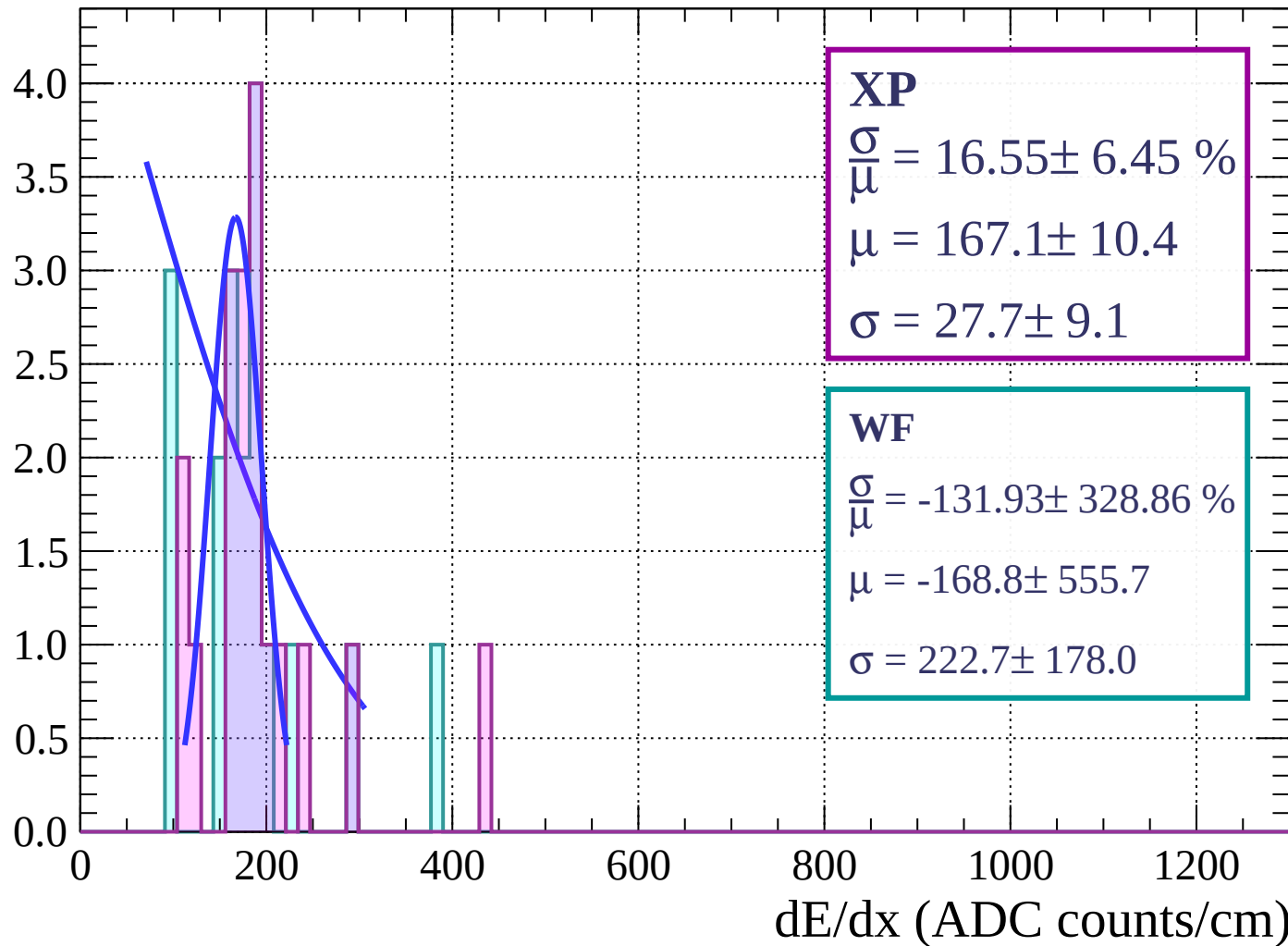
Energy loss | $63 < \theta < 66$

Count



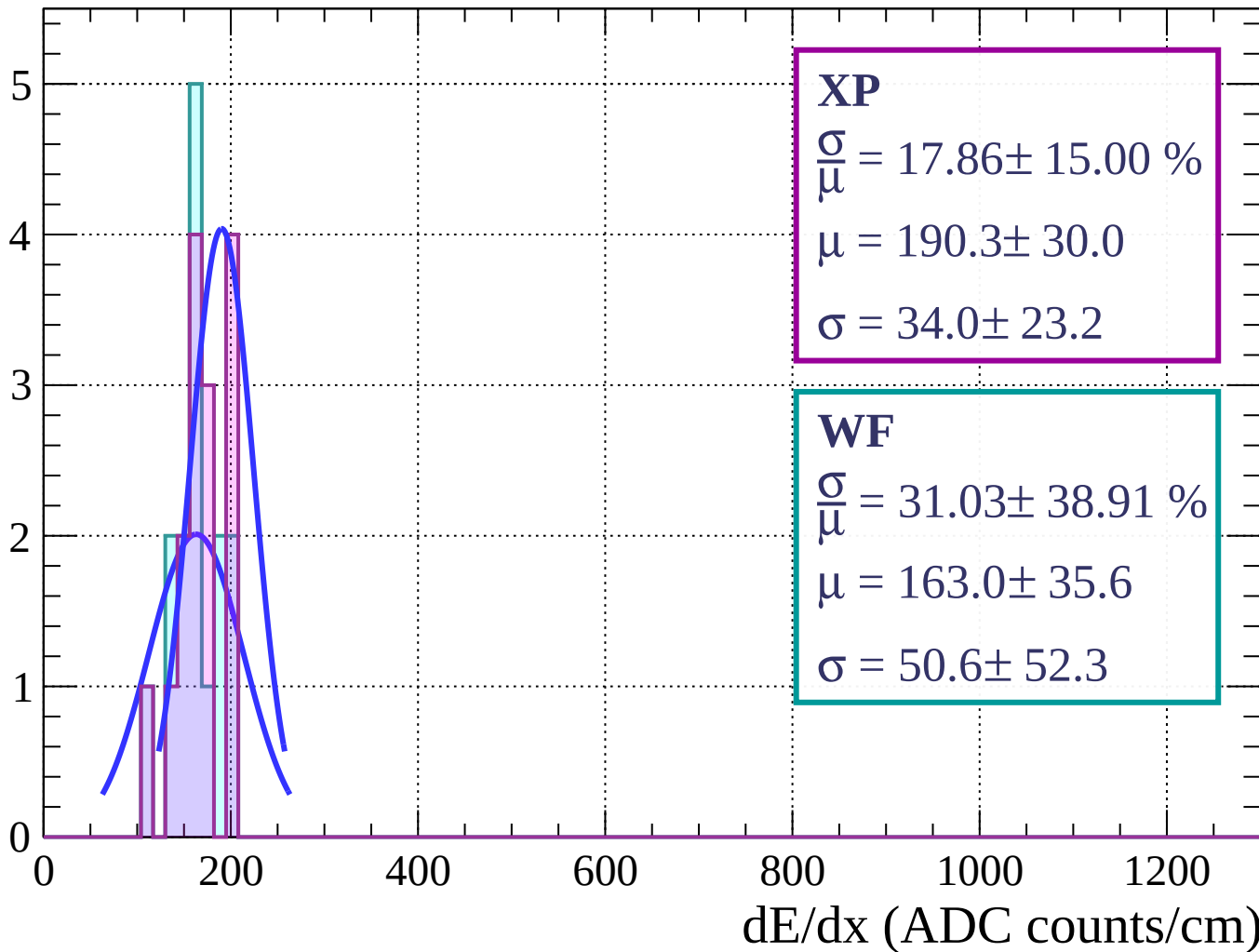
Energy loss | $66 < \theta < 69$

Count



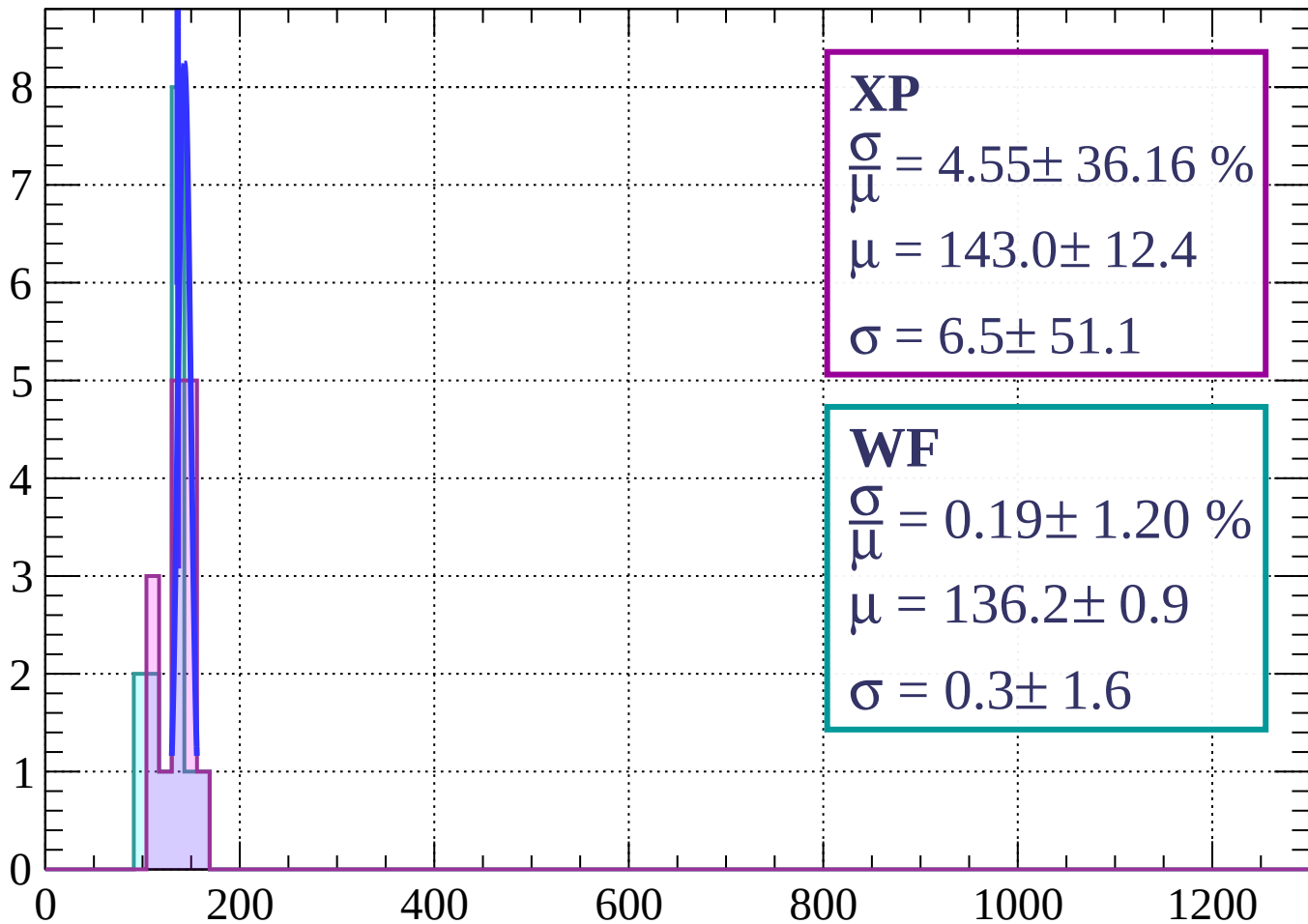
Energy loss | $69 < \theta < 72$

Count



Energy loss | $72 < \theta < 75$

Count



XP

$$\frac{\sigma}{\mu} = 4.55 \pm 36.16 \%$$

$$\mu = 143.0 \pm 12.4$$

$$\sigma = 6.5 \pm 51.1$$

WF

$$\frac{\sigma}{\mu} = 0.19 \pm 1.20 \%$$

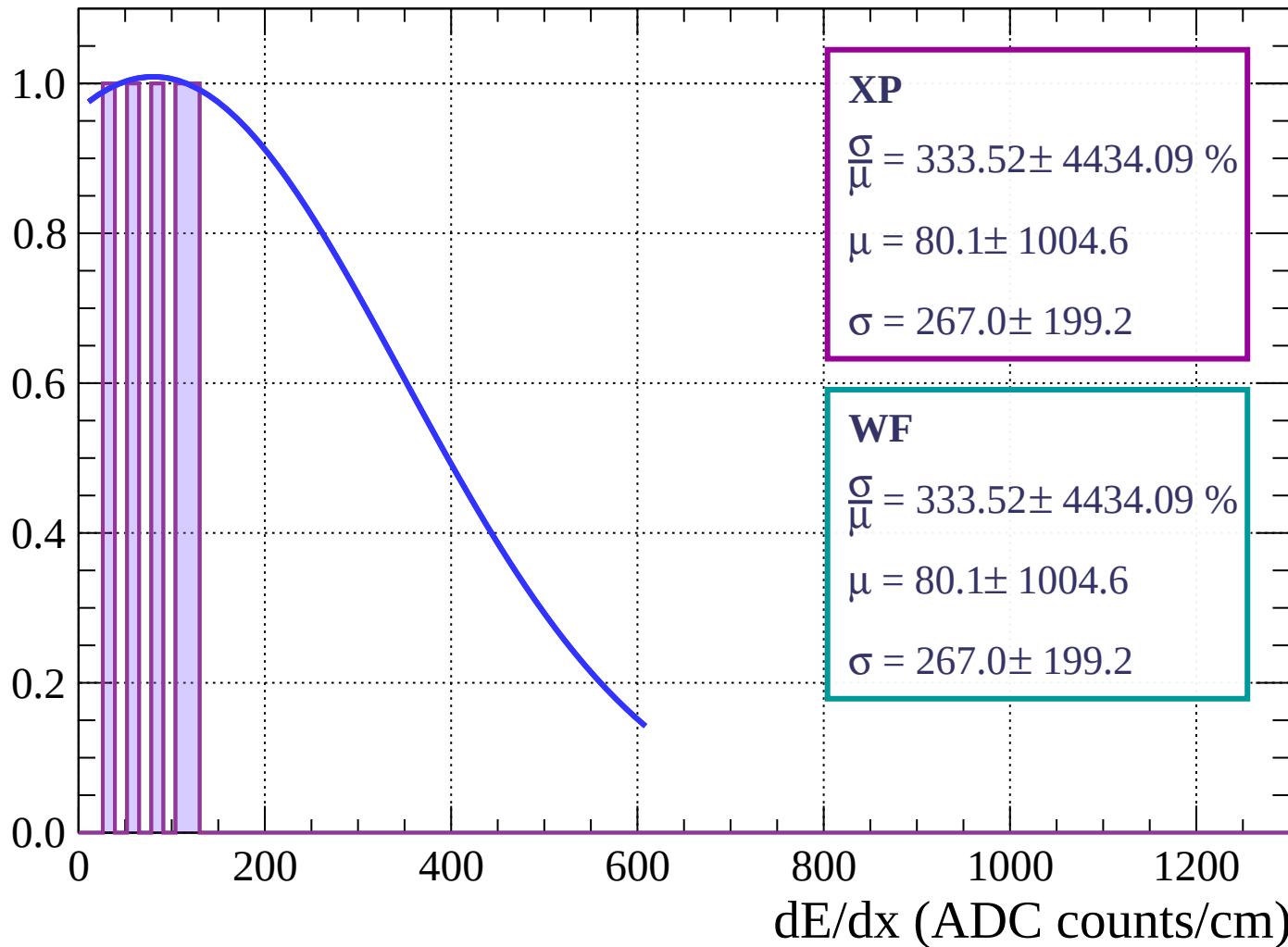
$$\mu = 136.2 \pm 0.9$$

$$\sigma = 0.3 \pm 1.6$$

dE/dx (ADC counts/cm)

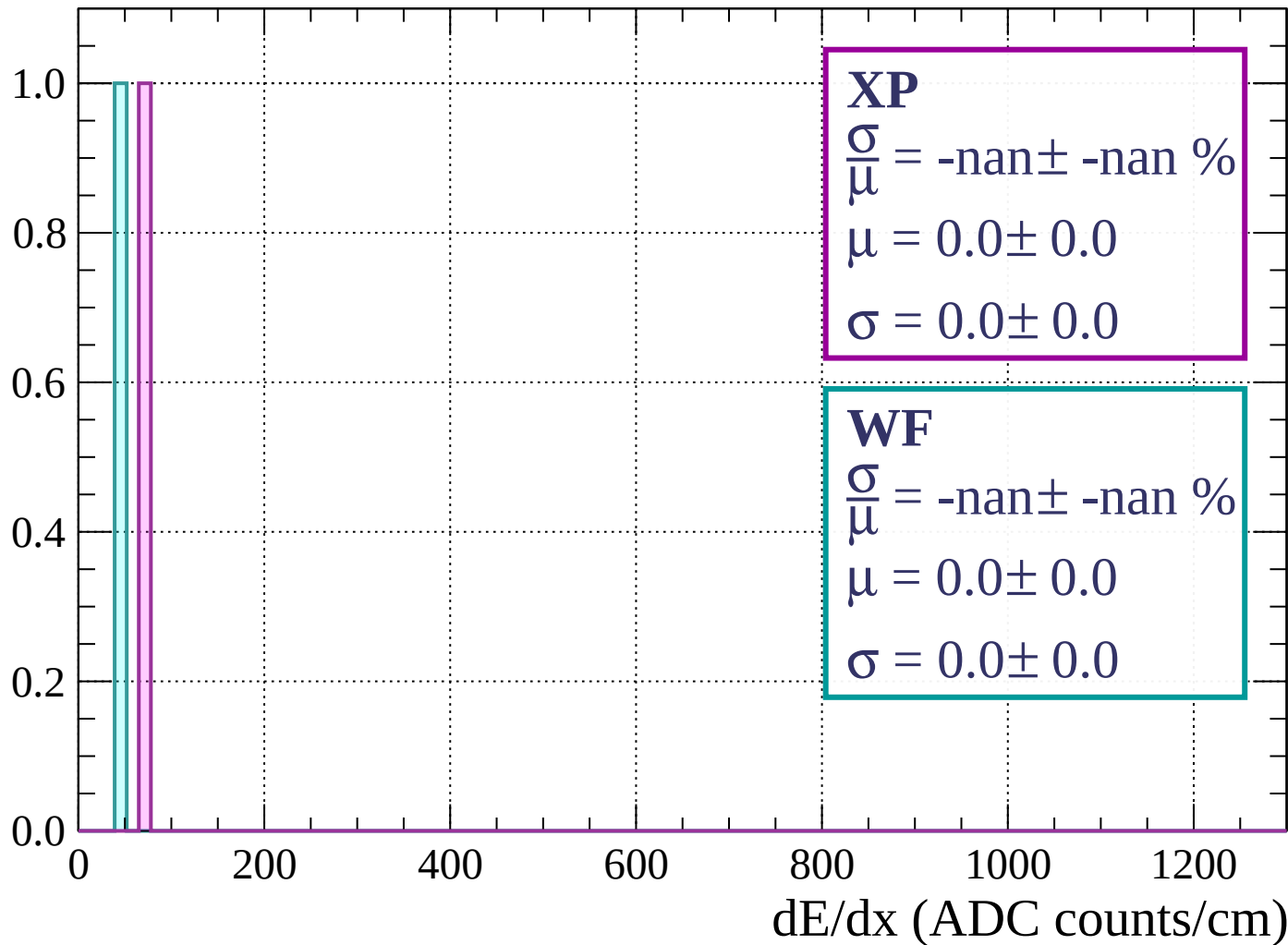
Energy loss | $75 < \theta < 78$

Count



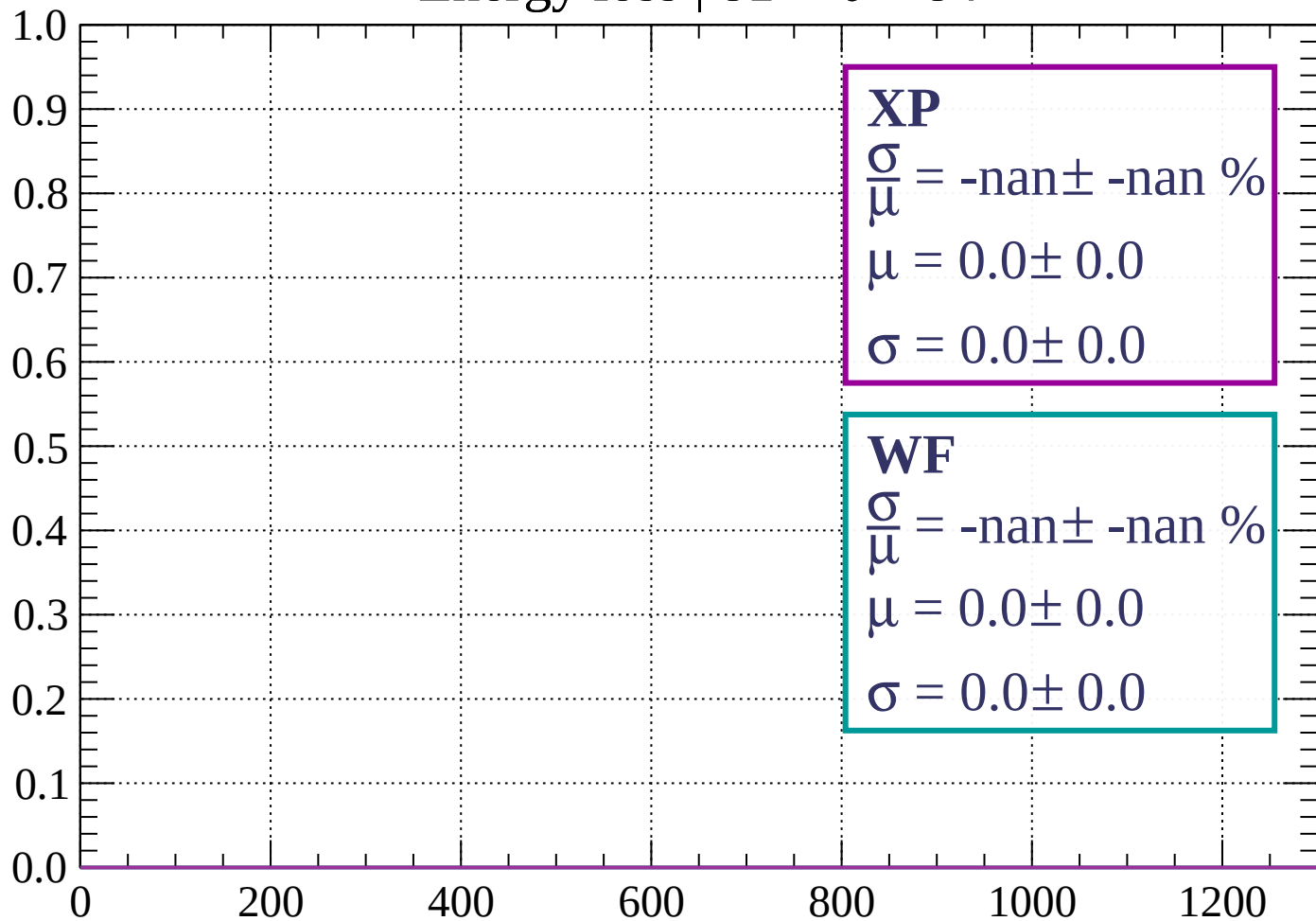
Energy loss | $78 < \theta < 81$

Count



Energy loss | $81 < \theta < 84$

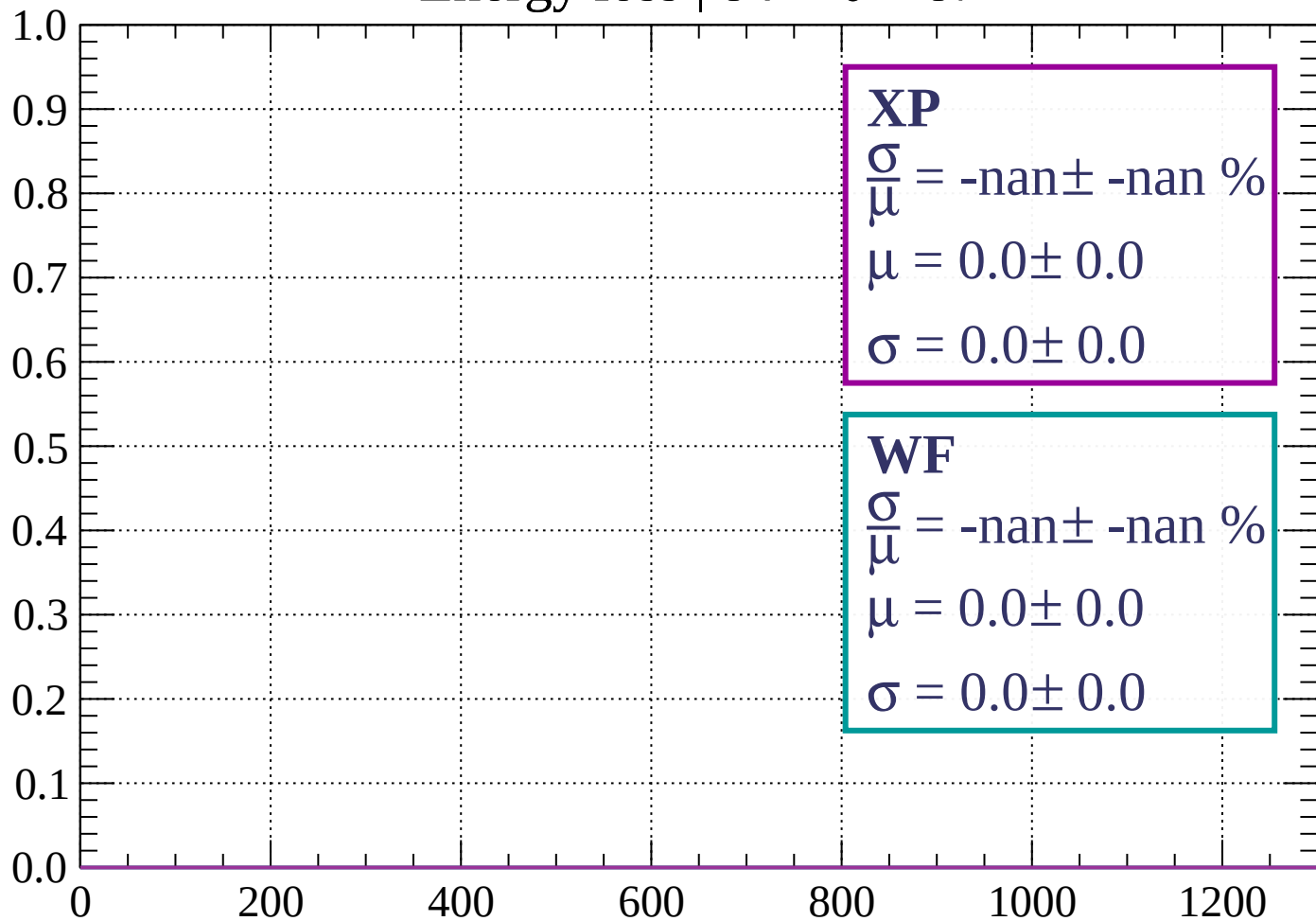
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

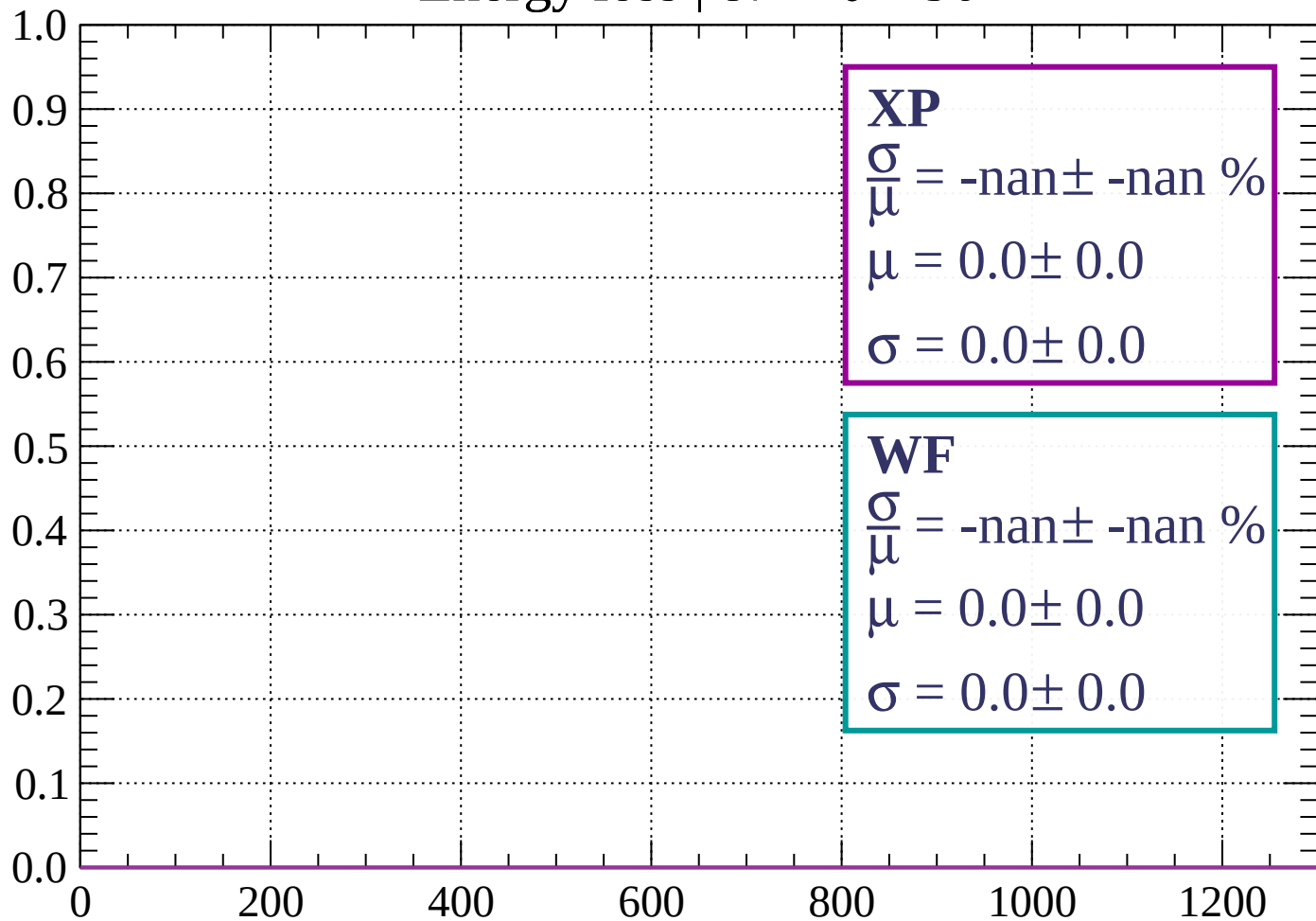
Count



dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

